



NMR Parameters Determination through ACE Committee Machine with Genetic Implanted Fuzzy Logic and Genetic Implanted Neural Network

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A b s t r a c t

Free fluid porosity and rock permeability, undoubtedly the most critical parameters of hydrocarbon reservoir, could be obtained by processing of nuclear magnetic resonance (NMR) log. Despite conventional well logs (CWLs), NMR logging is very expensive and time-consuming. Therefore, idea of synthesizing NMR log from CWLs would be of a great appeal among reservoir engineers. For this purpose, three optimization strategies are followed. Firstly, artificial neural network (ANN) is optimized by virtue of hybrid genetic algorithm-pattern search (GA-PS) technique, then fuzzy logic (FL) is optimized by means of GA-PS, and eventually an alternative condition expectation (ACE) model is constructed using the concept of committee machine to combine outputs of optimized and non-optimized FL and ANN models. Results indicated that optimization of traditional ANN and FL model using GA-PS technique significantly enhances their performances. Furthermore, the ACE committee of aforementioned models produces more accurate and reliable results compared with a singular model performing alone.

Keywords: NMR, rock physics, petrophysics, integrated intelligent systems.

1. INTRODUCTION

Nuclear magnetic resonance (NMR) log contains priceless information about the *in situ* characteristics of hydrocarbon-bearing intervals. Free fluid porosity and permeability are two invaluable parameters of reservoir formations which could be determined by interpreting transverse relaxation time data (T_2 distribution) of NMR log. NMR logging becomes impractical in cased-hole wells and where insufficient deal of money and time exists. However, invaluable information obtained from NMR log has tempted several researchers to devise quick, cheap, and practical methods of gaining NMR log parameters. Conventional well log data have been used by some researchers for synthesizing NMR log parameters through the use of intelligent systems (Mohaghegh 2000a, Malki and Baldwin 2002, Ogilvie *et al.* 2002, Labani *et al.* 2010). These methodologies enable reservoir engineers to extract free fluid porosity and permeability for both cased holes and open holes in a cheap and fast way. Since conventional well logs are cheap and available for almost all wells, many researchers are interested in extracting petrophysical, geomechanical, and geophysical properties from conventional well logs using intelligent systems (Kadkhodaie-Ilkhchi *et al.* 2009, Asoodeh and Bagheripour 2012a, 2013a; Asoodeh 2013, Bagheripour and Asoodeh 2013). A growing tendency among researchers has been emerged in recent years to utilized integrated approaches for enhancing accuracy of final prediction (Sharkey 1996, Chen and Lin 2006, Asoodeh *et al.* 2014a, b, Gholami *et al.* 2013, Asoodeh and Bagheripour 2013b). However, no optimization on individual intelligent systems has been done till now. In this study, artificial neural network (ANN) and fuzzy logic (FL) are optimized stochastically by means of hybrid genetic algorithm-pattern search (GA-PS) technique to estimate NMR log parameters. This strategy significantly enhances efficiencies of ANN and FL models. Owing to further increase in accuracy of final prediction, an alternative condition expectation (ACE) committee with GA-PS optimized FL and ANN models is constructed. This model is capable of satisfyingly improving accuracy of NMR log prediction using the concept of committee machine. Several papers have investigated performance of committee machine compared with individual intelligent systems performing alone (Asoodeh and Bagheripour 2013b, Afshar *et al.* 2014, Asoodeh and Kazemi 2013). The proposed strategy was applied in one of Iranian hydrocarbon fields with carbonate reservoir formations. A satisfying improvement was observed in accuracy of predicted NMR parameters, including free fluid porosity and permeability after three-step optimization approach.

2. METHODS

This study follows a three-step strategy for formulating conventional well logs to NMR log parameters. The first step is devoted to optimization of arti-

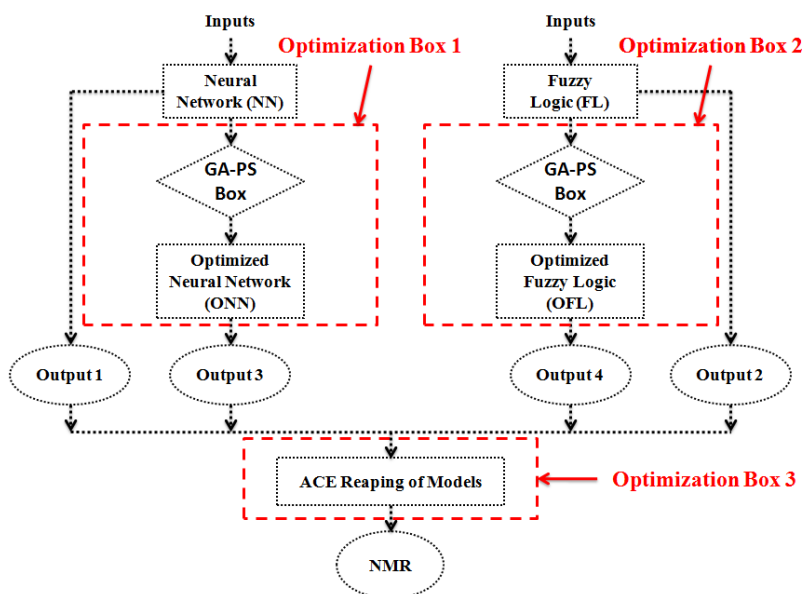


Fig. 1. General flowchart followed in this study.

ficial neural network (ANN) by means of hybrid genetic algorithm-pattern search (GA-PS) technique. Next, fuzzy logic (FL) model is optimized with GA-PS tool. Up to here, four models, including optimized FL, optimized ANN, non-optimized FL, and non-optimized ANN exist. In last stage of this study, an alternative condition expectation (ACE) committee of four models is constructed to combine their results. This step will reap benefits of all models and will enhance accuracy of final prediction. A flowchart of this study is provided in Fig. 1.

2.1 Artificial neural network

Artificial neural networks emulate the biological neural storage and analytical operations of brain through non-linear processing elements (neurons) connected by fixed (Lippmann 1987), variable (Barhen *et al.* 1989), or fuzzy (Gupta and Ding 1994) weights. Each neural network is made up of three layers (parallel arrangement of neurons): an input layer, one or more hidden layers depending on complexity of the problem, and an output layer. Back-propagation is a popular learning method for neural networks. It requires a set of input/output training data to gain knowledge from them and then applying that knowledge to new situations and problems. This method performs in a way that initial weights are randomly assigned to the neural network and its output is then calculated through random weights and biases.

Differences between neural network's outputs and desired outputs are evaluated and the obtained error is then propagated back to the neural network to adjust connection weights and biases. This process is conducted iteratively until network's output locates in close proximity of desired output.

2.2 Fuzzy logic

The basic theory of fuzzy logic was proposed by Zadeh in 1965 as an extension of traditional Boolean logic (Zadeh 1965). A fuzzy set is characterized by a membership function which assigns to each object a grade of membership ranging between zero and one (Zadeh 1965). The main part of fuzzy logic is the fuzzy inference system, which formulates inputs to an output. A fuzzy inference system consists of five major steps: fuzzification of input variables, application of fuzzy operators (AND, OR, and NOT) in the rule's antecedent, implication from the antecedent to the consequent, aggregation of consequent across the rules, and defuzzification. These steps have been discussed in a number of papers (Mohaghegh 2000c, Kadkhodaie-Ilkhchi *et al.* 2006). In the Takagi Sugeno fuzzy inference system (the most popular type of fuzzy inference system), membership functions are defined by subtractive clustering process, which divides the input-output data space into fuzzy clusters, each representing a specific part of the system behavior. Each membership function generates a set of fuzzy "if-then" rule for formulating inputs to output. The fuzzy system makes a sum of all "then" parts and uses a defuzzification method to give the final output.

2.3 Hybrid genetic algorithm-pattern search technique

Genetic algorithm is a stochastic global optimization tool which emulates the biological process of natural evolution for solving problems in widespread fields. It utilizes a specific terminology for optimization purpose, *i.e.*, the function meant to optimize is called fitness function, the probable solutions of the function are called chromosomes, involved parameters in a solution are called genes, and mathematical functions which provide a stochastic generation of new solutions are called genetic operations. Genetic algorithm starts with a population of randomly generated chromosomes and evolves toward better chromosomes by applying genetic operators. This strategy provides a stochastic search capability for genetic algorithm. Evolutionary process by use of genetic operations continues until it finds the best chromosome (or solution) in which global minimum of fitness function has been occurred. More description about genetic algorithm can be found in Mohaghegh (2000b).

In the pattern search technique, the algorithm searches a set of points, called a mesh, around the current point (the point computed at the previous

step of the algorithm). The mesh is formed by adding the current point to a scalar multiple of a set of vectors called a pattern. If the pattern search algorithm finds a point in the mesh that improves the objective function at the current point, the new point becomes the current point at the next step of the algorithm (MATLAB user's guide; Mathworks 2011).

2.4 Alternative condition expectation (ACE)

Alternating conditional expectation (ACE) is a nonparametric method, which was first established by Breiman and Friedman (1985). ACE is robust data driven technique widely utilized for identifying the complicated relationship between independent variables and response variable, especially in the situation when there are no prior assumptions of functional form for expressing relation between input/output data space. This method seeks the optimal transformations of response and a set of predictor variables, thereby the maximal multiple correlation between independent variables and the response variable is achieved. The optimal transformation is determined through minimizing the error variance between the transformed dependent variable and the sum of the transformed predicted variables. The value of error variance is minimized through employing of a series of single-function minimizations. Through iterative minimization of error variance, the optimal transformation of input and output is computed. Indeed, this method involved two operations for solving regression problems including conditional expectation and iterative minimization process. This technique gained increasing attention in both science and engineering fields due to its simplicity and its successful results in uncovering the nonlinearity relationship between dependent and independent variables (Malallah *et al.* 2006, Xue *et al.* 1996, Shokir 2007). For comprehensive study of ACE formulation principles, readers are referred to original paper of Breiman and Friedman (1985). In this study ACE is employed as nonlinear combiner to computing the contribution of different predictive models in overall prediction.

3. WHY OPTIMIZATION IS NEEDED?

3.1 For fuzzy logic

Subtracting clustering is an effective approach for determining the number of rules in Takagi Sugeno fuzzy model (Al-Jarrah and Halawani 2001), which reduces computation compared with mountain clustering (Bataineh *et al.* 2011). However, it has the flaw of being affected by constant clustering radius. It means, by use of subtractive clustering, no modification is done over spread of Gaussian membership functions and all extracted membership functions have the same variance value. Furthermore, subtractive clustering looks for cluster centers among the data points; however, the actual cluster

centers are not necessarily located at one of data points (Bagheripour and Asoodeh 2013). These shortcomings of subtractive clustering entail to propose an alternative for setting up fuzzy rules and clusters. Stochastic optimization capability of hybrid genetic algorithm-pattern search tool provides a powerful search engine which is capable of finding optimal values of cluster centers and spreads, regardless of the above-mentioned limitations of subtractive clustering.

3.2 For artificial neural network

As mentioned in Section 2.1, back-propagation algorithm is the widely-used method of determining connection weights. Performance of neural network is strongly dominated by initial weights and biases which are randomly assigned to it by back-propagation algorithm. It means, the final stage of convergence depends on where the back-propagation algorithm starts from. In other words, several runs of back-propagation algorithm with similar initial condition will converge to the same minimum. By change of the initial condition, final results of neural network might differ. Therefore, to find the optimal initial condition which leads to reaching global minimum, too many boring runs of back-propagation algorithm are needed. Nonetheless, during all these runs, neural network is highly at risk of trapping in local minima. Stochastic search capability of hybrid genetic algorithm-pattern search tool is independent of initial condition of its run. It will converge to global minimum regardless from where it starts (Conn *et al.* 1991). Therefore, the use of hybrid genetic algorithm-pattern search tool instead of back-propagation algorithm in the structure of neural network will improve the accuracy of modeling and eliminate the probability of sticking in local minima.

4. INPUTS-OUTPUTS DEPENDENCY

The first criterion in modeling by intelligent system is the quality of data. Therefore, low quality data should be removed before modeling. For this purpose, a quality control was done and bad hole intervals were removed by processing of well log data. Existence of logical relationship between inputs and outputs is another important criterion which should be kept in mind while modeling. Introducing some input parameters which have no dependency with outputs, makes the intelligent system confused and weakens the modeling. Outputs of this study are free fluid porosity and permeability. Free fluid porosity includes the pores containing fluids which are expected to flow. Permeability is defined as the capacity of a rock or sediment for fluid transmission, and is a measure of the relative ease of fluid flow under pressure gradients. Therefore, conventional well logs that have an underlying dependency with these two parameters are desired. Inputs for prediction of

free fluid porosity and permeability are the same. Because, behavioral physics of free fluid porosity and permeability are completely intertwined. Free fluid porosity increases as permeability increases and *vice versa*. Available well logs for this study were including DT, RHOB, NPHI, RT, PEF, GR, and Caliper. Caliper log was firstly used to identify bad hole intervals where standoff is higher than 1.5 inches in 8.5" hole. Bad hole intervals were removed because they contain erroneous recordings. Photoelectric factor (PEF) is a litho-log which determines composition of reservoir rocks. NMR log is independent of lithology. Therefore, PEF was not used as input. Gamma ray (GR) is an indicator of shale content. However, in unconventional carbonate reservoirs, GR is contaminated by diagenesis factors and is misleading. In this study, four conventional well logs, including sonic transit time (DT), bulk density (RHOB), neutron porosity (NPHI), and true formation resistivity (RT) were chosen as inputs. These are known appropriate in-

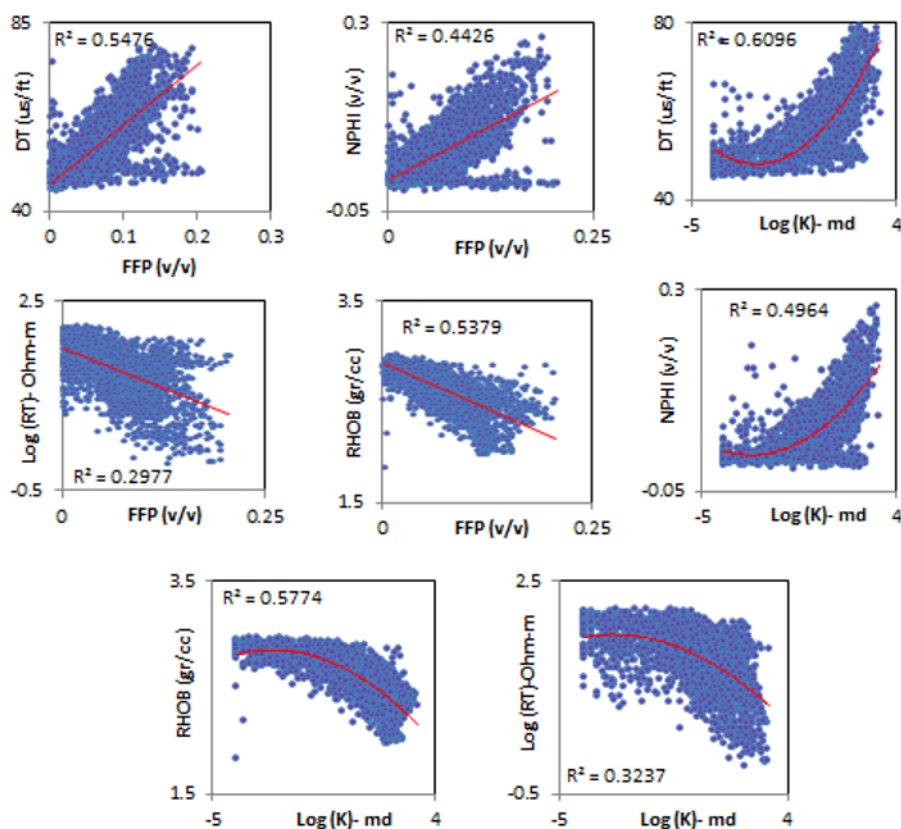


Fig. 2. Crossplots showing relationship between conventional well log data and NMR log parameters, including free fluid porosity and permeability.

put logs having logical relationship with outputs, because the first three logs are porosity logs and the last one is attributed to fluid flow pass (tortuosity) (Asoodeh and Bagheripour 2012a). Figure 2 shows the logical dependency of chosen conventional well log data and nuclear magnetic resonance log parameters using the concept of correlation coefficient. Appendix provides conversion factors from field units to SI units for permeability and sonic transit time. Conventional well logs and NMR logs which have been used for both training and testing of models are originated from Asmari formation. Asmari formation is composed of fossiliferous limestone, dolomitic limestone, argillous limestone, sandstone, and shale and constitutes the major reservoir rock for several carbonate oil fields of Iran.

5. MODELING, RESULTS, AND DISCUSSION

5.1 Optimization of artificial neural network

At the first stage, a three-layered structure of ANN was used for estimation of free fluid porosity and permeability from conventional well logs. This structure utilizes TANSIG and PURELIN transfer functions for its hidden and output neurons, correspondingly. TANSIG is mathematically equivalent to tangent hyperbolic function and PURELIN is a simple linear function of $f(x) = x$ (Asoodeh and Bagheripour 2012b). “Trainlm” training function was used for ANN to learn from training data (for more details about Trainlm refer to Hagan and Menhaj (1994)). To achieve optimal number of hidden neurons, different ANNs with different number of hidden neurons were constructed and performance of each was evaluated using mean square error function as shown in Fig. 3. Results showed that if seven and three nodes were inserted in the hidden layer of neural network for permeability and free fluid porosity, respectively, the highest performances are achieved. The constructed model was then evaluated by means of blind testing. A set of unseen well log data was used to assess performance of constructed ANN. Results showed that the correlation coefficient between predicted values using neural networks and NMR parameters was equal to 0.84 and 0.857 for permeability and free fluid porosity, respectively. Mean square error of prediction was equal to 0.73 [md²] and 0.000501 for permeability and free fluid porosity, correspondingly. In the next stage, the neural network is optimized through the search capability of hybrid genetic algorithm-pattern search tool. This procedure is discussed in the following paragraphs.

Stochastic search capability of hybrid genetic algorithm-pattern search tool enables it to escape from local minima and converge to global minimum. This feature of hybrid genetic algorithm-pattern search tool could be employed to eliminate the flaw of neural networks which use back-propagation algorithm for finding their weights. To extract the optimum val-

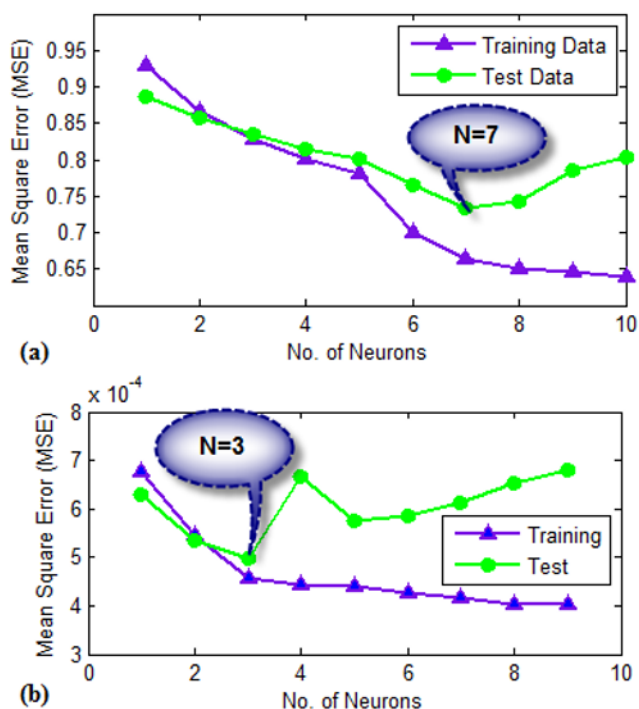


Fig. 3. Graph showing the mean square error (MSE) performance of back-propagation neural network in test data *versus* different number of nodes of hidden layer for: (a) permeability, and (b) free fluid porosity. Optimal models are achieved when 3 and 7 nodes are inserted in hidden layer of neural network for free fluid porosity and permeability, respectively.

ues of ANN's weights, the following mean square error function was introduced into hybrid genetic algorithm-pattern search tool:

$$MSE_{ONN} = \frac{1}{K} \sum_{\ell=1}^k (\text{Output}_{NMR\ell} - \text{Output}_{ONN\ell})^2, \quad (1)$$

where O_{estNN} is ANN output. Output_{ONN} is estimated values of nuclear magnetic resonance (NMR) log parameters from optimized neural network, Output_{NMR} is the NMR log parameter, MSE_{ONN} is the mean square error function of optimized neural network, and k is the number of training data. The process of producing an output from neural network is discussed by Asoodeh and Bagheripour (2013c). Hybrid genetic algorithm-pattern search technique looks for optimum values of connection weights of ANN such that Eq. 1 converges to its global minimum. Blue lines of Fig. 4 indicate mean value of mean square error (MSE) for a population of chromosomes (solutions) dur-

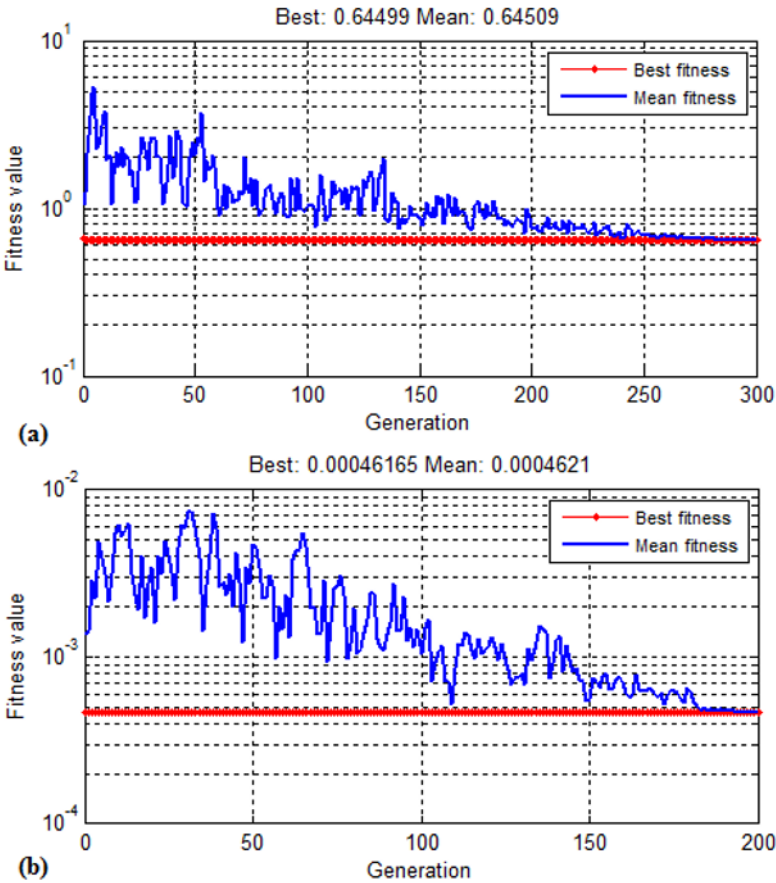


Fig. 4. Run of hybrid genetic algorithm-pattern search tool for stochastic optimization of neural network. This plot shows the best and mean fitness values of fitness functions after 300 generations for: (a) permeability and (b) free fluid porosity.

ing 300 generations for permeability model and 200 generations for free fluid porosity (FFP) models. The red lines in Fig. 4 show mean square error corresponding to best chromosomes. After stochastic optimization of neural network, performances of optimized models were evaluated by test data, *i.e.*, test data were introduced to optimized model and quality of estimation was assessed. Figure 5 indicates error distributions for optimized and non-optimized neural network models. It is obvious that stochastic optimization of neural networks led mean of error distributions to approach zero and standard deviation of error distributions to decrease. Furthermore, correlation coefficient of prediction has improved after stochastic optimization of neural networks.

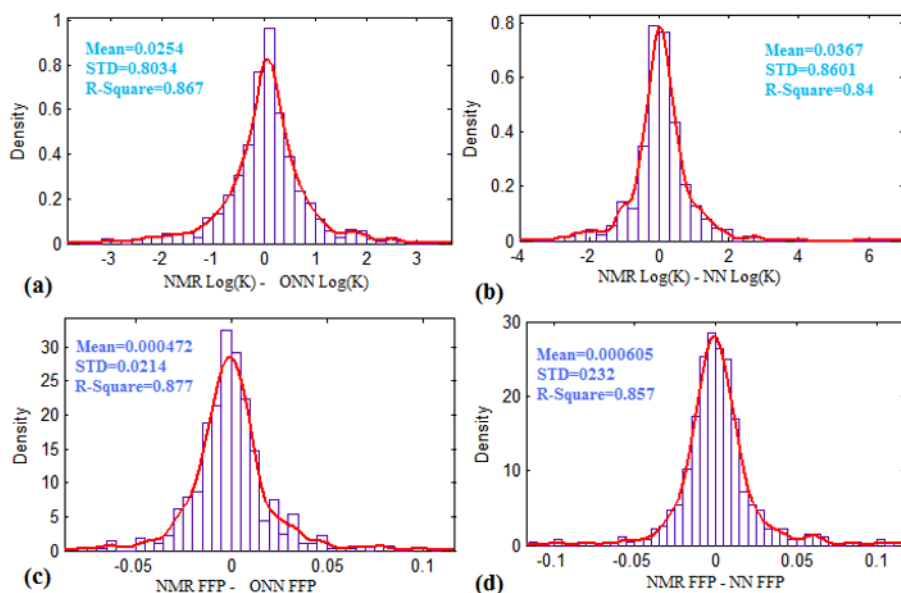


Fig. 5. Error distribution of: (a) optimized neural network for permeability, (b) non-optimized neural network for permeability, (c) optimized neural network for free fluid porosity, and (d) non-optimized neural network for free fluid porosity. In optimized models, the mean of error distributions approach zero and their standard deviation decrease. It means, optimized models performed better than non-optimized models.

5.2 Optimization of fuzzy logic

The next stage of this study was initiated with construction of fuzzy model for estimation of nuclear magnetic resonance (NMR) log parameters, including free fluid porosity and permeability from conventional well log data. To find optimum fuzzy model which best describes quantitative formulation between inputs and outputs, several fuzzy inference systems with various values of clustering radius were constructed. Fuzzy inference systems which have the lowest mean square error for the test data were chosen as optimal ones. Figure 6 indicates variations of mean square error for test and training data in conjugation with variation of number of rules for different clustering radii. This figure indicates fuzzy inference systems with clustering radii of 0.6 and 0.9 for permeability and free fluid porosity, respectively, can extract the best quantitative relationship between inputs and outputs. Furthermore, this figure shows that four and two linguistic rules can in turn handle the relationship between inputs and outputs for permeability and free fluid porosity models. Figure 7 shows the extracted Gaussian membership functions

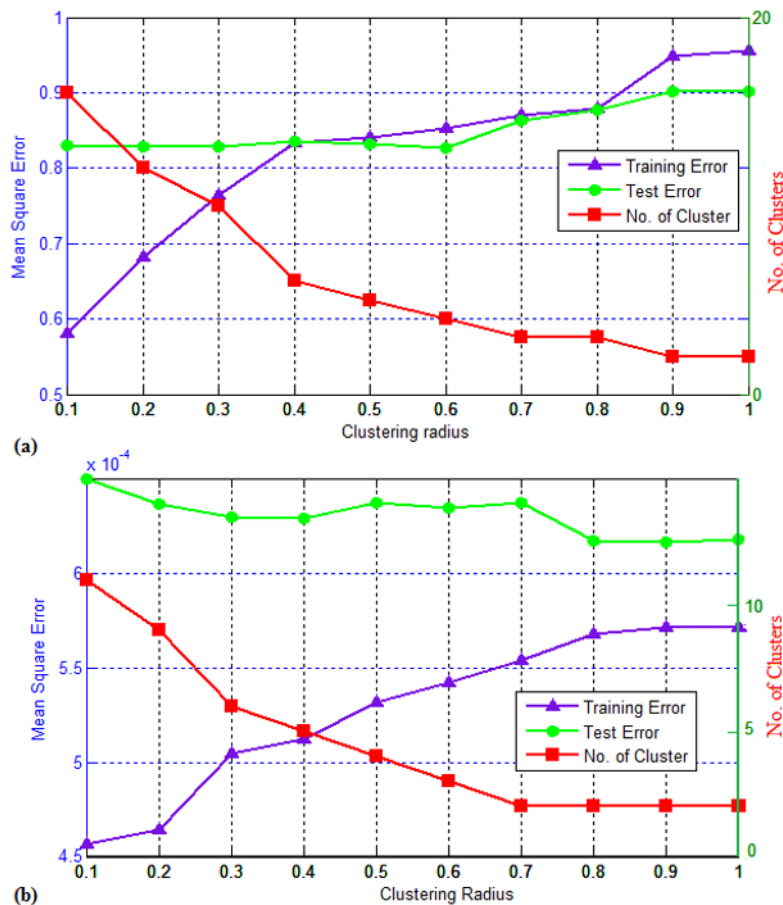


Fig. 6. Graphs showing the mean square error (MSE) performance of Takagi-Sugeno fuzzy inference system (TS-FIS) in test and training data in conjugation with number of rules *versus* clustering radius for: (a) permeability, and (b) free fluid porosity. Optimal models are achieved when 0.6 and 0.9 are chosen for clustering radius of permeability and free fluid porosity models, respectively.

corresponding to aforementioned linguistic rules. As this figure shows, the subtractive clustering implies no optimization over variances (spreads) of fuzzy clusters corresponding to each input. This flaw is eliminated in next step. Evaluating with test data, the results showed that correlation coefficient between fuzzy logic predicted free fluid porosity and NMR values was equal to 0.852. This value for permeability was equal to 0.845. Mean square errors of fuzzy logic predictions are equal to 0.000521 and 0.7201 [md²] for free fluid porosity and permeability, correspondingly.

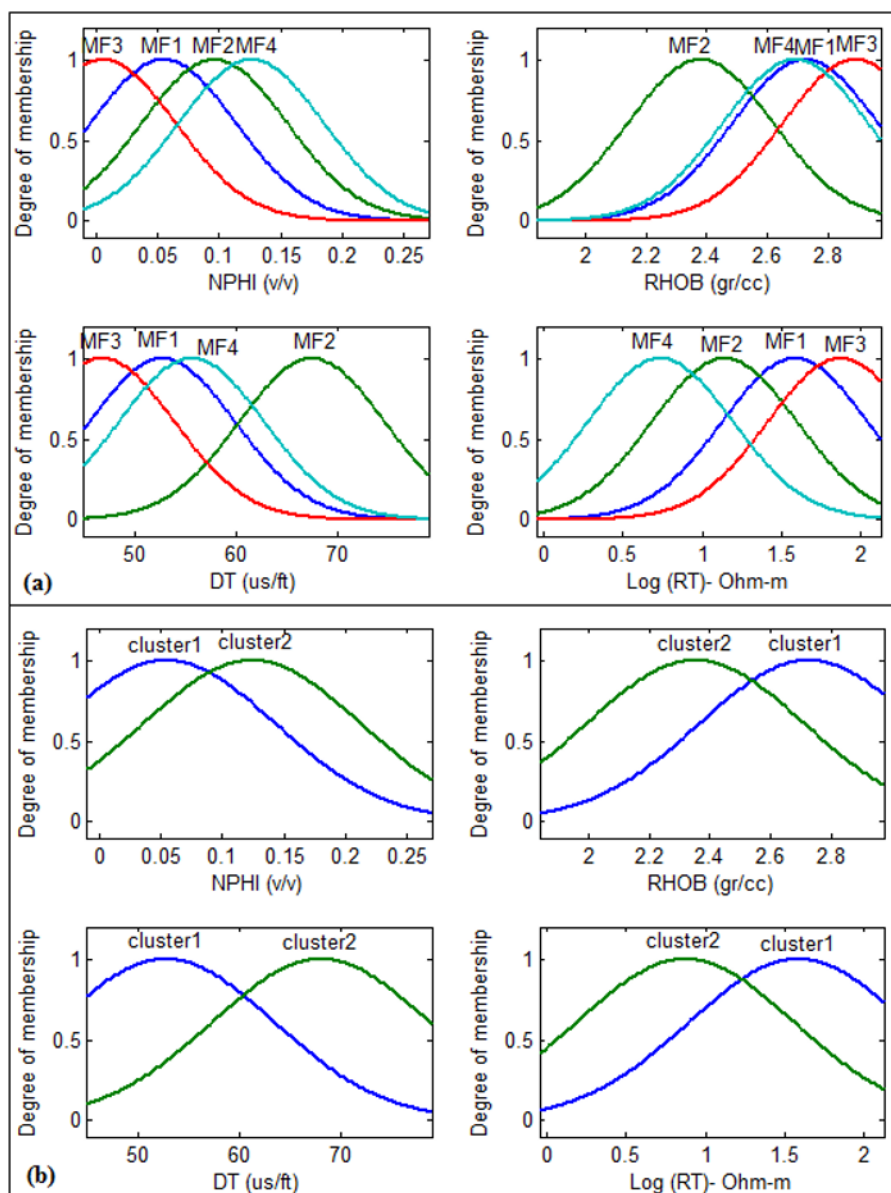


Fig. 7. TS-FIS generated Gaussian membership functions (fuzzy clusters) for the inputs of model meant to predict: (a) permeability, and (b) free fluid porosity.

Since conventional well log data, which constitute input data space, cover various ranges of data, all inputs were normalized to improve the performance of search capability of hybrid genetic algorithm-pattern search tool

for optimizing fuzzy logic model. Since this type of normalization does not destroy the consistency of input data, shape of membership functions will just be scaled and normalization would not have any effect in the overall shape of membership functions. Similar to ANN, a mean square error (MSE) function was used as fitness function of hybrid genetic algorithm-pattern search to find optimal membership functions such that MSE function reaches its global minimum. Stochastic search capability of hybrid genetic algorithm-pattern search tool makes it possible to optimize both mean and variance (spread) of input Gaussian membership functions along with optimization of output linear membership functions. Figure 8 shows results of hybrid genetic algorithm-pattern search runs for permeability and free fluid porosity. Extracted fuzzy Gaussian membership functions are illustrated in Fig. 9. A comparison between Figs. 7 and 9 shows that the proposed

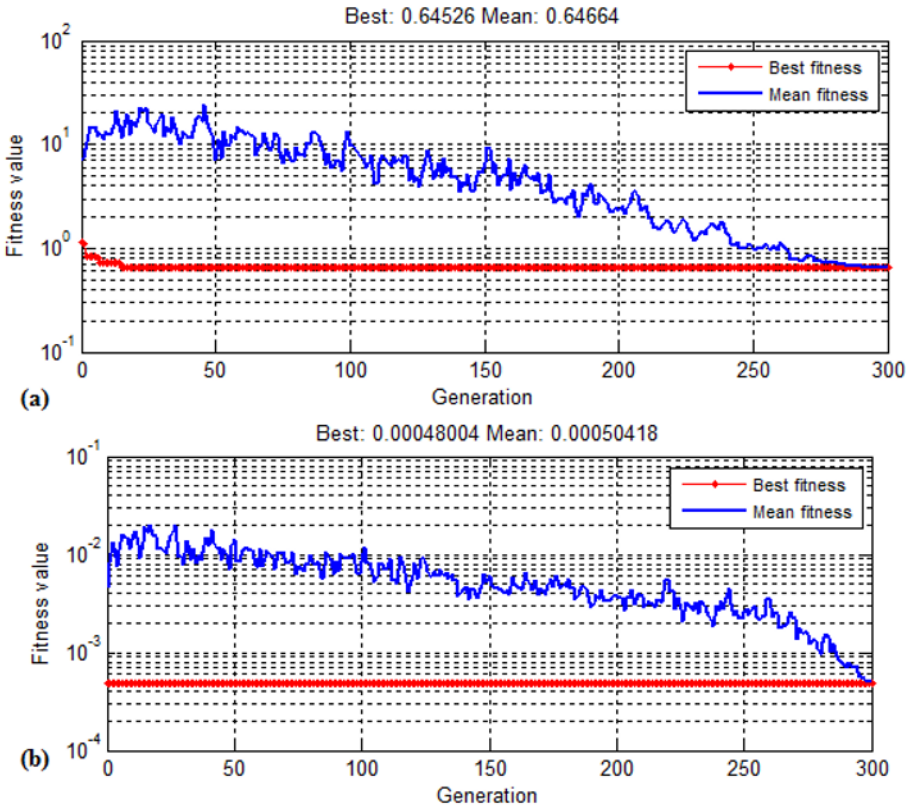


Fig. 8. Run of hybrid genetic algorithm-pattern search tool for stochastic optimization of fuzzy logic. This plot shows the best and mean fitness values of fitness functions after 300 generations for: (a) permeability, and (b) free fluid porosity.

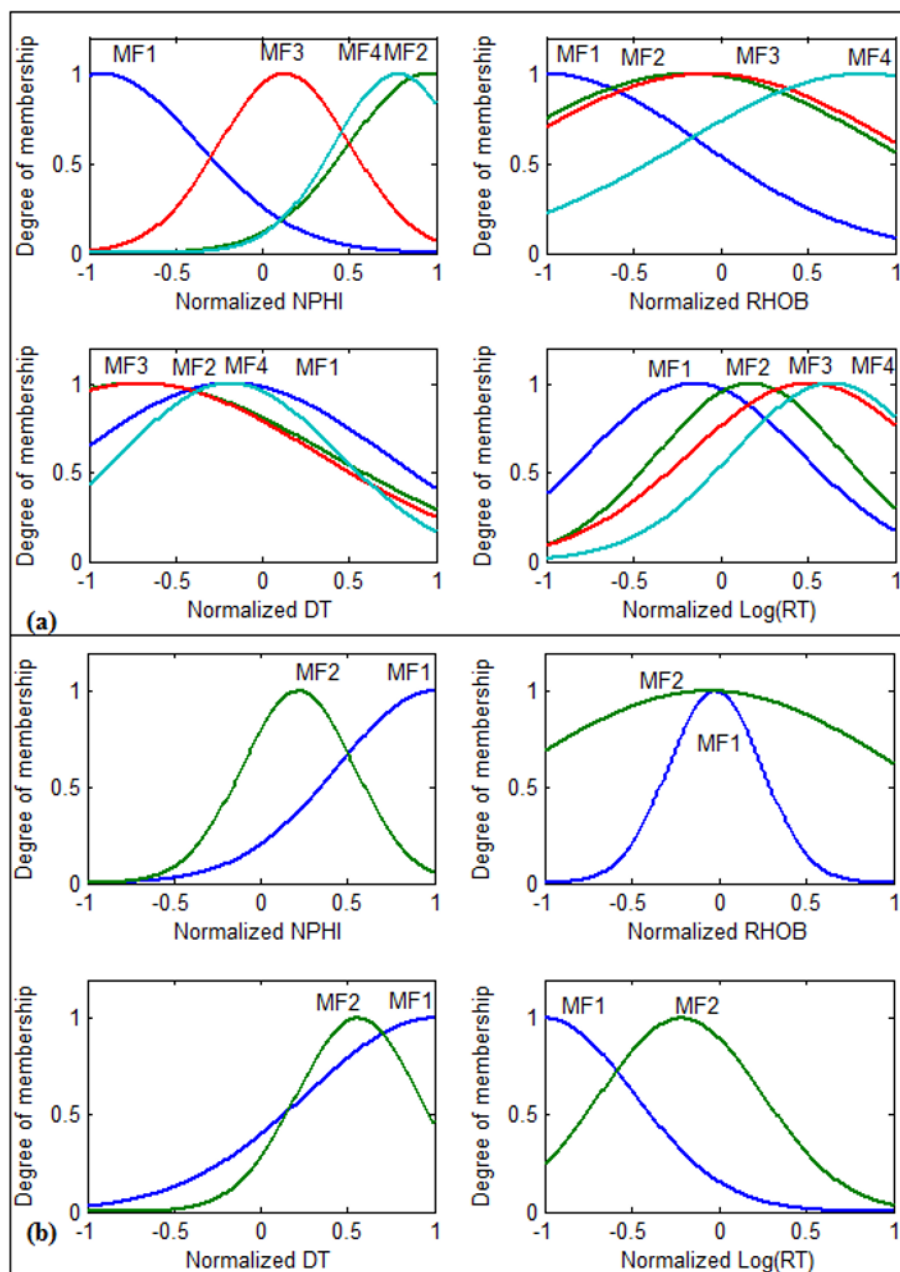


Fig. 9. Stochastically generated Gaussian membership functions (fuzzy clusters) for the inputs of model meant to predict: (a) permeability, and (b) free fluid porosity.

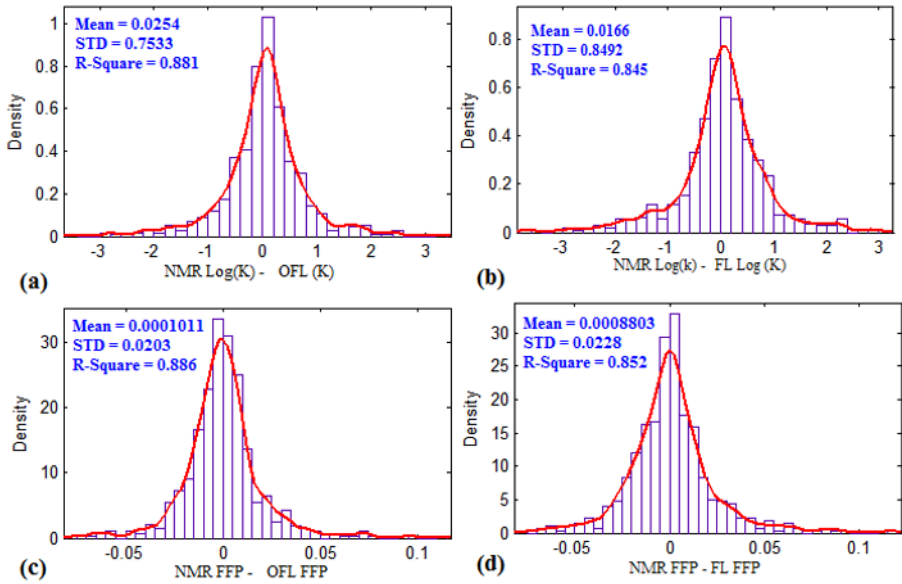


Fig. 10. Error distribution of: (a) optimized fuzzy model for permeability, (b) non-optimized fuzzy model for permeability, (c) optimized fuzzy model for free fluid porosity, and (d) non-optimized fuzzy model for free fluid porosity. In optimized models, mean of error distributions approach to zero and their standard deviation decrease. It means, optimized models performed better than non-optimized models.

model was capable of optimizing variances (spreads) of Gaussian membership function in addition to optimizing mean of Gaussian membership functions. Figure 10 illustrates error distributions of optimized and non-optimized fuzzy logic models. It is obvious that stochastic optimization of fuzzy logic models has reduced the absolute mean and standard deviation of error distributions. Furthermore, correlation coefficient of prediction improved after stochastic optimization of fuzzy logic models.

5.3 ACE committee of models

As mentioned previously, ACE is employed as a combiner to integrate outputs of different models and consequently reaping their benefits through constructing a more accurate model for estimating free fluid porosity and permeability. For this purpose, ACE transfers input parameters and target values into a desired space where maximum correlation exists between them. Thus, outputs of predictive models, including fuzzy logic, optimized fuzzy logic, neural network, and optimized neural network are employed as inputs of ACE and free fluid porosity and permeability are considered as ACE out-

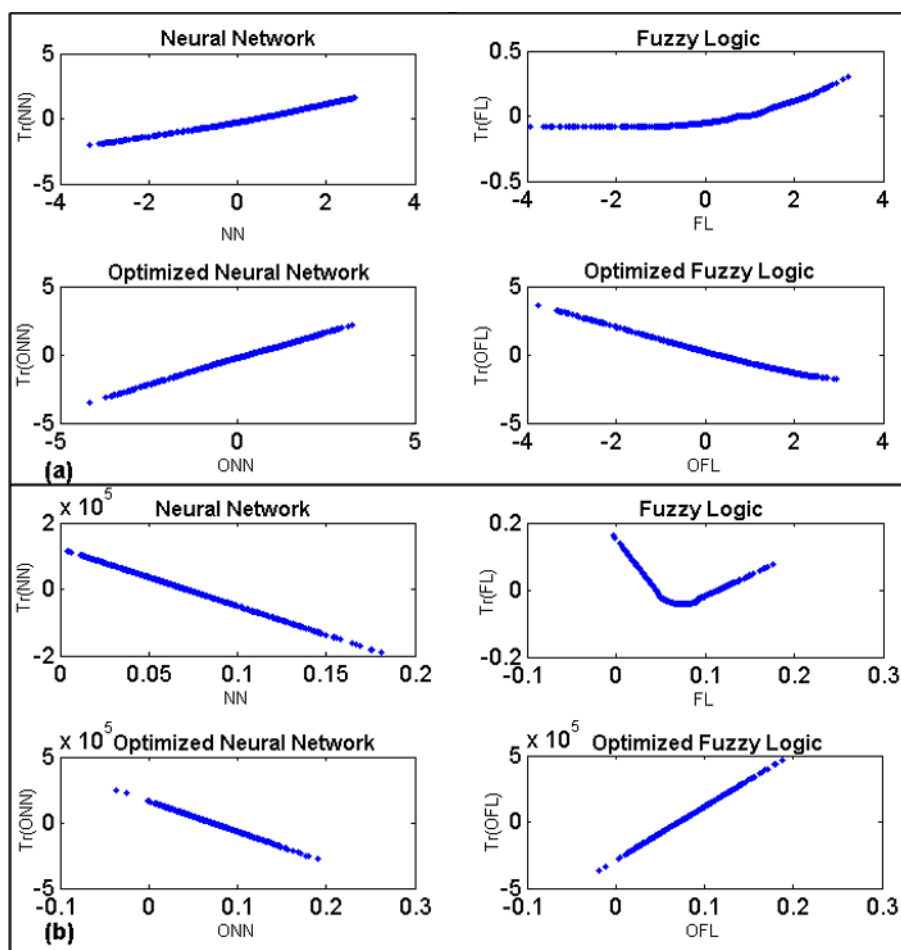


Fig. 11. Optimal transformation of input parameters which determined by ACE for: (a) permeability, and (b) free fluid porosity.

puts. Figure 11 demonstrates ACE extracted optimal transformation of input parameters for free fluid porosity and permeability, respectively. Although ACE algorithm approximates optimal transformations of input/output data space, it does not present a computational model for relating parameters and their optimal transformations. Hence, a simple polynomial function is fitted for making formulation between the parameters and their optimal transformations. The resulting polynomial functions with associated parameters for transforming neural network, optimized neural network, fuzzy logic, and optimized fuzzy logic are shown in Eqs. 2-9.

(a) Transformations for permeability modeling

$$\varphi^*(NN) = a_2(NN)^2 + a_1(NN) + a_0, \quad (2)$$

where $a_2 = 0.034882108437520$, $a_1 = 0.0626864203614067$, and $a_0 = -0.28077401469906$.

$$\varphi^*(FL) = a_3(FL)^3 + a_2(FL)^2 + a_1(FL) + a_0, \quad (3)$$

where $a_3 = 0.001772160478232$, $a_2 = 0.016384002242114$, $a_1 = 0.042023493356332$, and $a_0 = -0.050567771173985$.

$$\varphi^*(ONN) = a_1(ONN) + a_0, \quad (4)$$

where $a_1 = 0.756690942653011$, and $a_0 = -0.263928319591898$.

$$\varphi^*(OFL) = a_3(OFL)^3 + a_2(OFL)^2 + a_1(OFL) + a_0, \quad (5)$$

where $a_3 = 0.007540599087205$, $a_2 = 0.034647276929447$, $a_1 = -0.871553938494339$, and $a_0 = 0.211074313346156$.

(b) Transformations for FFP modeling

$$\varphi^*(NN) = a_1(NN) + a_0, \quad (6)$$

where $a_1 = -1735498.432421821$, and $a_0 = 122908.214346089$.

$$\varphi^*(FL) = a_4(FL)^4 + a_3(FL)^3 + a_2(FL)^2 + a_1(FL) + a_0, \quad (7)$$

where $a_4 = -1179.828742552175$, $a_3 = 244.364360922208$, $a_2 = 18.984460901200$, $a_1 = -5.166180545100$, and $a_0 = -0.050567771173985$.

$$\varphi^*(ONN) = a_1(ONN) + a_0, \quad (8)$$

where $a_1 = -2324155.414675274$, and $a_0 = 163629.816053569$.

$$\varphi^*(OFL) = a_1(OFL) + a_0, \quad (9)$$

where $a_1 = 4013501.320037709$, and $a_0 = -286489.622011131$.

Summations of optimal transformation of input parameters are as follows:

$$\varphi^*(\text{Log}(K)) = \varphi^*(NN) + \varphi^*(FL) + \varphi^*(ONN) + \varphi^*(OFL), \quad (10)$$

$$\varphi^*(FFP) = \varphi^*(NN) + \varphi^*(FL) + \varphi^*(ONN) + \varphi^*(OFL). \quad (11)$$

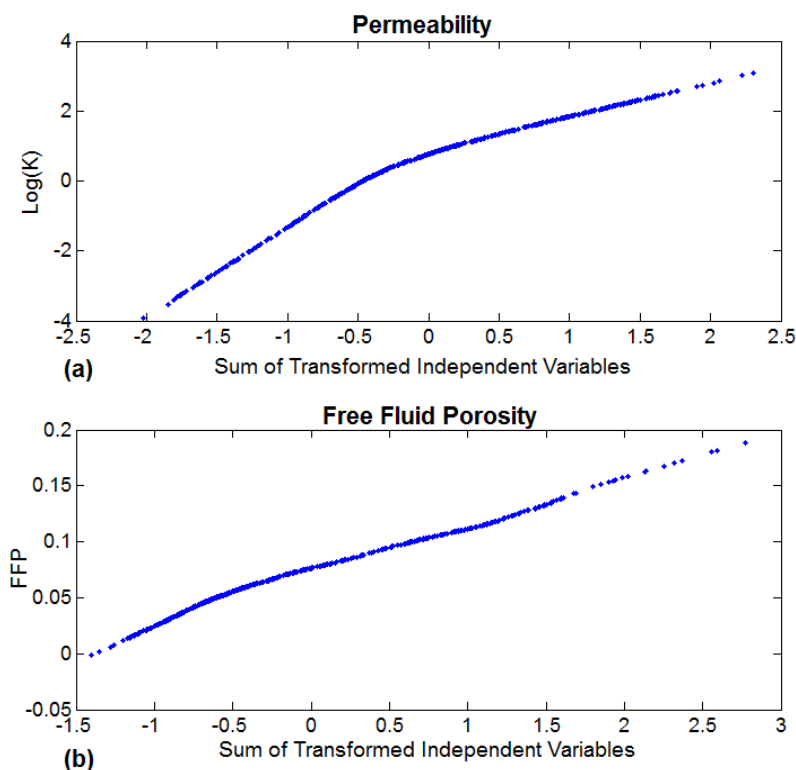


Fig. 12. Relation between sum of transformed independent variables and: (a) permeability, and (b) free fluid porosity determined by ACE.

Figure 12 shows the relationship between permeability and free fluid porosity and sum of optimal transformation of input parameters. A simple curve fitting can result in permeability and free fluid porosity through following equations.

$$\text{Log}(K) = a_3 \left(\varphi^*(\text{Log}(K)) \right)^3 + a_2 \left(\varphi^*(\text{Log}(K)) \right)^2 + a_1 \left(\varphi^*(\text{Log}(K)) \right) + a_0, \quad (12)$$

where $a_3 = 0.070938496352028$, $a_2 = -0.380116311467027$, $a_1 = 1.466278287151171$, and $a_0 = 0.727440481067394$.

$$\text{FFP} = a_3 \left(\varphi^*(\text{FFP}) \right)^3 + a_2 \left(\varphi^*(\text{FFP}) \right)^2 + a_1 \left(\varphi^*(\text{FFP}) \right) + a_0, \quad (13)$$

where $a_3 = 0.003322744745387$, $a_2 = -0.007556364428498$, $a_1 = 0.040164861128829$, and $a_0 = 0.076864737700259$.

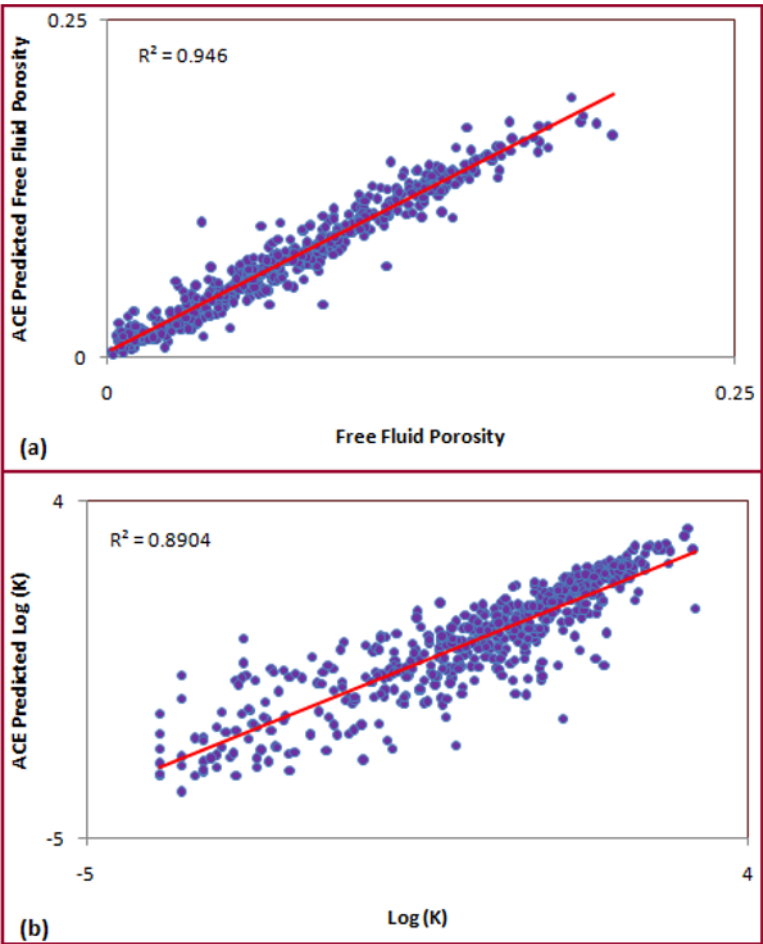


Fig. 13. Crossplots showing correlation coefficient between measured and predicted: (a) free fluid porosity, and (b) log (K).

After developing above formulation through the use of ACE method, performance of constructed model was assessed by means of test data. Figure 13 shows correlation coefficient between measured and predicted NMR parameters. This figure shows that the ACE model was capable of producing high accuracy results. For better illustration, results of ACE models are compared with individual models *viz.* neural network, fuzzy logic, optimized neural network, and optimized fuzzy logic in term of correlation coefficient and mean square error (Fig. 14). This figure shows that ACE is a good alternative for combining different models in order to enhance accuracy of final prediction.

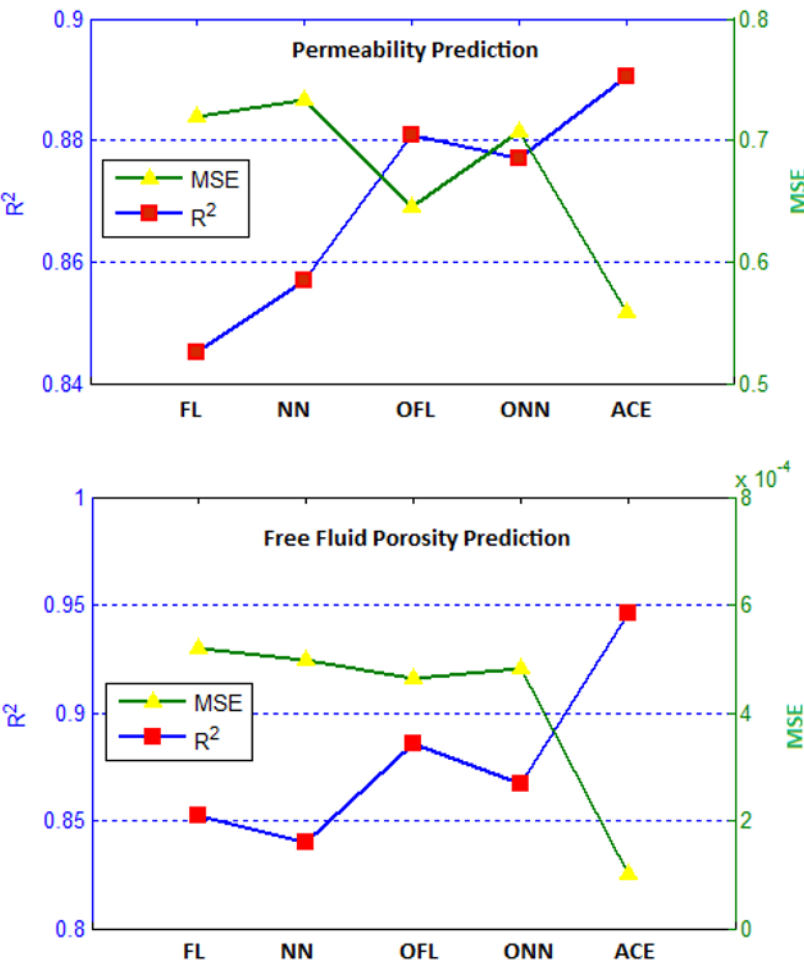


Fig. 14. A comparison between different models *versus* correlation coefficient and mean square error.

6. CONCLUSIONS

In this study, a stochastic optimization tool, called hybrid genetic algorithm-pattern search, was employed to strengthen fuzzy logic and artificial neural network. Strengthened models were then used to estimate nuclear magnetic resonance (NMR) log parameters, including free fluid porosity and rock permeability. At the final step of this study, an ACE committee with optimized and non-optimized models was constructed. Some noticeable points deduced from this study can be expressed as follows:

- ❑ In spite of complex behavior of carbonate reservoirs, neural network and fuzzy logic are able to extract the formulation between conventional well log data and NMR log parameters, including free fluid porosity and permeability.
- ❑ Performance of back-propagation neural network is greatly dominated by initial weights and biases which are randomly assigned to the neural network. Furthermore, the back-propagation algorithm is highly at risk of trapping in local minima.
- ❑ Fuzzy clustering is strongly governed by heterogeneity of training data. This flaw might result in non-smooth multi-dimensional formulation. Therefore, need of comprehensive training database is inevitable.
- ❑ Subtractive clustering has just some control on means of Gaussian membership functions, *i.e.*, no modification on variances (spread) of membership functions is done by it.
- ❑ Hybrid genetic algorithm-pattern search as a global optimization tool provides a stochastic search capability which is not affected by initial condition of algorithm run. It converges to global minimum regardless from where it starts.
- ❑ Extracting optimal weights and biases of neural network through the use of stochastic search of hybrid genetic algorithm-pattern search tool is more confident and efficient than the use of back-propagation algorithm.
- ❑ By extracting fuzzy formulation through the use hybrid genetic algorithm-pattern search tool, it is possible to have a control on both means and variances (spread) of Gaussian membership functions.
- ❑ Stochastically optimized fuzzy logic and neural network performed better than non-optimized ones.
- ❑ Convergence speed of hybrid genetic algorithm-pattern search is low. Substituting faster algorithms for future works are suggested.
- ❑ ACE committee machine with optimized and non-optimized models can significantly enhance accuracy of final prediction.
- ❑ In situations where there are multiple choices to solve a problem, by little additional computation it is possible to construct an ACE committee machine to improve precision of prediction.
- ❑ Implementation of the propounded strategy provides an accurate, quick and cost-effective way of estimating NMR log parameters, including free fluid porosity and rock permeability from conventional well log data for wells that have no NMR run.

Appendix

Conversion factors for transferring permeability and sonic transit time units from field units to SI units

Parameter	Conversion factor	
	Field units	SI units
Permeability	mili-darcy [md]	$= 9.86923 \times 10^{-4} \mu\text{m}^2$
Sonic transit time	microsecond per foot [$\mu\text{s}/\text{ft}$]	$= 3.289474 \times 10^{-6} \text{ s}/\text{m}$

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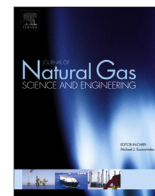
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Simulation of NMR response from micro-CT images using artificial neural networks



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ABSTRACT

The Nuclear Magnetic Resonance (NMR) log is amongst the functional techniques in petroleum investigation to segregating the reservoir and non-reservoir horizons precisely; furthermore, the NMR log provides an improved method to determine reservoir petrophysical parameters. Unfortunately, these data are usually sparse since acquiring NMR logs in producing cased wells is not possible and it is one of the most expensive tools in the logging industry thus its associated costs are the major limitation of its usage. Consequently, researchers have recently studied to virtually extract the NMR parameters via other routes. One such route, which we propose here is the possibility of estimating the T_2 distribution curve and magnetization decay by establishing a relationship between micro-CT images and NMR parameters by means of artificial neural networks (ANN) and image analysis algorithms. Specifically, two ANN networks were designed, which numerically image features from micro-CT images as inputs, while the amplitude of the magnetization and relaxation time were output parameters. We assessed the procedure by taking the error rate and correlation coefficient into consideration and we conclude that the ANN model is capable of finding logical patterns between image features and NMR responses, and is thus able to reliably predict NMR response behavior. Furthermore, we quantitatively compared ANN and random walk (RW) NMR predictions, and we demonstrate that ANN readily outperforms RW in terms of accuracy.

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1. Introduction

The quality of a reservoir system controls the economic profitability of a field. Typically this quality is assessed by conventional porosity and resistivity logs, however these logs have serious limitations (Asquith and Krygowski, 2004); for example the presence of clay leads to errors in porosity estimation, since porosity logs (neutron, density and sonic) strongly relies upon rock matrix than reservoir fluids properties (Asquith and Krygowski, 2004). Furthermore, the resistivity tools are unable to distinguish between clay-bound, capillary-bound and reservoir's moveable water, which ends in an overestimation of the amount of water saturation and ignoring the reservoir units in some cases. These limitations were tackled by NMR logging (which introduced in 1978 by

Schlumberger (Maute, 1992)), which played an important role in the specification of reservoir pore architecture and characterization and measurement of oil and gas in the formation. NMR logging is non-destructive, continuous and fast, which is certainly superior to conventional methods, providing reservoir appraisalment without the environmental impacts possible (Coates et al., 1999). Technically the NMR measurement responds only to hydrogen nuclei (in water, oil and gas) in the pore space (Darling, 2005). For this reason, NMR logging provides the feasibility of immediate quantification of porosity and the differentiation of fluid types and other important petrophysical parameters being connected to the pore morphology (Darling, 2005).

Since NMR logging is one of the most expensive tools in the logging industry (Blümich, 2005), researchers have recently studied to virtually extract the NMR parameters via different routes. Up to now, most studies focused on the exploitation of Artificial intelligence (AI) as an estimator to simulate the NMR parameters from conventional well log data, e.g. Mohaghegh et al. (2001) applied AI methods to predict the NMR-derived permeability,

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effective porosity and irreducible water saturation in a field located in East Texas; while Labani et al. (2010) struggled to estimate the NMR free fluid porosity and permeability from well log data using AI. Golsanami et al. (2014) used fuzzy logic, neuro-fuzzy and neural networks to estimate the magnetic resonance T_2 distribution and the T_2 logarithmic mean from well log data in the Zagros Basin, Iran. Furthermore, Talabi et al. (2009) simulated T_2 distribution and magnetization decay by establishing a relationship between micro-CT images and NMR parameters using a random walk (RW) method, this stochastic process consists of successive random steps (Lawler and Limic, 2010). Talabi et al. (2009) synthesized the NMR parameters using random walk theory in which the pore space voxels of sand packs were successively surveyed. The current study proposes a new numerical procedure for estimation of the T_2 curve and magnetization decay from micro-CT images integrating image analysis and neural networks. The chief section of the suggested process is a voxel-based feature extraction as a branch of image analysis techniques. The main steps of the methodology are shown in Fig. 1. The reliability of the technique is verified by the application of the samples displayed in Talabi et al. (2009).

2. Theoretical background

2.1. NMR data

The NMR log uses a radio frequency (RF) transmitter, receiver, and a permanent magnet. Before the tool generates magnetic fields that align the spin axes of the protons in a specific route, hydrogen protons pertaining to formation fluids was found to have random orientation throughout the pore space. To generate a measurable signal, the polarized protons must be at a condition of resonance. Resonance is achieved by pulsing the formation with an oscillating field to force these protons out of their new equilibrium condition. Once the oscillatory magnetic field is taken away, the magnetic signal emitted by spinning protons relaxing toward the original condition. These protons generate a series of so-called spin echoes by applying a sequence of repetitive pulses with the NMR tool. To synthesize a spin-echo train (which constitutes the raw NMR data), an NMR tool measures the amplitude of the spin echoes as a function of time.

The (transverse magnetization) amplitude of the spin-echo train at time t is given by (Kenyon, 1997)

$$M_x(t) = M_{0x} e^{-\frac{t}{T_2}} \quad (1)$$

Wherein the equivalent magnitude of the transverse magnetization at $t = 0$ is M_{0x} , T_2 is the time constant of the transverse magnetization decay (the transverse relaxation time).

Determining the T_2 distribution is of paramount importance in

NMR data analysis. This step is a mathematical inversion process called echo-fit or mapping. Due to the T_2 distribution of rock being a continuous function, a multi-exponential model is being utilized by the mapping process that assumes that the T_2 distribution comprises of m discrete relaxation times T_{2i} with corresponding porosity components ϕ_i . Eq. (2) displays the system of equations that present the individual echoes (Coates et al., 1999).

$$\begin{aligned} \text{echo}(1) &= \phi_1 e^{-\left[\frac{t(1)}{T_{2,1}}\right]} + \phi_2 e^{-\left[\frac{t(1)}{T_{2,2}}\right]} + \phi_3 e^{-\left[\frac{t(1)}{T_{2,3}}\right]} + \dots \\ &+ \phi_m e^{-\left[\frac{t(1)}{T_{2,m}}\right]} + \text{noise} \\ \text{echo}(2) &= \phi_1 e^{-\left[\frac{t(2)}{T_{2,1}}\right]} + \phi_2 e^{-\left[\frac{t(2)}{T_{2,2}}\right]} + \phi_3 e^{-\left[\frac{t(2)}{T_{2,3}}\right]} + \dots \\ &+ \phi_m e^{-\left[\frac{t(2)}{T_{2,m}}\right]} + \text{noise} \\ \text{echo}(n) &= \phi_1 e^{-\left[\frac{t(n)}{T_{2,1}}\right]} + \phi_2 e^{-\left[\frac{t(n)}{T_{2,2}}\right]} + \phi_3 e^{-\left[\frac{t(n)}{T_{2,3}}\right]} + \dots \\ &+ \phi_m e^{-\left[\frac{t(n)}{T_{2,m}}\right]} + \text{noise} \end{aligned} \quad (2)$$

Where through $t(i)$ is the pertaining time as the i th echo was obtained.

2.2. Image features extraction

In brief, Feature extraction is a low-level image analysis operation and composed of a sequence of operations on the image matrix (Marmo et al., 2005). An image feature standardly is a part of an image that encompasses fascinating attributes or a property of the image which we are interested in, depending on the overall goal and what the application is trying to achieve. Correspondingly, in this study several tests have been performed on micro-CT images prepared in binary mode (black voxels are the grains and white voxels represent pore space) to acquire 21 features (f1-f21) stated in Marmo et al. (2005) as below:

- f1 = ratio of edge voxels to the whole voxel;
- f2 = number of white areas;
- f3 = number of voxel of white areas;

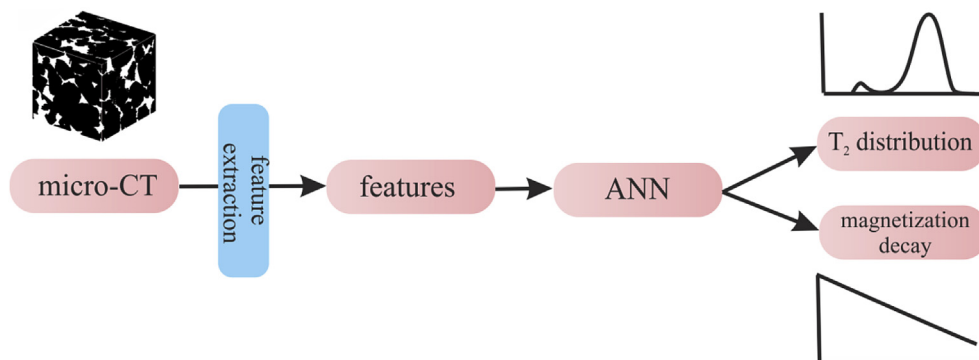


Fig. 1. Flowchart of proposed methodology.

f4 = number of white areas composed of more than 150 voxels;
 f5 = number of voxels within the white areas composed of more than 150 voxels; These previous five features (1–5) were extracted from the whole image, while the 16 remaining

features were obtained out of the each four largest selected white areas.

f6–f9 = number of voxel in the white area;
 f10–f13 = length in voxel of the white area;

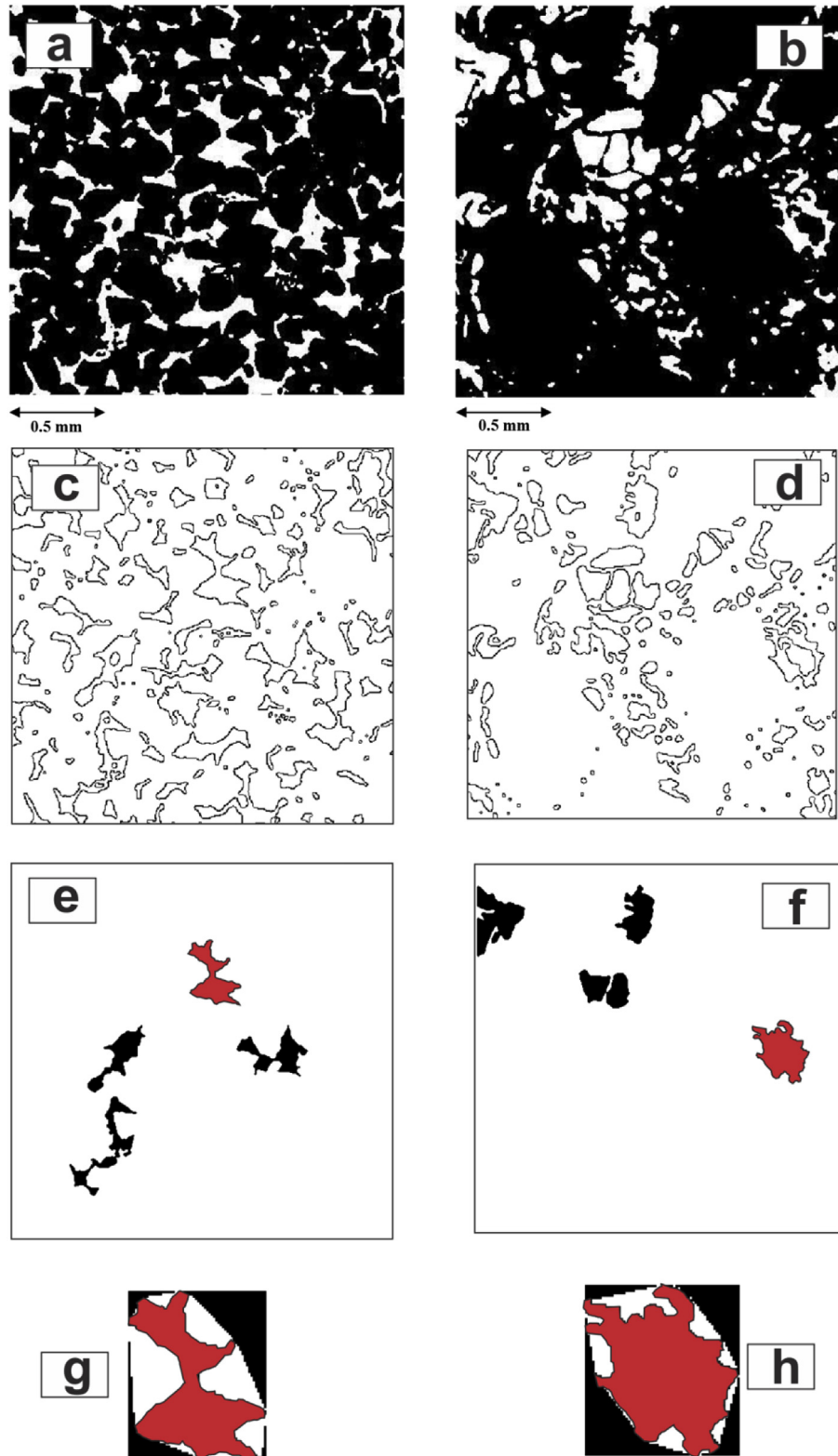


Fig. 2. (A) Binary micro-CT image slice through a sandstone (white voxels show pore space), (b) Binary micro-CT image slice through a carbonate rock (white voxels show pore space), (c–d) Edge voxels of a and b, (e–f) four selected white areas with more than 150 voxels from c and d, (g–h) Convex deficiency of one of the selected area in e and f.

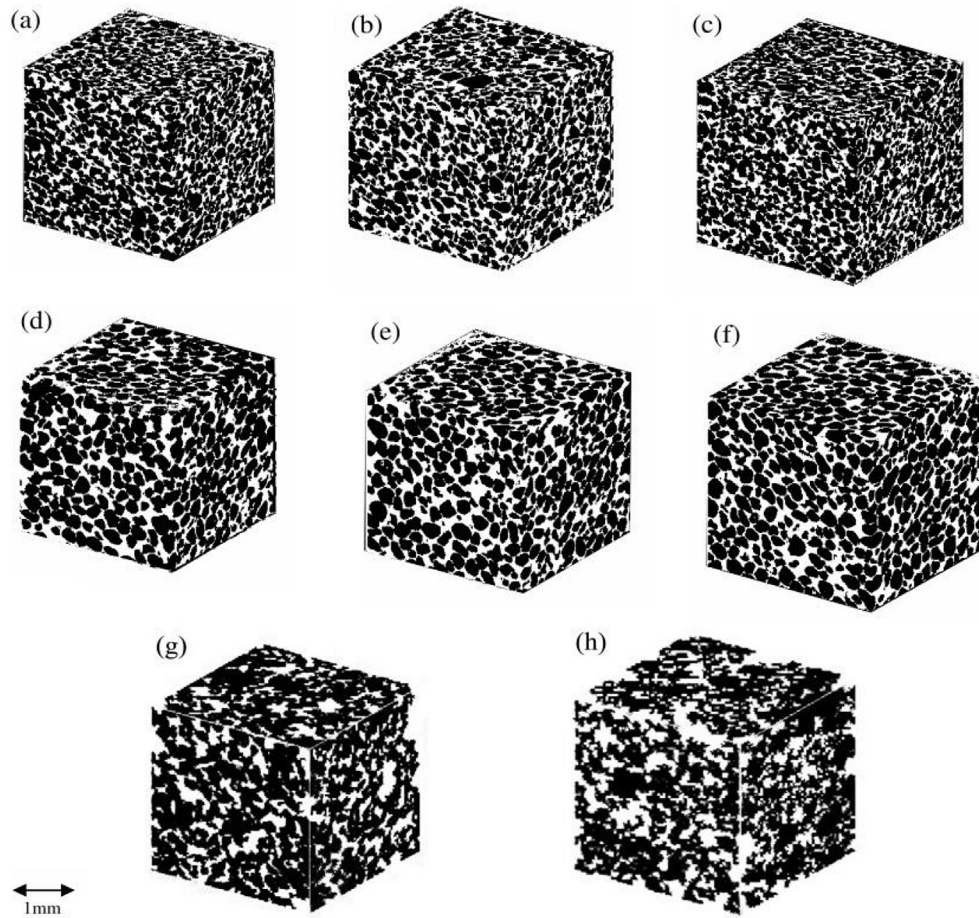


Fig. 3. Binarized 3D micro-CT images of (a) LV60A, (b) LV60B, (c) LV60C, (d) F42A, (e) F42B, (f) F42C, (g) Berea and (h) S3.

f_{14} – f_{21} = number of small and large convex deficiencies;

The smallest convex polygon is called convex hull, embodying the binary object which the corresponding convex deficiency are computed as important features (Liu-Yu and Thonnat, 1997). A

labeling algorithm detects and quantifies the convex deficiencies. The values 100 and 1000 are cut off criteria for a number of voxels to detect convex deficiencies. Hence, the large and small convex deficiencies were determined with more than 1000 voxels and less than 100 voxels, respectively.

Table 1
Numerical image features of micro-CT images.

	F42A	F42B	F42C	LV60A	LV60B	LV60C	Berea	S3
f1	0.08422962	0.08889211	0.09030005	0.11858438	0.10871405	0.12689751	0.07270977	0.08815466
f2	1054	819	610	537	515	1037	7578	5807
f3	9 144 359	11 002 886	11 381 890	13 668 674	13 549 403	11 775 246	4 388 720	1 780 428
f4	357	174	167	69	97	211	1071	73
f5	8 859 557	10 766 833	11 211 463	13 512 938	13 404 436	11 484 926	1 853 851	97 976
f6	716 090	2 570 125	1 470 676	3 092 727	1 807 632	575 299	5204	2353
f7	443 154	1 976 595	1 454 209	1 423 447	2 897 759	555 017	5250	1806
f8	597 560	786 359	1 268 110	1 633 467	1 740 786	630 153	6070	1826
f9	1 022 169	809 340	1 548 441	1 333 164	1 696 100	926 706	5155	2009
f10	5733.3502	15 918.4959	7733.85187	15 838.957	9506.22644	3768.21575	173.01441	84.914056
f11	4016.35353	12 495.616	8658.49262	7811.13648	16 183.5286	3276.58616	164.976562	76.1896413
f12	5581.02482	6369.15523	7799.50832	8543.755	9998.68019	4682.28745	208.685193	88.733396
f13	7115.33542	5571.35117	9378.1143	7938.1651	8617.41387	5837.59785	216.723343	142.895237
f14	43	105	49	248	98	13	2	9
f15	17	68	60	117	180	53	5	8
f16	30	27	60	130	144	54	5	1
f17	17	39	53	105	119	41	5	3
f18	95	349	209	433	267	114	5	1
f19	92	267	223	218	401	117	4	2
f20	101	124	198	252	243	103	4	4
f21	166	117	215	220	267	173	5	4

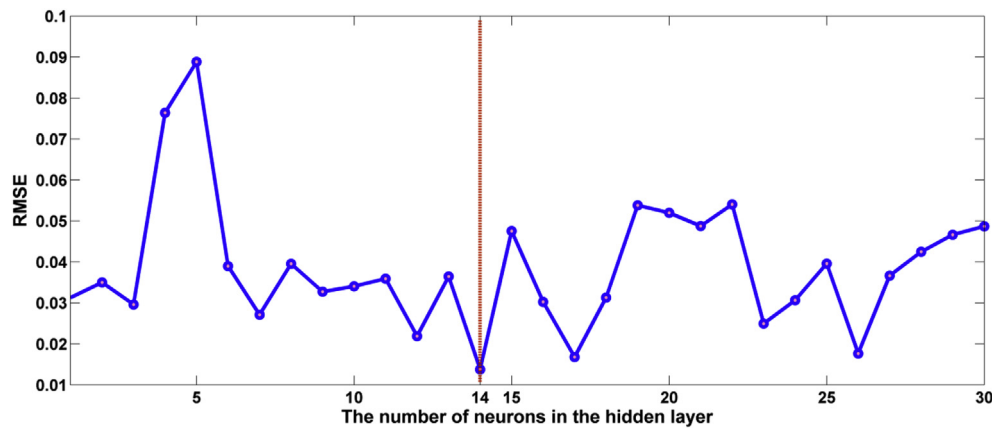


Fig. 4. Finding the optimum number of neurons via root mean square error (RMSE) analysis: the RMSE is plotted against the number of neurons in the hidden layer.

Fig. 2 displays the disparity between the features for two samples of sandstone and carbonate rock with different sedimentological and petrophysical parameters.

3. Results and discussion

3.1. Sample preparation

In the current investigation, previously achieved data from the samples displayed in Talabi (2008) and Talabi et al. (2009) was analyzed. In summary, eight samples (six sand packs (F42A, F42B, F42C, LV60A, LV60B and LV60C) and two sandstones (Berea and S3)) were selected for micro-CT imaging and NMR measurements. To determining the grain size distributions of the sands British standard meshes were used on an electric shaker sieving the samples for 80 min (Talabi, 2008). The sands were then poured into a 10 cm long thermoplastic heat shrink sleeve (diameter 3.81 cm) fitted with plastic end caps and pieces of circular filter paper to ensure a tight fit and so that no sand was lost. An electric vibrator used for tapping and vibrating to ensure compaction, then the samples were placed in an oven at 80 °C for 20 min and then the samples were left to cool (Talabi et al., 2009).

3.2. NMR measurements

Brine saturated sand packs (3.81 cm in diameter by 10 cm in length) were used for the NMR measurement. The brine was a solution of de-ionized water with 5 wt % Sodium Chloride (NaCl) and 1 wt % Potassium Chloride (KCl). The reason for adding NaCl was to increase the ionic strength of the water (to better

approximate formation brine) and KCl was used to prevent the sand from swelling (clay stabilization). NMR relaxation measurements were then performed on a MARAN2 bench top spectrometer at a temperature of 308 K and at 2 MHz. The magnetization decay (2 μ s inter-echo spacing) then generated the T_2 relaxation time distribution applying a curvature-smoothing regularization method (Chen et al., 1999; Toumelin et al., 2003).

3.3. Micro-CT imaging

The micro-CT images (the samples had following dimensions: 0.65 cm diameter and length of 4 cm) were acquired on a commercial XMT unit (Phoenix-X-ray System and Services GmbH). These greyscale micro-CT images were then filtered and converted to a binary format with the purpose of simulating the NMR parameters. Binarized 3D micro-CT images of F42A-C and LV60A-C sand packs and Berea and S3 sandstones are illustrated in Fig. 3 (also compare Talabi, 2008).

3.4. Image features extraction

The image analysis techniques allow conversion of an object into a set of numbers, i.e., some information extraction; the outcome of image analysis is a number or a bunch of numbers or a graph, it is not an image (Heilbronner and Barrett, 2014). This methodology was utilized on the binarized images, created databases were prepared as input parameters into the ANN model. Accordingly, scanned images of $750 \times 750 \times 450$ voxels, with resolutions ranging from 8 to 10 μ m were collected and analyzed in MATLAB®. Table 1 illustrates the numerical values of aforementioned features

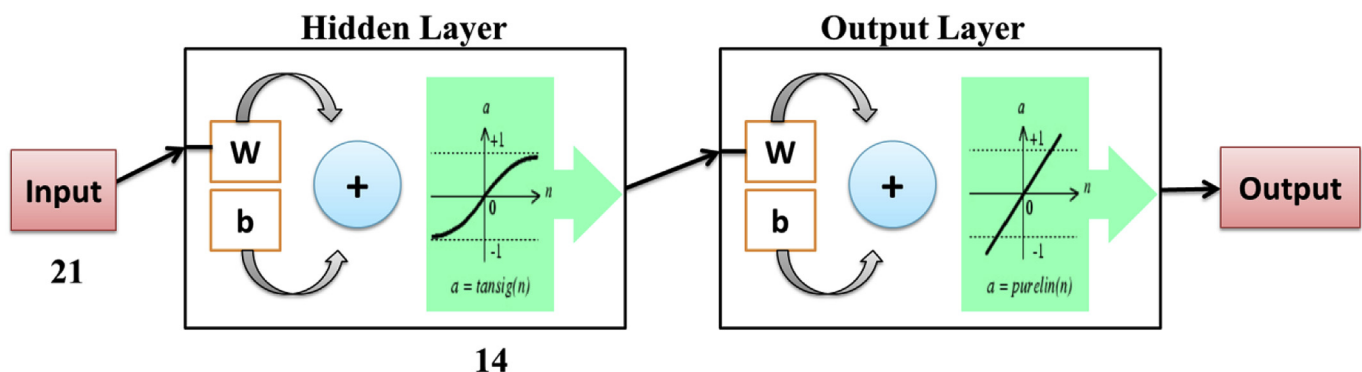


Fig. 5. Schematic diagram of constructed network with transfer functions. w and b are the weight and bias, respectively.

Table 2
Comparison of RMSE for ANN and RW in predicting T_2 distributions.

	ANN		RW	
	RMSE	R^2	RMSE	R^2
F42B	0.0121	0.9748	0.0121	0.9747
LV60B	0.0037	0.9980	0.0162	0.9606

of each sample.

An essential issue that should be considered is rock heterogeneity. A significant size difference between samples used for NMR measurement and micro-CT images leads to uncertainty. In order to avoid its effect on the results, the scale of the NMR measurements and micro-CT imaging must be similar.

3.5. NMR response simulation by using ANN

The proposed ANN approach was implemented on a PC with an Intel® Core™ i3 2.27 GHz CPU, 4 GB RAM, the Windows 7 operating system. Using MATLAB® software, a Feed-Forward Back-propagation type network was designed within three layer for NMR Response estimation from micro-CT images, which is widely believed to outperform those with more layers (Balkin and Ord, 2000; Khan et al., 2008; Kadkhodaie-Ilkhchi et al., 2008, 2010; Farzi, 2012; Bolandi et al., 2015). The network was then trained with the Levenberg Marquardt training algorithm (TrainLM), details of which is conveniently accessible in Bishop (1995) and Boadu (1997, 1998). Shifting from input layer to hidden layer is performed within Tangent logistic (TANSIG) transfer function whilst linear (PURLIN) transfer function preferentially connects the hidden and output layers. The input layer of the trained network consisted of 21 neurons, based on the 21 inputs (i.e. the number of extracted

features), together with 14 neurons in the hidden layer. This was the optimal network setup as verified by the root mean squared error (RMSE) analysis, Fig. 4. Note that the network succeeded in minimizing the RMSE value in 11 epochs. Essentially, the performance of the model was measured by RMSE analysis; and this performance is equal to the expected value of the square of the difference between the value taken by the estimator and the parameter's true values, exploited to optimizing the value of weights and default bias. Thus, if a network runs multiple times, the optimum value of the parameters result in steady output with low uncertainty range. Alongside the performance, processing time is another issue which relevant to computational power. The architecture of the designed networks is demonstrated in Fig. 5.

To predict the NMR parameters from image features using the ANN network, four sand packs (F42A, F42C, LV60A and LV60C), as well as two sandstones (Berea and S3) were selected. Two ANN models were constructed for estimation of T_2 distribution and magnetization decay, in both models numerical image features were used as input values. In the final step, the ANN models were applied to simulating the NMR response of two additional sand packs (F42B and LV60B).

So far, Talabi (2008) simulated NMR responses using the RW method resulting in a good agreement between the simulated and the experimental measurements. This technique attempts to simulate T_2 and magnetization decay by establishing a relationship between micro-CT images and NMR parameters. In this study, ANN systems were used to develop a new correlation for NMR parameter prediction based on micro-CT images. All the laboratory data set utilized for our analyses throughout this paper are taken from Talabi (2008) and Talabi et al. (2009). The comparisons between predicted and measured values utilizing RW and neural networks (Table 2) show both techniques were successful in generating the

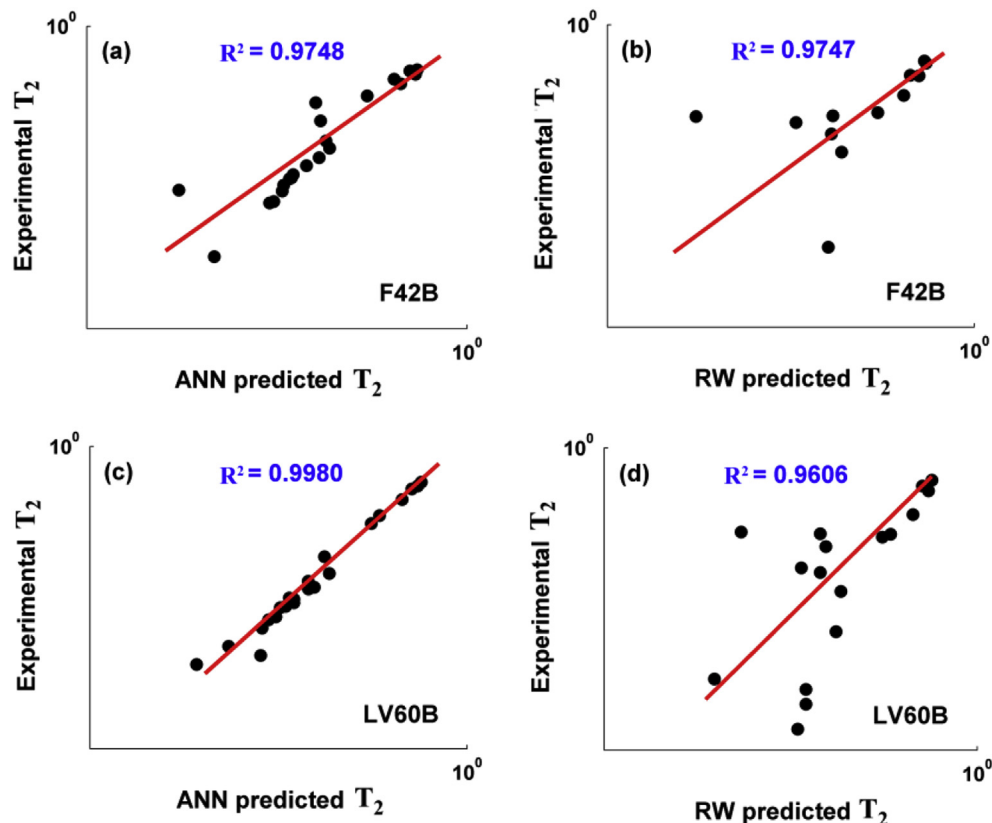


Fig. 6. Crossplots showing correlation coefficient between Experimental and predicted T_2 distributions by ANN and RW for F42B and LV60B sand packs.

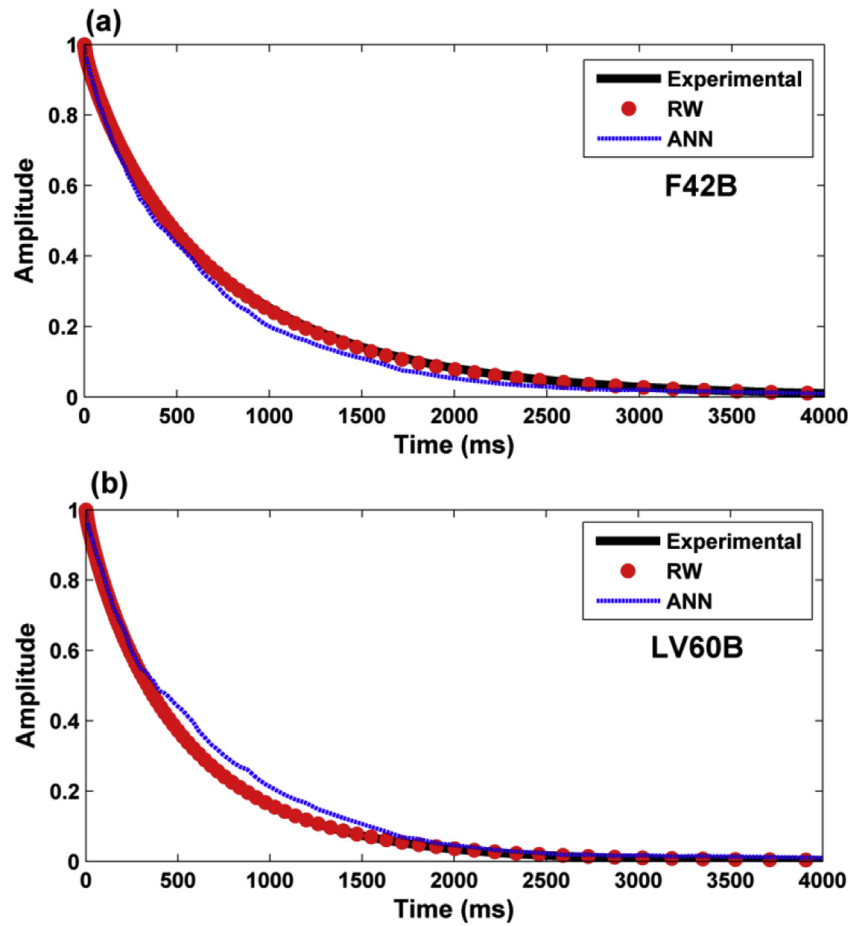


Fig. 7. a. Comparison of the experimental magnetization decay with RW and ANN predictions for F42B. b. Comparison of the experimental magnetization decay with RW and ANN predictions for LV60B.

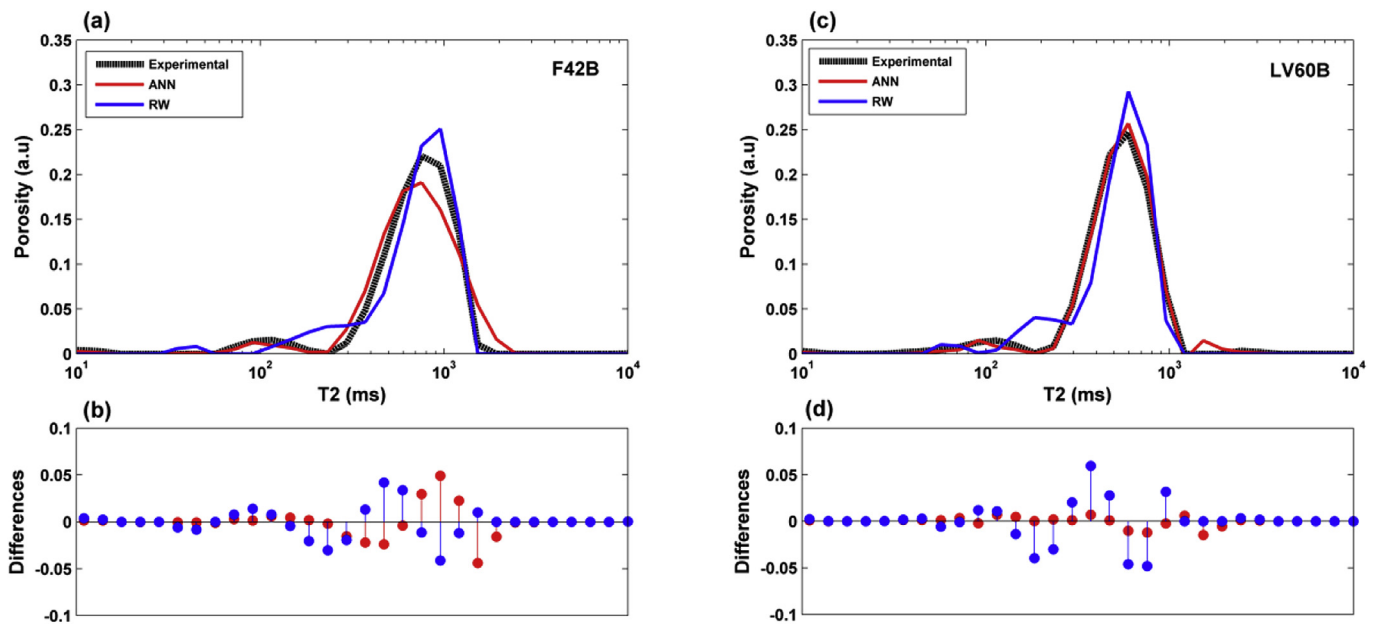


Fig. 8. a. Comparison of the experimental T₂ distribution with RW and ANN predictions for F42B. b. Stem diagram of the differences in the porosity values of RW and ANN predictions with experimental T₂ distribution for F42B. c. Comparison of the experimental T₂ distribution with RW and ANN predictions for LV60B. d. Stem diagram of the differences in the porosity values of RW and ANN predictions with experimental T₂ distribution for LV60B.

NMR parameters. Also, drawing an analogy between ANN and RW results shows that ANN has had an improvement on predicting the amplitude of the transverse magnetization and relaxation time. RMSE and correlation coefficient (R) between measured and estimated data in test samples were considered as a criterion to evaluate the models accuracy. According to Fig. 6, the ANN simulated T_2 curve and magnetization decay has approximately been fitted with experimental data. The ANN model predicted T_2 curve for F42B and LV60B sand packs with RMSE of 0.0121 and 0.0037 and correlation coefficients; $R^2 = 0.9748$ and 0.9980 , respectively; which are the evidence of the best performance of the ANN model.

The ANN simulated magnetization decays and T_2 distributions of the F42B and LV60B are compared with the experimental data and RW simulated curves in Figs. 7 and 8. The results prove that ANN outperforms the RW method and it can be considered as a powerful tool for NMR response estimation, especially in cases where an accurate estimation criterion is critical.

4. Conclusion

The Nuclear Magnetic Resonance (NMR) response is one of the most noteworthy tools which is used to precisely characterize the reservoir without environmental effects. In the present study, we have predicted the NMR response from microCT numerical image features via an artificial intelligence algorithm with a good degree of accuracy. The micro-CT images of six sand packs were used for the construction of the neural network (ANN) model and two sand packs were taken to evaluate the satisfaction of the developed model. Accordingly, a Feed-Forward Back-propagation multilayered network took an input set of 21 numerical features predicted the T_2 distribution together with the magnetization decay. Quantitative comparisons of the results and errors from ANN and RW predictions and experimental data demonstrate that for the estimation problem of NMR T_2 curve ANN readily outperforms RW in terms of accuracy ($R^2 \sim 0.98$ for the ANN model; while $R^2 \sim 0.97$ for the RW method on average). One of the other advantages of this method is that the solid phase was only considered and fluid phase does not play a role in this prediction technique. The main conclusion to be drawn is that the NMR response can be reliably and precisely simulated by using image features based neural network models.

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Estimating NMR T_2 distribution data from well log data with the use of a committee machine approach: A case study from the Asmari formation in the Zagros Basin, Iran

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ABSTRACT

The Nuclear Magnetic Resonance (NMR) log is one of the most valuable logs in petroleum exploration which is used to precisely evaluate the reservoir and non-reservoir horizons. Along with porosity logs (neutron, density, sonic), NMR log is used to estimate the porosity and permeability of the hydrocarbon bearing intervals. The current study focuses on estimating NMR T_2 distribution data from conventional well log data with the use of artificial intelligent systems. The eight bin porosities of the combinable magnetic resonance (CMR) T_2 distribution alongside with the T_2 logarithmic mean (T_2 LM) values are predicted using the intelligent models developed in this study. The methodology applied here combines the results of the individual models in a committee machine with intelligent systems (CMIS) for estimating the NMR T_2 distribution and T_2 logarithmic mean data. The Fuzzy logic (FL), the adaptive neuro fuzzy system (ANFIS) and artificial neural networks (ANNs) are utilized as intelligent experts of the CMIS. The NN models are developed with four different training algorithms (Levenberg–Marquardt (LM), scaled conjugate gradient (SCG), one step secant (OSS) and resilient back-propagation (RP)) and the best one is chosen as the optimal NN expert of the CMIS. The CMIS assigns a weight factor to each individual expert by the simple averaging and weighted averaging methods. A genetic algorithm (GA) optimization technique is used to derive the weighted averaging coefficients. The results indicate that the GA optimized CMIS performs better than the individual experts acting alone for synthesizing the NMR T_2 curve and T_2 LM data from one specific set of conventional well logs.

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1. Introduction

NMR log provides an improved method to determine the reservoir petrophysical parameters. As an example since the NMR porosity is not affected by lithology and underground radioactive materials, it provides meticulous information about the porosity and pore size distribution of rocks. NMR logs have the possibility of in-situ measurement of the permeability, free fluid index, irreducible water and irreducible oil saturation. Moreover, identification of the fluid types and volumes is possible via the NMR T_2 data interpretation. Nevertheless, running NMR log in producing cased wells is not possible. On the other hand, it is one of the most expensive tools in the logging industry and its associated costs are the major limitation of its usage. Therefore,

there are only a few wells for which this log is run. In opposition to such problems, artificial intelligence has proved to be a cost-effective and time saving method to virtually extract the NMR log parameters. Recent studies have investigated the use of intelligent estimators such as artificial neural networks, fuzzy and adaptive neuro-fuzzy inference systems and the intelligent optimizers such as genetic algorithms to estimate the NMR log parameters from conventional well log data in many different cases.

Wong et al. (1995, 1998) used backpropagation neural networks to predict porosity and permeability of the oil wells. Mohaghegh et al. (1996) used ANNs for petroleum reservoir characterization. Mohaghegh et al. (2001) applied intelligent systems to synthesize artificial NMR logs in a field located in East Texas. Ogilvie et al. (2002) used GA and FL techniques to predict permeability from the NMR tool data. Chen and Lin (2006) used a committee machine with empirical formulas (CMEFs) for permeability prediction. Kadkhodaie-Ilkhchi et al. (2008, 2009) developed committee machines for prediction of total organic carbon and oil content from the well log data in the South Pars Gas Field,

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Iran. Rolon et al. (2009) used artificial neural networks to synthesize specific petrophysical logs from other logs.

Most of the previous works are in relation to estimating the parameters derived from NMR log (e.g. porosity and permeability). The current study focuses on estimating the original NMR T_2 and T_2 LM distribution alongside with the CMR bin porosities. Since T_2 and other NMR logs can be in relation to many other petrophysical and geological properties (e.g. facies, rock types, pore throat network, fluid saturation, wettability) it is worth synthesizing virtual NMR logs directly from one specific set of conventional well log data which are available for almost all wells. This study suggests that using the different neural network training algorithms and combining the best NN model with the FL and NF models can lead to the high accuracy of the results for construction of the synthetic T_2 curves.

The study area is located in Zagros basin, western Iran. The hydrocarbon bearing units of this area are Pabdeh, Asmari, Gajbaran, Agajari, etc. In the area that this research was carried out, the Paleocene to Pliocene Asmari formation is the most important reservoir unit that mainly consists of limestone, dolomite, shaly limestone and shale. The low-porosity and high-permeability Asmari formation undergoes minor lithological changes throughout the area. The latter fact provides capability of employing developed models of the current research in the neighboring fields. It is assumed that the formation contains no ions of chromium, manganese, iron, nickel or other paramagnetic ions and also no iron containing minerals which could considerably influence NMR measurements. Drilling fluid did not contain these ions neither.

2. NMR log

The nuclear magnetic resonance log is used as a tool to study the structure of porous media and their various rock–fluid interactions. This technique allows the determination of rock and fluid properties such as porosity and pore size distribution, permeability, water saturation, wettability, viscosity, etc. NMR logging exploits the large magnetic moment of hydrogen, which is abundant in rocks in the form of oil, gas and water. The NMR signal amplitude is proportional to the quantity of hydrogen nuclei present in a formation and can be calibrated to give a value for porosity that is free from lithology effects (Allen et al., 1997). The most important mechanism affecting the NMR relaxation is the grain-surface relaxation. Molecules in fluids are in constant Brownian motion, diffusing about the pore space and bouncing off the grain surfaces. Upon interaction with the grain surface, hydrogen protons can transfer some nuclear spin energy to the grain (contributing to T_1 relaxation) or irreversibly dephase (contributing to T_2 relaxation). Therefore the speed of relaxation most significantly depends on how often the hydrogen nuclei collide with the grain surface and this is controlled mainly by the surface-to-volume ratio of the pore in which the nuclei are located. Collisions are less frequent in larger pores resulting in a slower decay of the NMR signal amplitude and allowing a petrophysicist to understand the distribution of pore sizes (Kenyon et al., 1995).

3. Theory and methodology

The methodology applied in this work is based on selection of the appropriate input data. This step leads to selection of six well logs as the input data. The selection process will be explained in detail in the forthcoming section. The second step is to develop the different fuzzy, neuro-fuzzy and NN models. The NN models were developed with the four training algorithms including Levenberg–

Marquardt (Hagan and Menhaj, 1994), Resilient Back-Propagation (Riedmiller, 1993), Scaled Conjugate Gradient (Moller, 1993) and One Step Secant (Battiti, 1992). The next step is selection of the

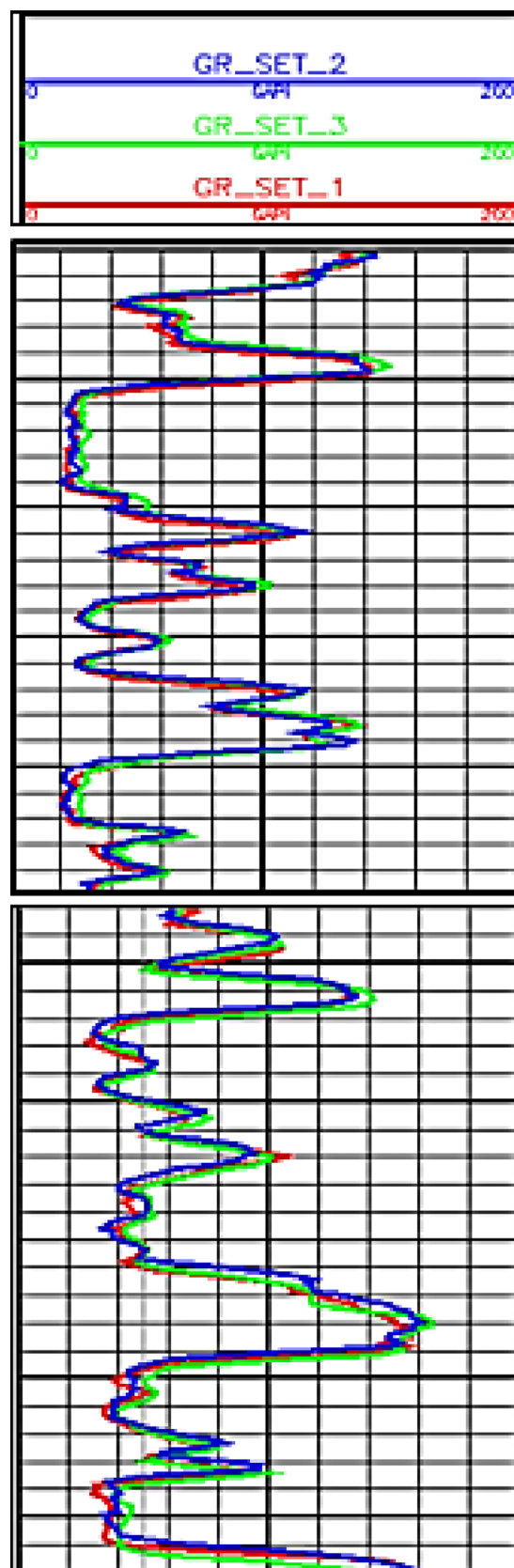


Fig. 1. GR logs from three sets of raw data used as depth controllers.

best FL, NF and NN models among the numerous intelligent models. The selection procedure was based on the minimum prediction error. The fourth step is to construct the CMIS. The individual experts of the CMIS were combined by the simple and weighted averaging methods. The optimal weigh values for the weighted averaging CMIS were extracted by the genetic algorithm optimization technique.

Artificial Neural Networks are inspired by biological nervous systems and are composed of parallel operating elements. These networks are trained for function fitting, pattern recognition and data clustering. Back propagation Neural Network (BP-NN) is a type of supervised-training networks. A three layered network with a tangent sigmoid (TANSIG) transfer function in hidden layer and a linear transfer function (PURELIN) in output layer was satisfactory for the aim of the present research. This kind of network is capable of fitting approximately any practical function having a finite number of discontinuities (Matlab user's guide,

2008). In this study four different BP-NNs with four neurons in hidden layer but Levenberg–Marquardt (LM), scaled conjugate gradient (SCG), one step secant (OSS) and resilient back-propagation (RP) training algorithms were utilized and the network with the best performance (with minimum error) was selected as the final NN model for estimating T_2 distribution data.

Fuzzy sets were introduced by Zadeh (1965) to treat data that were rather fuzzy than precise. It maps an input space to an output space through a fuzzy inference system (FIS). Two important types of fuzzy inference systems are the Mamdani-type FIS and Takagi-Sugeno-type FIS (Matlab user's guide, 2008). The Sugeno-type model which is implemented in this study uses the subtractive clustering in the fuzzy inference engine. The subtractive clustering is an effective, robust and quick approach to estimate the number of fuzzy clusters and cluster centers (Jarrah and Halawani, 2001).

Neural networks have proved to act well in a wide variety of applications but they cannot explain the rules and decisions they

Table 1

Correlation coefficients between CBPs and available well log data.

	CBP1	CBP2	CBP3	CBP4	CBP5	CBP6	CBP7	CBP8	T_2 LM
NPHI (pu)	0.0549	0.095	0.3113	0.5006	0.3589	0.15	0.4037	0.4913	0.1648
RHOZ (kg/m ³)	5E – 5	0.0073	0.4763	0.4968	0.4576	0.635	0.7585	0.468	0.1234
SPHI (pu)	0.0267	0.0679	0.4802	0.5111	0.4435	0.4012	0.7167	0.4538	0.1091
GR (GAPI)	0.5558	0.7103	0.0492	0.0142	0.0446	0.1564	0.0991	0.016	0.0323
PEFZ (barns/electron)	0.0062	0.0222	0.4073	0.3307	0.3109	0.5413	0.595	0.3108	0.0482
PHIE (pu)	0.0797	0.0593	0.3207	0.6734	0.66	0.8164	0.8681	0.6686	0.2457

Table 2

Parameters of input membership functions for fuzzy models.

	NPHI		RHOZ		SPHI		GR		PEFZ		PHIE	
	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ
CBP1												
mf1	0.1129	0.1961	267.2000	2290.0000	0.1012	0.1886	51.6900	33.5200	1.2540	1.9880	0.0899	0.2178
mf2	0.1129	0.1168	267.2000	2719.0000	0.1012	–0.0188	51.6900	39.4500	1.2540	3.8150	0.0899	0.0714
mf3	0.1129	0.0021	267.2000	3007.0000	0.1012	–0.0624	51.6900	2.6350	1.2540	5.4490	0.0899	0.0000
CBP3												
mf1	0.0627	0.0875	144.6000	2740.0000	0.0577	–0.0389	28.6300	34.2600	0.6577	3.8820	0.0488	0.0577
mf2	0.0627	0.1901	144.6000	2266.0000	0.0577	0.1981	28.6300	28.5300	0.6577	1.8430	0.0488	0.2357
mf3	0.0627	0.0012	144.6000	2926.0000	0.0577	–0.0572	28.6300	3.3640	0.6577	5.2900	0.0488	0.0000
mf4	0.0627	0.1294	144.6000	2583.0000	0.0577	0.0387	28.6300	30.2100	0.6577	4.4890	0.0488	0.1020
mf5	0.0627	0.2248	144.6000	2539.0000	0.0577	0.0847	28.6300	36.6400	0.6577	3.5560	0.0488	0.1620
mf6	0.0627	0.2080	144.6000	2313.0000	0.0577	0.1971	28.6300	81.4500	0.6577	2.3600	0.0488	0.1046
CBP5												
mf1	0.1127	0.1147	257.9000	2738.0000	0.1005	–0.0184	50.7900	46.6800	1.1820	3.9410	0.0869	0.0669
mf2	0.1127	0.1927	257.9000	2302.0000	0.1005	0.1936	50.7900	30.5500	1.1820	1.8750	0.0869	0.2135
mf3	0.1127	0.1763	257.9000	2407.0000	0.1005	0.1176	50.7900	37.9800	1.1820	4.6030	0.0869	0.1569
CBP7												
mf1	0.0439	0.1901	104.0000	2266.0000	0.0460	0.1981	21.8500	28.5300	0.5106	1.8430	0.0399	0.2357
mf2	0.0439	0.1133	104.0000	2752.0000	0.0460	–0.0273	21.8500	48.5200	0.5106	3.8810	0.0399	0.0584
mf3	0.0439	0.0501	104.0000	2664.0000	0.0460	–0.0290	21.8500	18.4300	0.5106	4.8540	0.0399	0.0465
mf4	0.0439	0.1454	104.0000	2489.0000	0.0460	0.0792	21.8500	31.3900	0.5106	4.4330	0.0399	0.1232
mf5	0.0439	0.1822	104.0000	2318.0000	0.0460	0.1710	21.8500	46.8000	0.5106	2.0670	0.0399	0.1713
mf6	0.0439	0.2030	104.0000	2543.0000	0.0460	0.0648	21.8500	29.4800	0.5106	3.4220	0.0399	0.1572
mf7	0.0439	0.0463	104.0000	2807.0000	0.0460	–0.0883	21.8500	17.3100	0.5106	3.9920	0.0399	0.0361
mf8	0.0439	0.1640	104.0000	2641.0000	0.0460	0.0186	21.8500	50.6600	0.5106	3.2310	0.0399	0.0944
T_2 LM												
mf1	0.0502	0.1867	120.5000	2267.0000	0.0469	0.1992	22.6100	29.9800	0.5262	1.8940	0.0413	0.2306
mf2	0.0502	0.0875	120.5000	2740.0000	0.0469	–0.0389	22.6100	34.2600	0.5262	3.8820	0.0413	0.0577
mf3	0.0502	0.1969	120.5000	2370.0000	0.0469	0.1904	22.6100	77.8100	0.5262	2.2480	0.0413	0.0981
mf4	0.0502	0.0364	120.5000	2614.0000	0.0469	–0.0426	22.6100	29.1100	0.5262	4.7600	0.0413	0.0424
mf5	0.0502	0.1921	120.5000	2591.0000	0.0469	0.0494	22.6100	34.0800	0.5262	3.7780	0.0413	0.1434
mf6	0.0502	0.0094	120.5000	2921.0000	0.0469	–0.0467	22.6100	12.1300	0.5262	5.1170	0.0413	0.0000
mf7	0.0502	0.1887	120.5000	2440.0000	0.0469	0.1746	22.6100	113.5000	0.5262	2.6330	0.0413	0.0270
mf8	0.0502	0.1279	120.5000	2592.0000	0.0469	0.0495	22.6100	32.2700	0.5262	4.9480	0.0413	0.0988

Table 3

Parameters of output membership functions for fuzzy models.

	C1	C2	C3	C4	C5	C6	C7
CBP1							
mf1	−0.0379	0.0000	0.0141	0.0004	0.0002	0.1031	−0.1025
mf2	0.0429	0.0000	−0.0079	0.0001	0.0007	−0.0331	0.0076
mf3	−0.2687	0.0000	0.0989	0.0000	−0.0075	0.3965	−0.0219
CBP3							
mf1	−0.0227	0.0000	0.0214	0.0000	−0.0049	−0.0361	0.0259
mf2	−0.0732	−0.0001	−0.0419	0.0000	−0.0073	−0.1090	0.2399
mf3	−0.0724	0.0000	−0.0123	0.0002	0.0009	−0.0757	0.0422
mf4	0.0715	0.0000	−0.1054	−0.0001	−0.0006	−0.1387	0.1425
mf5	0.0203	−0.0001	−0.0313	−0.0001	0.0037	−0.0410	0.2211
mf6	−0.0679	−0.0001	−0.0034	−0.0001	0.0033	−0.1300	0.2157
CBP5							
mf1	0.0200	0.0000	0.0082	0.0000	0.0008	0.0300	−0.0160
mf2	0.0339	−0.0001	−0.1033	−0.0002	−0.0008	−0.0035	0.1628
mf3	−0.0017	0.0000	−0.0178	0.0001	−0.0030	0.2296	0.0169
CBP7							
mf1	0.2083	0.0001	0.1424	0.0004	−0.0458	0.2271	−0.1290
mf2	−0.0070	0.0001	−0.0058	0.0001	0.0085	0.2039	−0.1696
mf3	0.1173	−0.0001	0.0873	−0.0001	0.0046	−0.0910	0.1413
mf4	0.0449	0.0000	0.0836	−0.0002	0.0092	−0.0369	0.0419
mf5	−0.1437	0.0000	−0.1321	0.0005	−0.0036	0.5821	0.0287
mf6	−0.2236	0.0000	−0.0175	0.0008	−0.0047	0.5912	−0.0418
mf7	−0.0541	0.0000	0.0129	0.0002	0.0035	0.1402	−0.1366
mf8	−0.1191	0.0000	−0.0094	0.0000	0.0084	0.0728	0.1057
T2LM							
mf1	−0.5638	−0.0015	−0.2887	−0.0050	0.4275	−1.2250	3.4400
mf2	1.2900	0.0005	−1.4410	0.0024	0.2664	3.4730	−2.7360
mf3	2.7260	−0.0011	−1.8600	−0.0095	0.1583	−1.7430	3.0430
mf4	1.3050	−0.0009	0.6618	0.0004	0.2102	−2.5090	1.5030
mf5	−0.8971	0.0001	1.0550	−0.0014	0.1461	0.3975	−0.4744
mf6	0.7154	0.0004	−0.2974	−0.0020	−0.1467	−0.0543	−0.5049
mf7	0.4377	−0.0003	−1.0790	0.0013	−0.0787	2.6550	0.8095
mf8	0.5518	−0.0015	2.2400	−0.0009	0.2585	−1.4580	2.9480

Table 4

Parameters of input membership functions for neuro–fuzzy models.

	NPHI		RHOZ		SPHI		GR		PEFZ		PHIE	
	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ
CBP1												
mf1	0.1334	0.2437	267.2000	2290.0000	0.0868	0.2432	51.6900	33.5200	1.2590	1.9900	0.1353	0.2250
mf2	−0.0829	0.1184	267.2000	2719.0000	−0.0667	0.1010	51.6900	39.4500	1.2620	3.8120	0.1155	0.0575
mf3	0.0937	0.0111	267.2000	3007.0000	0.0963	−0.0640	51.6900	2.6350	1.2630	5.4430	0.1216	0.0334
CBP3												
mf1	0.1095	0.0935	144.6000	2740.0000	0.0764	−0.0389	28.6300	34.2600	0.6593	3.8810	0.0486	0.0651
mf2	0.0733	0.1988	144.6000	2266.0000	0.0705	0.1869	28.6300	28.5300	0.6605	1.8450	0.1088	0.2091
mf3	0.0616	0.0027	144.6000	2926.0000	0.0529	−0.0529	28.6300	3.3640	0.6578	5.2900	0.0492	0.0016
mf4	0.1099	0.2040	144.6000	2583.0000	0.0441	0.1069	28.6300	30.2100	0.6662	4.4840	0.0640	0.1230
mf5	0.0724	0.2554	144.6000	2539.0000	0.0808	0.0529	28.6300	36.6400	0.6549	3.5590	0.0940	0.1373
mf6	0.0486	0.1970	144.6000	2313.0000	−0.0014	0.2350	28.6300	81.4500	0.6505	2.3560	0.0211	0.1072
CBP5												
mf1	0.0419	0.0908	200.6000	2738.0000	0.0340	−0.0183	39.5000	46.6800	0.9206	3.9450	0.0904	0.0625
mf2	0.1485	0.1999	200.6000	2287.0000	0.0940	0.1896	39.5000	31.0300	0.9160	1.8700	0.1259	0.1541
mf3	0.0532	0.1755	200.6000	2544.0000	0.0766	0.1095	39.5000	28.6300	0.9029	3.6090	0.1067	0.1450
mf4	0.1498	0.0653	200.6000	2884.0000	0.1056	0.0123	39.5000	13.5200	0.9251	5.1220	0.1249	0.0483
CBP7												
mf1	0.0553	0.0937	182.0000	2719.0000	0.0524	−0.0670	38.2400	39.4500	0.8908	3.8140	0.0374	0.0975
mf2	0.0716	0.2051	182.0000	2266.0000	0.0744	0.2139	38.2400	28.5300	0.8880	1.8410	0.0220	0.2508
mf3	0.0987	0.1481	182.0000	2457.0000	0.0777	0.0795	38.2400	36.5200	0.8947	4.3750	0.1403	0.0978
T2LM												
mf1	0.0417	0.1837	90.3900	2251.0000	0.0084	0.2232	16.9600	29.6200	0.3885	1.8240	0.0130	0.2562
mf2	0.1074	0.1968	90.3900	2285.0000	0.0428	0.1936	16.9600	50.8400	0.3708	2.0630	0.0362	0.1967
mf3	0.0366	0.0901	90.3900	2740.0000	0.0300	−0.0593	16.9600	34.2600	0.3935	3.8810	0.0665	0.0900
mf4	0.0721	0.2281	90.3900	2370.0000	0.0964	0.1719	16.9600	90.2900	0.4237	2.3530	0.0689	0.0951
mf5	0.0377	0.0068	90.3900	2965.0000	0.0366	−0.0522	16.9600	4.7300	0.3946	5.3570	0.0310	0.0000
mf6	0.0745	0.0433	90.3900	2634.0000	0.0677	−0.0317	16.9600	29.7900	0.3944	4.2290	0.0674	0.0512
mf7	0.0506	0.1196	90.3900	2770.0000	0.0491	−0.0153	16.9600	54.5600	0.3972	3.8020	0.0712	0.0382
mf8	0.0566	0.1969	90.3900	2440.0000	0.0514	0.1730	16.9600	113.5000	0.3868	2.6310	0.0484	0.0361

Table 5
Parameters of output membership functions for neuro-fuzzy models.

	C1	C2	C3	C4	C5	C6	C7
CBP1							
mf1	−0.0431	0.0001	0.0156	0.0005	−0.0015	0.1512	−0.1772
mf2	−0.0109	0.0000	−0.0072	0.0002	0.0012	0.0439	−0.0315
mf3	−0.0039	0.0000	0.0237	0.0000	−0.0010	−0.0077	0.0152
CBP3							
mf1	−0.0168	0.0000	0.0175	0.0000	−0.0008	0.0192	0.0159
mf2	−0.0764	0.0000	−0.0613	0.0002	−0.0044	−0.0029	0.1258
mf3	−0.1625	0.0000	−0.0203	0.0002	−0.0012	−0.0411	0.0233
mf4	0.0485	0.0000	0.0899	−0.0002	0.0113	−0.1467	0.0114
mf5	−0.0198	−0.0001	0.0266	−0.0002	0.0105	−0.0909	0.2142
mf6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CBP5							
mf1	0.0975	0.0000	−0.0055	−0.0001	0.0002	−0.0200	−0.0073
mf2	0.0096	0.0000	−0.0428	0.0000	−0.0002	0.0797	0.0580
mf3	0.0487	0.0000	−0.0008	0.0001	−0.0002	0.1900	0.0245
mf4	−0.1191	0.0000	0.0815	0.0001	0.0001	0.0939	−0.0225
CBP7							
mf1	0.0502	0.0000	0.0451	0.0001	0.0020	0.1505	−0.1391
mf2	0.0361	0.0000	−0.0114	0.0006	−0.0226	0.2510	−0.0371
mf3	−0.0205	0.0000	0.0335	−0.0001	−0.0026	0.2134	0.0697
T2LM							
mf1	0.9826	−0.0013	−12.5200	−0.0192	0.4061	6.5070	3.7730
mf2	2.1560	−0.0061	−2.4030	−0.0266	0.4250	−11.5800	16.9700
mf3	8.9070	0.0000	1.4990	−0.0022	0.2698	−7.0770	−0.8973
mf4	2.5240	−0.0029	−2.0150	−0.0162	0.1838	−10.5800	8.8640
mf5	−0.1289	0.0003	0.2836	−0.0028	−0.1275	5.0810	−0.0960
mf6	1.1890	−0.0007	1.6960	−0.0012	0.0968	−1.1920	1.5210
mf7	−2.1280	0.0007	−1.9050	0.0089	0.1577	9.3450	−3.0690
mf8	0.4120	−0.0003	−1.3740	−0.0023	0.0490	−2.3040	1.0420

Table 6
Parameters of the final fuzzy and neuro-fuzzy models.

FL models				NF models				
	r	No. of MFs	Type of MFs	r	No. of MFs	Type of MFs	Max. number of training epochs	The best model occurrence epoch
CBP1	0.9	3	Gaussian	0.9	3	Gaussian	200	9
CBP2	0.3	7	Gaussian	0.8	4	Gaussian	300	6
CBP3	0.5	6	Gaussian	0.5	6	Gaussian	300	4
CBP4	0.9	3	Gaussian	0.6	5	Gaussian	300	5
CBP5	0.9	3	Gaussian	0.7	4	Gaussian	500	7
CBP6	0.5	6	Gaussian	0.9	3	Gaussian	150	24
CBP7	0.4	8	Gaussian	0.7	3	Gaussian	250	3
CBP8	0.5	7	Gaussian	0.5	7	Gaussian	200	3
T ₂ LM	0.4	8	Gaussian	0.3	8	Gaussian	100	47

Table 7
No. of training data and the best training algorithm for neural network models.

	Total no. of training data points	No. of training data points	No. of validation data points	Training algorithm
CBP1	698	559	139	SCG
CBP2	732	586	146	RP
CBP3	668	535	133	LM
CBP4	702	562	140	SCG
CBP5	680	544	136	LM
CBP6	760	608	152	SCG
CBP7	728	583	145	OSS
CBP8	660	528	132	OSS
T ₂ LM	1069	856	213	SCG

make. Fuzzy logic is good at this case but it cannot automatically acquire the rules it uses to make those decisions. The ANFIS or adaptive neuro fuzzy inference system is a hybrid system which

combines the neural networks and fuzzy logic to overcome these problems. ANFIS allows the membership function parameters to be chosen in a way that the given input–output data is tracked in the best way. This is because these parameters are chosen so as to tailor the membership functions to the input/output data in order to account for the types of variations in the data values (Matlab user's guide, 2008). In this study 10 groups of NF models were constructed based on the ANFIS step sizes of 0.1–1. Among each group a single model was picked as the best and the final NF model was chosen from among these 10 models.

The genetic algorithm (GA) introduced by John Holland (1975) is an optimization and search technique based on the principles of genetics and natural selection (Haupt and Haupt, 1998). It starts with a number (population) of numerical arrays known as chromosomes (individuals) that represent possible answers of the problem. At the next step, the fitness value of these chromosomes is computed and the genetic operator (crossover or mutation) is

applied on those with better fitness values to produce new individuals (generation). GA is fundamentally based on the survival of the best chromosomes.

A committee machine is an ensemble of estimators consisting typically of neural networks or decision trees. It is helpful if mapping an input space to an output space is not approximated by a single estimator but by several estimators, where each estimator can focus on a particular region in the input space. Each individual estimator is called an expert. In this study, experts of the committee machine

were FL, BP-NN and NF. These experts could be combined by the simple averaging or weighted averaging methods. The different optimization algorithms such as the genetic algorithm and pattern search method could be used to determine the optimal contribution of weights.

4. Case study

4.1. Selection of appropriate input log data

The conventional well logs and NMR data used in this study as input–output data pairs came from one of the wells in the Zagros Basin penetrated into the Asmari Formation. Fortunately, the NMR log spanned over a large interval of about 345 m which provided enough measurements to develop and test the intelligent models in the same well. Approximately 235 m of this interval was used for training the intelligent models and the remaining 110 m-interval was used as unforeseen part of the well to evaluate the performance of the intelligent models. The reliability of the constructed models was measured on 78 testing depth samples. For the available data the first step was to process and remove the bad hole intervals. Gamma Ray log was employed as a depth

Table 8

Optimal CMIS weight values produced by genetic algorithm.

	NN model	FL model	NF model
CBP1	0.5964	0.4023	0.0004
CBP2	0.8380	0.0036	0.1584
CBP3	0.8216	0.1045	0.0739
CBP4	0.9500	1.521E–05	0.0490
CBP5	0.6674	0.0004	0.3330
CBP6	0.9878	0.0063	0.0069
CBP7	0.9975	3.715E–06	0.0015
CBP8	0.9990	5.376E–07	2.107E–07
T ₂ LM	0.7703	0.2287	9.815E–06

Table 9

Performance of intelligent models for synthesizing CBP1, CBP2 and CBP3.

Intelligent system	CBP1		CBP2		CBP3	
	MSE (pu) ²	R ²	MSE (pu) ²	R ²	MSE (pu) ²	R ²
FL	2.20E–05	0.7555	2.763E–05	0.7588	3.490E–05	0.5032
NF	3.42E–05	0.648	2.887E–05	0.7654	3.517E–05	0.4923
NN	2.09E–05	0.8117	1.890E–05	0.8054	3.207E–05	0.5350
GA Optimized CMIS	1.95E–05	0.8002	1.833E–05	0.8073	3.206E–05	0.5360
Simple Averaging CMIS	2.22E–05	0.7621	2.160E–05	0.7873	3.264E–05	0.5309

Table 10

Performance of intelligent models for synthesizing CBP4, CBP5 and CBP6.

Intelligent system	CBP4		CBP5		CBP6	
	MSE (pu) ²	R ²	MSE (pu) ²	R ²	MSE (pu) ²	R ²
FL	4.227E–05	0.6802	3.100E–05	0.8309	4.143E–05	0.9186
NF	3.685E–05	0.6901	3.030E–05	0.8226	3.701E–05	0.9456
NN	3.046E–05	0.8043	2.798E–05	0.8275	3.013E–05	0.9488
GA optimized CMIS	3.044E–05	0.8008	2.735E–05	0.8366	3.007E–05	0.9490
Simple averaging CMIS	3.456E–05	0.7405	2.834E–05	0.8388	3.400E–05	0.9483

Table 11

Performance of intelligent models for synthesizing CBP7, CBP8 and T₂LM.

Intelligent system	CBP7		CBP8		T ₂ LM	
	MSE (pu) ²	R ²	MSE (pu) ²	R ²	MSE	R ²
FL	2.270E–04	0.8691	1.666E–04	0.4822	1.025E–02	0.5788
NF	2.013E–04	0.8929	2.444E–04	0.3123	1.201E–02	0.5492
NN	1.612E–04	0.9075	9.964E–05	0.6422	7.560E–03	0.8777
GA optimized CMIS	1.606E–04	0.9075	9.945E–05	0.6422	7.270E–03	0.8430
Simple averaging CMIS	1.875E–04	0.9000	1.482E–04	0.5141	8.560E–03	0.7184

controlling tool as shown in Fig. 1. The gamma ray logs from the three different sets of raw data were used as depth controllers between the conventional and NMR logs. As known, the NMR T_2 distribution curves that produce observed magnetizations are correlated to measured spin echo trains through a mathematical inversion process. The echo trains are presented in the following equation.

$$\begin{aligned} \text{echo}(1) &= \phi_1 e^{-[t(1)/T_{2,1}]} + \phi_2 e^{-[t(1)/T_{2,2}]} + \phi_3 e^{-[t(1)/T_{2,3}]} + \dots + \phi_m e^{-[t(1)/T_{2,m}]} + \text{noise} \\ \text{echo}(2) &= \phi_1 e^{-[t(2)/T_{2,1}]} + \phi_2 e^{-[t(2)/T_{2,2}]} + \phi_3 e^{-[t(2)/T_{2,3}]} + \dots + \phi_m e^{-[t(2)/T_{2,m}]} + \text{noise} \\ &\vdots \\ \text{echo}(n) &= \phi_1 e^{-[t(n)/T_{2,1}]} + \phi_2 e^{-[t(n)/T_{2,2}]} + \phi_3 e^{-[t(n)/T_{2,3}]} + \dots + \phi_m e^{-[t(n)/T_{2,m}]} + \text{noise} \end{aligned}$$

where $t(i) = i$ TE and $i = 1, \dots, n$, is the time when the i th echo was acquired. In the inversion process, a set of n linear equations of m unknowns, $\phi_1, \dots, \phi_m$ are inverted to set of ϕ_i . The inversion process must involve regularization to stabilize the solution; therefore the set

of ϕ_i is not unique. The area under the curves that represents porosity is generally robust though (Coates et al., 1999). Hence an equivalent normalization approach for all ϕ_i s was employed in the current study. Even though the T_2 distribution of rocks is normally a continuous function, the mapping process uses a multi-exponential model that assumes that the T_2 distribution consists of m discrete relaxation times T_{2i} with corresponding porosity components ϕ_i (Coates et al., 1999). The NMR T_2 distribution data were divided into eight equally sized bins. These bins or T_2 arrays are referred to as CBP (Combinable Magnetic Resonance, CMR Bin Porosity) and are represented by CBP1 to CBP8, respectively. Each bin is referred to as the amount of porosity from the T_2 distribution within a specific time range. Developing a strong intelligent model capable of making accurate predictions vitally needs using noise-free input arguments, as a single noisy input value disturbs the weight matrix in neural networks or the fuzzy rules in fuzzy inference systems. With this in mind and the fact that simultaneous modeling of all CBPs proved to be displeasing, cross plot and regression analysis were implemented and narrowly

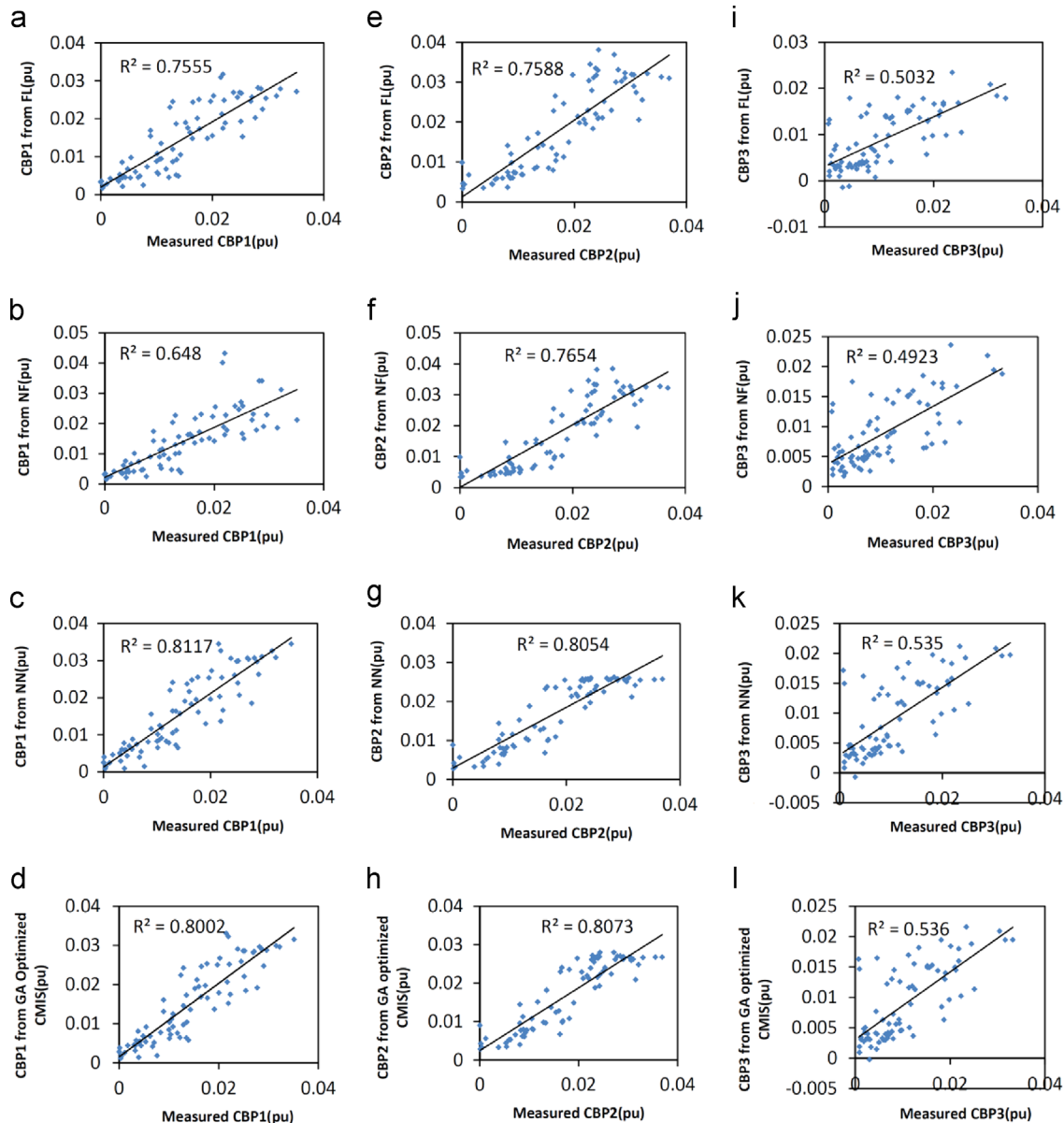


Fig. 2. Cross plots showing correlation coefficients between measured and predicted (a–d) CBP1, (e–h) CBP2 and (i–l) CBP3 values in test data.

distinguished input values were chosen to develop the intelligent model of each CBP independently. According to the cross plot and regression analysis, neutron porosity (NPHI), bulk density (RHOZ), sonic porosity (SPHI), gamma ray (GR), standard resolution photo-electric factor (PEFZ) and effective porosity data from the high resolution integrated logging tool (PHIEHILT) showed stronger relationships with CBPs. The relationships (correlation coefficients) between CBPs and available well log data are presented in Table 1. The stronger the relationships between input and target logs are, the more accurate the predictions are. Wide-range variations of the T_2LM values i.e. from 0.6151 to 586.8975 ms could cause lower performance of the constructed models; hence the T_2LM data were normalized between 0 and 1 before constructing models and

therefore the performance of the T_2LM models was represented in normal scale (i.e. without unit).

4.2. Fuzzy models

The fuzzy models used in this research are the Sugeno-type FIS developed in the programming environment of MATLAB software. The membership functions (MFs), their parameters and fuzzy “if-then” rules were determined by the subtractive clustering method. This sort of clustering offers the capability of tracking the input-output data in the best way. The clustering radius (r) was the major factor controlling the number of clusters, fuzzy rules and accuracy of the fuzzy models. The optimal FIS with minimum error

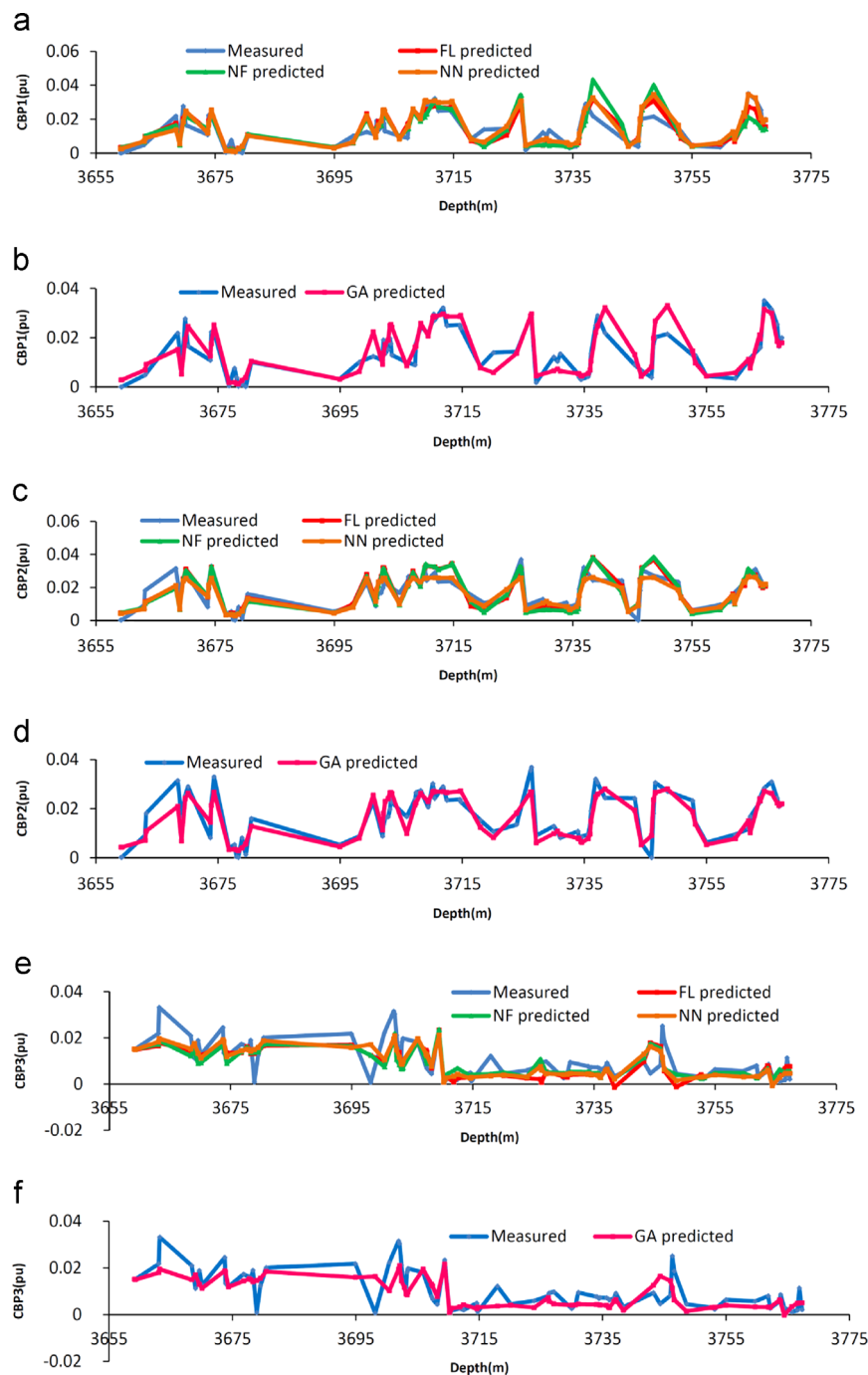


Fig. 3. Comparison between measured and predicted (a, b) CBP1, (c, d) CBP2 and (e, f) CBP3 values in test data versus depth.

was extracted by assigning a set of values between 0.1 and 1 to the clustering radius. The parameters of input and output membership functions (mf_1 , mf_2 , ..., mf_n) are shown in Tables 2 and 3, respectively.

4.3. Neuro-fuzzy models

The NF models are quite similar to the fuzzy model except for the parameters of the membership functions that are adjusted by a BP-NN. The proper step size values and number of training epochs were obtained using a trial and error method. A step size value of 0.1 led to satisfactory results in all cases. Models were trained with the use of back propagation neural network and least squares error algorithm. For all NF models, the number of training epochs was set to values greater than 100. For each case, the maximum limit of this parameter was selected and no further change in the validation error was observed. Among all models the one with the best performance was selected as the final NF model for that case. For all NF models the error tolerance was set to

0.1 and MSE (mean squared error) was used as the performance criterion. The parameters of the input and output membership functions are shown in Tables 4 and 5, respectively. The optimal values of the clustering radius, the number and type of the membership functions and training epochs are presented in Table 6.

4.4. Neural network models

Four three layered back propagation neural networks including LM, RP, SCG and OSS training algorithms with four neurons in their hidden layer and TANSIG-PURELIN transfer functions were designed. The best performing NN model was chosen as the final NN model for estimating the T_2 distribution and T_2 LM data. The number of training and validation data and the best performing training algorithm for each neural network model are presented in Table 7. About 80% of the total training data pairs were used for training the networks and the remaining 20% were used to validate them. Finally, 78 data points, selected from different

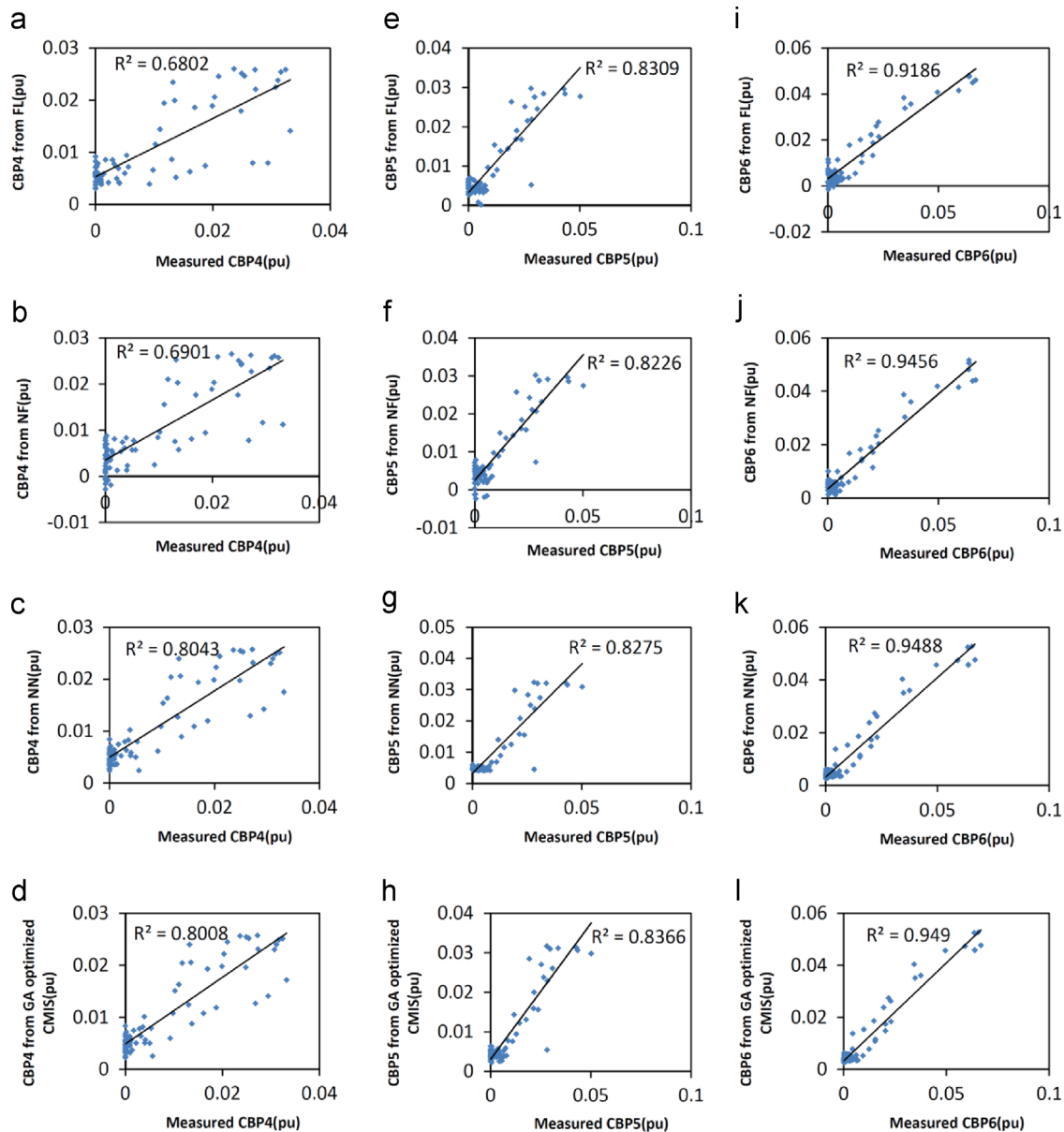


Fig. 4. Cross plots showing correlation coefficients between measured and predicted (a–d) CBP4, (e–h) CBP5 and (i–l) CBP6 values in test data.

lithologies inside the well were used to measure the reliability of the models. The results of neural network models for the CBPs and T₂LM data are presented in the coming sections.

4.5. CMIS models

The final FL, NF and NN models for CBPs and T₂LM data were combined in a CMIS through two different approaches. In the first approach, the equal weigh factors were assigned to each CMIS expert. In this method (known as the simple averaging CM), all experts have equal contribution in predicting the overall output. The final results of this approach were not satisfactory in comparison to the weighted averaging method. In the second approach, a genetic algorithm optimization technique was employed to obtain a better

combination of the individual experts. The optimized weight values led to the acceptable results and the constructed CMIS showed a better performance in comparison to the individual experts acting alone. For the genetic algorithm model, the initial population size was set to 30 and 20 for the CBPs and T₂LM data, respectively. The next generation was produced with a crossover fraction of 0.8 and elite count of 2. The optimized weight values for the CBPs and T₂LM models are presented in Table 8. The objective function that should be optimized with GA is defined as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (w_1 \times FL_i + w_2 \times NF_i + w_3 \times NN_i - T_i)^2$$

This function shows MSE of the CMIS prediction where w_1 , w_2 and w_3 are the weight factors corresponding to the FL, NF and NN

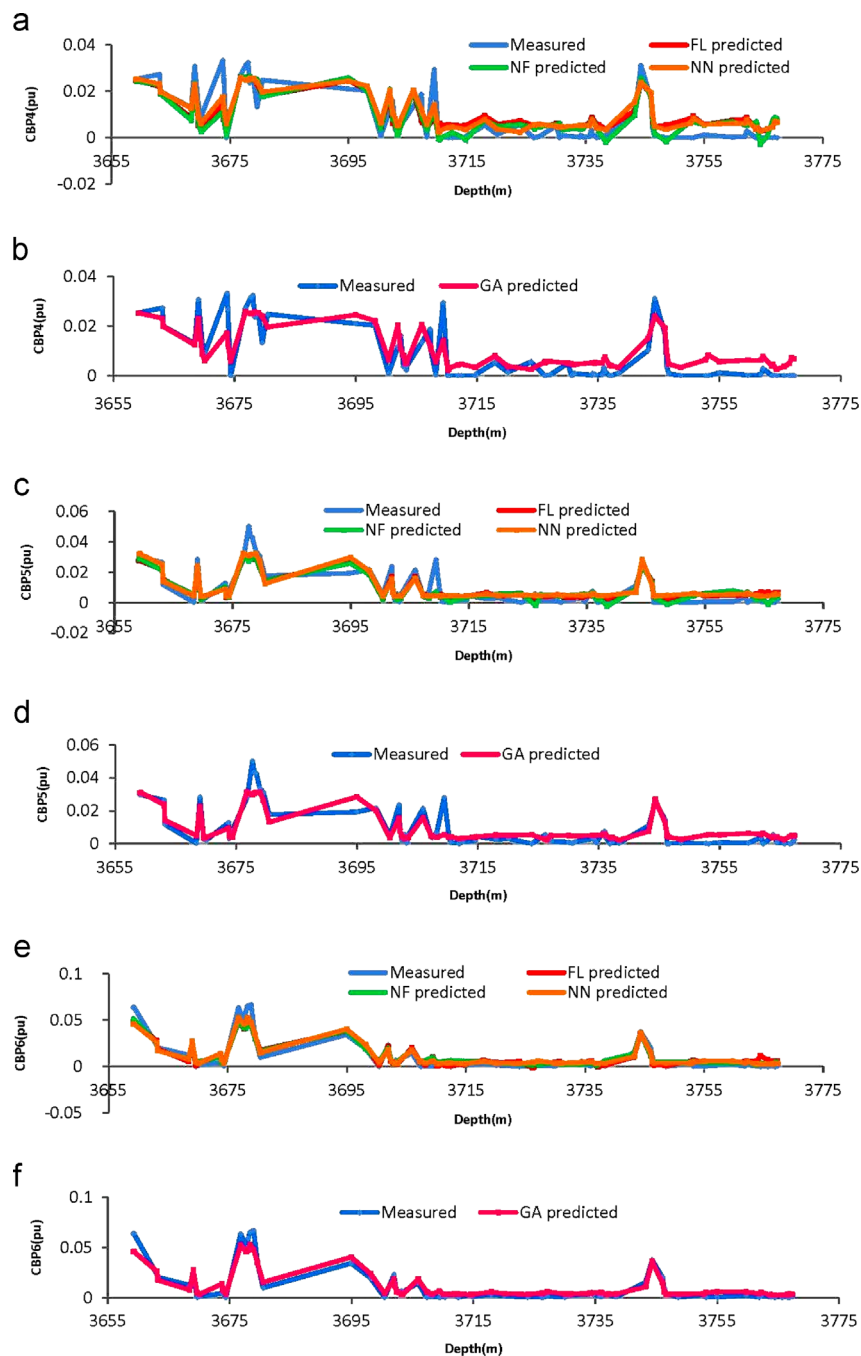


Fig. 5. Comparison between measured and predicted (a, b) CBP4, (c, d) CBP5 and (e, f) CBP6 values in test data versus depth.

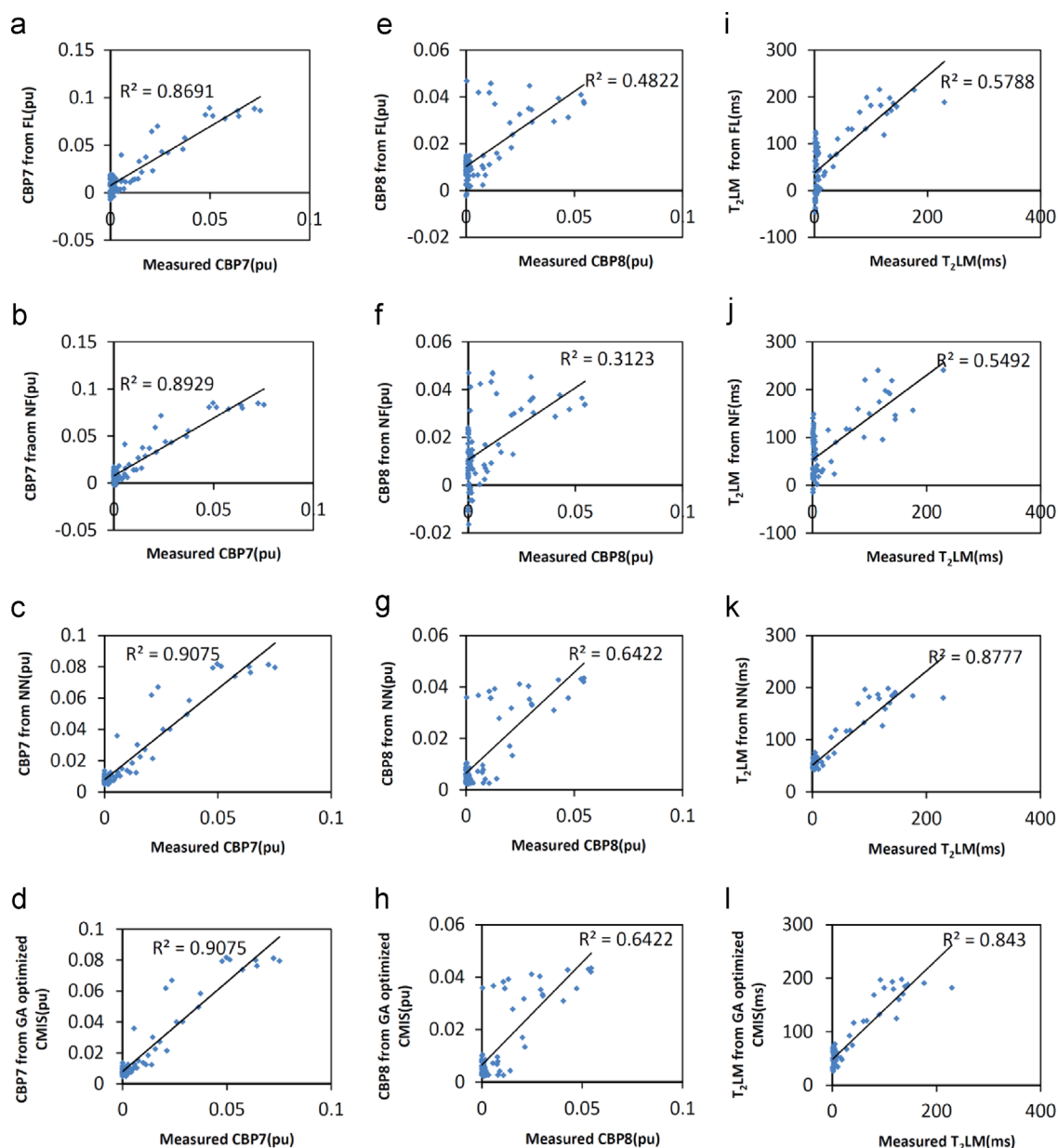


Fig. 6. Cross plots showing correlation coefficients between measured and predicted (a–d) CBP7, (e–h) CBP8 and (i–l) T_2LM values in test data.

predictions, respectively. T_i is the target value (measured CBP or T_2LM using NMR log) and n is the number of training data. A comparison of the performance of the different intelligent models for the CBPs and T_2LM data is shown in Tables 9–11. The correlation coefficient between the NMR measurements and predictions from the intelligent models are displayed in Figs. 2–7. Finally, in the light of the satisfactory results of the CMIS model described in this study the virtual NMR logs are constructed. Graphical comparisons between the measured and estimated T_2 distribution for some T_2 distribution test samples are illustrated in Fig. 8(a–p).

5. Discussion

In this study, the high performance CMIS models were developed for synthesizing the CBP2, CBP1, CBP5, CBP6, CBP4, CBP3, CBP8, CBP7 and T_2LM data. The correlation coefficient between

the measured and estimated NMR curves was more than 0.8 for all models except CBP3 and CBP8. However, in spite of the low correlation values, the average difference between the measured and predicted values or mean squared error is in the same range for all models. For the field studied here, the NN models had superior performance compared with the FL and NF models in all cases. The later statement does not include all NN models constructed from the four training algorithms. However, in some cases the individual NN training algorithms even performed weaker than the FL and NF models. That is, why a higher weight was attributed to the NN models in the weighted averaging optimization. As a result, the optimized models were much closer to the NN models in comparison to the other FL and NF ones. The simple averaging CM was also applied to combine the individual intelligent models. It showed a better performance than either the FL or NF models in all cases. The neuro-fuzzy models performed better than the fuzzy inference system for synthesizing CBP3 through CBP7, while the FL models were more

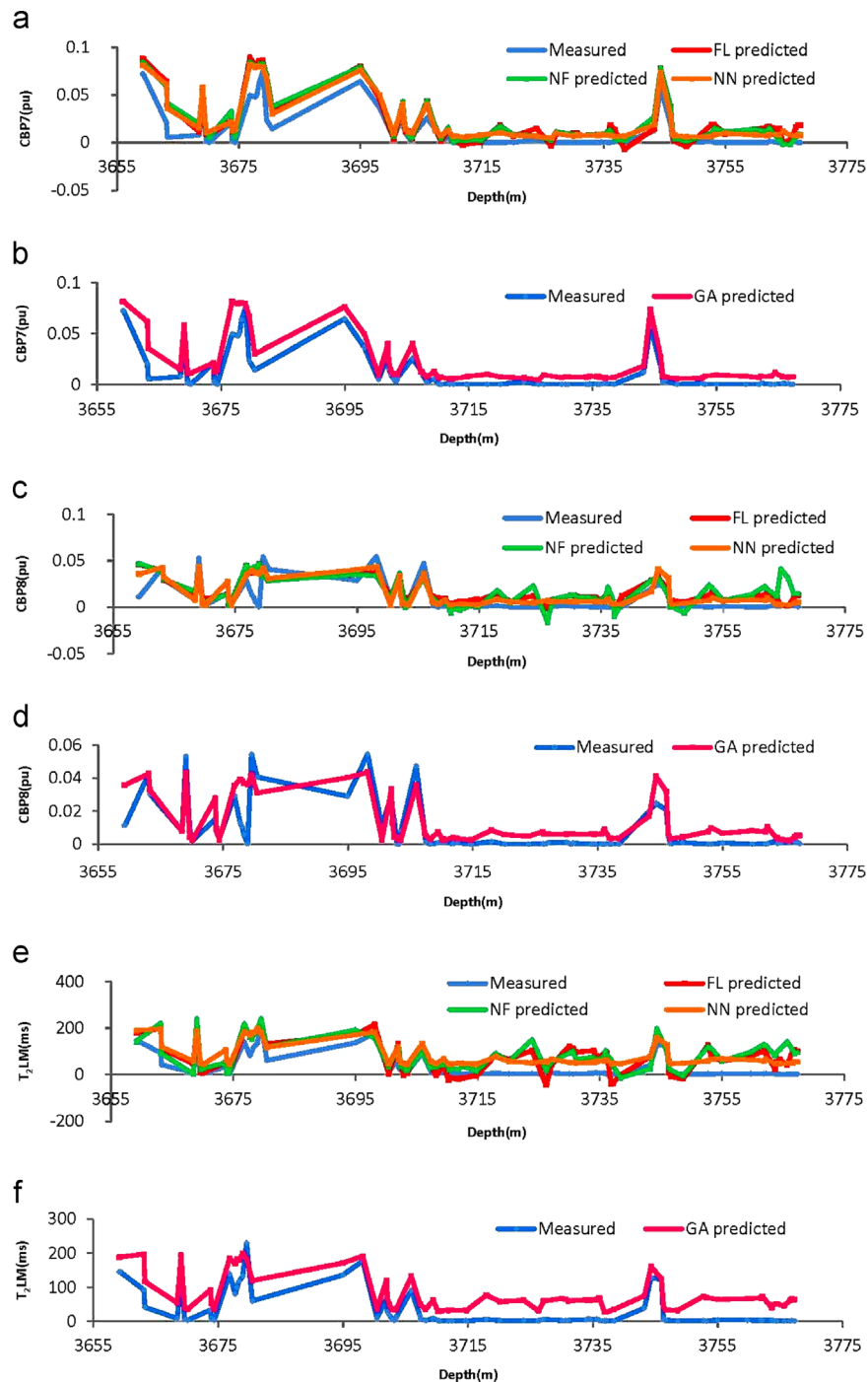


Fig. 7. Comparison between measured and predicted (a, b) CBP7, (c, d) CBP8 and (e, f) T₂LM values in test data versus depth.

successful for estimation of the other CMR bin porosities. By investigating the effects of neural network training parameters on the developed models, it was observed that the number of neurons in the hidden layer did not affect the models' performances considerably. That is why all NN models were developed with 4 neurons in their hidden layer. Even though the number of NF training epochs was greater than 100 in all cases, the optimal NF models were achieved with epochs lower than 50. The CMIS is a cost-effective, time saving and reasonably accurate method that performs better than the individual experts acting alone to estimate the NMR T₂ distribution and T₂ logarithmic mean data from one specific set of conventional well logs.

6. Conclusion

The aim of the present paper was to examine the possibility of synthesizing the T₂ distribution curves from a set of predefined available well log data. The results of this study showed that a committee machine with intelligent systems is capable of predicting the T₂ distribution data and T₂LM values from one specific set of conventional well log data with a reasonable degree of accuracy. Considering the required time, effort and costs of NMR logging in a field-wide extent, this methodology is far superior to practically logging all wells of the field. This work demonstrates that using neutron, density, sonic, lithology, gamma ray and effective

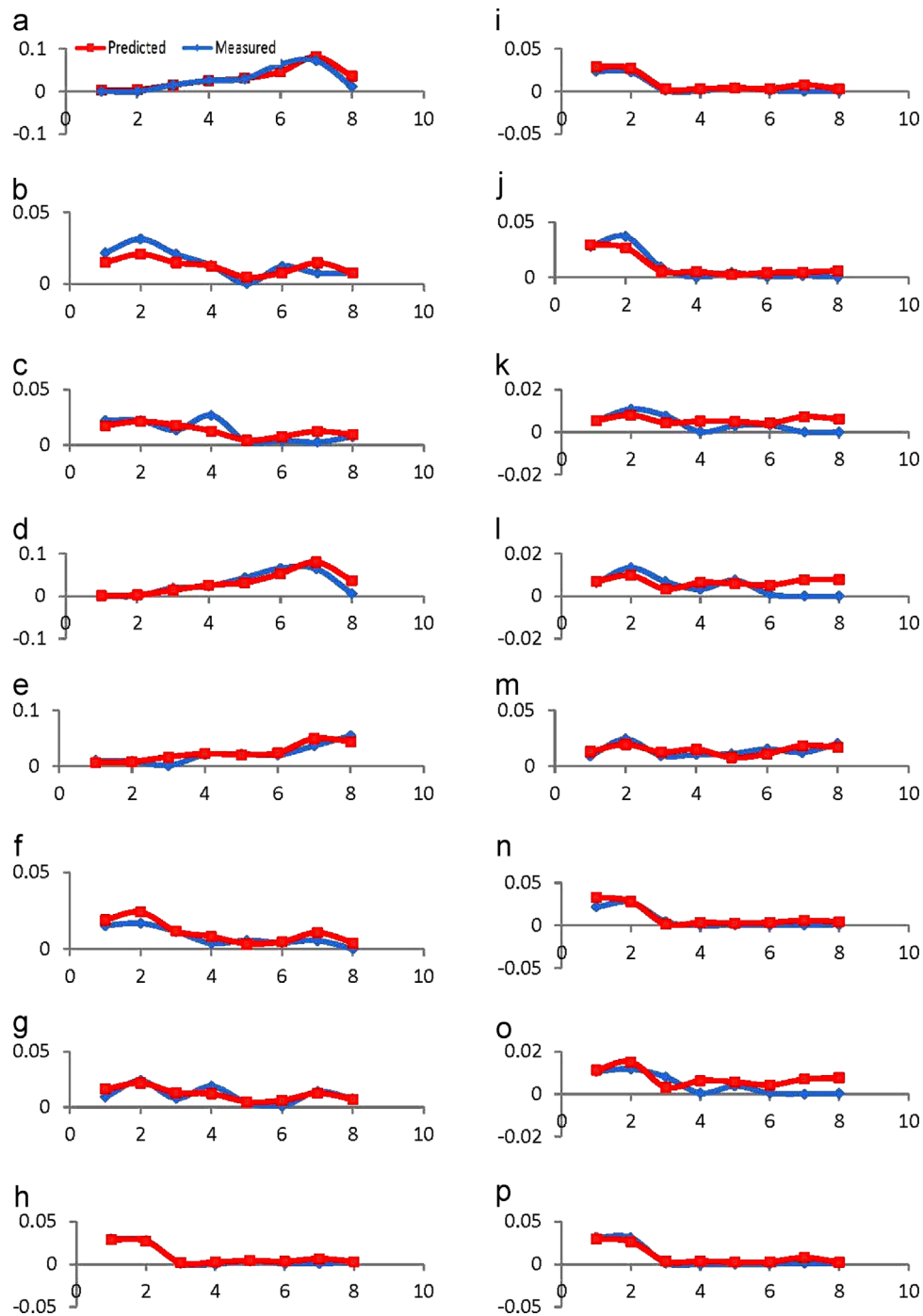


Fig. 8. Comparison between measured and predicted T_2 curves for some test samples (a–p).

porosity logs can offer prediction of T_2 data in a convenient way. A specific set of input well log data with both adequacy and accuracy possesses the NMR related features of the Earth; hence the intelligent models are able to find logical patterns between well logs and NMR T_2 distribution data.

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Article

NMR-Based Study of the Pore Types' Contribution to the Elastic Response of the Reservoir Rock

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Abstract: Seismic data and nuclear magnetic resonance (NMR) data are two of the highly trustable kinds of information in hydrocarbon reservoir engineering. Reservoir fluids influence the elastic wave velocity and also determine the NMR response of the reservoir. The current study investigates different pore types, i.e., micro, meso, and macropores' contribution to the elastic wave velocity using the laboratory NMR and elastic experiments on coal core samples under different fluid saturations. Once a meaningful relationship was observed in the lab, the idea was applied in the field scale and the NMR transverse relaxation time (T_2) curves were synthesized artificially. This task was done by dividing the area under the T_2 curve into eight porosity bins and estimating each bin's value from the seismic attributes using neural networks (NN). Moreover, the functionality of two statistical ensembles, i.e., Bag and LSBoost, was investigated as an alternative tool to conventional estimation techniques of the petrophysical characteristics; and the results were compared with those from a deep learning network. Herein, NMR permeability was used as the estimation target and porosity was used as a benchmark to assess the reliability of the models. The final results indicated that by using the incremental porosity under the T_2 curve, this curve could be synthesized using the seismic attributes. The results also proved the functionality of the selected statistical ensembles as reliable tools in the petrophysical characterization of the hydrocarbon reservoirs.

Keywords: NMR relaxation; elastic response; statistical ensembles; deep learning; coalbed methane

1. Introduction

Nuclear magnetic resonance as a characterization technique of rock media is one of the best techniques to evaluate the presence, quality, and specifications of the fluids in different geological settings. While the conventional logging methods [1–3] strongly suffer from the interferences of the inevitable lithological variations, NMR logging simply acts based on the fluids within the pores. This property brings many superiorities to NMR when compared to other logging techniques which are deeply discussed in the published

[4–11]. Moreover, despite the complexity of multiple-porosity systems, NMR has shown the capability of producing high-resolution information in these media. On the other hand, nowadays petroleum exploration industry relies on the reflection seismic surveying data as the only tool that could provide the required information of the very deep subsurface formations with a reliable degree of accuracy [12–15]. From such a point of view, NMR data and seismic data, are considered two of the most precise techniques of exploring hydrocarbon-bearing formations [16,17]. Nevertheless, both these techniques come up with high technical requirements and the subsequent costs are their primary bottleneck. Hence, once these two kinds of data are recorded, researchers and engineers implement various approaches to obtain the most possible amount of information hidden within them. Herein, a notable research direction has been employing artificial intelligence techniques to obtain various information on reservoir properties from either kind of the above-mentioned data. These studies have investigated obtaining permeability and porosity from NMR and conventional logs [8,10,18–21] or estimating petrophysical characteristics from the combination of seismic attributes and well-log data [22–25]. Meanwhile, novel types of neural networks and deep learning have been introduced for feature engineering of geological data [26–29].

Considering that the seismic data are recorded in almost all hydrocarbon exploration projects, this research investigates the feasibility of employing seismic data for estimating the NMR bin porosities and then synthesizing NMR T_2 relaxation curves. A bin refers to a subsection of the area under the T_2 curve whilst the entire area represents the overall porosity of the sample. This porosity could also be estimated from the elastic response of the sample in the lab or the equivalent seismic response in the field-scale. Considering that NMR enables us to distinguish between various pore types, i.e., micro, meso, and macropores, in the present study, we carried out elastic and NMR measurements on the same samples in the lab and investigated the contribution of each pore type to the velocity of the compressional wave. We also observed how the fluid saturation affects the elastic and NMR response of an individual core sample and determined the relationship between the changes of the T_2 curve and compressional P-wave velocity under decreasing water saturation. Based on these observations, we then used the intelligent networks to estimate the incremental NMR porosity values from seismic attributes and synthesized the NMR curves in the field scale. To the authors' best knowledge, neither the relationship between the elastic and NMR response under the effect of fluid saturation nor the introduced method of synthesizing the NMR T_2 curves have not been reported in the existing literature. In the next step, we explored the functionality of statistical ensembles for determining the petrophysical properties of the reservoir and compared their performance with that of the deep learning neural networks. We focused on Bag and LSBoost ensembles. We believe they are also a suitable tool for petrophysical estimations and are equally efficient and handy as neural networks. The performance of these two ensembles was assessed by estimating NMR permeability values. Throughout the modeling procedure, porosity was adopted as a benchmark to confirm reliability. The studied statistical ensembles proved to be of similar strength as the deep learning network.

Reservoir fluids are the shared link between the elastic and NMR responses of the reservoir rock. Fluid properties such as type, saturation, and density considerably affect both these properties. The existing studies have already proved that the pore fluids have a significant effect on the acoustic impedance and the Poisson's ratio which is directly correlated with seismic amplitudes [30]. These studies have also verified that the permeability of the reservoir is directly related to its seismic response. A comprehensive discussion on how permeability affects the seismic response and also using seismic attribute analysis for mapping lateral permeability variations of the reservoir can be found in [31] and [32]. Meanwhile, many studies have determined the relationship between the elastic properties, confining pressure, saturation, and microstructure of the coal and non-coal rocks [33–40]. Even though all these studies highlight that the fluids impact the acoustic and NMR measurements, no study has deeply investigated the relationship between these

two different impacts in the laboratory. Meanwhile, even though many researchers have tried to predict various NMR logs from the well-logging data, no study has used the seismic data to estimate the incremental porosity under the T_2 curve and synthesize artificial T_2 distribution curves.

2. Laboratory Experiments of Elastic and NMR Response

2.1. Core Samples

The studied reservoir was the Qinnan coal reservoir located at the southern Qinshui Basin in northern China. The methane-bearing formations of this reservoir are of Carboniferous and Permian ages. The structure of the coal seam in the study area is simple and the coal seam thickness is rather stable with a high gas content. Figure 1 represents the locations of the wells in the studied field.

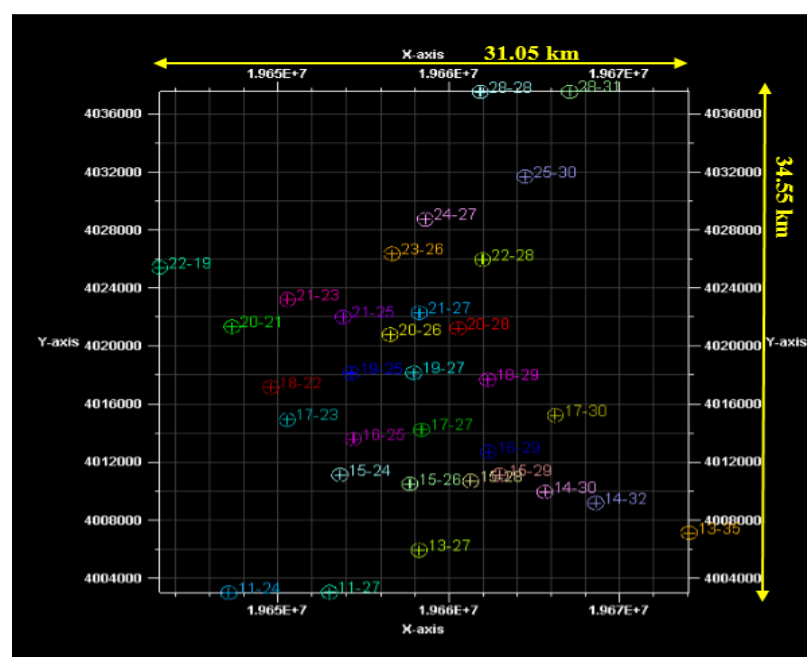


Figure 1. The drilled wells in the coalbed methane (CBM) gas field from where the seismic data and core samples were retrieved.

2.2. Relationship between the Elastic and NMR Responses

Two core samples extracted from the study area were prepared in the laboratory and the elastic wave velocity and NMR T_2 relaxation curves were measured at different degrees of water saturation. First of all, the samples were dried in the oven for 72 h at 80 °C, and then saturated in a vacuum saturation tank. After the full saturation of the samples was achieved, the compressional wave velocity (V_p), as well as the NMR transverse relaxation (T_2) curves, were measured. Next, it was time to desaturate the samples and conduct these two tests again. The desaturation was conducted using centrifuging technique. The measurements were conducted on the samples at three different states, i.e., fully saturated, partially saturated, and irreducible saturation states. For getting the partially saturated state, the two samples were centrifuged at 2000 rpm (revolution per minute) for 45 min, and then the measurements were conducted. Considering that the NMR machine applies a rather high temperature on the samples, we first conducted the elastic experiments and then carried out the NMR tests. After the first set of measurements, the samples were not put back into the centrifuge machine as it was not certain how was the fluid situation inside the pore space after all the procedures applied to the samples. Therefore, the samples were moved back to the saturation tank and were pressurized for two days. Then, the second round of centrifugation was carried out which lasted for 90 min under

3000 rpm. To prevent the breaking of coal samples' weak skeleton under the centrifuge force, a low revolution speed was applied and the experiment time was increased in order to get to the irreducible fluid saturation. After 1.5 h at this stage, no significant change was observed in the samples' weight and NMR relaxation curves. This implied that the water in the meso and macropores had been removed from the pore space. Finally, the third set of measurements was conducted. It is worth noting that the elastic wave velocity and T_2 relaxation curves were obtained respectively under the room temperature (24–25 °C) and 32 °C. The relaxation phenomenon was monitored using the CPMG (Carr Purcell Meiboom Gill) pulse sequence where the waiting time was 10 s, inter-echo spacing time was 0.3 ms, the number of scans was 64, and the number of peaks was set to 10,000 on the NMR machine. The results of these experiments are shown in Figure 2. It is worth noting that in this figure, the number of points on the relaxation curve is 100. The inversion algorithm for transforming the free induction decay (FID) to the T_2 relaxation curve was the simultaneous iterative reconstruction technique (SIRT).

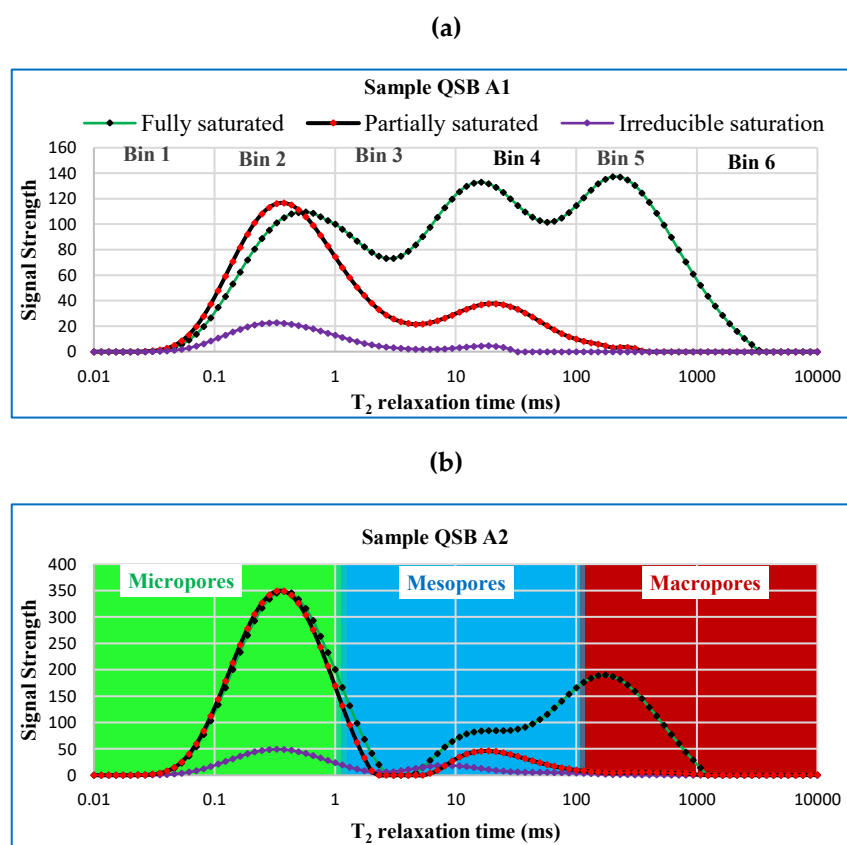


Figure 2. Nuclear magnetic resonance (NMR) T_2 relaxation curves obtained in the laboratory for Sample QSB A1 (a), and Sample QSB A2 (b) under the three states of water saturation, i.e., fully saturated, partially saturated, and irreducible saturation.

While inverting the FID data, the matrix-vector form of the ill-posed inverse Laplace transform problem could be written as:

$$M = Ea$$

where M is the measured decay amplitude of size $m \times 1$, E indicates the kernel matrix with the size of $m \times n$, and a is the T_2 spectrum with the size of $n \times 1$. m and n are the numbers of echo trains and predefined T_2 value, respectively. SIRT algorithm is an optimization technique and functions on the premise of algebraic reconstruction. It begins with a simple back-projection, after which the reconstruction is improved iteratively starting from the layergram [41]. The matrix-vector form of the update equation is derived by:

$$a^{t+1} = a^t + CEE_q^T R(m - E_q a^t)$$

where t represents the steps, E_q is the compressed kernel matrix, and C and R are diagonal matrices that contain the inverse of the sum of the columns and rows of the system matrix represented by:

$$C(j, j) = \frac{1}{\sum_{i=1}^m E_q(i, j)} \quad (j = 1, 2, \dots, n)$$

$$R(i, i) = \frac{1}{\sum_{j=1}^n E_q(i, j)} \quad (i = 1, 2, \dots, m)$$

More details on this subject could be found in (Ge. Et al, 2017) [42].

The area under the NMR curve was divided into six subsections called ‘bins’. The idea is obtained from the Schlumberger CMR tool which divided the T_2 curve into eight bins called CBP (CMR Bin Porosity) and reports the porosity of each bin separately. Bin porosity is equal to the area under one piece of the relaxation curve. The bin porosity of the two samples and the elastic velocities were measured as provided in Table 1. Considering that relaxation occurs faster in smaller pores, they appear at short T_2 times. Hence, the peaks from smaller to larger times respectively represent smaller to larger pores. A peak represents a group of pores with similar sizes. On our test results, we observed three peaks indicating the presence of micropores ($x < 2$ nm), mesopores ($2 < x < 50$ nm), and macropores ($x > 50$ nm) where x indicates the pore size. It is worth mentioning that the microfractures were classified as macropores as well.

Table 1. The results of the laboratory measurements of the elastic and magnetic resonance responses of the coal samples at three states of fully saturated, partially saturated, and irreducible saturation.

	Sample QSB A1			Sample QSB A2		
	Fully Saturated	Partially Saturated	Irreducible Saturation	Fully Saturated	Partially Saturated	Irreducible Saturation
Bin 1	54.83	59.04	17.76	216.22	220.74	34.31
Bin 2	1407.12	1550.77	305.33	4518.45	4534.76	637.77
Bin 3	1520.36	610.10	81.81	934.53	555.25	221.21
Bin 4	1877.52	462.29	29.38	1569.87	516.82	135.66
Bin 5	1919.41	49.09	0.00	2290.21	88.20	19.95
Bin 6	214.95	0.00	0.00	14.12	1.68	0.00
V_P (m/s)	2813.14	1209.59	1174.65	2735	1430.45	1398.60
Bins summed	6994.19	2731.29	434.28	9543.4	5917.45	1048.9
Decrease in the overall bin area	-	-4262.9	-2297.01	-	-3625.95	-4868.55
Saturation	100%	39%	6%	100%	62%	11%
Decrease in V_P (m/s)	-	-1603.55	-34.94	-	-1304.55	-31.85

We observed that the peak of micropores was located in bin 2. Likewise, the peaks of mesopores and macropores respectively appeared in bins 3–4 and bin 5. Therefore, we classified the pore space into three sections of the micropores, mesopores and macropores as shown in Figure 2. Then we analyzed the contribution of each bin and each pore type to the elastic wave velocity as shown in Figure 3 (top).

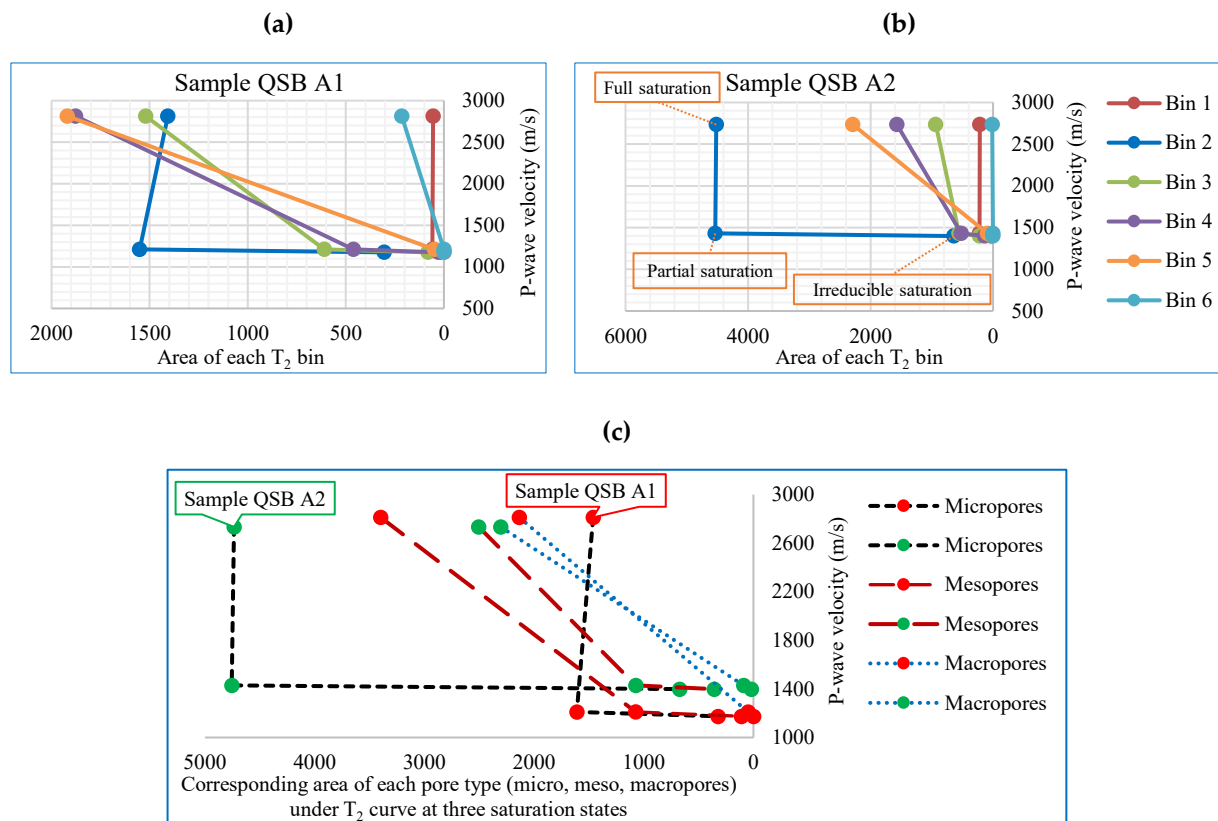


Figure 3. (a, b) Analyzing the contribution of each pore type (micro, meso, and macropores) to the decrease of elastic P-wave velocity for Sample QSB A1 and Sample QSB A2. The three data points on each graph represent the three saturation states, from left to right: full saturation, partial saturation, and irreducible saturation, respectively; (c) The green and red colors indicate the corresponding data points of the two samples. For each sample, there are three curves, which show the changes of microporosity (black dashed line), mesoporosity (dark red long-dashed line), and macroporosity (blue dotted line) of that sample at three saturation states, from left to right: full saturation, partial saturation, and irreducible saturation, respectively. The porosity value for each pore type at each saturation state was calculated by summing the values of bins 1–2 (for microporosity), bins 3–4 (for mesoporosity), and bins 5–6 (for macroporosity).

When centrifuging the samples, the fluid in the larger pores escapes the sample first. This is highlighted by the disappearance of the corresponding peak on the T_2 curve as we observed in the tests as well. When the fluid runs out and larger pores get filled with air, the acoustic velocity drastically decreases. Such a decrease can be seen in Figure 3 (bottom). This phenomenon followed the same trend in each pore type (micro, meso, and macropores) on both samples. Moreover, the rate of decrease (slope of the curve) was the fastest for micropores, then for mesopores, and the slowest for macropores. This is again because of the influence of the pore size on the fluid's capability to move. These observations are in agreement with our fundamental understanding of the NMR and waves' traveling phenomena within the porous media and verify the reliability of the results. Figure 4 illustrates the two adopted samples and also indicates the various pore types observed in the scanning electron microscopy (SEM) images of the samples. As can be seen, pores of different sizes existed inside the samples and the range of the pore size distribution was rather wide. Meanwhile, the existing pores appeared as pore-clusters with different sizes or as independent pores being dispersed at different locations.

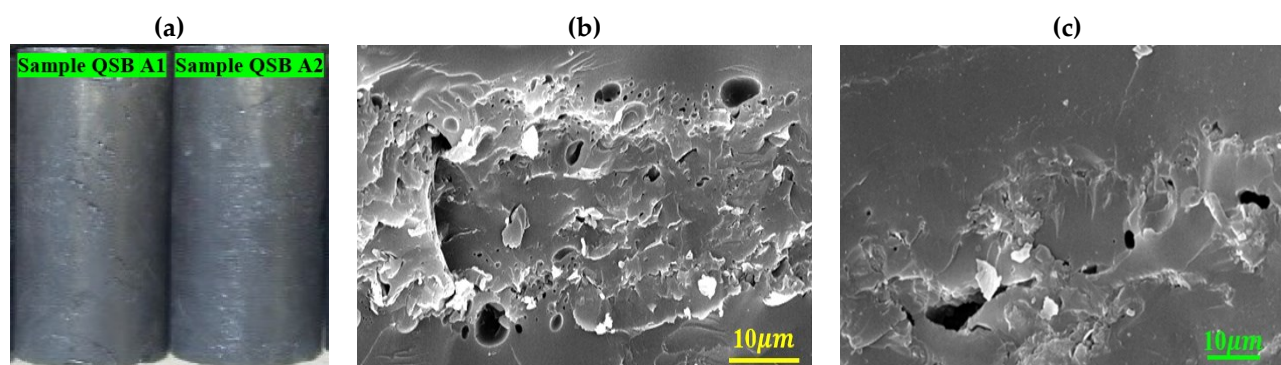


Figure 4. (a) The two studied samples; (b, c) Various pore types, i.e., micro, meso, and macropores inside the Sample QSB A1 and Sample QSB A2, respectively, under the scanning electron microscope. These SEM images only represent the qualitative situation of the internal pore space of the samples and the real bulk porosity was quantitatively measured by the NMR machine.

Figure 5 shows the relationship between the changes in the elastic and NMR responses of the two coal core samples for the cumulative amount of area under the T_2 curve. As follows from this figure, the decreasing saturation causes a decrease in both responses of the core samples. This decrease is fast in higher saturation values (from full to partial saturation) and slows down in the lower saturation values (from partial saturation to irreducible saturation state). This is because, in higher saturations, the fluids considerably stimulate the compressional wave [43–45]. Moreover, since the combined amount of meso and macropores were larger for Sample QSB A1 than the Sample QSB A2 (79.02% vs. 50.4%), the decrease was faster in this sample.

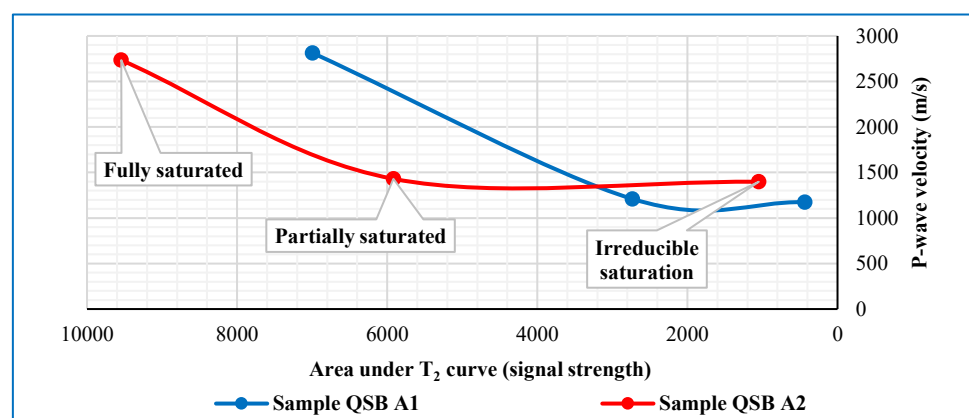


Figure 5. The relationship between the changes in the elastic and NMR responses of the two coal core samples in the laboratory. Three data points from left to right respectively indicate the fully saturated, partially saturated, and irreducible saturation states.

Figure 6 represents the relationship between water saturation with the area under the T_2 curve and P-wave velocity. As expected, and described above, the area under the T_2 curve and P-wave velocity should decrease with the decreasing fluid saturation. We also compared our results with observations of other researchers in the literature [33,43] and found an acceptable agreement.

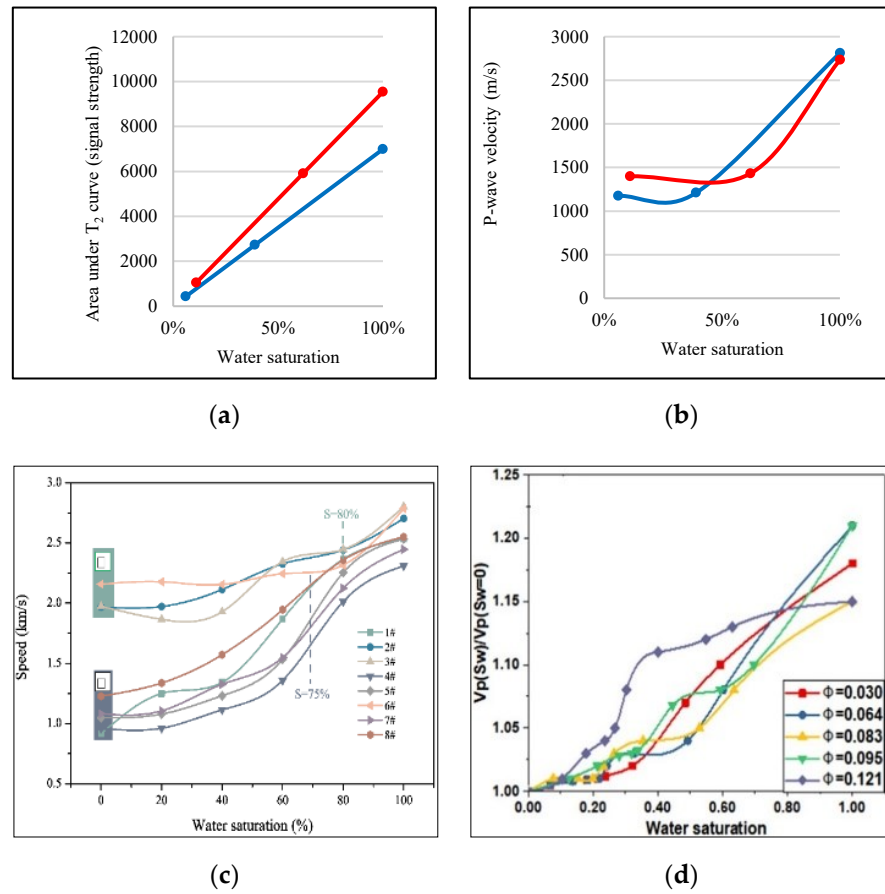


Figure 6. The relationship between water saturation and area under T₂ curve (a), and P-wave velocity (b). Our results were compared with the existing literature: (c) after [33] and (d) after [43].

In our tests, the cut-off values for micro, meso, and macropores were respectively 1, 10, and 100 ms. Finally, we could calculate the contribution of each pore type to the elastic wave velocity using the following relationships. These relationships are for full saturation status. In well logging practices, Wyllie's time equation is generally used to define the relationship between the porosity and sonic transit time as [46–49]:

$$\Delta t = \Delta t_{ma}(1 - \phi_s) + \Delta t_f \phi_s \quad (1)$$

$$\phi_s = \frac{\Delta t - \Delta t_{ma}}{\Delta t_f - \Delta t_{ma}} \quad (2)$$

where Δt , Δt_{ma} , and Δt_f , all in ($\mu s/ft$) respectively indicate is the transit time of elastic wave in the rock sample, the transit time of the matrix, and the transit time of the fluid. Moreover, ϕ_s is the porosity derived from sonic measurements.

Since the porosity of each pore type could be calculated as:

$$\phi_{mic} = \phi_s \frac{\int_{T_2=0.01}^{T_2=1} T_2 dt}{\int_{T_2=0.01}^{T_2=10000} T_2 dt} \quad (3)$$

$$\phi_{mes} = \phi_s \frac{\int_{T_2=1}^{T_2=100} T_2 dt}{\int_{T_2=0.01}^{T_2=10000} T_2 dt} \quad (4)$$

$$\phi_{mac} = \phi_s \frac{\int_{T_2=100}^{T_2=10000} T_2 dt}{\int_{T_2=0.01}^{T_2=10000} T_2 dt} \quad (5)$$

By using the above-described incremental area under the T_2 curve., we could finally determine the contribution of each pore type to the elastic wave's transit time as:

$$\Delta t_{mic} = \phi_{mic}(\Delta t_f - \Delta t_{ma}) + \Delta t_{ma} \quad (6)$$

$$\Delta t_{mes} = \phi_{mes}(\Delta t_f - \Delta t_{ma}) + \Delta t_{ma} \quad (7)$$

$$\Delta t_{mac} = \phi_{mac}(\Delta t_f - \Delta t_{ma}) + \Delta t_{ma} \quad (8)$$

According to the NMR results, micro, meso, and macropores respectively contributed to 21%, 48.5%, and 30.5% of the total porosity of the Sample QSB A1 and to the 49.6%, 26.2%, 24.1% of the total porosity of the Sample QSB A2, respectively. The overall porosities of these two samples were respectively 4.3% and 5.6%. This implies that the micro, meso, and macro porosities of the samples were respectively 0.9%, 2.09%, 1.31% for Sample QSB A1 and 2.78%, 1.47%, and 1.35% for Sample QSB A2. Considering that different pore types occupied dissimilar portions of the pore space in the two samples, their impact on the elastic wave velocity was also different. The length of the samples was 5 cm, so the overall transit times in the two samples were respectively 17.8 μ s and 18.3 μ s, for which respectively 0.76 μ s and 1.03 μ s referred to the pore space. Herein, contributions of the micro, meso, and macropores to the transit time of the elastic wave in Sample QSB A1 were respectively 0.16, 0.37, 0.23 μ s, and in Sample QSB A2 were respectively 0.51, 0.27, 0.25 μ s. It is worth noting that these values represent the average value as the wave propagation inside the pore space happens simultaneously in all pore types.

3. Field-Scale Estimation of Relaxation Curves

3.1. Seismic and NMR Data from the Field

After the impact of the fluid saturation, as well as the contribution of micro, meso, and macropores to the overall transit time of the elastic wave, were determined, the idea was used to estimate NMR bin porosities from the seismic attributes and synthesize artificial T_2 relaxation curves. This stage of the study started by selecting the NMR well-logging data and the seismic data cube. The data was preprocessed and the seismic attributes at the well locations were extracted. These attributes were calculated as described in Table 2.

Table 2. A brief explanation of the most important seismic attributes employed in the current study [50].

Instantaneous Frequency	Instantaneous frequency is the time derivative of the phase. This attribute should not be confused with the frequency of the wavelet. It is often used to estimate seismic attenuation. Oil and gas reservoirs usually cause drop-off of high-frequency components. It helps to measure cyclicity of geological intervals and can be useful for cross-correlation across faults. It could also identify contacts between gas and water or gas and oil. Instantaneous frequency tends to be unstable in the presence of noise and is sometimes difficult to interpret. This attribute is calculated as: $\omega = \frac{\delta\theta}{\delta t}$
Instantaneous Phase	The attribute is calculated on a sample by sample basis without regard of the waveform. The attribute provides an amplitude independent display. It is commonly used to find continuity of weak events and to distinguish small faults and dipping events. Instantaneous phase is calculated as: $\theta = \tan^{-1}(\frac{g}{f})$
Frequency Filter	The frequency-filter attribute is useful for enhancing particular seismic events or to reduce unwanted noise in the data. This can improve the accuracy of interpretation.
Apparent Polarity	The polarity of the instantaneous phase, calculated at the local amplitude extreme. The apparent polarity reveals the sign of the reflection coefficient and therefore indicates features that would change it, e.g., unconformities. It is useful for checking the lateral variation of polarity along a reflection layer. On a noisy seismic section, event continuity can be clearer on the apparent polarity than the original seismic section. This attribute reveals reflection details without the waveform effects and can help detect thick beds when the data is of good quality (less noise).
Amplitude Envelope	The total instantaneous energy of the analytic signal (the complex trace), independent of phase, calculated as: $\sqrt{f^2 + g^2}$ where f and g are the “real” and “imaginary” components of the seismic trace. So, if f is the real part, which are just the original seismic trace samples, g will be the samples from the Hilbert transform (also called quadrature amplitude) of the seismic trace.

Relative Acoustic Impedance	Relative acoustic impedance is a running sum of regularly sampled amplitude values. This attribute is calculated by integrating the seismic trace and passing the result through a high-pass Butterworth filter with a hard-coded cut-off at (10*sample rate) Hz.
Reflection Intensity	Reflection Intensity is the average amplitude over a specified window (9 samples in the present study) multiplied with the sample interval. Reflection Intensity is useful for the delineation of amplitude features while retaining the frequency appearance of the original seismic data.
RMS Amplitude	RMS amplitude is the square root of the sum of the squared amplitudes divided by the number of live samples as shown in the following formula: $\sqrt{\frac{\sum_i^n amp^2}{K}}$
Second Derivative	The second time derivative of the input seismic volume. The combination of the original amplitude, first derivative, and second derivative allows you to express seismic interpretation in relationship to maximum, minimums, greatest descents, and descent polarity. The second derivative can be used to help guide the pick by providing continuity in areas of where reflections are poorly resolved on the raw amplitude. Lateral amplitude variations are visibly diminished, which will make auto-tracking regional events more difficult.

The intelligent models learned to predict CMR bin porosities from the seismic attributes. In the study area, the NMR data was presented by eight bin values, namely CBP1-CBP8. The three most significant components of a seismic trace include its amplitude, frequency, and phase. Therefore, in order to have robust models, the input attributes of the neural network models should include a representative of every property (Figure 7). Eventually, five attributes from among more than 13 seismic attributes including seismic inversion results, integrate of the amplitude values, two-way-time (TWT), derivative instantaneous amplitude, squared inversion, amplitude weighted cosine phase, frequency filter, cosine instantaneous phase, instantaneous frequency, amplitude weighted phase, integrated absolute amplitude, and inverted inversion (inversion⁻¹) etc. were selected as the input elements of the modeling task. These attributes were chosen based on the researchers' knowledge of the study area and also the analysis of cross-correlation values. Alongside the seismic section, Figure 8 illustrates four examples of these attributes at the location of one studied well. Table 3 shows the linear relationship of the inputs and targets of our modeling process that were CBPs and each of the input attributes. It is worth mentioning that for successful modeling using the intelligent estimators, there should be a meaningful relationship between the input and target values, implying that irregular distributions of data on a crossplot would not lead to reliable results. The stronger this relationship is, the better the modeling results would be. While the liner relationship is the simplest form, the neural networks use high-order polynomial relationships, for which the cross-correlation values are much larger than the linear manner.

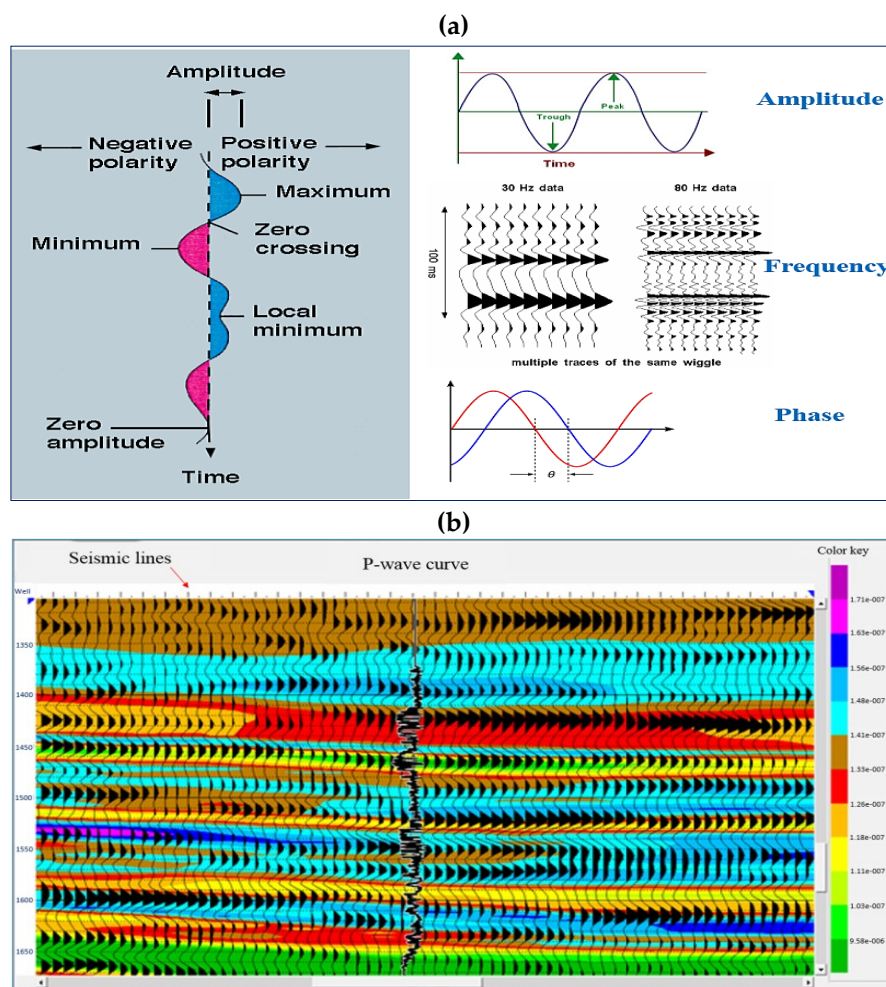


Figure 7. (a) The most significant components of a seismic trace; (b) Raw seismic data (Wiggle traces) and acoustic impedance inversion (color attribute) in the location of one of the studied wells overlain by the P-wave velocity from logging curve.

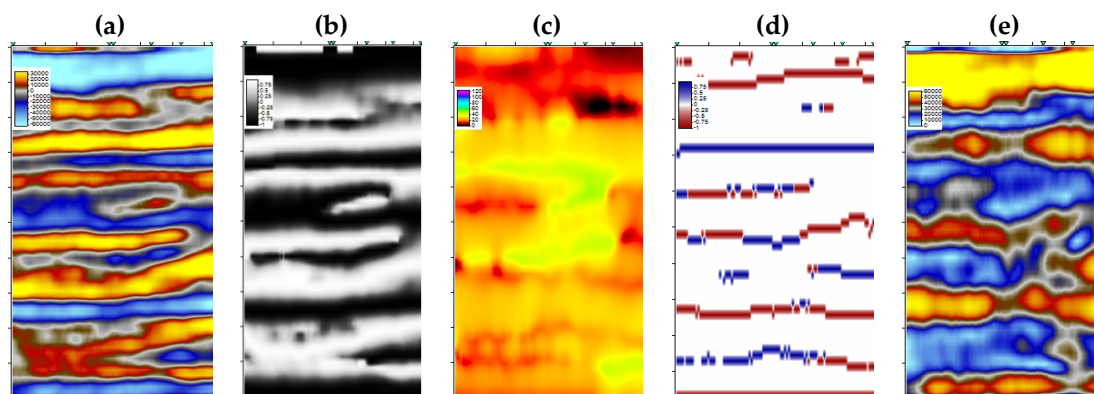


Figure 8. Two-dimensional (2D) seismic section passing through the employed well (a); and its four corresponding seismic attributes, including cosine of phase (b), instantaneous frequency (c), apparent polarity(d), and amplitude envelope (e).

Table 3. Linear relationships of the seismic attributes (inputs) and CBPs (outputs)

	Attribute	R ²		Attribute	R ²
CBP1	Derivative Instantaneous Amplitude	0.1315	CBP2	Inversion	0.1236
	Squared Inverted Inversion	0.1320		Derivative Instantaneous Amplitude	0.1070
	TWT	0.1586		Frequency Filter	0.1650
	Frequency Filter	0.2018		Instantaneous Phase	0.1112
	Amplitude Weighted Cosine Phase	0.1341		TWT	0.1446
CBP3	Squared Inversion	0.2728	CBP4	Amplitude Weighted Cosine Phase	0.2302
	Cosine Instantaneous Phase	0.3860		Instantaneous Frequency	0.2745
	Seismic Amplitudes	0.4238		Frequency Filter	0.4918
	Amplitude Weighted Cosine Phase	0.3140		Cosine Instantaneous Phase	0.5503
	Amplitude Integrate	0.3130		Seismic Amplitudes	0.6225
CBP5	Amplitude Weighted Cosine Phase	0.3602	CBP6	Instantaneous Frequency	0.4461
	TWT	0.3655		Frequency Filter	0.5483
	Squared Inversion	0.4747		Amplitude Weighted Phase	0.4575
	Cosine Instantaneous Phase	0.5098		Cosine Instantaneous Phase	0.4182
	Seismic Amplitudes	0.5747		Seismic Amplitudes	0.4654
CBP7	Derivative Instantaneous Amplitude	0.4885	CBP8	Amplitude Weighted Cosine Phase	0.3428
	Inverted Inversion	0.7804		Derivative Instantaneous Amplitude	0.3969
	Amplitude Integrate	0.6728		Inverted Inversion	0.7129
	Amplitude Weighted Cosine Phase	0.4530		TWT	0.5693
	TWT	0.7173		Integrated Absolute Amplitude	0.1402

3.2. Estimating NMR CBPs from Seismic Attributes

After providing the input seismic attributes and CMR data, the next step was defining the intelligent models. Herein, we used neural networks and fuzzy logic models. In order to have a more reliable estimation, several different neural network algorithms were adopted and the best-performing one was selected as the final estimator. This model produced the smallest error and the best correlation. The input arguments of the constructed models were seismic attributes and the outputs were CBP values of the T_2 curves. Herein, the backpropagation neural networks (BP-NN) were adopted. BP-NN is a type of supervised-training network. We made use of feedforward BP-NN with 20 or 40 neurons in their hidden layers. The tangent sigmoid (TANSIG) function was used to introduce non-linearity to the system. These backpropagation neural networks were trained respectively by Levenberg–Marquardt (LM), resilient back-propagation (RP), one step secant (OSS), scaled conjugate gradient (SCG), Conjugate Gradient with Powell/Beale Restarts (CGB), and back-propagation with adaptive learning rate (GDA). Finally, the network with the best performance was selected as the final intelligent model for estimating each CBP value. The performance was determined using the mean square error (MSE) and the determination coefficient (R^2).

Table 4 represents the number of data points used either for constructing or for testing the accuracy of the final estimators. This means that we selected about 80% of our data for making the models and used the other 20%, which was unforeseen by the models during their construction, to test the models' accuracy and prove their reliability. It is worth mentioning that for each of the employed algorithms, an adequately large number of models were developed. Since fuzzy logic and ANFIS (hybrid neuro-fuzzy) models did not perform as well as neural networks, the results are not represented.

Table 4. Properties of the training data and performance of the intelligent models.

Target Curve	CBP1	CBP2	CBP3	CBP4	CBP5	CBP6	CBP7	CBP8
No. of input attributes	5	5	5	5	5	5	5	5
Total No. of available data pairs	520	277	476	536	545	431	503	568
No. of training and validation data	442	235	405	456	463	366	428	483
No. of test data	78	42	71	80	82	65	75	85
No. of neurons in hidden layers	30	30	30	30	30	30	30	30
Iteration the best ANN model was achieved	321	203	311	260	288	346	198	169
Performance of the ANN model (MSE)	1.12 $\times 10^{-05}$	2.27 $\times 10^{-05}$	1.05 $\times 10^{-05}$	1.80 $\times 10^{-04}$	8.41 $\times 10^{-05}$	7.06 $\times 10^{-06}$	6.92 $\times 10^{-06}$	2.29 $\times 10^{-05}$
Correlation with measured data (R^2)	0.93	0.81	0.88	0.99	0.98	0.94	0.98	0.94

As mentioned earlier, the ANN models varied in the number of their internal neurons and their training algorithms. Each ANN model was retrained for 400 times and the optimal model was selected. The iteration at which the optimal was achieved is outlined in Table 4. Additionally, to guarantee the comprehensiveness of the developed estimators, the training and test data sets were chosen in a way that would track the trend of geological variations reflected in the NMR T_2 relaxation curves. Concerning the neural network modeling tasks, the training algorithm is the most crucial factor that would influence the quality of the prediction task. On account of this fact, in this research, we used six independent training algorithms and observed their estimation capability. This comparison is represented in Tables 5–8. These tables show the agreement of estimated CBP values with the measured ones.

Table 5. Comparison between the performance of various ANN training algorithms for bins 1 and 2.

Training Algorithm	Modeled Parameter	CBP1		CBP2	
		R^2	MSE	R^2	MSE
conjugate gradient with Powell/Beale restarts (CGB)		0.73	7.19×10^{-05}	0.67	6.95×10^{-05}
back-propagation with adaptive learning rate (GDA)		0.67	9.49×10^{-05}	0.66	5.32×10^{-05}
Levenberg–Marquardt (LM)		0.93	1.12×10^{-05}	0.81	2.27×10^{-05}
one step secant (OSS)		0.63	1.35×10^{-04}	0.71	1.70×10^{-05}
resilient back-propagation (RP)		0.62	9.21×10^{-05}	0.66	3.43×10^{-05}
scaled conjugate gradient (SCG)		0.76	4.65×10^{-05}	0.69	3.12×10^{-05}

Table 6. Comparison between the performance of various ANN training algorithms for bins 3 and 4.

Training Algorithm	Modeled Parameter	CBP3		CBP4	
		R^2	MSE	R^2	MSE
conjugate gradient with Powell/Beale restarts (CGB)		0.78	2.60×10^{-05}	0.94	1.60×10^{-03}
back-propagation with adaptive learning rate (GDA)		0.71	1.52×10^{-05}	0.87	3.20×10^{-03}
Levenberg–Marquardt (LM)		0.88	1.05×10^{-05}	0.99	2.00×10^{-04}
one step secant (OSS)		0.79	2.64×10^{-05}	0.92	1.20×10^{-03}
resilient back-propagation (RP)		0.76	3.02×10^{-05}	0.88	3.00×10^{-03}
scaled conjugate gradient (SCG)		0.83	3.66×10^{-05}	0.94	1.50×10^{-03}

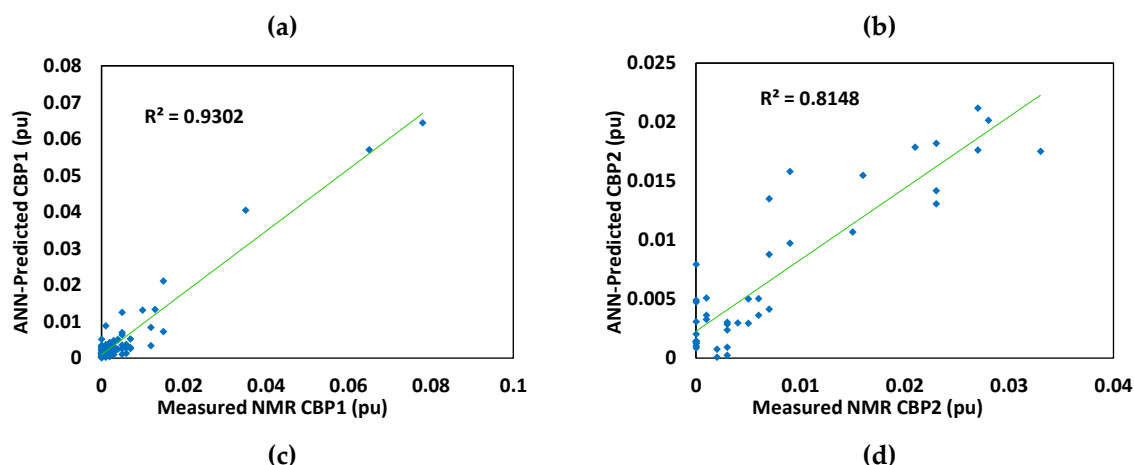
Table 7 Comparison between the performance of various ANN training algorithms for bins 5 and 6

Training Algorithm	Modeled Parameter	CBP5		CBP6	
		R ²	MSE	R ²	MSE
conjugate gradient with Powell/Beale restarts (CGB)		0.94	2.93×10 ⁻⁰⁴	0.91	1.12×10 ⁻⁰⁵
back-propagation with adaptive learning rate (GDA)		0.93	2.85×10 ⁻⁰⁴	0.83	3.17×10 ⁻⁰⁵
Levenberg–Marquardt (LM)		0.98	8.41×10 ⁻⁰⁵	0.94	7.06×10 ⁻⁰⁵
one step secant (OSS)		0.93	3.14×10 ⁻⁰⁴	0.84	2.09×10 ⁻⁰⁵
resilient back-propagation (RP)		0.91	3.99×10 ⁻⁰⁴	0.88	1.49×10 ⁻⁰⁵
scaled conjugate gradient (SCG)		0.92	2.62×10 ⁻⁰⁴	0.90	1.39×10 ⁻⁰⁵

Table 8. Comparison between the performance of various ANN training algorithms for bins 7 and 8.

Training Algorithm	Modeled Parameter	CBP7		CBP8	
		R ²	MSE	R ²	MSE
conjugate gradient with Powell/Beale restarts (CGB)		0.95	1.54×10 ⁻⁰⁵	0.90	4.04×10 ⁻⁰⁵
back-propagation with adaptive learning rate (GDA)		0.95	1.49×10 ⁻⁰⁵	0.90	4.11×10 ⁻⁰⁵
Levenberg–Marquardt (LM)		0.98	6.92×10 ⁻⁰⁶	0.94	2.29×10 ⁻⁰⁵
one step secant (OSS)		0.94	1.90×10 ⁻⁰⁵	0.91	4.14×10 ⁻⁰⁵
resilient back-propagation (RP)		0.93	1.95×10 ⁻⁰⁵	0.90	4.23×10 ⁻⁰⁵
scaled conjugate gradient (SCG)		0.95	1.62×10 ⁻⁰⁵	0.92	4.20×10 ⁻⁰⁵

According to the obtained results, among the adopted techniques, the Levenberg–Marquardt training algorithm was selected as the best performing model for estimating all CBP values. While for NN models, the training algorithm is the most influencing factor on the quality of the output results, for the fuzzy and neuro-fuzzy models, clustering radius is the most important factor. However, neither fuzzy nor neuro-fuzzy models did not lead to acceptable results. It is worth mentioning that in our study most neural networks came to relatively acceptable results. However, Table 4 only presents the performance of the best models rather than all developed models. After all, CBP2 and CBP3 were predicted weaker than the others. These two bins fall in the category of micropores and the reason for their lower prediction would be the complexity of the magnetization relaxation in smaller pores. Nevertheless, all other bin values were estimated with an accuracy of more than 93%. We believe the most important reason for such a good performance is that the input data comprehensively encompassed different petrophysical features of the geological formations. The cross-correlations of the predicted and measured CBP values are illustrated in Figure 9.



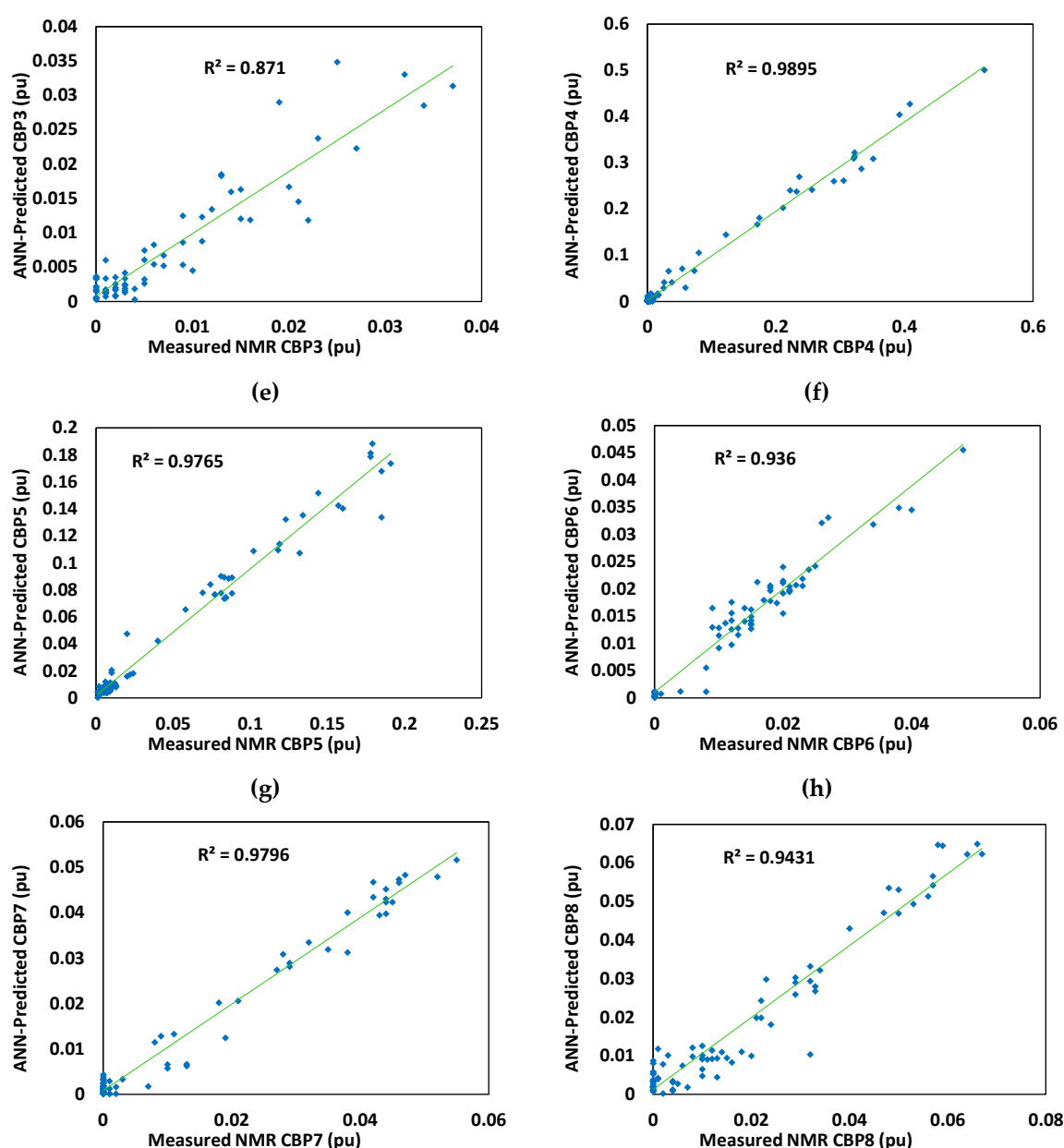


Figure 9. Correlations of the measured values of bin porosities with the predicted values (pu indicates the porosity unit): (a) CBP1; (b) CBP2; (c) CBP3; (d) CBP4; (e) CBP5; (f) CBP6; (g) CBP7; (h) CBP8.

Finally, the T_2 relaxation curves could be synthesized. Figure 10 depicts the comparison between the measured and predicted T_2 curves at ten different depths of the test well. These ten points serve as an example and, based on the established models, the simulation process would be done in any desired section of the well or the area surrounding the well as far as no considerable changes happen in the geological conditions of the objective area compared to the well locations where the models were developed. As can be seen from this figure, models performed satisfyingly in estimating T_2 relaxation curves and successfully draw out the general trend of relaxation variations that can be very useful in case of the quick, inexpensive, and in handy evaluation of the geological formations.

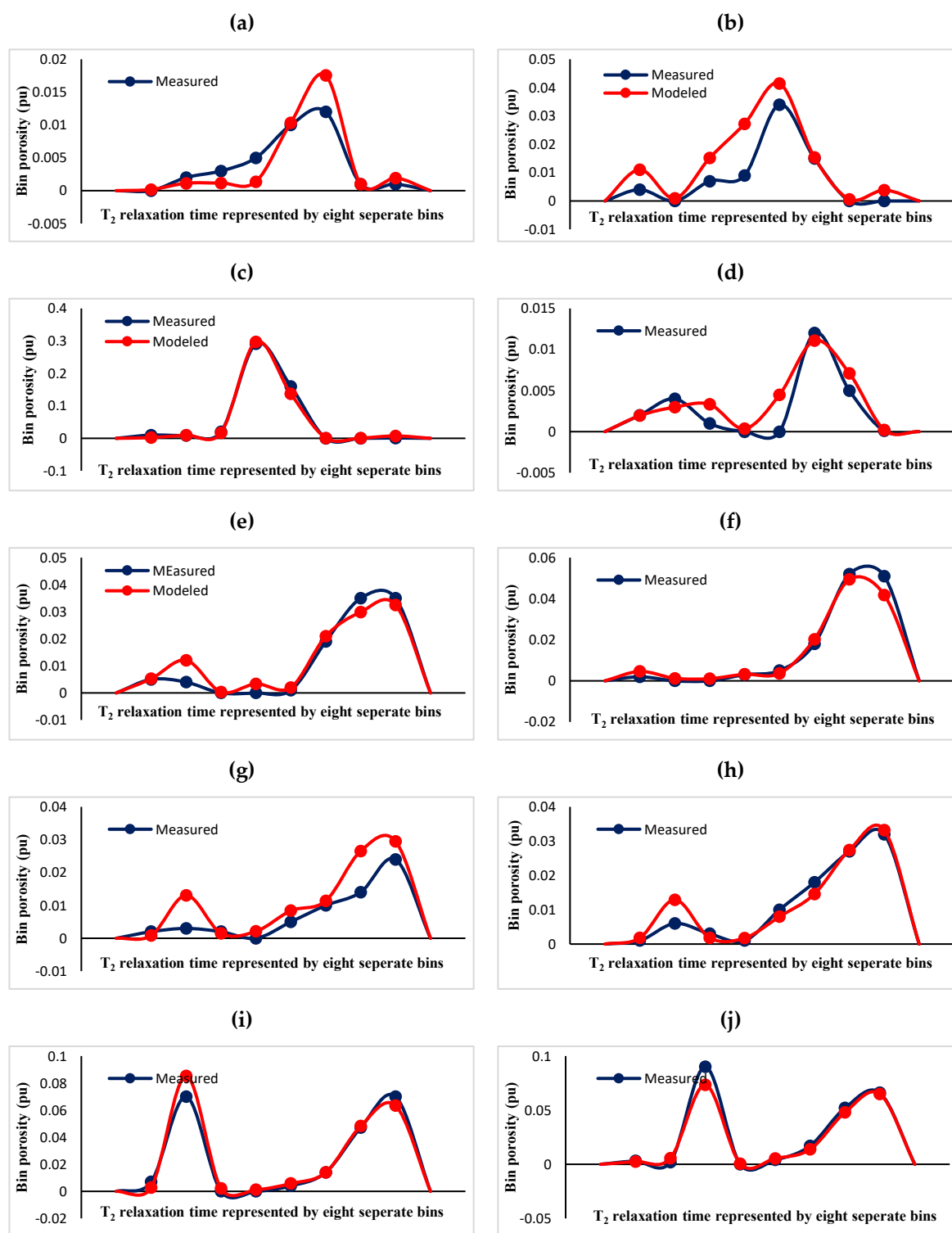


Figure 10. (a–j) Measured and predicted T₂ relaxation curves at ten example points of one studied well. Such relaxation curves could be synthesized for every desired part of the reservoir using the established models and the introduced method.

After we predicted NMR T₂ curves in the location of the studied wells, and since the models proved to be trustworthy, we tried to upscale our prediction model and estimate the T₂ curves for areas around one of the wells in the studied coal formation. Figure 11 illustrates the 3D structure of this formation in the entire coalbed methane field. Figure 12 shows the predicted curves at the corresponding depths of the coal formation, and also

illustrates the location of the simulation area and the position of the well around which the models were upscaled. Figure 13 represents the results of the upscaling procedure and shows the simulated NMR T_2 transverse relaxation curves at various depths.

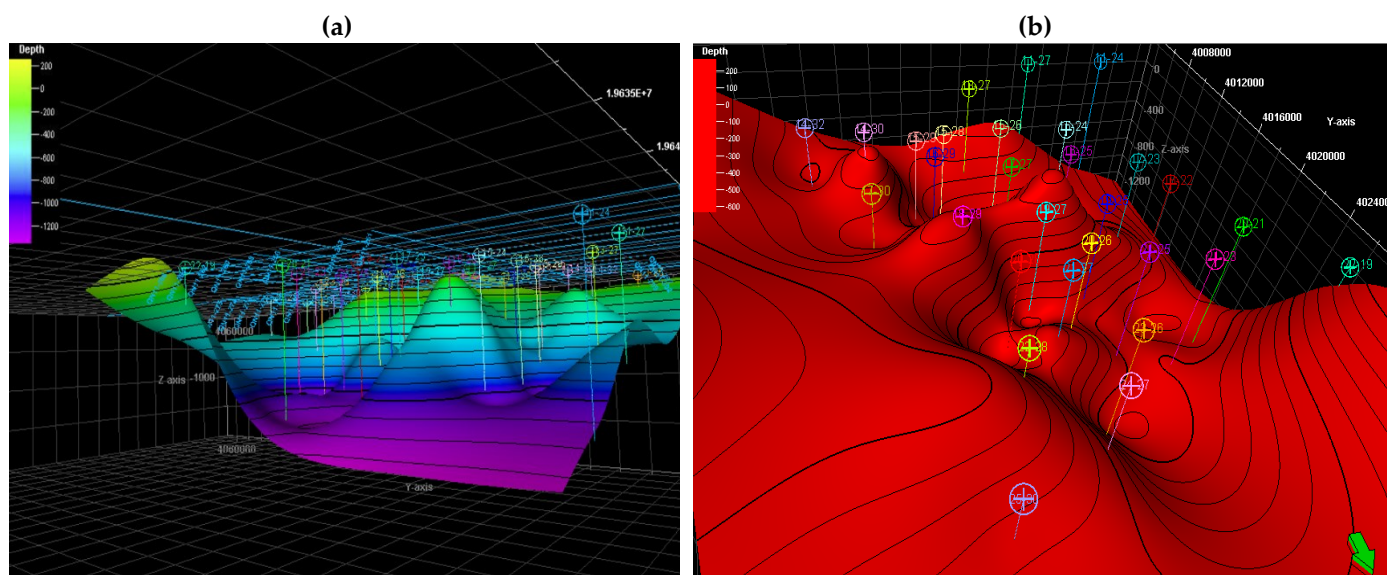


Figure 11. (a, b) The coal formation in the subject CBM gas field from two different views. The thickness of the coal layer is in the range of 2–8.3 m.

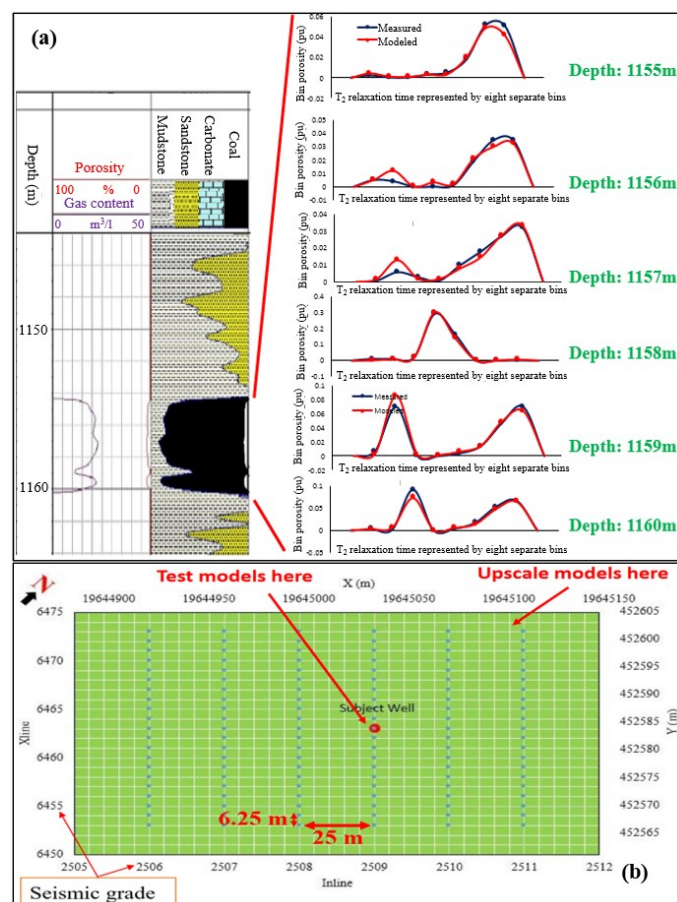


Figure 12. (a) Predicted T_2 curves in the corresponding depths on the well-logging section; (b) Map of the areas and position of the selected simulation nodes compared to the subject well. The spacing

of the seismic survey inlines and xlines are respectively 25 m and 6.25 m on the map. Therefore, the simulation covered an area of $125 \times 125 \text{ m}^2$ at the depth of 1157 m.

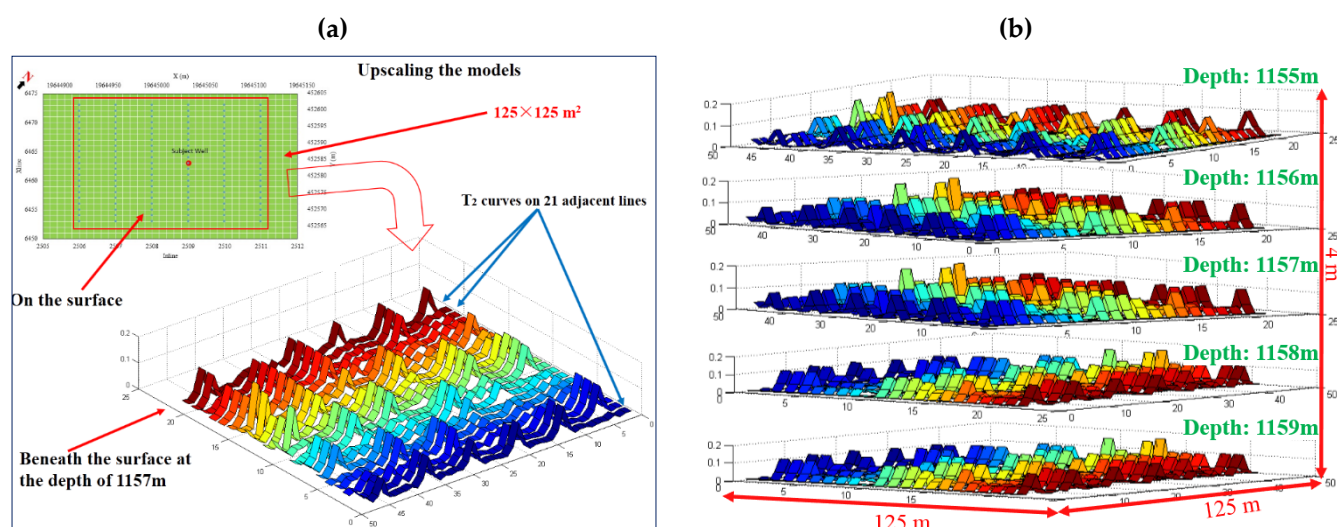


Figure 13. (a) The finally simulated T₂ curves in the depth of 1157 m and their corresponding position on the surface; (b) Simulated T₂ curves at various depths of the reservoir which could be estimated for any desired area around the well as far as no considerable geological changes happen.

4. Statistical Models and Deep Learning

4.1. Bag and LSBoost Ensembles

The present study examined the performance of the statistical Bag and Least Square Boosting (LSBoost) ensembles in predicting the petrophysical properties of the reservoir and compared their performance with those of deep learning regression networks. Statistical ensembles are user-friendly and in handy which could be of great applications in the assessment of the reservoir layers. They also require less memory than intelligent techniques when analyzing large-sized field data.

Statistical supervised learning splits into two broad categories namely classification, and regression [51]. The regression technique is used when the problem involves modeling and analyzing the relationship between a dependent and one or more independent variables. Within the regression category, ensemble learning is a framework of combining multiple weak learning algorithms to produce a strong learning algorithm [52]. The present research utilized regression ensembles, as the statistical supervised learning machines, to obtain permeability and porosity information from 3D seismic attributes. The ensemble algorithms used herein were regression Bag and Least Square Boosting (LSBoost) that were implemented in the MATLAB environment. Bagging and boosting algorithms both learn from weak learners, but Bagging trains weak learners in parallel while boosting trains them in a sequential manner (Figure 14) [53]. The ensembles in this research included decision trees as their weak learners to predict responses to the input data. The selection of the number of trees inside each type of ensemble algorithms was based on the performance analyses of 200 various ensembles. This means that by changing the number of trees in the range of [1–200], numerous Bag and LSBoost ensembles were created and their performance in predicting the new data was observed. The purpose of this step was to obtain the optimal number of weak learners (Figure 15). At the next step, the most reliable ensemble was selected as the optimal model for further usage and predicting permeability or porosity. The method which was used for evaluating this final ensemble's quality was using an independent test data set (30% of the available well data). It is worth mentioning that Bagging is oriented towards decreasing variance and boosting is oriented towards decreasing bias, and LSBoost is faster in training.

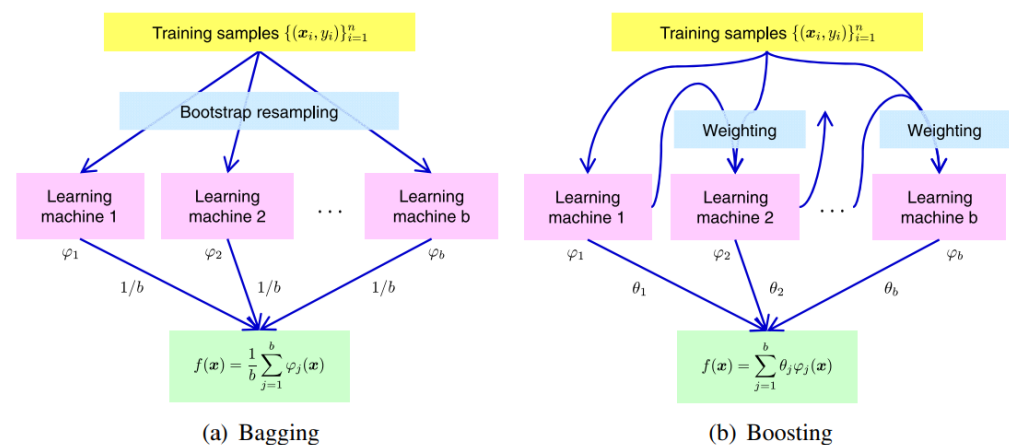


Figure 14. Ensemble learning. Bagging (a) trains weak learners in parallel while boosting (b) trains them sequentially. [53].

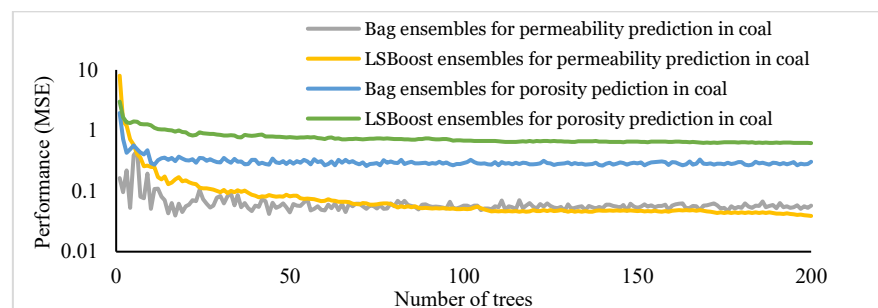


Figure 15. Prediction error versus the number of trees for statistical ensembles in the studied unconventional coal reservoir.

4.2. Deep Learning and Deep Belief Network

Deep learning is rather the latest achievement of artificial intelligence and machine learning that has captured the imagination of countless scientists and researchers. Deep learning refers to the in-depth feature engineering through successive layers of neural networks for increasingly meaningful representations of the data [54]. Deep herein doesn't mean any kind of deeper understanding of the data, but it rather means the larger number of the functioning layers. A deep learning network consists of an input layer, several hidden layers, and an output layer. Deep architectures would be constructed by stacking autoencoders or by using deep belief networks (DBNs) [55]. DBN is the most effective technique among all deep learning models and is the technique of stacking many individual unsupervised networks that use each network's hidden layer as the input for the next layer. The DBN implemented in the present study stacked restricted Boltzmann machines (RBM) in the hidden layers. The deep hierarchy of a DBF transforms the input features into nonlinear higher-level abstractions to extract more significant patterns and follow the trends in the input space [56]. These layers are first trained in an unsupervised manner. The structure of our adopted DBN consisted of four hidden layers with the sizes of 200, 100, 50, and 25 nodes, with eight input neurons (seismic attributes) and one output neuron (permeability or porosity). The RBMs were trained with a sigmoid activation function. They would also be trained using rectified linear activation units (ReLU). It is worth noting that in a deep learning network, it is also feasible to keep the size of all hidden layers the same. However, this approach was not implemented in this study. An RBM consists of a visible and a hidden layer where the hidden units are not connected to each other but have undirected, symmetrical connections to the visible units [56]. In other words, every node in the visible layer is connected to every node in the hidden layer, but

no nodes in the same group are connected as depicted in Figure 16. The details on RBM could be found in [57,58].

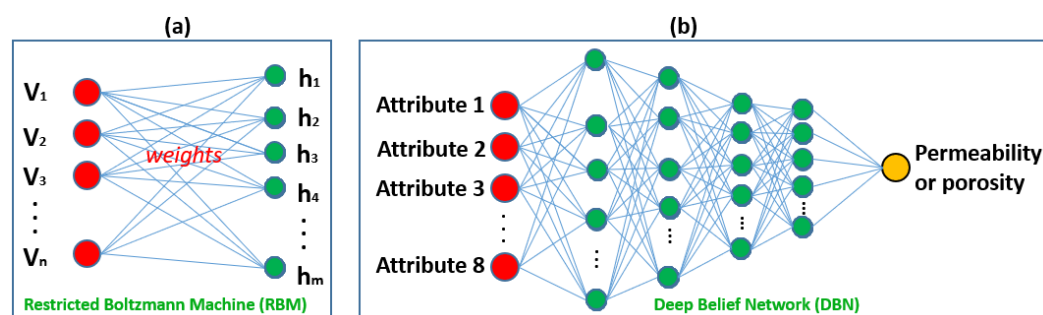


Figure 16. (a) Graphical description of restricted Boltzmann machine (RBM) with n visible and m hidden units; (b) deep belief network (DBN) adopted in the present study with four hidden layers, eight input neurons, and one output neuron.

If the sigmoid activation function is:

$$S(y) = \frac{1}{1 + e^{-y}} \quad (9)$$

Then, the output of the neuron will be:

$$p(S_i = 1) = \frac{1}{1 + \exp(-b_i - \sum_j S_j w_{ij})} \quad (10)$$

where w_{ij} is the weight value that the j^{th} hidden layer assigns to the i^{th} input value [56].

The selected attributes for developing the statistical and artificial neural network (ANN) models of the reservoir are represented in Table 9, and the seismic cube from which the attributes were extracted is shown in Figure 17. This table also shows the dependence of each target parameter, namely porosity and permeability, on each of the input parameters. For this reservoir, the permeability and porosity models were developed respectively using 231, and 355 data points. Both these parameters were estimated using 8 input attributes. Herein, approximately 70% of the entire data set was employed to train the models. The rest of the available data was used to assess and test the robustness of the models when feeding them with unforeseen data. Respectively 46 and 60 data pairs were implemented in this step to evaluate the built permeability and porosity models of the coal reservoir. The data were randomly selected from varying depths of the reservoir to guarantee the inclusiveness of the models. The portion of the data used for constructing statistical models is provided in Table 10. Table 11 represents the performance of the three final models in predicting permeability and porosity values. Figures 18 and 19 graphically illustrate the agreement of the synthesized permeability and porosity curves with the measured log data. As can be seen, permeability and porosity were all estimated with an accuracy of more than 96%. The obtained results of this section confirmed that the statistical ensembles could perform as strong as neural networks in modeling the petrophysical properties of the reservoirs and could be considered as a reliable tool. The authors believe that such tools further facilitate manipulation of the large-sized seismic and well logging data that usually demand a big size of computer memory for their processing and interpretations. It is worth noting that while there has been a great deal of hype surrounding neural networks (NN) and making them seem magical, these networks are just nonlinear statistical models [59–63]. In early NN models, the neuron fired when the total signal passed to it exceeded a certain threshold. This threshold value determined if the neuron would be activated inside the model or not. This principle corresponded to the use of a step function as the activation function of the neuron. Nevertheless, later NN models were recognized as a useful tool for nonlinear statistical modeling, and for this purpose, the step function which was not smooth enough for optimization was replaced by a smoother sigmoid function [59].

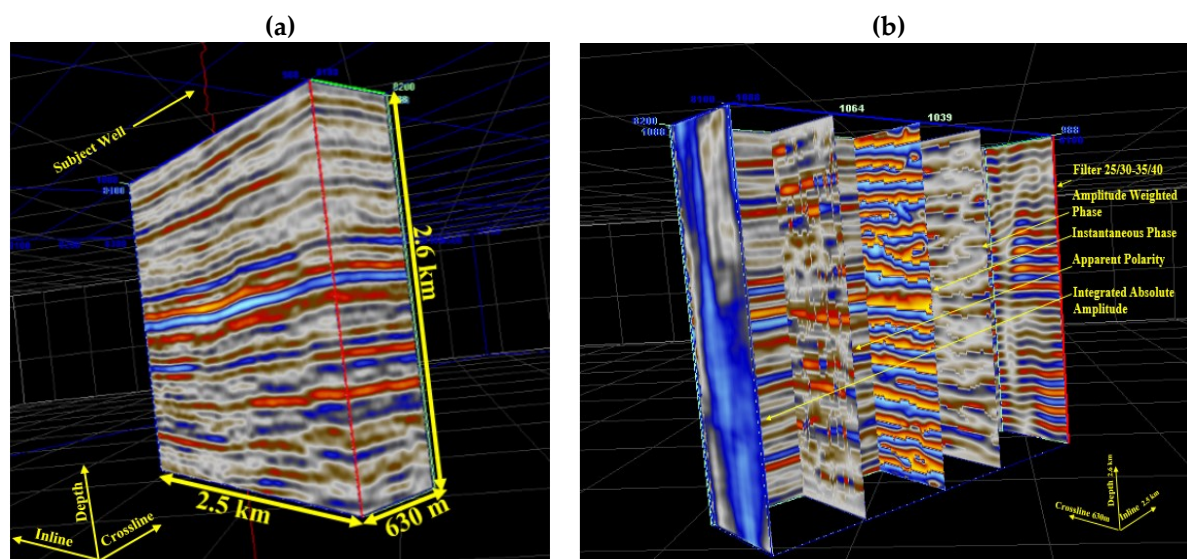


Table 9. The selected input attributes for estimating permeability and porosity.

Permeability		Porosity	
Attribute	R ²	Attribute	R ²
Amplitude Envelope	0.4160	Amplitude Envelope	0.5872
RMS Amplitude	0.4639	RMS Amplitude	0.5826
Integrated Absolute Amplitude	0.3369	Integrated Absolute Amplitude	0.5497
Filter 25/30–35–40	0.5240	Filter 25/30–35–40	0.6121
Reflection Intensity	0.5142	Reflection Intensity	0.5614
Instantaneous Phase	0.3701	Instantaneous Phase	0.5947
Derivative Instantaneous Amplitude	0.4669	Relative Acoustic Impedance	0.4433
Cosine Instantaneous Phase	0.7454	Instantaneous Frequency	0.2635

Figure 17. (a) 3D seismic cube used for extracting the employed attributes at the well location; (b) Employed seismic attributes.

Table 10. Properties of the employed data pairs for estimating petrophysical properties from the seismic attributes by both statistical and deep learning models.

Target Curve	Permeability	Porosity
Number of input variables	8	8
Total No. of available data pairs	231	355
No. of training data	185	295
No. of test data	46	60
No. of neurons in the hidden layer	20	20
Iteration the best ANN model was achieved	265	94
Optimal number of trees for statistical Bag ensembles	17	72
Optimal number of trees for statistical LSBoost ensembles	200	200

Table 11. Performance of the employed algorithms in estimating petrophysical properties from the post-stack seismic data.

Target NMR Curve	Permeability		Porosity	
Model performance	MSE	R ²	MSE	R ²
Bag ensemble	3.95×10^{-02}	0.9965	2.59×10^{-01}	0.9807
LSBoost ensemble	3.89×10^{-02}	0.9950	6.19×10^{-01}	0.9525

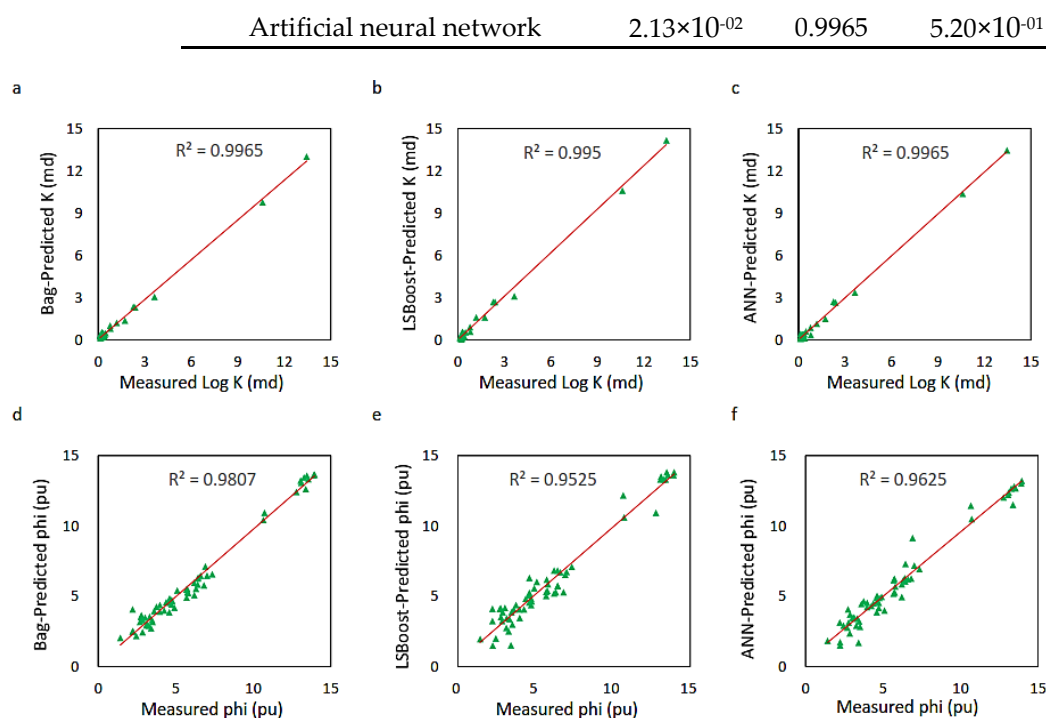


Figure 18. Correlations of the statistically and intelligently estimated permeability (a–c), and porosity (d–f) with the measured values in the unconventional gas field of the Qinshui Basin of China.

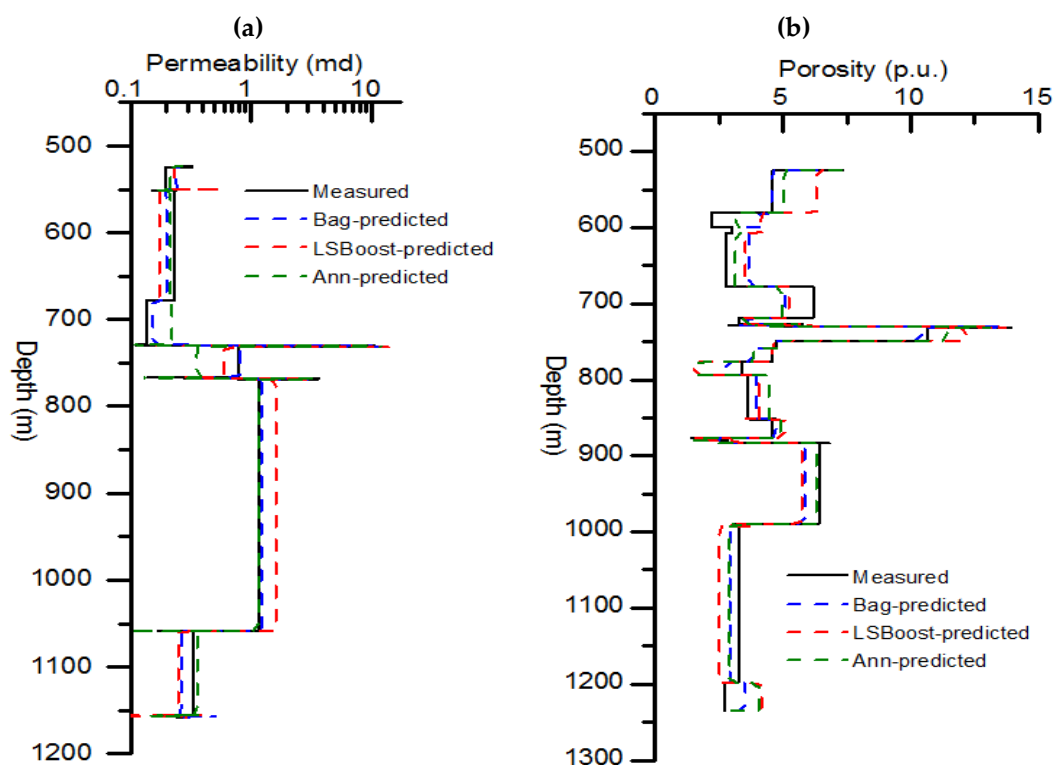


Figure 19. Measured (black), Bag-predicted (blue), LSBoost-predicted (red), and deep learning-predicted (green) curves of permeability (a) and porosity (b) in the unconventional coal reservoir. The stepped representation provides the comparison among the average permeability at different sections of the well, i.e., low permeable and highly permeable sections.

As mentioned earlier, throughout the modeling procedure, porosity was used as the benchmark to guarantee the reliability of the models. Figure 20 represents the 3D porosity cube around one of the subject wells estimated using the statistical Bag ensemble which was evaluated using the original seismic attributes. The agreement of the measured data and the Bag-predicted porosity cube verified the performance of the statistical methods.

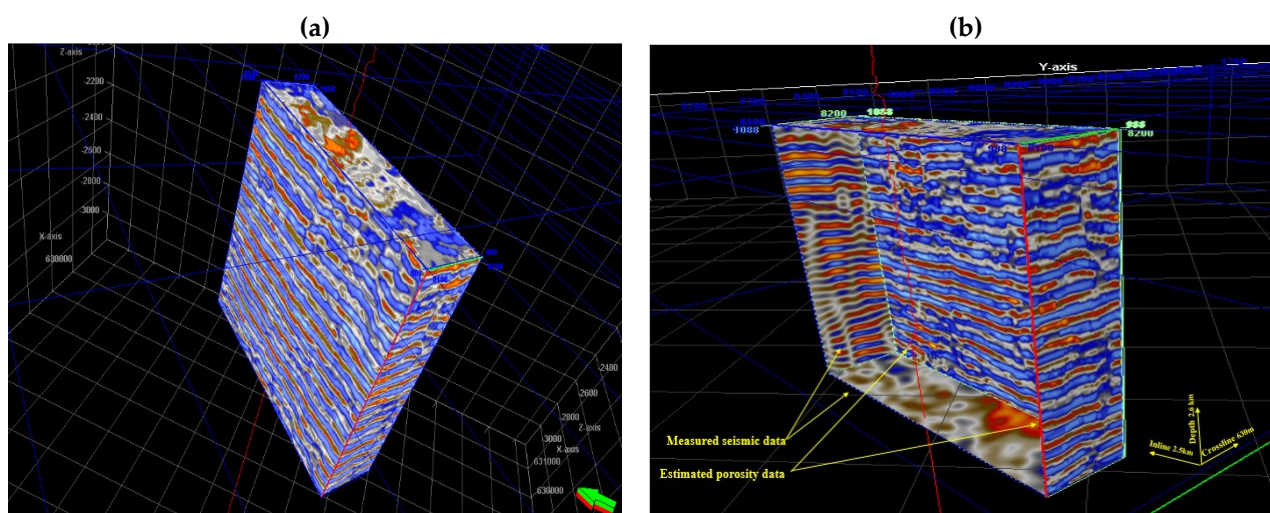


Figure 20. (a) 3D porosity cube around one of the subject wells estimated using the statistical Bag ensemble; (b) Evaluation of the estimated porosity cube (right, front, and surface walls) using a measured 3D seismic attribute, i.e., frequency filter. The satisfying agreement of the statistical model's result with the measured data proves the reliability of the technique.

4.3. Limitations of the Present Study

The most notable limitation of the present study was applying the research idea merely to coal samples. Nevertheless, other kinds of conventional and unconventional reservoir rocks should also be studied to characterize the detailed contribution of the pore types to the elastic wave velocity. Moreover, a higher number of core samples and a higher number of saturation states would also increase the accuracy of the obtained results. Meanwhile, the present study only investigated two kinds of statistical algorithms and the others are yet to be studied.

5. Conclusions

In the present study, the contribution of different pore types to the elastic P-wave's velocity was characterized. These results were used to determine the relationship between the bin porosities of the NMR test and the elastic wave velocity under the effect of water saturation. Accordingly, the seismic attributes were successfully used to produce artificial NMT T_2 curves with acceptable accuracy. Moreover, the statistical ensembles were investigated as an alternative tool for the estimation of petrophysical properties and proved as good as the deep learning approach. After all, the research came to the following conclusions:

- Both statistical ensembles and intelligent techniques proved efficient in the case of this research study.
- The performance of the soft-computing techniques strongly depends on the quality and type of input data. Such a conclusion highlights the importance of choosing the best input data when using these techniques.
- When using soft-computing procedures for reservoir characterization, more sensitivity should be shown to unconventional reservoirs according to their more complicated petrophysical properties. However, Geological complexity does not necessarily act as an obstacle to weaken the functionality of these algorithms.

- The seismic wave is composed of three main components which are phase, frequency, and amplitude. Therefore, in any study involving the seismic attributes, it is better to consider the attributes in such a way that one representative of each class is included inside the attribute category. This approach could help to obtain reliable results.
- When making the final decision as to select a model for upscaling purposes, comprehensive consideration of its features and variables including error range, correlation, availability of the inputs, and the requisite amount of the training data and memory is required.

Finally, the authors believe that the ideas introduced in the present study provide new insights for a deeper understanding of the hydrocarbon reservoirs and facilitate comprehending the reciprocal interactions of seismic and magnetic resonance phenomenon within the porous media of the reservoir rock. These ideas are more highlighted in complex pore systems as carbonate reservoirs, which could be suggested as the topic of a future study. Moreover, despite all the advancements, deep learning in the service of petrophysical modeling is still at the beginning of the way and has got its particular challenges that need to be solved.

6. Nomenclature

E_q	Compressed kernel matrix
C	Diagonal matrix that contains the inverse of the sum of the columns
R	Diagonal matrix that contains the inverse of the sum of the rows
T_2	NMR transverse relaxation time
x	Pore size
Δt	Transit time of elastic wave in the rock sample
Δt_{ma}	Transit time of the matrix
Δt_f	Transit time of the fluid
ϕ_s	Porosity derived from sonic measurements.
ϕ_{mic}	Porosity of micropores
ϕ_{mes}	Porosity of mesopores
ϕ_{mac}	Porosity of macropores
Δt_{mic}	Transit time of micropores
Δt_{mes}	Transit time of mesopores
Δt_{mac}	Transit time of macropores
CBP	CMR bin porosity
R^2	Correlation coefficient
MSE	Mean square error
pu	Porosity unit

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Neural network modeling of in situ fluid-filled pore size distributions in subsurface shale reservoirs under data constraints

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Abstract

Subsurface nuclear magnetic resonance (NMR) logs acquired in the wellbore environment are sensitive to fluid-filled pore size distribution, fluid mobility, permeability, and porosity in the near-wellbore reservoir volume. NMR response of a formation layer is processed to extract the T2 distribution, which approximates the fluid-filled pore size distribution. NMR logs are acquired in limited number of wells due to financial and operational challenges, which adversely affects reservoir characterization. We developed two neural-network-based machine learning techniques, long short-term memory (LSTM) network and variational autoencoder with a convolutional layer (VAEc) network, to process the ‘easy-to-acquire’ formation mineral and fluid saturation logs to generate synthetic NMR T2 distributions in the absence of ‘hard-to-acquire’ NMR T2 distribution log. Both the predictive models are trained and tested on limited wireline log measurements randomly selected from a 300-ft depth interval of the Bakken shale formation. Synthesis performances of LSTM and VAEc models in terms of R^2 are 0.78 and 0.75, respectively. Noise is inevitable in logging data due to the complex wellbore and formation conditions. Notably, both the predictive models robustly synthesize the fluid-filled pore size distributions in the presence of 50% noise in input logs and 30% noise in training T2 data. The performance of the proposed methodology improves with access to larger volume of training data from other formation types. The proposed method is critical to the synthesis of in situ fluid-filled pore size distributions in shale formations under data constraints due to financial and operational challenges.

Keywords Convolution · Recurrent · Neural network · Shale · Nuclear magnetic resonance · Deep learning

1 Introduction

A well log is a measurement of certain physical property of subsurface geological formation located at a depth of several thousand feet below the ground. Well logs are acquired through wellbore, 4 to 6 inches in diameter, drilled into these formations to access them for purposes of energy and earth resource extraction. The well logs are

interpreted using empirical and mechanistic models for near-wellbore reservoir characterization. Well logging tools are run in the wellbore during drilling or after drilling in an open-hole or a cased-hole environment. The borehole and formation conditions in which the well logging tools operate are uncertain, complex, and hostile that give rise to noise-prone measurements.

One advanced well log is NMR measurement, which is sensitive to hydrocarbon and water volumes present in the connected pore volumes. Subsurface NMR log is valuable for reservoir characterization because it contains physical information about the pore-filling fluid distribution, type, and mobility. Porosity and permeability of a reservoir are routinely estimated from NMR-T2 distribution, computed from the NMR log. However, the acquisition of NMR log is limited due to the high cost and operational challenge in deploying and running these tools in the uncertain, complex, and hostile borehole environment. In shale

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formations, NMR log acquisition is not common due to operational and financial challenges. In such data-constrained environment, new methods are required to determine the fluid-filled pore size distribution in hydrocarbon-bearing shale reservoirs.

Machine learning (ML) aims to extract patterns and hidden relationships in data. ML technique formulates a data-driven predictive model, which is useful when the patterns and relationships among data entities are governed by complex processes and mechanisms. Applications of CNN are common for image processing, which may involve object detection, face recognition, and image captioning. RNN is assisted by an internal memory to processes a sequential data. Long short-term memory (LSTM) is a popular type of RNN due to its ability to ‘memorize’ relatively longer-term dependencies in data. LSTM has been applied to speech recognition, language translation, and image captioning [1]. Basic artificial neural networks (ANN) and shallow predictive models have been applied in oil production prediction [2–4], pressure transient analysis [5, 6], and history matching [7]. However, the advanced machine learning methods, involving RNN and CNN, have not yet been widely applied in the petroleum industry. In this paper, two neural-network-based machine learning techniques, LSTM network and variational autoencoder with a convolutional layer (VAEc) network, process easy-to-acquire formation mineral and fluid saturation logs to generate the NMR T₂ distributions, which approximate fluid-filled pore size distributions in the Bakken shale (BS) formation.

A few studies have been conducted to implement advanced ML methods in well-log-based reservoir characterization. Mohaghegh et al. [8] used a five-layer ANN to generate synthetic irreducible water saturation, effective porosity and permeability from gamma ray, caliper, resistivity, and spontaneous-potential logs from 4 different wells. Jamshidian et al. [9] used ANN model to determine the NMR-derived free-fluid porosity and permeability from conventional well logs of sonic, neutron porosity, bulk density, and resistivity from the South Pars gas field. Labani et al. [10] estimated NMR-derived free-fluid porosity and permeability from conventional logs, such as sonic, resistivity, and gamma ray. Instead of predicting reservoir properties, a few studies predicted well log by processing conventional logs. Golsanami et al. [11] combined models of fuzzy logic and ANN to predict the eight-bin porosities of NMR T₂ distribution from neutron, density, sonic, lithology, gamma ray, and effective porosity logs acquired from a reservoir unit consisting of limestone, dolomite, and shale. Farzi et al. [12] simulated NMR response from micro-CT images of six sand packs and two sandstone samples using an ANN. Asoodeh and Bagheripour [13] applied a committee machine to generate

compressional, shear, and Stoneley wave velocities from conventional log data. Recent studies show promising results about applying deep learning models for NMR T₂ spectral generations [14, 15].

In this study, we implement variational autoencoder with a convolutional layer (VAEc) network as the first model. CNN is inspired by the complex arrangement of the biological visual cortex, where a set of neurons is sensitive to a specific visual feature, such as angle, curvature, or edge. A convolutional layer in CNN contains a combination of mathematical filters, in which each filter is sensitive to a specific feature of an image. When multiple convolutional layers are combined, the CNN can learn features of varying visual complexity. Variational autoencoder is a special type of generative neural network that aims to reproduce the input of the network. It comprises two computational frameworks, an encoder followed by a decoder. The encoder encodes the input into a latent layer, whereas the decoder decodes the information in the latent layer to reconstruct the input. With this architecture, the VAE learns the data distribution in the input and generates the input data that are similar to the training data, and can project a high-dimensional data to a lower-dimensional space. As a generative model, the VAE has been applied in various generative modeling problems like image generation [16] and text generation [17]. We implement long short-term memory (LSTM) network as the second model. LSTM is a type of RNN with an improved ability to solve the gradient vanishing problem in RNN. With the ability of ‘memorizing’ values for a long time, the LSTM is superior in dealing with sequences, like text, where values in a different position may have a strong dependency. LSTM applications include machine translation [18, 19], natural language generation [20], and time series prediction. Joint application of LSTM and CNN can accomplish the work of video description [21], image captioning [1], and text to image generation [22].

Well logs have complex hidden relationships among each other. Although different well logs sense different physical phenomena and formation properties, certain combinations of well logs tend to correlate because they are controlled by certain combinations of formation properties and physical processes, such as mineral composition, fluid type, fluid saturations, pore size distribution, fracture distribution, to name a few. Under data-constrained environments due to financial or operational challenges, the goal of this study is to utilize LSTM and VAEc models to extract the complex hidden relationships between the easy-to-acquire well logs and the NMR T₂ distribution logs, which approximate the fluid-filled pore size distribution. In doing so, fluid-filled pore size distribution in Bakken shale formation can be synthesized by processing 10 easy-to-acquire formation mineral and fluid saturation logs.

The paper is organized as follows: Sect. 2 explains the theory NMR measurements, LSTM, VAE, and CNN; Sect. 3 explains the data preprocessing; Sect. 4 presents the results of data-driven synthesis of NMR; and Sect. 5 presents the performance and robustness of the models.

2 Theory and methods

2.1 NMR measurement theories

NMR measurement utilizes the polarization and relaxation phenomena of hydrogen nuclei in the directional magnetic field to investigate the fluid properties and fluid distributions in porous materials. In oil and gas reservoirs, hydrocarbon and water contain the hydrogen nuclei that respond to the external magnetic excitation. Therefore, subsurface NMR logs measure the mobility and distribution of reservoir fluid in the subsurface formations. Hydrogen nuclei in the formation fluids are first polarized by a strong magnetic field exerted by the NMR logging tool in the borehole. Subsequently, the magnetic field is removed resulting in the polarized nuclei to lose energy and relax to an equilibrium state. This transition emits magnetic signals and is detected and recorded as NMR raw logs. NMR raw logs can be fitted by a set of exponentials with different decay constants [23]. The decay constants in these exponentials are the T2 times, which is a function of the bulk fluid relaxation ($T_{2\text{bulk}}$), surface relaxation ($T_{2\text{surface}}$), and diffusion relaxation ($T_{2\text{diffusion}}$) mechanisms. The relationship can be expressed as:

$$\frac{1}{T_2} = \frac{1}{T_{2\text{bulk}}} + \frac{1}{T_{2\text{surface}}} + \frac{1}{T_{2\text{diffusion}}} \quad (1)$$

Bulk relaxation is controlled by volumetric properties, such as pore volume, fluid saturation, fluid mobility, and fluid composition. Surface relaxation occurs at the fluid–solid interface when hydrogen nuclei interact with the grain surface, it is influenced by grain surface area, grain surface to pore volume ratio, grain mineralogy, and wettability [23]. Raw NMR logging data are processed to derive the NMR-T2 distribution at each depth along the borehole. NMR-T2 distribution is the relationship between NMR signal amplitude and T2 relaxation time. The T2 distribution represents near-wellbore pore-filling fluid distribution and mobility. T2 distribution is a 1D spectral response as a function of T2 time, where larger T2 time is correlated with larger pore volumes and larger fluid mobility. NMR log is the only log that can provide reservoir engineers with pore size distribution information of the reservoir. Pore size distribution describes the distribution of the pore diameter. Pore size distribution is critical in reservoir characterization. Pore size distribution can be used to calculate the

formation permeability, porosity, and rock typing essential to the reservoir characterization procedure.

2.2 Petrophysical basis for the proposed log generation

NMR logging tool is usually run at relatively slower speed of 2000 ft/h in bigger diameter uniform boreholes around 6.5-in in diameter. NMR signals due to the inherent physics have poor signal-to-noise ratio in the complex borehole conditions. Acquisition of high-quality NMR log in subsurface borehole environment requires highly trained wireline field engineers. Oil and gas companies do not deploy NMR logging tool in every well in a reservoir due to the financial and operational challenges involved in running the NMR tool. One alternative is to synthesize the entire NMR-T2 distribution spanning 0.3 ms to 3000 ms by processing conventional logs. An accurate prediction of NMR T2 distribution will assist the geoscientists and engineers to quantify the pore size distribution, permeability, and bound fluid saturation in hydrocarbon-bearing reservoirs, thereby improving project economics and reservoir characterization capabilities. Synthesis of the entire NMR T2 without core data is a challenging and novel task.

Generation of NMR T2 distribution log using conventional “easy-to-acquire” logs is feasible because these logs are sensitive to fluid saturations, pore size distribution, and mineralogy. Resistivity measurement and the mud-filtrate invasion effect on resistivity logs are influenced by the pore sizes, pore throat sizes, tortuosity, and Maxwell–Wagner polarization, which depends on the pore size distribution. Neutron log is sensitive to hydrogen index of fluids in the formation, which correlates with average porosity of the formation. GR is sensitive to clay volume that determines the clay-bound water and capillary-bound water, which is sensed by the NMR T2 distribution. Density log is sensitive to mineral densities and fluid densities in the formation. Mineral density is controlled by mineralogy that controls the lithology and pore size distribution of the formation. For example, sandstone formation with predominantly quartz mineralogy will have a distinct pore size distribution compared to carbonate formation with predominantly calcite mineralogy, which will be distinct from shale formation with predominantly clay mineralogy. NMR T2 distribution depends on fluid saturations, fluid mobility, pore size distribution, and surface relaxation, which depends on the mineralogy of the formation. These physical properties that govern NMR T2 also influence the logs, namely neutron, density, resistivity, sonic, and GR logs. The petrophysical basis for choosing these logs is because each of these logs has complex relationship with pore size distribution, fluid saturation, and mineralogy. We use

artificial neural networks to identify and extract these complex relationships that cannot be easily obtained by any other mechanistic models. The original goal of this study is not to develop a precise mechanistic equation to describe the physical relationships between the conventional logs and NMR T2 distribution. Rather, we train neural networks to learn these hidden patterns that relate conventional logs to the NMR T2 distribution. Another important concept that makes the NMR T2 generation possible is because most of these logs are influenced by water saturation and hydrocarbon saturation, and the vertical distribution of fluid saturations along the depth is controlled by capillary pressure, which is governed by the pore size distribution.

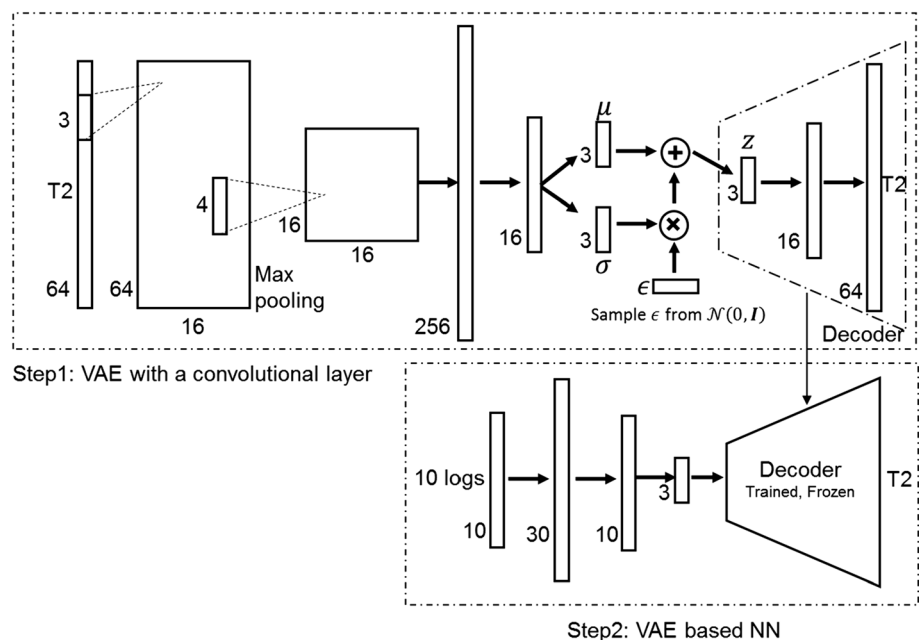
2.3 Architecture and training process of the VAEc model

A convolutional layer is combined with a VAE-based neural network (NN) to better extract features of NMR T2 distribution, which is processed as a 1D vector. The architecture of the generative model for purposes of VAEc-based NN model training is presented in Fig. 1. In the first part of training, VAEc learns to reproduce the measured T2 distributions. The encoder part of the VAE compresses the 64-dimension T2 data to 3-dimensional latent space. Subsequently, the decoder decompresses the latent variables to reconstruct the measured T2 input data. The first step of training, which involves the encoder and decoder, comprises 8 layers: 1 convolutional layer, 1 max pooling layer, 1 latent layer with 3 neurons, and 5 dense fully connected layers. The first convolutional layer is $64 \times 16 \times 1$ and is generated by filtering the 64×1 T2 distribution of a

specific depth with 16 filters of size 3×1 . The max pooling layer processes the $64 \times 16 \times 1$ output with a filter size of 4×1 to obtain a $16 \times 16 \times 1$ output. The output of the max pooling layer is then flattened and fed to two fully connected layers with 256 and 16 neurons each. The fifth layer of the VAEc is a 3-dimensional latent layer, which samples from the output of the previous layer. This layer is chosen to be 3 dimensional based on extensive numerical experiments. The 64-dimensional NMR T2 data are compressed to 3-dimensions through these five layers. After that, the decoder decodes the latent vector to 64-dimension through 3 fully connected layers containing 3, 16, and 64 neurons, respectively. The loss function in the VAEc model is comprised of two components: reconstruction loss and the Kullback–Leibler (KL) divergence loss. The reconstruction loss emphasizes the difference between the input and output NMR T2 data when forcing the VAEc to reproduce the input as precise as possible, whereas the KL divergence loss evaluates the difference between the distribution of the latent vector and the unit Gaussian distribution.

In the second step of training, a NN model comprising 4 layers is connected to the decoder trained in the first step. The trained decoder is frozen to preserve the generalization of the T2 distribution learned from the training data. The first three layers process 10 mineral composition and fluid saturation logs acquired at a specific depth in the subsurface to produce a 3-dimensional vector, which can be decoded by the trained decoder to generate the T2 distribution that best matches the T2 distribution measured at that depth. The architecture of the 4-layered NN model used in the second part of the training is shown Fig. 1.

Fig. 1 The architecture of the VAEc model during the training phase



2.4 Training process and architecture of LSTM

In the second predictive model, the 10 mineral composition and fluid saturation logs and NMR-T2 distribution measured in the subsurface along the length of a well are processed using LSTM as sequences. LSTM adopts a recurrent neural network architecture that remembers patterns over a sequence. LSTM is applied for sequence-to-sequence mapping, where a sequence is a list of ordered elements. LSTM excels other deep neural networks in that it can deal with long-term dependency and unfixed length of input data. This sequence-to-sequence learning model is successfully applied to language translation. Language translation relies on the fact that sentences in different languages are distinct representations of one common meaning. In a similar manner, different subsurface logs acquired at a specific depth are distinct responses to one common material. We implement LSTM to capture the dependency in NMR T2 distribution, which usually exhibits a smooth distribution. The smooth shape of NMR-T2 distribution illustrates the dependency between the data point in NMR T2.

The architecture of the LSTM model is demonstrated in Fig. 2. Like VAEc model, this model comprises 1 encoder followed by 1 decoder. Both encoder and decoder are collections of LSTM layers. The encoder and decoder LSTM layers are made up of 10 and 64 chained LSTM modules, respectively. Each module is made of gates controlling data flow inside the LSTM modules. Controlled by various gates, the LSTM can choose to forget or update the information flowing through the modules. The encoder compresses the 10 formation mineral and fluid saturation logs into a single vector. Then, the decoder decodes the vector into target sequences. The 10 input logs are taken as sequence, and the encoder of the model takes one of the ten logs at each time step. After all of the 10 input logs are read by the encoder, the encoder generates a vector v of size 15 in the last step. Size 15 is found based on experience with the LSTM predictions of T2 data. The vector v generated by the encoder is repeated 64 times and fed to the decoder.

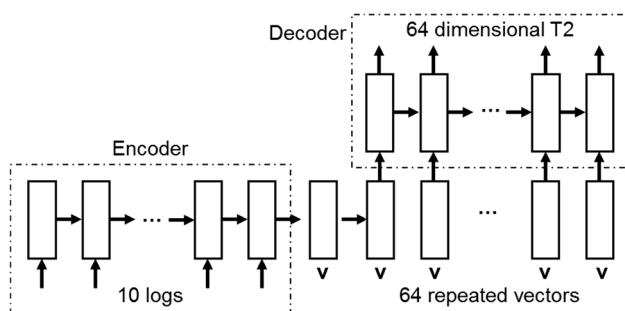


Fig. 2 The architecture of the LSTM model during training and testing

The decoder takes the 15-dimensional vector v in 64 time steps and tries to reconstruct a 64-dimensional T2 distribution for a single depth corresponding to the 10 input logs relevant to that depth.

The loss function used for the LSTM model is mean squared error function. The optimizer used to train the LSTM model is RMSprop. During the training process, the prediction loss is evaluated using the mean squared error function and RMSprop is used to update the weights during backpropagation.

2.5 VAEc-based NN model versus LSTM model

The convolutional layers in VAEc-based NN model tend to capture the features of the NMR T2 distribution better than the LSTM model. VAEc-based NN model is trained in a two-step process, which is more complicated than one-step training process of LSTM. During the VAEc model design process, we need to test different number of layers, different neurons, different number of filters, etc., in both training Step 1 and Step 2, whereas the LSTM model is trained in one step. Consequently, VAEc-based NN tends to require more time to tune the model parameters. Nonetheless, VAEc performs more accurate synthesis. In the Step 2 of the training, formation mineral and fluid compositions are processed using the trained decoder, which means the model is forced to synthesize NMR T2 distribution similar to those that were learnt in Step 1, thereby resulting in better accuracy. The decoder of LSTM is not pretrained during the training process, which means more effort is required to prevent overfitting through efficient architecture and training step selection. The LSTM model has an advantage of long sequence processing and can capture the dependency in the conventional logs as well as that in NMR T2. Usefulness of LSTM for the proposed objective will increase with an increase in the dependency in the log data. In our implementations, the LSTM model is deeper, whereas the VAEc-based NN model is wider compared to the other.

3 Data preprocessing

3.1 Bakken shale formation characteristics and data

The training and test data used in this study were derived from subsurface measurements acquired by various downhole logging tools deployed in 5- to 7-inch-diameter borehole drilled in the Bakken shale (BS) formation. The borehole intersected 7 distinct formation lithologies (Formation 1–7). Formations 1 and 3 are organic-rich hydrocarbon source shale rocks, and Formation 2 is a dolomite

reservoir. Total organic carbon (TOC) of Formations 1 and 3 ranges from 12 to 26 wt%, whereas the TOC of Formation 2 ranges from 0.1 to 0.3 wt%. Formations 4 to 7 are primarily dolomite and shale of poor to moderate reservoir quality.

Resistivity, neutron, density, natural gamma ray (GR), dielectric, and NMR logs were acquired in the Bakken shale formation. A few of these logs were inverted using Techlog mineral inversion software to obtain 10 formation mineral content and fluid saturation logs. The fluid-filled pore size distribution synthesis methodology proposed in this study is developed using the 10 inversion-derived formation mineral content and fluid saturation logs for training and testing of the NN models along with the NMR-T2 distributions for training the NN models.

3.2 Data preprocessing

Data preprocessing of the logs improves the efficiency of the proposed synthesis. Data cleaning and data normalization techniques were used in this study. Well log data are affected by environmental noise generated when the logging tool is run in the borehole and is also affected by the uncertainty due to physical variabilities in the surrounding volume. Thin beds and formation layers with unrepresented formation properties will decrease the training efficiency and synthesis accuracy. During data preprocessing, we remove the data corresponding to these thin layers that act as outliers. Data were acquired from 597 sequential discrete depths along the borehole intersecting the 300-ft depth interval of Bakken shale formation. Measured T2 distributions predominantly exhibit 1 or 2 peaks. Less than 4% of measured T2 distributions exhibit more than 2 peaks. Twenty-two data points that have 3-peak NMR T2 data were screened out from this study. The formation mineral content and fluid saturation logs are normalized to have zero mean and unit variance, which ensures the logs have the same scale and contribute equally to the loss function. After data cleaning and normalization, data from 592 depth points make up the training and test dataset. Four hundred and seventy-five depths were randomly selected as the training data, and the remaining 100 depths were used as the testing data.

4 VAEc-based NN model

4.1 Training efficiency

The VAEc model constructed in Sect. 2.3 is trained on a desktop computer with 3.5 GHz CPU and 32 GB RAM. It takes 41.25 s for the two-stage training process to train the model 1000 times with training data, including 10 logs and

64-bin T2 distribution, from 460 discrete depths. Training loss and prediction accuracy of the model are monitored to select appropriate number of steps for the training. The training steps are selected by monitoring the mean square error (MSE) of the learning on both training and testing datasets. In the initial training stage, MSEs for training and testing decrease as the model learns from the training dataset. After certain number of steps of training, the testing MSE tends to increase, and this is the point where the model starts overfitting the datasets. We stop training at this training step to avoid overfitting. We set the training steps to 2000, for which the model has a low loss and high prediction accuracy. In the first step of training, VAEc generalizes the NMR T2 in the training dataset and the decoder generates a typical NMR T2 from the 3-dimensional latent vector. Figure 3 is a 2D sample from the 3-dimensional latent space generated by the encoder. The figure represents the learning achieved by the VAEc-based NN model.

Figure 3 is a slice plane from the 3D latent space. It demonstrates how the latent vectors are related to the NMR T2 distributions. The VAEc learns from the training dataset and generalizes the typical NMR T2 shape in the latent space. The decoder can decode a latent vector into an NMR T2 distribution. Each NMR T2 in Fig. 3 corresponds to a 3-dimensional vector, which represents a coordinate in the space. By sampling from the latent space, we can visualize VAEc's understanding of a typical NMR T2 distribution. In Fig. 3, the learned T2 distribution changes gradually in the 3D latent space with slight consecutive variations in the number of peaks, smoothness of the distribution, the position of the peaks, and height and width of the peak. The VAEc learning ensures that the gradual changing characteristic of the learned NMR T2 plotted in Fig. 3 is in accordance with the characteristic of NMR T2 in the formation. NMR T2 in adjacent formations should be similar to each other with gradual changing characteristics. After the first stage of training, the second stage relates the learned features in NMR data to the input logs used for the synthesis.

4.2 Testing process and synthesis accuracy

The synthesis accuracy of the trained VAEc-based NN is tested using the 100 testing data. R^2 values are calculated using the predicted NMR T2 and measured NMR T2 distributions, such that

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

where $\hat{y}_i$ and y_i are predicted and actual NMR T2 amplitude at bin i , and n is the number of bins.

Fig. 3 Examples of the learned manifold from the training dataset. *x*-axis represents T2 relaxation time from 0.3 to 3000 ms; *y*-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25

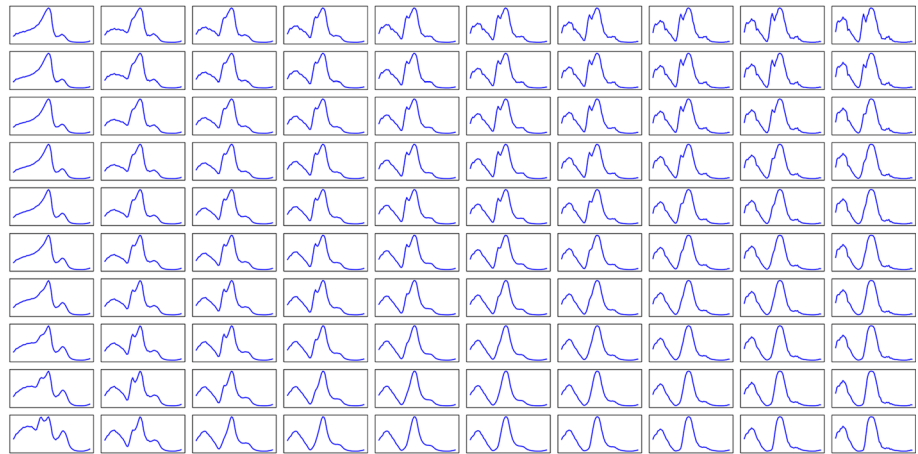
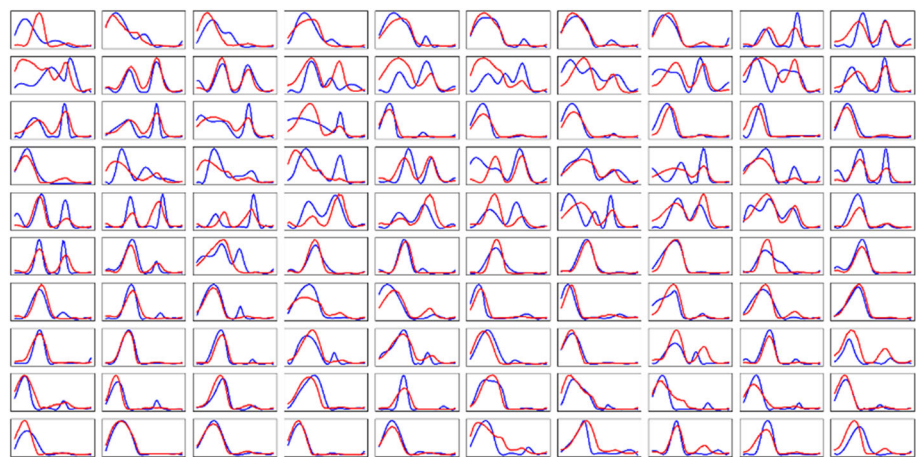


Fig. 4 True NMR T2 (blue) and predicted NMR T2 (red) in the testing phase. *x*-axis represents T2 relaxation time from 0.3 to 3000 ms; *y*-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25 (color figure online)



The results are shown in Fig. 4. The average R^2 value for the 100 predictions of T2 distributions in test data is 0.75. In Fig. 4, the randomly selected NMR T2 testing data show a variety of shapes. About 60% of 100 samples are single-peak NMR T2 and 40% are double-peak. For single-peak NMR T2, the predicted NMR T2 and true NMR T2 almost overlay with each other. For double-peak NMR T2, the accuracy is lower than single-peak data. Although the VAEc model can predict the peak position of double-peak NMR T2 with high accuracy, the model has low prediction accuracy for the height and shape of peaks.

Distribution of R^2 of prediction for the 100 testing data is shown in Fig. 5. R^2 achieved for most depths is above 0.7. However, VAEc model has lower synthesis accuracy at depths exhibiting 2-peaked T2 distribution. This is primarily due to the limited training data with 2-peaked NMR T2 distribution. In the 300-ft thick Bakken shale formation studied in this work, close to 1/3 of the depths have 2-peaked T2 distribution. Variation in the shape of T2 distribution indicates changes in formation characteristics and changes in the relationships between T2 and formation mineral content and fluid saturation. In the absence of large

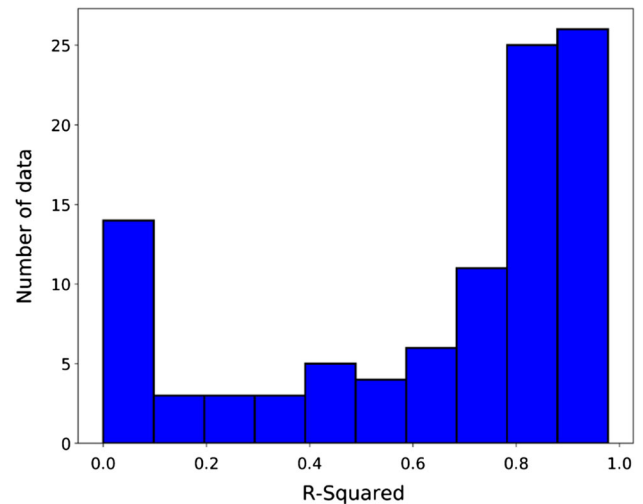


Fig. 5 R^2 distribution representing synthesis accuracy of the VAEc model during the testing phase

volume of suitable training data, the proposed data-driven predictive models will not be able to accurately determine the mathematical relationships between T2 distributions

and the input logs, thereby leading to poor synthesis accuracy for depths having physical characteristics that are not adequately represented in the training dataset.

4.3 Sensitivity to noise in input logs

Environmental noise and uncertainty due to physical variability affect well logs. We studied these adverse effects on the synthesis accuracy by adding 50% noise to each of the 10 input logs, one at a time, and then studying the corresponding variations in synthesis accuracy across the entire T2 distribution spectrum. Data preprocessing is performed to normalize each of the 10 input logs between -1 and 1 . Noise data are generated from a uniform distribution ranging between -0.5 and 0.5 and added to the normalized data resulting in addition of 50% noise to the input logs. VAEc model is trained 10 times, wherein 50% noise is added to one of the 10 logs, one at a time. During each time of training, the VAEc model is trained for 2000 training steps, during which the weights inside the network are updated by backpropagation of the loss. Our numerical experiments indicate overfitting when the training steps are higher than 2000.

Mean square error (MSE) in the synthesis of fluid-filled pore size distribution, approximated as T2 distribution, using noise-prone logs and original logs is calculated for this sensitivity study. MSE is defined as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (3)$$

where $\hat{y}_i$ and y_i are predicted and actual NMR T2 amplitude at bin i and n is the number of bins. MSE variation when performing the synthesis using the noise-prone logs with respect to the MSE of synthesis using the original logs is shown in Fig. 6. Positive MSE variation indicates that the

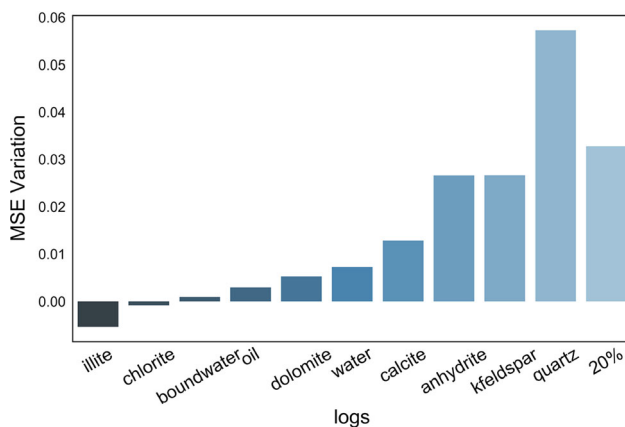


Fig. 6 Variation in MSE of the proposed synthesis when 50% noise is added to each of the 10 logs (one at a time). The bar on the right is the MSE variation when 20% noise is added simultaneously to each input log

addition of noise to logs lowers the synthesis accuracy. When 50% noise is added to the quartz and illite content logs, MSE increases by 5.72% and decreases by 0.54%, respectively. The robustness of the proposed methodology is demonstrated with this sensitivity study because the change in MSE due to noise is minor despite adding a large amount of noise to one log. Figure 6 indicates that the synthesis accuracy is sensitive to noise in quartz, feldspar, anhydrite, and calcite content logs, whereas the synthesis accuracy is insensitive to noise in oil saturation, bound water saturation, chlorite content, and illite content logs.

To evaluate the importance of input logs in the proposed synthesis, we study the change in synthesis accuracy when one of the 10 input logs is not used in the synthesis methodology. The model is trained 10 times; in each training, one of the 10 logs is removed. The model is trained for 2000 steps during each training with the 9 remaining input logs. MSE variation due to the removal of one log is shown in Fig. 7. The proposed synthesis methodology is most affected when bound water saturation and illite content logs are removed. MSE increases about 11% when bound water and illite logs are removed. Quartz, calcite, and feldspar content logs have intermediate adverse effects on the synthesis accuracy. In contrast, water saturation, anhydrite content, dolomite content, and oil saturation logs when removed result in slight improvement in the synthesis accuracy.

4.4 Sensitivity to noise in NMR T2

Noise contained in NMR T2 distributions used during the training phase will introduce inaccuracy in the synthesis. Simulating the effect of noise in T2 distribution requires a different approach compared to that used for studying the effect of noise in input logs. To add noise into T2 distribution while keeping its original characteristic, we first applied Gaussian fitting to fit each NMR T2 with 3

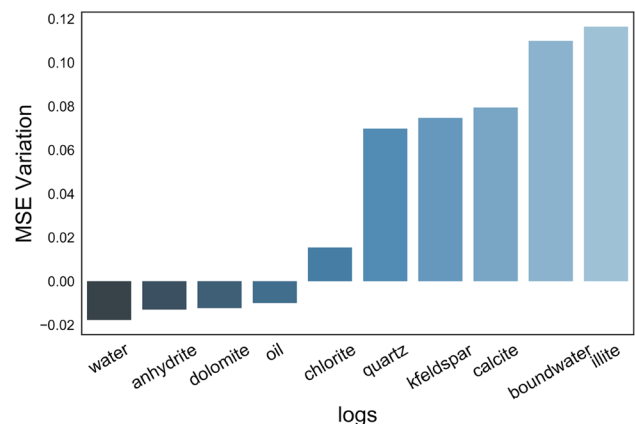


Fig. 7 MSE variation when one input log is removed one at a time

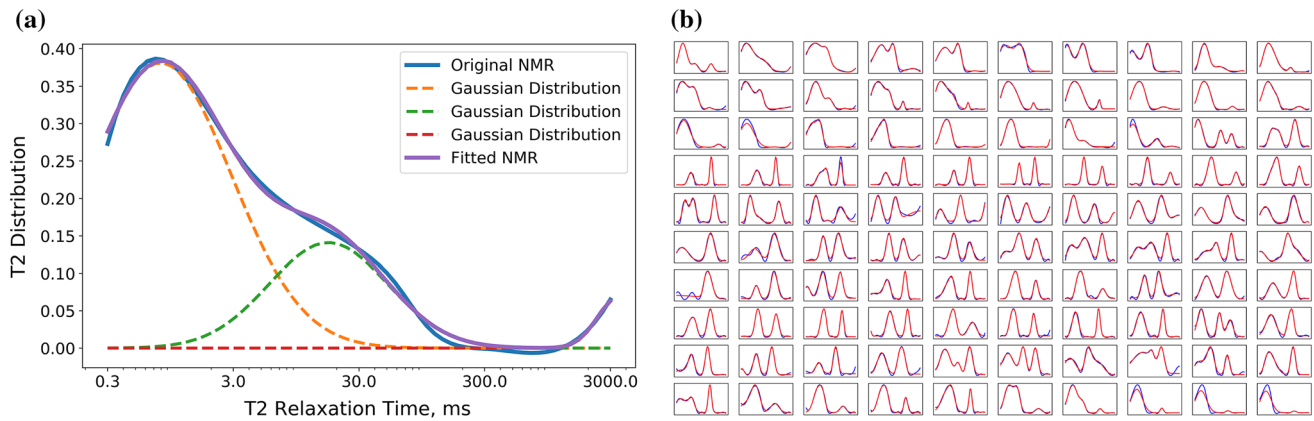


Fig. 8 **a** Demonstration of fitting NMR T2 with 3 Gaussian distributions, and **b** examples of multi-Gaussian fits (red) to the original (blue) NMR T2 without any additional noise (color figure online)

Gaussian distributions as shown in Fig. 8a. Each Gaussian distribution is controlled by 3 parameters of position, amplitude, and width. In this way, each NMR T2 is described by 9 parameters. By slightly changing the 9 controlling parameters, we can change the peak amplitude, position, and width of the measured T2 distribution used for the training. We added 30% of random noise to each of the 9 parameters to add noise to the measured NMR T2 while keeping its original characteristic. The multi-Gaussian fits to the measured T2 distributions are shown in Fig. 8b. The fit is near perfect. The noise-prone T2 distributions generated using the aforementioned approach are shown in Fig. 9. By adding noise to the controlling parameters, the noisy T2 keeps the overall characteristics and the median R^2 value between the original NMR T2 and noisy NMR T2 is 0.69.

The VAEc model is trained with 10 original logs without additional noise and NMR T2 containing the synthetically added 30% noise to the 9 controlling parameters. The

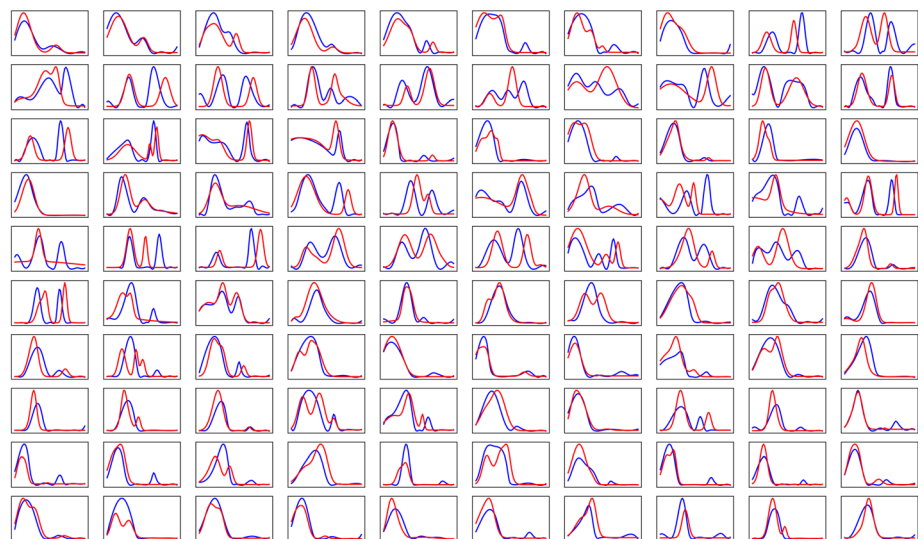
architecture and training steps remain the same. The synthesis accuracy of VAEc model decreases slightly due to the noise in the T2 data used for the training. Synthesis of T2 distribution with noise in the training T2 data has an R^2 of 0.72, which for training with noise-free original NMR T2 was 0.75. This demonstrates the robustness of the proposed synthesis methodology to noise in the training T2 data.

5 LSTM model

5.1 Training efficiency

LSTM model is trained and tested with the same dataset as the one used for the VAEc-based NN model, described in the above section. LSTM requires more time to train. It took 566.33 s to train the LSTM model with 1000 training steps, whereas it took only 41.25 s for the VAEc-based NN

Fig. 9 Original NMR T2 (blue) without additional noise and NMR T2 with synthetically added 30% noise (red). x-axis represents T2 relaxation time from 0.3 to 3000 ms; y-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25 (color figure online)



model training with the same steps. The two neural networks use different training processes. The VAEc-based model has 1825 parameters to train, whereas the LSTM model has 2071 parameters. Even though the two models have a similar number of parameters, the differences in calculation process cause one-order difference in the training times. The LSTM model takes one of the 10 logs at each time step. Besides, when predicting the NMR T2 distribution, the LSTM model predicts one value of the 64-dimensional NMR T2 at each time step, which means it takes 64 time steps to predict all values of the NMR T2. This calculation process is more complicated than the VAEc-based NN model.

5.2 Testing and accuracy

The average synthesis accuracy of the LSTM model in terms of R^2 is 0.78. The predicted T2 distributions and the R^2 of the synthesis performance are shown in Figs. 10 and 11. LSTM model better predicts single-peak T2 distribution compared to those with double peaks. In Fig. 11, the synthesis performance exhibits higher frequency of occurrence above R^2 of 0.9 or below 0.1, as compared to the performance of VAEc-based NN model, shown in Fig. 5. We claim that the LSTM model predicts single-peak T2 more accurately than the VAEc-based model, whereas the LSTM performance in predicting the double-peak T2 distribution is worse than that of VAEc-based NN model. Several LSTM synthesis for one-peak NMR do not demonstrate the typical T2 shape, for example, the 5th T2 distribution in the last row and the 6th distribution in the fourth row.

Compared to the VAEc-based NN model, the LSTM model has more freedom making it easier to overfit. First, the decoder part of the VAEc model is pretrained, which constrains the capacity of the model. Second, the LSTM model has more parameters compared with the VAEc

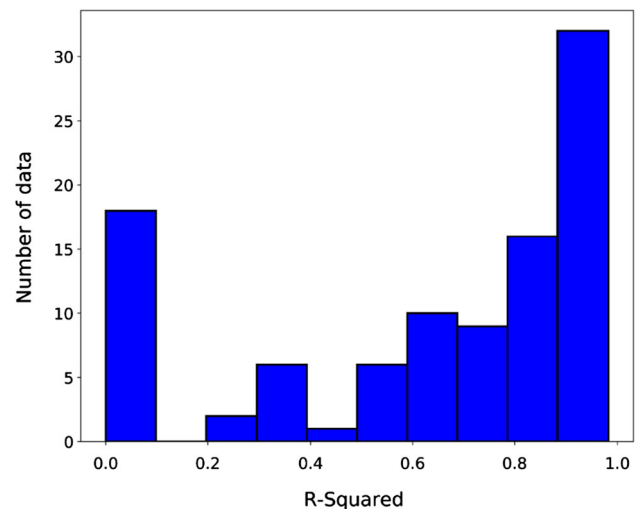


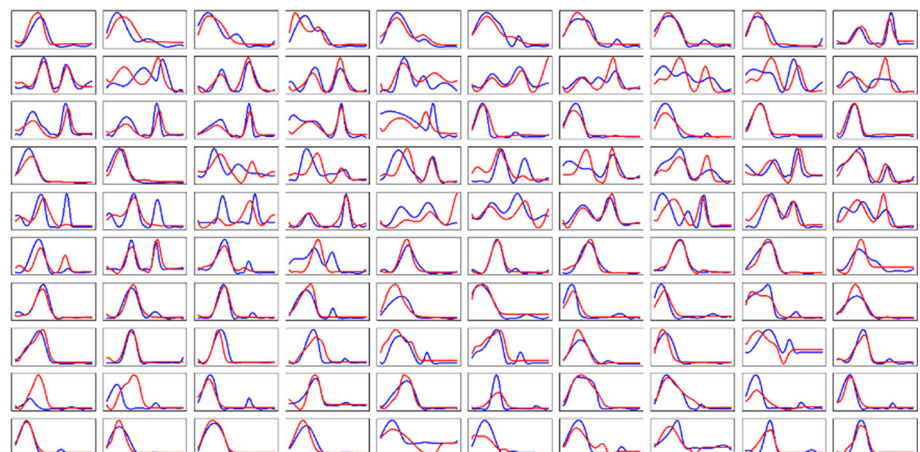
Fig. 11 R^2 distribution representing synthesis accuracy of the LSTM model during the testing phase

model. The decoder trained in the first stage of VAEc training when used in the second stage of the training constrains the capacity of the VAEc-based NN model. The VAEc architecture forces reconstruction of the NMR T2 distribution through the decoder. The freedom of the LSTM model may lead to overfitting to some extent, and overfitting leads to more dispersed synthesis accuracy in the entire formation. LSTM predicts more accurately with more freedom, but it also overfits samples.

5.3 Sensitivity to noise analysis

The model sensitivity is also tested for LSTM model using same procedure demonstrated in Sect. 4.3, by adding 50% noise to each log. Synthetically added noise in water saturation, chlorite content, k-feldspar content, and anhydrite content reduces the synthesis accuracy in comparison with that achieved using LSTM to process input logs without additional noise. Notably, the addition of noise to dolomite

Fig. 10 True NMR T2 (blue) and predicted NMR T2 (red) in testing phase. x-axis represents T2 relaxation time from 0.3 to 3000 ms; y-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25 (color figure online)



content log and bound water-saturation log improves the synthesis accuracy. This result is different with that of the VAEc model. In VAEc model, the noise in quartz content, k-feldspar content, and anhydrite content drastically decreases the synthesis accuracy. The difference of the MSE variation between VAEc and LSTM model indicate that the two models relies on different hidden relationship to predict NMR T2 from formation composition logs. Both LSTM and VAEc-based NN models are adversely affected when k-feldspar content, anhydrite content, calcite content, and oils saturation logs contain noise. VAEc model is more sensitive to noise in logs compared to the LSTM model, especially for the quartz content log (Fig. 12).

Changes in synthesis accuracy when 1 of the 10 input logs is removed from the LSTM model is studied in this section. Overall, the chlorite, anhydrite, and illite content logs when individually removed have the largest adverse effect on the synthesis. The bound water saturation and dolomite content logs when individually removed result in the improvement of the synthesis accuracy. For LSTM and VAEc-based NN models, the effects of removal of one of the 10 logs are completely opposite, except for the dolomite content log, which when removed results in a small improvement (Fig. 13).

5.4 Sensitivity to noise in NMR T2

Similar to the procedure in Sect. 4.4, we studied the effect of noise in the measured T2 data used for training on the synthesis accuracy. Contrary to the VAEc model, which shows a good stability to noise in the training T2 data, the synthesis accuracy of LSTM model significantly deteriorates on adding noise to the T2 data. Synthesis accuracy quantified in terms of R^2 drops from 0.78 to 0.43 when the training T2 data contain 30% noise. The drop in the

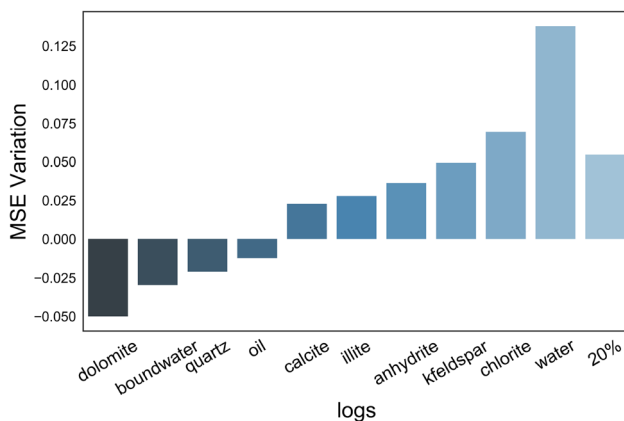


Fig. 12 Variation in MSE of the proposed synthesis when 50% noise is added to each of the 10 logs (one at a time). The bar on the right is the MSE variation when 20% noise is simultaneously added to each of the 10 input logs

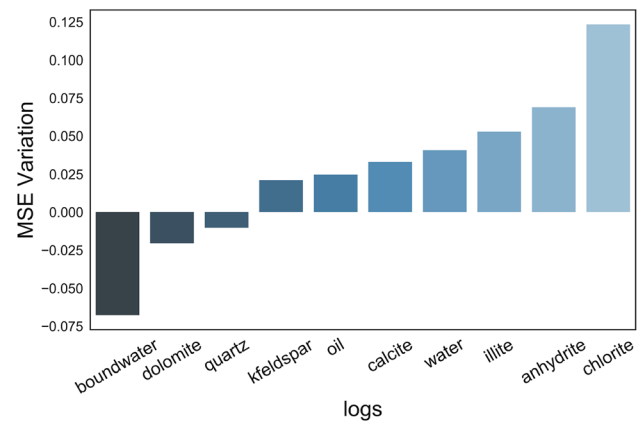


Fig. 13 MSE variation when one input log is removed one at a time

prediction accuracy indicates that the LSTM architecture is sensitive to noise in NMR T2. Unlike the VAEc model, the LSTM model takes NMR T2 as sequences. The LSTM model has the ability to find long- and short-term dependency in input and in output data. The prediction of NMR T2 values at higher relaxivity time is affected by the NMR T2 values at lower relaxivity time when using the LSTM model. Adding noise in the NMR T2 changes the dependency among the 64 bins of the NMR T2 distribution; thus, the LSTM model fails to capture the dependencies. VAEc-based NN model has a pretrained decoder, which ensures the most dominant features in the noisy NMR distributions are captured for purposes of synthesis during the testing phase.

5.5 Originality of the proposed research methods and findings

- Downhole nuclear magnetic resonance (NMR) logs acquired in the borehole environment are valuable for subsurface characterization because they contain information about the pore size distribution, fluid composition, fluid saturation, fluid mobility, formation permeability, and porosity. NMR log acquisition can be challenging due to operational and financial constraints. NMR T2 distributions of the subsurface earth formations when generated by processing conventional well logs using deep learning neural network (NN) models improve the accessibility to subsurface pore size distributions, residual saturations, movable fluid content, and permeability.
- Compared to other conventional well logs, NMR log acquisition encounters financial and operational constraints during the subsurface deployment of the NMR tool in the borehole environment. Consequently, NMR tool is deployed in very few wells, whereas conventional logging tools are run in almost

all the wells. To overcome these constraints, NMR T2 distribution of the subsurface earth is generated by processing easy-to-acquire conventional logs using deep learning neural network (NN) models.

- (c) Recent advances in machine learning and deep learning enable extraction of complex hidden relationships in a dataset and perform predictions based on the learned relationships. The ability of conventional multilayer neural network to find relationships in high-dimensional data is limited. Our prediction problem involves establishing relationships between 10 scalar measurements and 64-dimensional spectral response. To solve such a high-dimensional prediction problem, we implemented deep neural networks with specially designed architecture that honors the physical phenomena influencing the data.
- (d) For the first time, 10 conventional logs are integrated for the desired NMR T2 generation which is very different from the typical CNN and LSTM applications on images and audio.
- (e) CNN honors the image-like spectral behavior of the T2 distribution at a given depth that requires multi-scale filters for efficient extraction of features, whereas LSTM honors the time series-like short-term and long-term dependencies of conventional log inputs and T2 distributions along the depth of a geological formation. There has not been any research work on using these techniques for improving subsurface characterization under data constraints.
- (f) A comparative study of VAEc versus LSTM capabilities in generation of spectral response of geomaterial sheds light on the limitations and advantages of the two distinct architectures, one meant for image processing and the other meant for audio processing. We found that when there is noise in T2 distribution used for training, VAEc model has stable performance, whereas the R^2 of LSTM model drops from 0.78 to 0.43.

5.6 Limitations and challenges of proposed research methods and findings

- (a) In this pioneering work, the data being used for the training/testing are limited in quality and quantity due to proprietary concerns of the oil and gas company that donated the data. Lack of NMR data is primarily due to the high cost of the NMR logs and the proprietary nature of this data that is not easily shared by the oil and gas companies. Access to a larger dataset is required for validating the prediction performance of the proposed methodology.

- (b) The 10 conventional logs and NMR T2 distribution logs used for training the neural network model were recorded during different logging runs performed in a 5-inch-diameter borehole intersecting the geological formation under investigation. Consequently, the training dataset is prone to noise and adversely affected by the heterogeneity, complexity, and uncertainty due to the surrounding borehole and near-wellbore environments. This will affect the prediction performance of the deep learning neural networks for the proposed application.

We test the methodology on well logs from a single well. Cross-well validation is needed to test the robustness of the methodology. Due to lack availability of NMR logs and their proprietary nature, cross-well validation has not been performed in this study.

6 Summary of results

A neural network model based on variational autoencoder with convolutional layers (VAEc-based NN model) and a ‘long short-term memory’ recurrent neural network model (LSTM model) are trained and tested to synthesize in situ fluid-filled pore size distribution in the Bakken shale formation. To that end, the two models process 7 formation mineral content and 3 fluid saturation logs. The training/testing data were measured in an open borehole environment across a 300-ft depth interval of the Bakken shale formation. LSTM and VAEc-based NN models have a similar number of parameters and are trained/tested on the same dataset; however, LSTM incurs one order of magnitude higher training time. The synthesis accuracy for VAEc-based NN and LSTM models is 0.75 and 0.78, respectively. VAEc-based NN model can synthesize NMR T2 distributions, which approximates the in situ fluid-filled pore size distribution, in the presence of 50% noise in input logs and 30% noise in training T2 data without significant deterioration in the synthesis accuracy. On the other hand, LSTM model is significantly affected by noise in the training T2 data and robust in the presence of 50% noise in the input logs. Synthesis accuracy of LSTM model quantified in terms of R^2 drops from 0.78 to 0.43 when the training T2 data contain 30% noise. LSTM model is more sensitive to noise in chlorite content, k-feldspar content, and water saturation logs, whereas VAEc model is more sensitive to noise in quartz, anhydrite, and k-feldspar content logs. The information contained in 7 formation mineral content and 3 fluid saturation logs has redundancy, such that removing one of the 10 logs does not significantly affect the synthesis accuracy. LSTM and VAEc-based NN models exhibit completely opposite sensitivity to removal

of anyone of the 10 input logs, indicating that the two models extract different hidden log relationships for purposes of generating NMR T2 distributions, which approximates the in situ fluid-filled pore size distribution.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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Use of intelligent systems in reservoir characterization via synthetic magnetic resonance logs

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Abstract

Magnetic Resonance Imaging (MRI) logs are well logs that use nuclear magnetic resonance to accurately measure free fluid, irreducible water (MBVI), and effective porosity (MPHI). Permeability is then calculated using a mathematical function that incorporates these measured properties. This paper describes the methodology developed to generate synthetic MRI logs using data obtained by conventional well logs such as SP, Gamma Ray, Caliper, and Resistivity. The synthetically generated logs are named Virtual Magnetic Imaging Logs or “VMRI” logs for short. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Reservoir; MRI logs; Free fluid

1. Summary

This methodology incorporates artificial neural networks as its main tool. Virtual Magnetic Resonance Imaging (MRI) logs for irreducible water saturation (MBVI) and effective porosity (MPHI) as well as permeability (MPERM) were generated for four wells. These wells are located in East Texas, Gulf of Mexico, Utah, and New Mexico. The results are quite encouraging.

It is shown that MPHI, MBVI, and MPERM logs can be generated with a high degree of accuracy. For

each case, 30% of the data were used to develop the neural model. The model was then tested on the remaining 70% of the data for verification. The models provide VMRI logs with approximately 80% to 97% accuracy using data not employed during model development.

This methodology does not supersede the need for performing MRI logging in a field. It is designed to supplement the process by reducing the cost of using MRI logging on an entire field. The natural application of this process is in fields that have conventional logs for all of the wells but MRI logs for only a few wells. To generate the Virtual MRI logs for every well in a field, data from wells that have both conventional and MRI logs are used in the model development and verification. The model is then applied to the other wells in the field to generate the virtual MRI logs for these wells.

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Using this process, the operator can log a few strategically chosen wells using physical MRI tools and produce virtual MRI logs for the entire field. This will allow the development of an accurate representation of effective porosity, free fluid, irreducible water saturation, and permeability for the entire field.

2. Introduction

John M. Austin and Tom L. Faulkner (Austin and Faulkner, 1993) published a superb paper in August 1993 in “The American Oil and Gas Reporter” providing some valuable information about the MRI logging technique and its benefits to low resistivity reservoirs.

The MRI log measures effective porosity—total porosity minus the clay bound porosity—as well as irreducible water saturation. The irreducible water saturation is the combinations of clay bound water and water held due to the surface tension of the matrix material. The difference between effective porosity (MPHI) and the irreducible water saturation (MBVI) is called the free fluid index. This is the producible fluid in the reservoir. This demonstrates how valuable MRI log is to low resistivity reservoirs. In many low resistivity reservoirs, matrix irreducible water rather than producible water may cause a drop in resistivity. While producible water can seriously hamper production and make the pay quite unattractive, the same cannot be said for the irreducible water. Therefore, a reservoir that seems to be a poor candidate for further development—looking only at the conventional logs—may prove to be an attractive prospect once MRI log is employed.

As it will be shown in this paper, MRI logs may provide information that results in an increase in the recoverable reserves. This takes place simply by including the portions of the pay zone into the recoverable reserve calculations that were excluded during the analysis using only the conventional well logs.

General background about neural networks has been published in numerous papers and will not be repeated in this paper. An example of such publications is included in the references (Mohaghegh et al., 1995, 1996a,b, 1997).

3. Methodology

Four wells from different locations in the United States are used to demonstrate the development of Virtual MRI logs. These wells are from East Texas, New Mexico, Utah, and Gulf of Mexico. These wells are from different formations and since the virtual MRI methodology is a formation specific process, the authors had no options but to test the methodology using single wells. In this paper, the authors will use a part of the pay zone to develop the model and verify it using the remainder of the pay. The ideal way to show the actual potential of this methodology is to use several wells from the same formation. The prerequisite is that both conventional and MRI logs for the wells should be available. In such a situation, a few of the wells would be used to develop the model and the remaining wells would be used for verification purposes. The authors are in the process of locating data to demonstrate the accuracy of the virtual MRI methodology in these situations.

For each well in this study, Gamma Ray, Spontaneous Potential, Caliper, and Resistivity logs were available. These logs were digitized with a resolution of 6 in. for the entire pay zone. Thirty percent of the data were chosen randomly for the model development and the remaining 70% of the pay were used for verification. In all four cases, the model was able to generate synthetic MRI logs with correlation coefficients of up to 0.97 for data that was not used during the model development process.

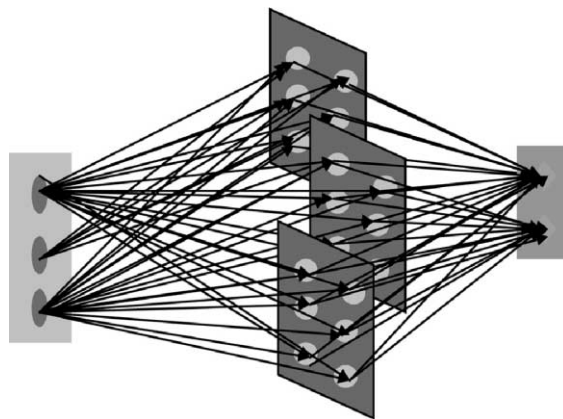


Fig. 1. Schematic diagram of the neural networks used in this study.

The model development process was implemented using a fully connected, five-layer neural network architecture with different activation functions in each layer. These layers included one input layer, three hidden layers and one output layer. Each of the hidden layers has been designed to detect distinct features of the model. A schematic diagram of the network architecture is shown in Fig. 1.

Please note that in this figure all of the neurons and/or connections are shown. The purpose of the figure is to show the general architecture of the

network used for this study. A supervised gradient descent backpropagation of error method was used to train the neural networks. The input layer has six neurons representing Depth, Gamma Ray, SP, Caliper, Medium and Deep Resistivity. Each hidden layer included five neurons. Upon completion of the training process, each neural network contained six weight matrices. Three of the weight matrices had 30 elements while the remaining matrices each had five. In most cases, acceptable generalization was achieved in less than 500 visits to the entire training data.

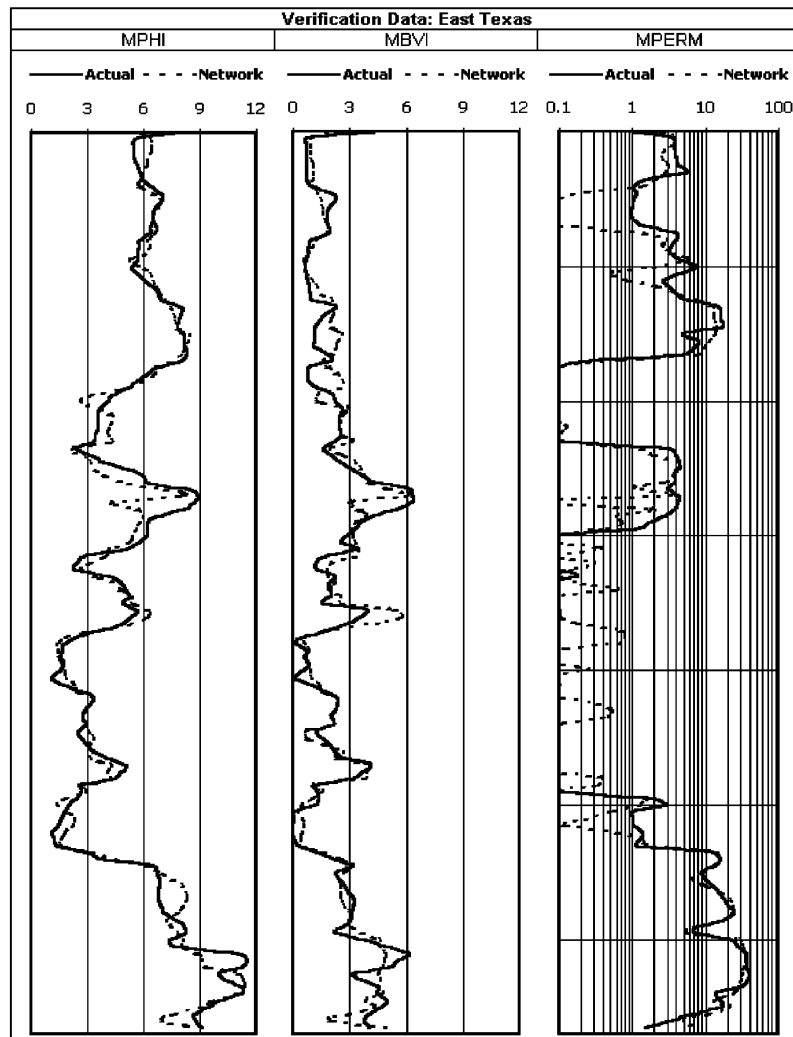


Fig. 2. Virtual and actual MRI logs for the verification data set for the well in East Texas.

Once the network was trained, it was used to generate the virtual MPHI, MBVI and MPERM logs. The MPHI and MBVI logs were then used to estimate the reserves. In one case—the well in New Mexico—that was a tight reservoir, the resolution of the permeability data made it impossible to train an adequate network.

It should be noted that it might be more effective not to use a neural network to replicate the MPERM

log. Since this log is derived from MPHI and MBVI, it would be better to calculate the virtual MPERM log from the virtual MPHI and MBVI.

4. Results and discussion

Figs. 2 and 3 show the actual and virtual MPHI, MBVI, and MPERM logs for the well in East Texas.

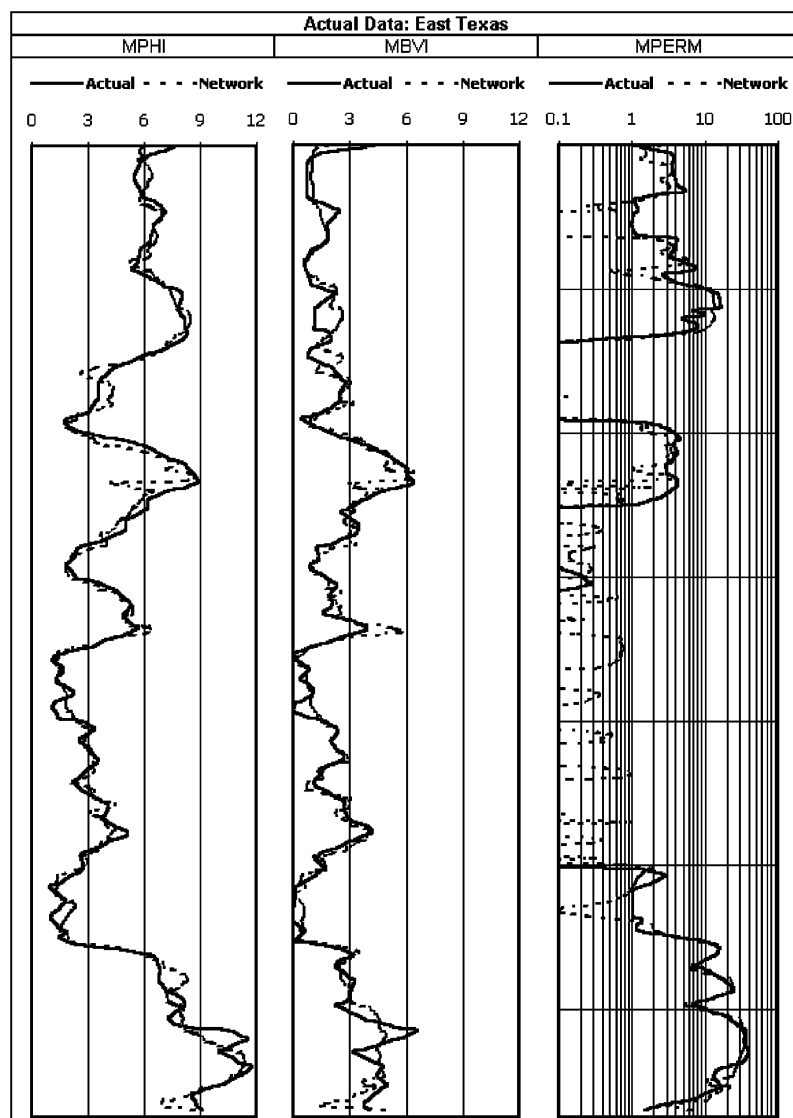


Fig. 3. Virtual and actual MRI logs for the entire pay zone for the well in East Texas.

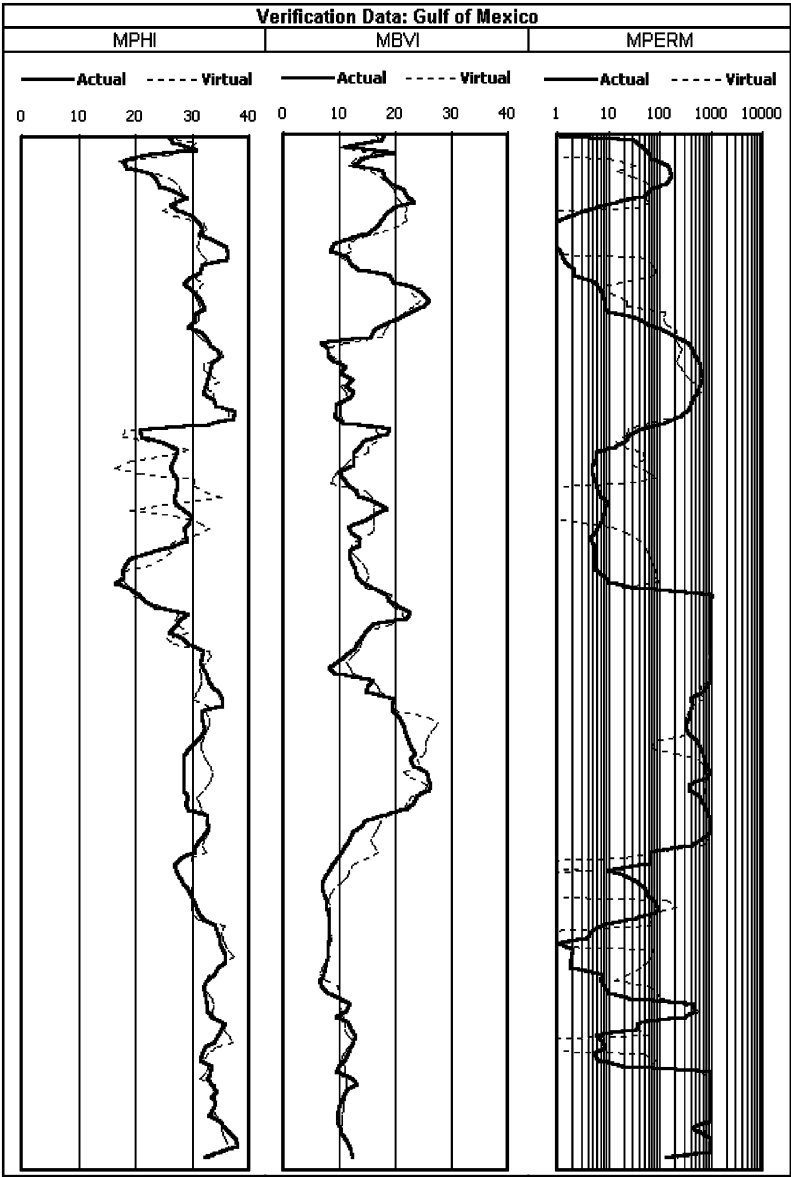


Fig. 4. Virtual and actual MRI logs for the verification data set for the well in Gulf of Mexico.

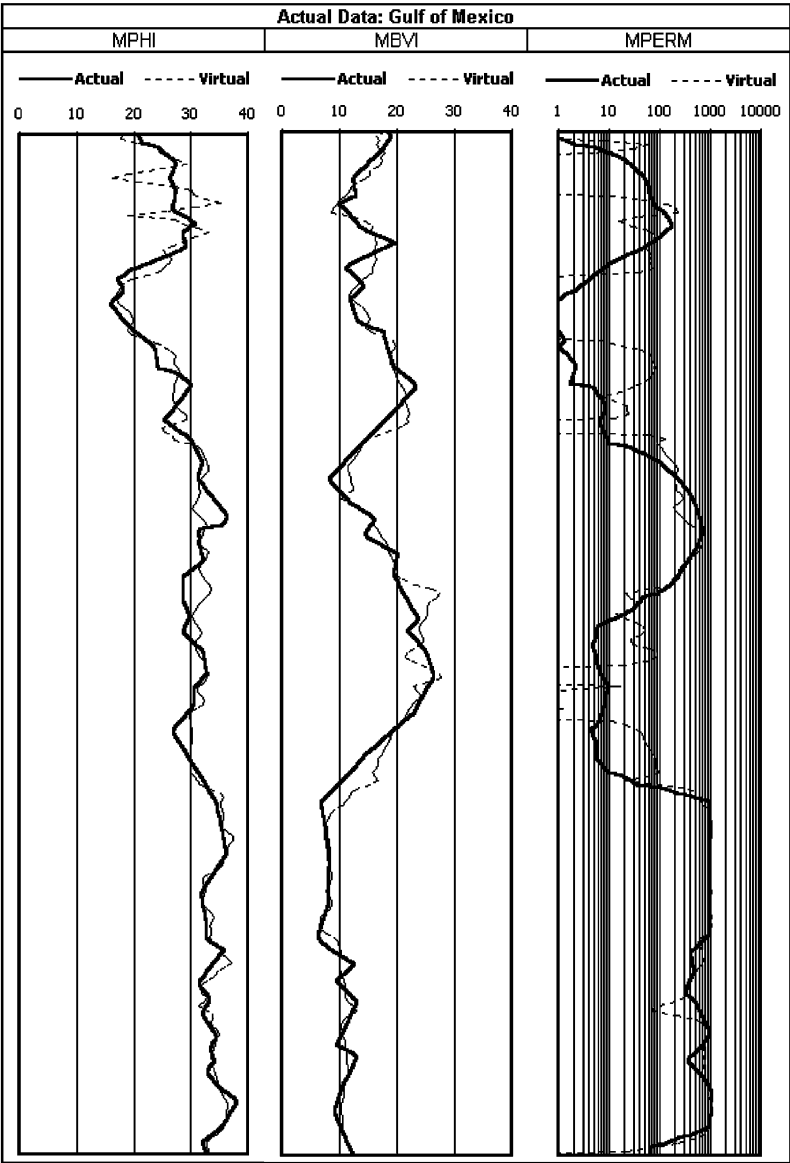


Fig. 5. Virtual and actual MRI logs for the entire pay zone for the well in Gulf of Mexico.

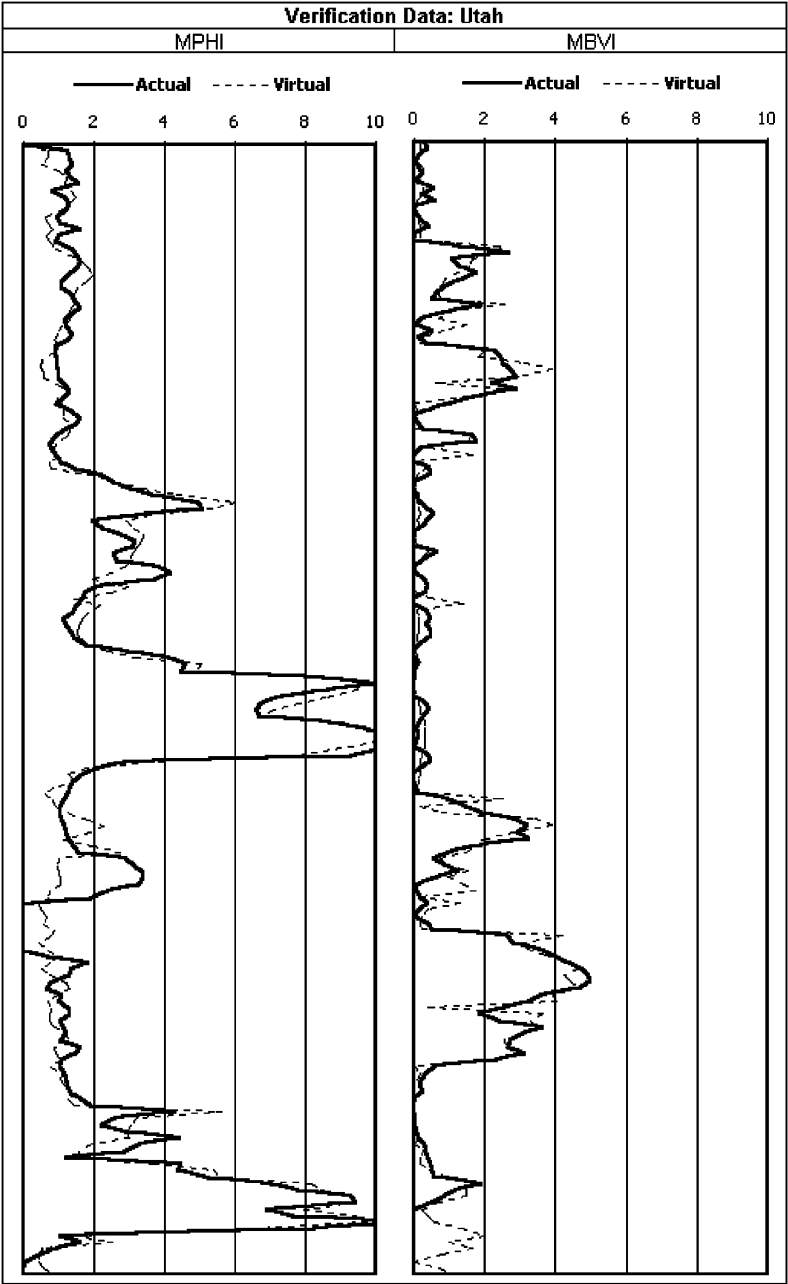


Fig. 6. Virtual and actual MRI logs for the verification data set for the well in Utah.

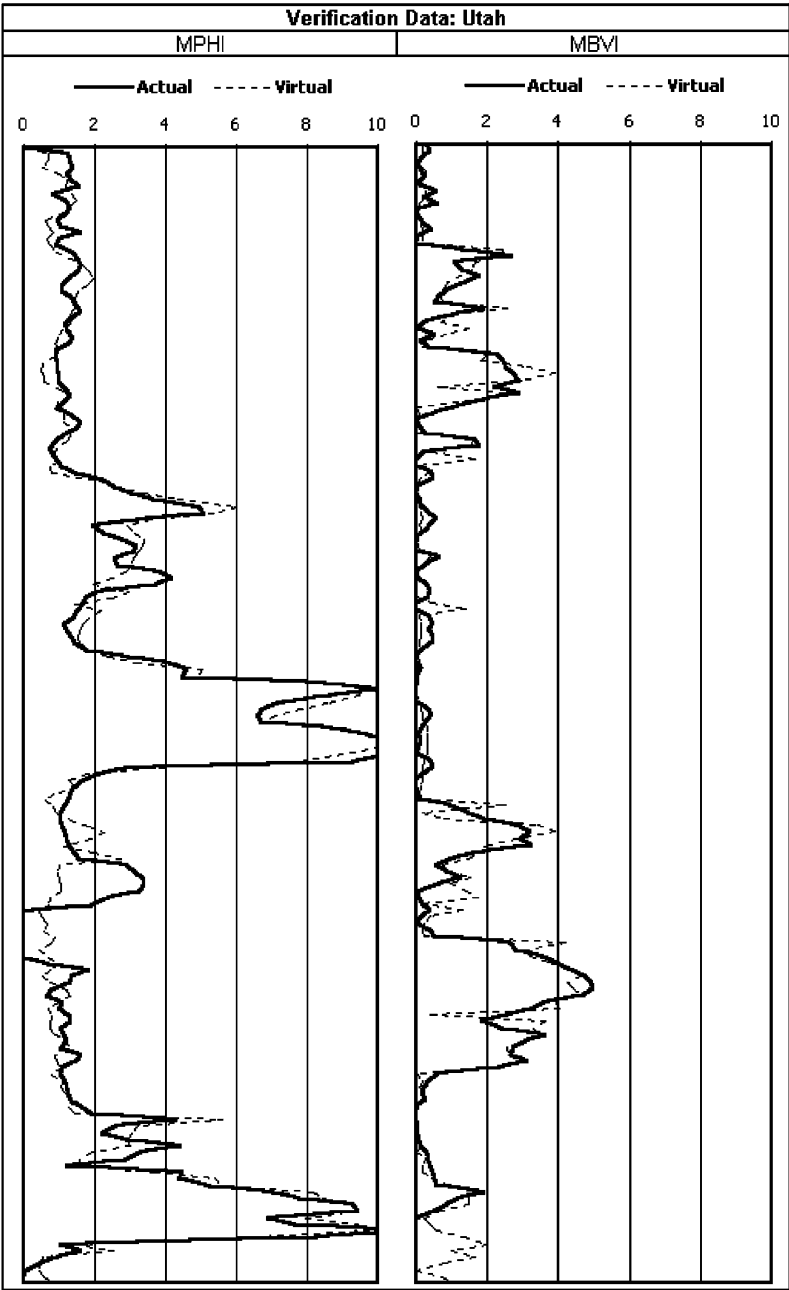


Fig. 7. Virtual and actual MRI logs for the entire pay zone for the well in Utah.

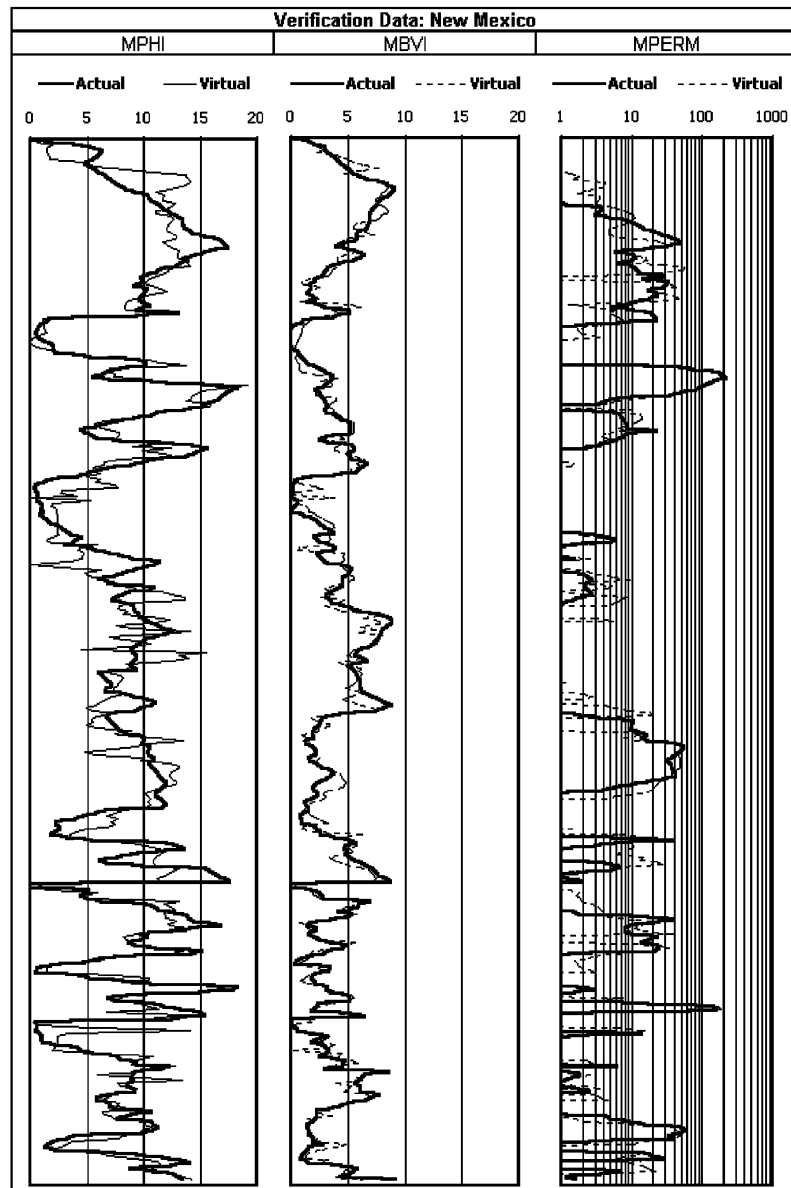


Fig. 8. Virtual and actual MRI logs for the verification data set for the well in New Mexico.

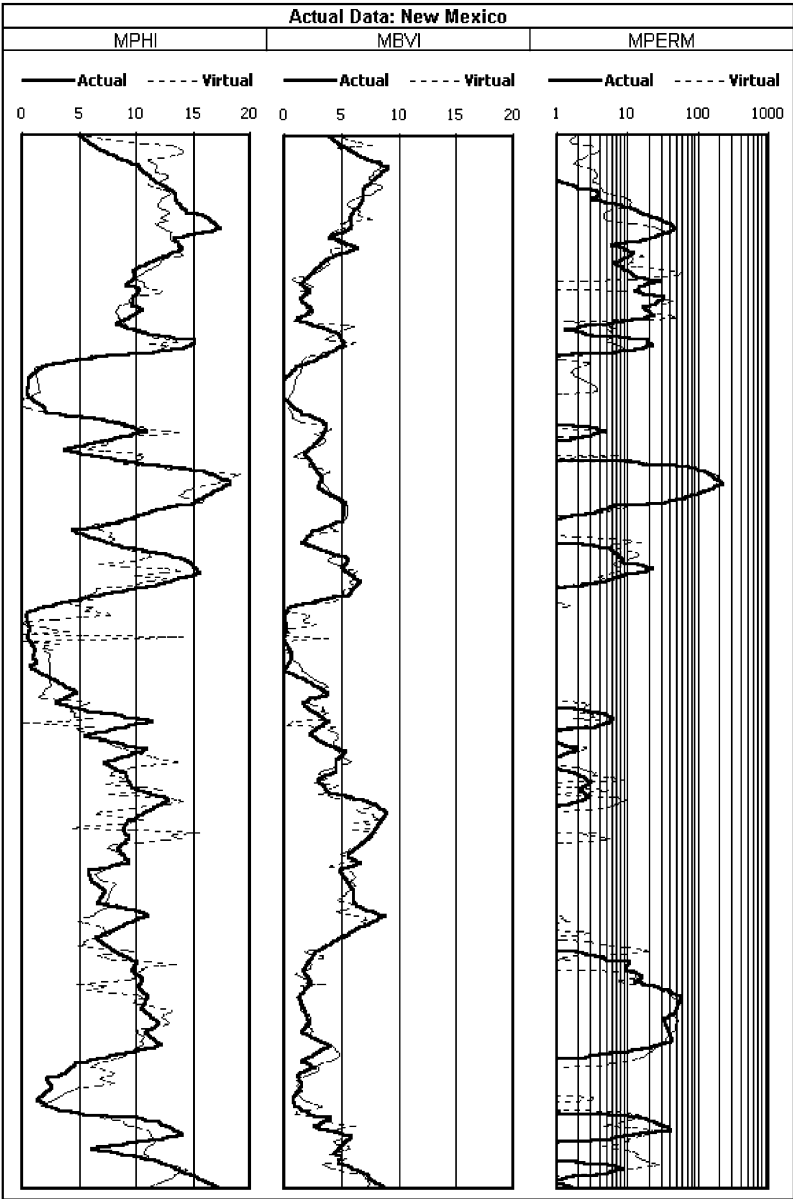


Fig. 9. Virtual and actual MRI logs for the entire pay zone for the well in New Mexico.

Fig. 2 shows only the verification data set—data never seen by the network before—while Fig. 3 contains the virtual and actual logs for the entire pay zone. Virtual effective porosity log (MPHI) has a correlation coefficient of 0.941 for the verification data set and a 0.967 correlation coefficient for the entire pay zone. The values for virtual MBVI log are 0.853 and 0.894, respectively. The virtual permeability log for this well also shows a strong correlation, 0.966 for the verification data set and 0.967 for the entire pay zone.

Figs. 4–9 show similar results for the wells from Utah, Gulf of Mexico, and New Mexico, respectively. In all cases shown in these figures, virtual MRI logs closely follow the general trends of the actual MRI logs. Please note that MPERM logs are shown in logarithmic scale and therefore the difference in the lower values of the permeability can be misleading. The correlation coefficient provides a more realistic mathematical measure of closeness of these curves to one another.

Table 1 is a summary of the analysis done on all four wells. This table contains the correlation coefficients for all the logs that were generated. This table

Table 1
Correlation coefficient between actual and virtual MRI logs

Well location	MRI log type	Data set	Correlation coefficient
Texas	MPHI	Verification	0.941
		Entire well	0.967
	MBVI	Verification	0.853
		Entire well	0.894
	MPERM	Verification	0.966
		Entire well	0.967
Utah	MPHI	Verification	0.800
		Entire well	0.831
	MBVI	Verification	0.887
		Entire well	0.914
	MPERM	Verification	0.952
		Entire well	0.963
Gulf of Mexico	MPHI	Verification	0.858
		Entire well	0.893
	MBVI	Verification	0.930
		Entire well	0.940
	MPERM	Verification	0.945
		Entire well	0.947
New Mexico	MPHI	Verification	0.957
		Entire well	0.960
	MBVI	Verification	0.884
		Entire well	0.926

Table 2

A per-acre estimate of the recoverable reserves using actual and virtual MRI logs

Well location	MRI type	Reserve Bbls/acre	Percent difference
Texas	Actual	52,368	– 1.4
	Virtual	51,529	
New Mexico	Actual	24,346	– 1.9
	Virtual	23,876	
Gulf of Mexico	Actual	240,616	+ 0.3
	Virtual	241,345	
Utah	Actual	172,295	– 1.8
	Virtual	169,194	

shows the accuracy of the virtual MRI log methodology on wells from different locations in the United States. The lowest correlation coefficient belongs to virtual MPHI log for the well located in Utah—0.800—while the best correlation coefficient belongs to virtual MPERM log for the well located in East Texas—0.966.

It should be noted that in this study absence of ideal data forced authors to use only 30% of the available data for construction and training of the neural networks. Authors believe that if more data were available from a particular field, the accuracy of the networks would increase.

Although the correlation coefficients for all the virtual logs are quite satisfactory, it should be noted that once these logs are used to calculate estimated recoverable reserves, the results are even more promising. This is due to the fact that many times the effective porosity and saturation is averaged. After all, MRI logs are used in two different ways. One way is to locate and complete portions of the pay zone that have been missed due to the conventional log analysis. This is more a qualitative analysis than a quantitative one since the engineer will look for an increase in the difference between MBVI and MPHI that correspond to a high permeability interval. The second use of these logs is to estimate the recoverable reserves more realistically.

The reserve estimates calculated using virtual MRI logs when compared to estimates calculated using actual MRI logs were quite accurate. As shown in Table 2, the reserve estimates using virtual MRI logs ranged from underestimating the recoverable reserves by 1.8% to over estimating it by 0.3%.

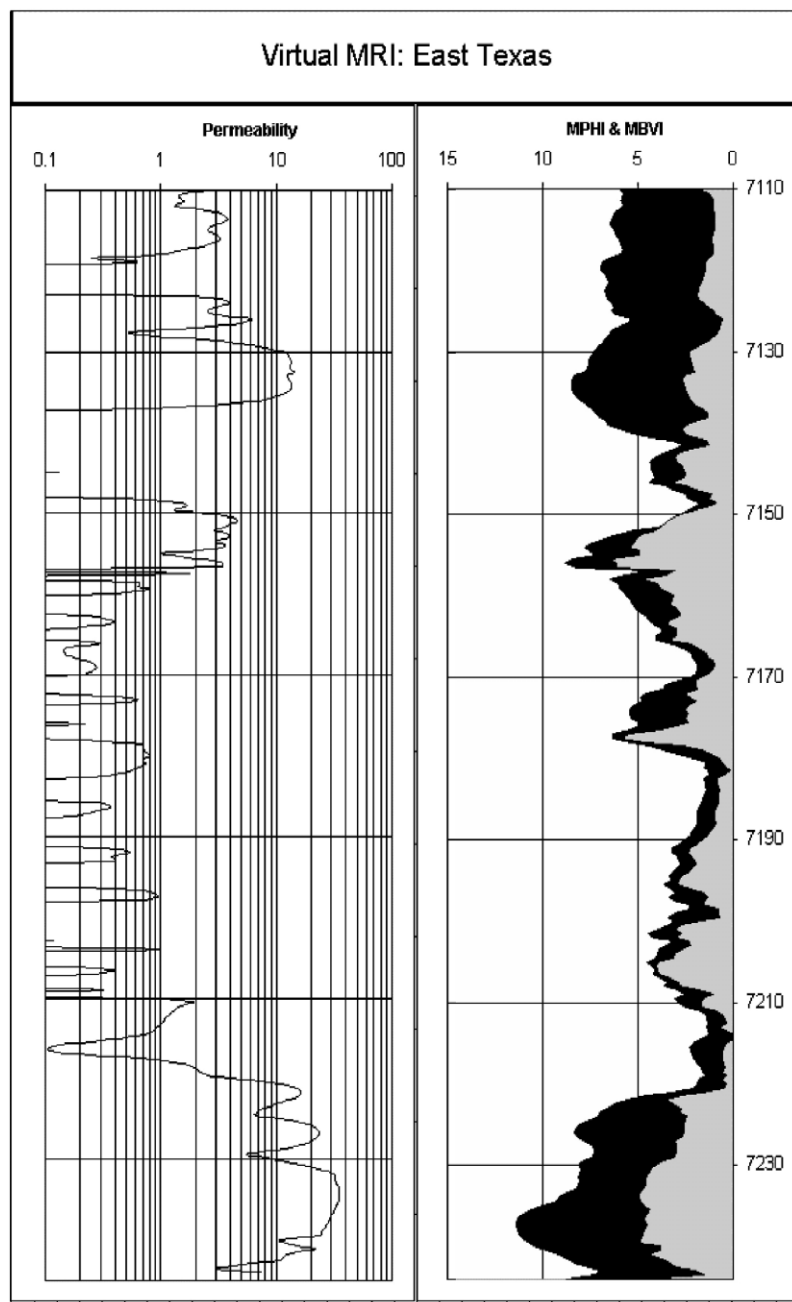


Fig. 10. Virtual MRI logs for the well in East Texas.

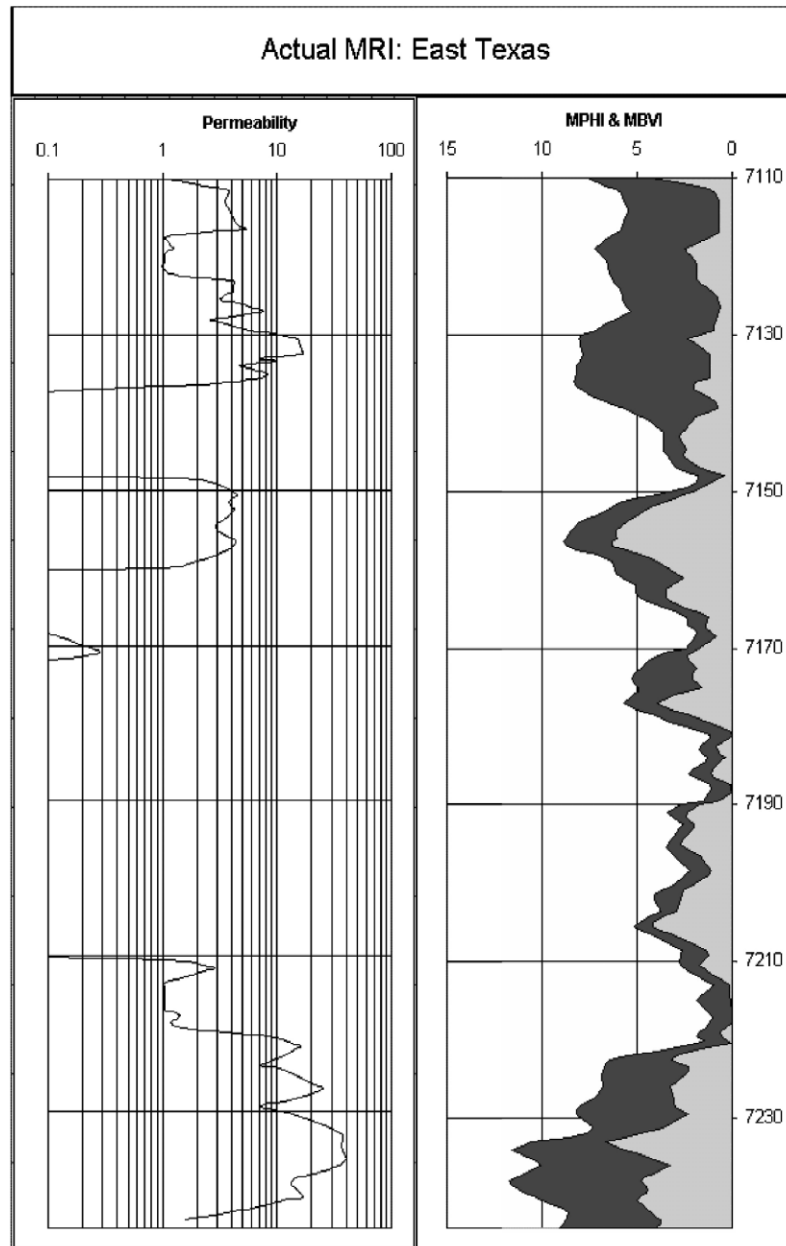


Fig. 11. Actual MRI logs for well in East Texas.

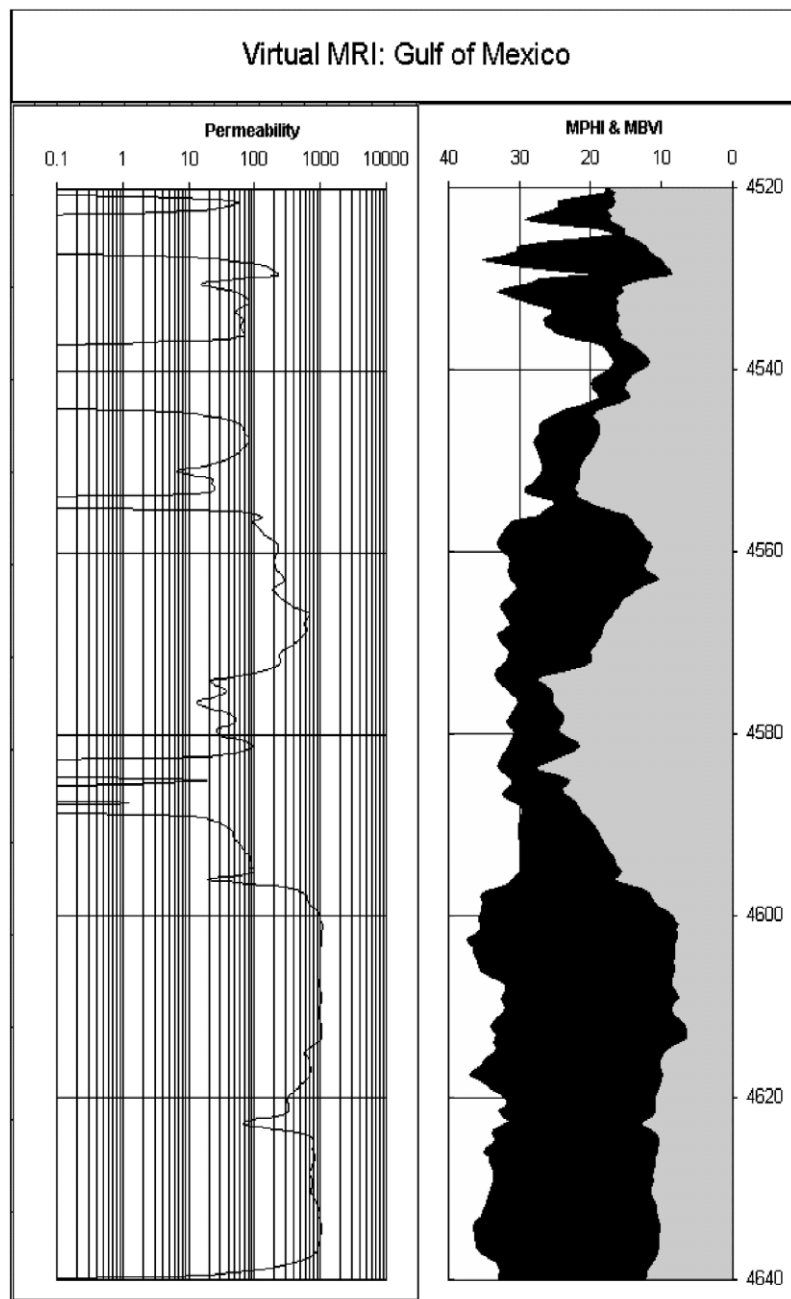


Fig. 12. Virtual MRI logs for the well in Gulf of Mexico.

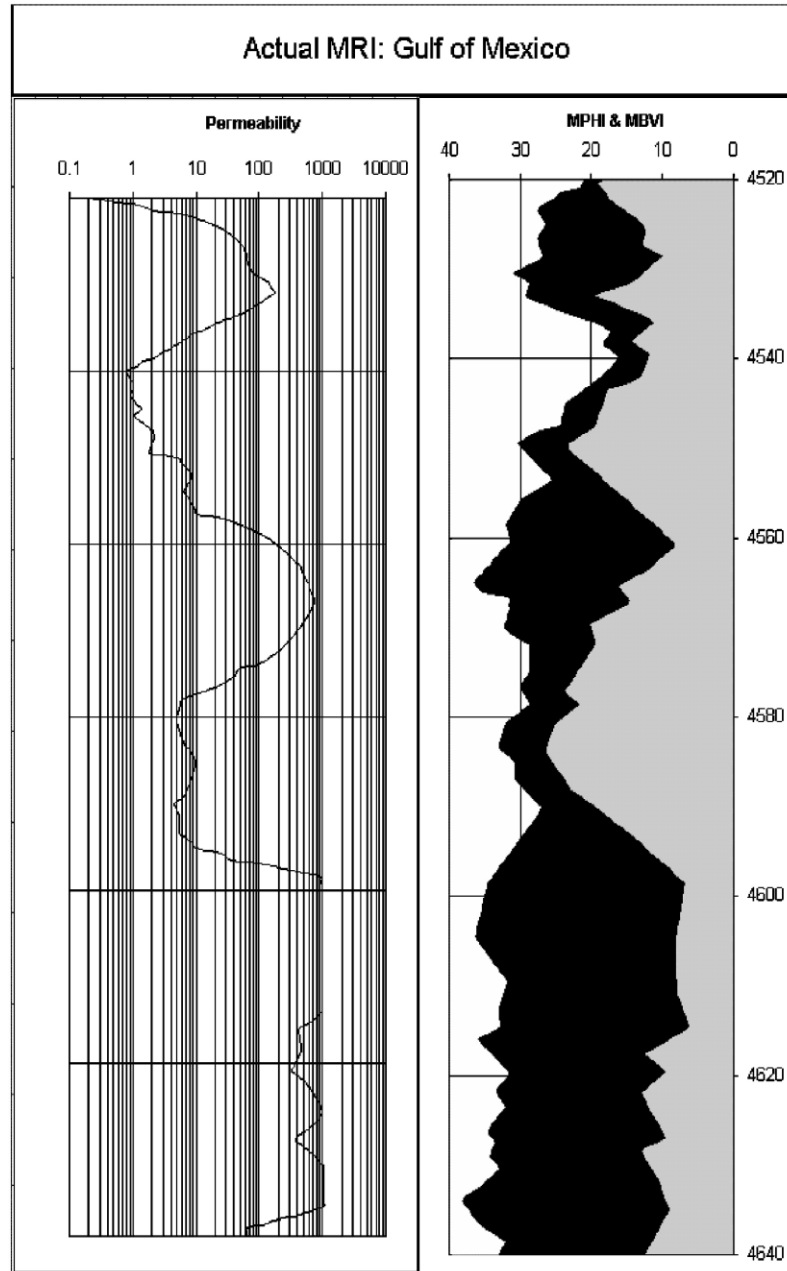


Fig. 13. Actual MRI logs for the well in Gulf of Mexico.

Figs. 10–13 show the virtual and actual MRI logs for wells in East Texas and the Gulf of Mexico. These logs are shown in the fashion that MRI logs are usually presented. These logs clearly show the free fluid index—difference between MBVI and MPHI logs—and the corresponding permeability values. This particular representation of the MRI logs is very useful to locate the portions of the pay zone that should be completed. The parts of the pay that has a high free fluid index and corresponds to a reasonably high permeability value are excellent candidates for completion.

5. Conclusions

In this paper, authors demonstrated that development of reasonably accurate virtual magnetic imaging logs is quite possible. The neural networks that are constructed and trained for a particular formation may not be used to generate virtual MRI logs for other formations. This conclusion is similar to the prior experience of the authors (Mohaghegh et al., 1996a). As in the case of virtual measurement of formation permeability, the methodology is formation-dependent.

Virtual MRI logs are valuable tools to accurately and realistically estimate recoverable reserves—especially in low resistivity formations—as well as identification of portions of the pay zone that should be completed. The most effective way of employing the virtual MRI logging method is to:

- Physically perform MRI logs on several strategically located wells in the field,

- Use the data to develop a virtual MRI tool for the formation, and
- Use the tool to generate virtual MRI logs for the remaining wells in the field.

Using virtual MRI logs methodology in field-wide bases can cut the cost considerably when compared to performing actual MRI logging on all the wells in a field. Authors also conclude—from our preliminary work that has not been shown in this paper—that similar methodology can be employed to generate many different kinds of virtual well logs such as SP, density porosity, etc., from other available logs.

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Magnetization Evolution in Network Models of Porous Rock under Conditions of Drainage and Imbibition

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Surface-enhanced relaxation of nuclear magnetization in fully and partially saturated water-wet porous media is studied using a pore network simulator. The simulator is based on a description of the pore space in terms of a regular cubic lattice of pores and throats following respective size distributions. The latter are obtained from geometric characterization of 3D stochastic replicas of real porous rock samples. Concepts of percolation theory are used to simulate the pore-scale distribution of a strongly wetting phase under conditions of quasistatic drainage by or imbibition against a nonwetting phase. The results of these simulations compare favorably with experimental mercury intrusion/retraction curves. The equations governing magnetization evolution in a connected pore system are then solved by matrix diagonalization for different values of wetting-phase saturation along the primary drainage and secondary imbibition paths. Analysis of the corresponding spectra of decay rates provides new insights regarding the influence of pore structure (pore and throat size distributions, spatial correlation), surface relaxation strength, and fluid distribution on diffusive coupling between pores. For pore and throat size distributions and surface relaxation strength representative of sandstones, diffusive coupling is quite important, especially under conditions of partial saturation. Remarkably, simulations show that correlated heterogeneity is the main reason pores appear poorly coupled with respect to NMR relaxation—an assumption underlying the correspondence between pore size and relaxation time distributions. Finally, for a given value of wetting-phase saturation, the history of saturation change (drainage or imbibition) is shown to have a profound effect on the spectrum of decay rates. © 2002 Elsevier Science (USA)

Key Words: magnetic resonance; relaxation; spatial correlation; diffusion; drainage; imbibition; network model; pore size distribution.

1. INTRODUCTION

The sensitivity of nuclear magnetic resonance (NMR) to molecular-level events within fluids provides important opportunities for determining fluid states and properties within porous media (1). NMR methods have recently found important practical applications in downhole well logging for hydrocarbon formation evaluation (2). In a NMR measurement, polarized

nuclear spins in fluid molecules probe the pore space by diffusive motion. The magnetic moment carried by these spins relaxes due to dipolar interactions with other fluid molecules and, for a wetting fluid, also due to interactions at the fluid-solid interface. The total magnetization evolves by diffusion and relaxation processes and is sensitive to the geometry of the pore space (3). Measurements of the decay of nuclear magnetization are relatively straightforward. However, interpretation of these measurements to determine structural properties of porous media and characterize the distribution and flow of fluids therein is a task that remains partly unresolved (4).

In a seminal contribution, Brownstein and Tarr (5) established that the effect of surface-enhanced relaxation on magnetization decay within a single isolated pore of simple geometry is described, to an excellent approximation, by a single relaxation time, T_s , depending on the value of the dimensionless parameter $\rho\alpha/D$,

$$\frac{1}{T_s} = \begin{cases} \frac{\rho}{\alpha}, & \frac{\rho\alpha}{D} \ll 1 \\ \frac{D}{\alpha^2}, & \frac{\rho\alpha}{D} \gg 1 \end{cases}, \quad [1]$$

where D is the self-diffusion coefficient of the saturating fluid, ρ is a parameter characterizing the strength of surface relaxation, and α is a characteristic pore size corresponding to V_p/A_p , the ratio of pore volume to pore surface area. The condition $\rho\alpha/D \ll 1$ corresponds to the so-called “fast-diffusion” regime—a situation for which the relaxation rate is controlled by pore surface interactions. The condition $\rho\alpha/D \gg 1$ corresponds to the “slow-diffusion” regime—a situation for which the rate of surface relaxation equals the rate of molecular diffusion to the fluid-solid interface.

Bulk relaxation processes within the fluid contribute in parallel to surface relaxation, so that the observed relaxation rate is given by (1)

$$\frac{1}{T} = \frac{1}{T_s} + \frac{1}{T_b}, \quad [2]$$

where $1/T_b$ is the bulk relaxation rate. Bulk relaxation is often, but not always (6), negligible by comparison to surface relaxation (i.e., $T_b \gg T_s$). In any case, its effect can be easily

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separated from the effect of surface relaxation on magnetization decay and, for this reason, it is not discussed further.

The interpretation of magnetization decay observed in porous rock is based on two fundamental, and seemingly contradictory, assumptions (2). The first assumption is that each pore is in the “fast-diffusion” regime; i.e., diffusion is fast enough to maintain a uniform magnetization across space. This assumption is essential for establishing a direct relationship between relaxation time and pore size, as shown in Eq. [1]. Sufficiently large pores may be in the slow-diffusion regime, although in such cases it is likely that the longer surface relaxation will be swamped by T_b (2). For a variety of rock samples, Latour *et al.* (6) obtained relaxation time distributions that were independent of temperature over the range 25–175°C. Since the relaxation time in the diffusion-limited case depends explicitly on the bulk diffusion coefficient (cf. Eq. [1]), itself a function of temperature, the results of Latour *et al.* (6) suggest that the samples studied were in the fast-diffusion regime.

The second assumption is that diffusion between pores is unimportant, so that magnetization in each pore decays independently of other pores. On this assumption, the broad distribution of relaxation times usually observed in samples of porous rock (6–8),

$$M(t) = M_0 \sum_i f_i e^{-t/T_i}, \quad [3]$$

is interpreted as a volume-based distribution of pore sizes, where f_i is the fraction of pore volume associated with pores of characteristic size $(V_p/A_p)_i = \rho T_i$ (9). Since, however, the pore space is highly connected, it is reasonable to expect that nuclear spins will probe more than one pore during NMR observation times (3), in apparent contradiction with the assumption of weak coupling.

Following Cohen and Mendelson (10), McCall *et al.* (11) investigated the effects of diffusive coupling between pores under conditions of fast diffusion. These authors formulated and solved the coupled magnetization balance equations for a fully saturated network of pores connected with pore throats. They demonstrated that the breadth of the decay rate distribution decreases and the distribution shifts to higher rates, as coupling increases. For very strong coupling, the spectrum is reduced to a narrow peak at a rate determined by the ratio of the total surface to total volume of the pore system. This situation corresponds to a magnetization that is uniform within the entire sample and decays as a single exponential.

McCall *et al.* (11) argued that pore coupling in sandstone rocks is weak, and therefore the assumption that the spectrum of decay rates reflects the density of pore sizes is probably a good one. Their conclusions, however, were based on simulations using random pore networks in which the size of pore throats was uniform. In real sandstone rock and other porous media, not only pore and throat sizes follow respective distributions, but also correlation between the size of a throat and the size of

adjacent pores (12, 13), as well as spatial correlation among the pore sizes (14–16), are very likely to exist. None of these correlations were included in the model of McCall *et al.* (11). Their theoretical prediction that a significant narrowing of the decay rate spectrum should take place as the coupling strength is increased by increasing the bulk diffusion coefficient (i.e., by increasing temperature) was not confirmed by the experiments of Latour *et al.* (6).

Importantly, the assumption that a relaxation time distribution corresponds to a pore size distribution also underlies attempts to estimate a drainage capillary pressure curve by simply rescaling a relaxation time distribution (17). This correspondence has received little theoretical attention. NMR relaxation measurements in porous samples that have undergone drainage-type displacement have revealed qualitative changes in the spectrum of decay rates, which are consistent with the accepted view that (i) relaxation time is proportional to pore size for water-saturated pores and (ii) the nonwetting fluid enters progressively smaller pores as the water saturation decreases (4).

Network models have been instrumental in understanding the distribution and flow of immiscible fluids in porous media (18, 19). However, with the exception of McCall *et al.* (11), who only considered fully saturated networks of very simple and somewhat unrealistic geometry, NMR relaxation in porous media has not been studied by network simulation. This task is undertaken here. We are particularly interested in network models which are consistent with structural information extracted by analysis of 3D images of the pore space and which can quantitatively describe experimental mercury porosimetry data (20). With such models at hand, our attention is then focused on information about the pore geometry and fluid distribution that may be garnered by NMR relaxation analysis.

This paper is structured as follows. In the first section we describe the construction of the pore network model, paying particular attention to its geometric and correlation aspects (pore and throat size distributions, pore/throat correlation, spatial structure, pore surface roughness). In the same section we briefly describe the algorithms used to simulate drainage and imbibition, develop the magnetization balance equations, and discuss their solution. In a subsequent section, we first demonstrate that the network model can successfully simulate experimental mercury-air drainage and imbibition capillary pressure curves obtained from Berea sandstone and North Sea chalk samples. We then analyze magnetization decay spectra obtained at different values of saturation along the primary drainage and secondary imbibition paths. We show that, in the absence of spatial correlation among the pores and for pore structure and relaxation parameters consistent with experimental data for Berea sandstone, diffusive coupling is quite pronounced. We then demonstrate that spatial correlation of the pore sizes masks the effects of diffusive coupling, so that the observed spectrum of decay rates at $S_w = 1$ corresponds well with the pore size distribution, even when diffusive coupling is significant. We also discuss the significance of these results for the interpretation of NMR relaxation spectra

at partial saturation under conditions of primary drainage and secondary imbibition. Finally, we use the network simulator to explore the correspondence between NMR relaxation spectra at $S_w = 1$ and primary drainage capillary pressure curves.

2. THEORY

2.1. Network Model of Rock Pore Structure

Following previous work (21, 22), we choose to represent the structure of porous rock with a regular cubic lattice of pores and throats. Pores are identified with sites in the lattice and throats with bonds between sites. We consider that pores are irregular in shape and choose to characterize them by an equivalent-cube dimension, R_p , so that their pore volume and surface area are given by $V_p = 8R_p^3$ and $S_p = 24R_p^2$, respectively. We also assume that pore sizes are randomly distributed on lattice sites according to a known pore size distribution (PSD), in an uncorrelated or spatially correlated fashion. In the latter case, we choose an isotropic exponential autocorrelation function with correlation length equal to five lattice sites (15, 22).

Consistent with recent advances in 3D pore space characterization (23, 24), we take pore throats to represent volumeless constrictions between pores. The throat cross sections may also be irregular. We choose to characterize pore throats in terms of a single dimension, R_t , such that $R_t = \sqrt{A_t}/2$, where A_t is the throat cross-sectional area. The distributions of both A_t and V_p can be measured, if a description of the pore space in the form of 3D volume data is available (23, 24). Determination of the characteristic dimensions R_p and R_t from such information is straightforward. Throat sizes from a respective distribution (TSD) are distributed on the bonds of the lattice according to the bond-site correlation scheme detailed elsewhere (21, 25). This scheme introduces positive correlation between the size of a throat and the size of the pores communicating through it, in agreement with the generally accepted view of rock pore structure (13). To account for resistance to molecular diffusion between pores in the model, we consider an interpore diffusion length distribution (DLD). Interpore diffusion lengths correspond to the lengths of distinct pore space channels. Their determination by morphological skeletonization of 3D volume data has been detailed elsewhere (23). Interpore diffusion lengths are also positively correlated to the size of the adjoining pores. Pore size, throat size, and interpore length distributions obtained from 3D stochastic models representative of Berea sandstone are shown in Fig. 1.

2.2. Percolation Modeling of Drainage and Imbibition

Assuming strongly water-wet conditions, we model quasistatic drainage as bond-correlated site percolation without trapping (12, 15, 25), and quasistatic imbibition as mixed-percolation (22) on a lattice. The reader is referred to the original publications for detailed descriptions of the algorithms used to determine the fluid occupancy of pores and throats as a function of capillary pressure during drainage and imbibition. Here, we

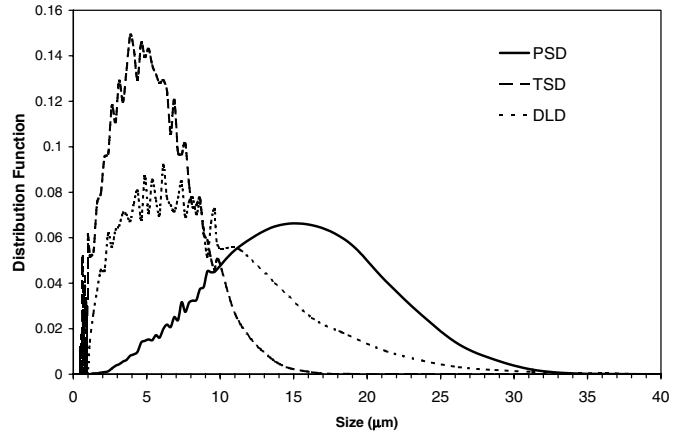


FIG. 1. Number-based pore size (PSD), throat size (TSD), and diffusion length (DLD) distributions for Berea sandstone.

note that the critical capillary pressure for piston-type invasion of the nonwetting phase into a pore or throat during drainage is computed from

$$P_c^{(piston)} = \frac{\sigma f(\theta_R)}{R}, \quad [4]$$

where θ_R is the receding contact angle, σ is the interfacial tension and the function $f(\theta_R)$ is given by (26)

$$f(\theta_R) = \frac{\theta_R + \cos^2 \theta_R - \pi/4 - \sin \theta_R \cos \theta_R}{\cos \theta_R - \sqrt{\pi/4 - \theta_R + \sin \theta_R \cos \theta_R}}. \quad [5]$$

The critical capillary pressure for piston-type retraction of the nonwetting phase from a pore or throat during imbibition is also computed from Eq. [4] and Eq. [5] using the advancing contact angle θ_A in place of θ_R . Cooperative pore filling (27, 28) is not taken into account.

The critical capillary pressure for snap-off in pore throats during imbibition is computed as

$$P_c^{(snap-off)} = \frac{P_c^{(piston)}}{\xi}, \quad [6]$$

where the critical pressure ratio, ξ , is defined as the ratio of capillary pressure for piston-type retraction to the capillary pressure for snap-off in a pore throat (29). For throats and pores of simple geometry, the critical pressure ratio ξ can be computed from interfacial stability considerations (22, 30). However, to avoid imposing arbitrary geometric restrictions and to preserve the flexibility of the network model, we choose to treat ξ as an adjustable parameter in the simulation (31). We expect that, if the model provides a reasonable representation of real media, the values of ξ which give good agreement with experimental data will also agree reasonably well with theoretical predictions of this parameter (22, 30).



FIG. 2. Illustration of irregular pore geometry in stochastically reconstructed porous media. Pore throats are shown in black (after (24)).

2.3. Modeling the Effect of Irregular Pore Geometry on Fluid Distribution

It is important to note that the rock-pore interface in real porous media is quite rough and often fractal (32–34). Accordingly, pore and throat sizes may be expected to span a broad range of length scales. In agreement with this expectation, experimental mercury porosimetry curves usually reveal that considerable fractions of the total pore volume are intruded over a very broad range of pressures. These observations have two important consequences for the simulation of drainage and imbibition using network models. First, PSD and TSD functions determined by analysis of 3D volume data of limited resolution (ca. 1–5 μm for sandstone and 0.1–0.3 μm for chalk) and subsequently used in network models do not represent the entire spectrum of pore and throat sizes present in real rock. Secondly, even if the pore volume distribution were perfectly known, mapping of irregular pores (see Fig. 2) onto regular, albeit angular, objects (e.g., cubes) would still result in poor approximation of the volume of the wetting fluid that remains in the pores after they are invaded by a nonwetting fluid. Tsakiroglou and Payatakes (35) recognized this and proposed to endow network models with a geometrically determined fractal model of the pore surface, in connection with simulations of mercury porosimetry.

Here, we propose to capture the effect of pore structure that is not explicitly represented in the network model, on the distribution of wetting fluid at the pore scale as follows. Consider that upon invasion of the nonwetting fluid into pore i at capillary pressure $P_{c,i}^*$, the remaining wetting fluid occupies a fraction S_w^* of the volume of the pore. At capillary pressure $P_c > P_{c,i}^*$, the fraction of pore volume occupied by the wetting fluid is assumed to vary according to the expression:

$$\frac{S_{w,i}}{S_w^*} = \left(\frac{P_{c,i}^*}{P_c} \right)^\eta. \quad [7]$$

The initial fraction S_w^* and “filling” exponent η are physically meaningful model parameters. For $0 < \eta < 1$, drainage of the wetting fluid within each pore at $P_c > P_{c,i}^*$ corresponds to filling with the nonwetting phase of a surface fractal with dimension $2 < D_s < 3$. This follows from consideration of a fractal distribution of length scales characterizing the geometry of the pore walls (34). For $\eta > 1$ the pore behaves as a rough, but not fractal, object. For example, $\eta = 3$ for wetting fluid in the corner of a cube and $\eta < 3$ for more irregular (but nonfractal) geometries. The parameter S_w^* may be identified with the fraction of pore volume contributed by length scales not explicitly represented in the network model, because they are lacking from the measured TSD and PSD and/or because the measured volume of each pore is assigned to a regular geometric object. The value of $P_{c,i}^*$ is different for each pore and depends on the adjoining throat sizes and pore accessibility. During primary imbibition, and for as long as piston-type retraction or disconnection of the nonwetting fluid in a pore has not taken place, the wetting fluid content of a pore increases according to Eq. [8]. This implies neglect of nonwetting phase trapping within the unresolved pore space. Note finally that, unlike the roughness model of Tsakiroglou and Payatakes (35), the model proposed here does not require explicit description of the geometry of roughness features.

2.4. Modeling of Magnetization Evolution in the Pore Network

Following McCall *et al.* (11), we assume that magnetization within each pore is uniform (fast-diffusion regime), but magnetization exchange by diffusion between pores is possible. A balance of magnetization M_i may be written for each pore i , expressing the fact that the net rate of magnetization decay in each pore is determined by magnetization relaxation in the vicinity of the fluid-matrix interface and by diffusion to and from nearest neighboring (NN) pores,

$$\frac{dM_i}{dt} = -\rho \cdot A_{p,i} \cdot \frac{M_i}{V_{w,i}} + D \cdot \sum_j^{NN} \frac{A_{w,ij}}{L_{ij}} \cdot \left(\frac{M_j}{V_{w,j}} - \frac{M_i}{V_{w,i}} \right), \quad [8]$$

where $A_{p,i}$, $V_{w,i}$ are the surface area and volume occupied by water within pore i , respectively, D is the self-diffusion coefficient of water, and $A_{w,ij}$, L_{ij} are the cross-sectional area and length associated with the diffusion of water between pores i and j . Note that, since pore surfaces are covered with wetting fluid at all times, $A_{p,i}$ is a constant for each pore, whereas $V_{w,i}$ and $A_{w,ij}$ depend on fluid occupancy and capillary pressure. Specifically, $V_{w,i} = S_{w,i} V_{p,i}$ and $A_{w,ij} = S_{w,ij} A_{t,ij}$, where $S_{w,i}$ and $S_{w,ij}$ for pores and throats are both computed using Eq. [7].

In compact form, Eq. [8] may be written as

$$\frac{dM_i}{dt} = -\omega_\rho \cdot r_i \cdot M_i + \omega_c \cdot \sum_j^{NN} \frac{s_{ij}}{l_{ij}} \cdot \left(\frac{M_j}{v_j} - \frac{M_i}{v_i} \right), \quad [9]$$

where

$$r_i = \frac{A_{p,i}}{V_{w,i}} \cdot \frac{\langle V_w \rangle}{\langle A_p \rangle}, \quad v_i \equiv \frac{V_{w,i}}{\langle V_w \rangle}, \quad s_{ij} \equiv \frac{A_{w,ij}}{\langle A_w \rangle}, \quad \text{and} \quad l_{ij} \equiv \frac{L_{ij}}{\langle L \rangle}$$

(angular brackets denote average values). The parameters

$$\omega_\rho \equiv \rho \cdot \frac{\langle A_p \rangle}{\langle V_w \rangle} = \frac{1}{\tau_\rho} \quad \text{and} \quad \omega_c \equiv \frac{D \langle A_w \rangle}{\langle V_w \rangle \langle L \rangle}$$

represent characteristic rates of relaxation at the fluid-solid interface and of molecular diffusion between pores, respectively. Application of Eq. [9] to each pore yields an initial-value, linear ordinary differential equation problem, which is most efficiently solved by matrix diagonalization methods (11). To simplify the solution, the following change of variables is made:

$$\theta_i = \frac{M_i}{\sqrt{v_i}}. \quad [10]$$

This change in variables modifies Eq. [10] as

$$\frac{d\theta_i}{dt} = -\omega_\rho \cdot r_i \cdot \theta_i + \omega_c \cdot \sum_j^{NN} \frac{s_{ij}}{l_{ij}} \cdot \left(\frac{\theta_j}{\sqrt{v_j \cdot v_i}} - \frac{\theta_i}{v_i} \right), \quad [11]$$

or in matrix form,

$$\frac{d}{dt}\{\theta\} = -[H] \cdot \{\theta\}, \quad [12]$$

where $[H]$ is a symmetric Hamiltonian matrix. The vector of initial conditions is given as

$$\theta(0) = \frac{1}{\sqrt{N}} \cdot \begin{pmatrix} \sqrt{v_1} \\ \sqrt{v_2} \\ \vdots \\ \sqrt{v_N} \end{pmatrix}, \quad [13]$$

where N is the total number of pores in the network. The solution of the problem defined by Eq. [12] and Eq. [13] is expressed in terms of the eigenvalues and eigenvectors of matrix $[H]$ as

$$\frac{M(t)}{M(0)} = \sum_n |\theta(0) \cdot \psi_n|^2 \cdot e^{-\lambda_n t}, \quad [14]$$

$$n(\lambda) = \sum_n |\theta(0) \cdot \psi_n|^2 \cdot \delta(\lambda - \lambda_n), \quad [15]$$

where ψ_n are the eigenvectors and λ_n are the corresponding eigenvalues. Equation [15] gives the volume-weighted distribution $n(\lambda)$ of decay rates λ . In the absence of coupling between pores ($\omega_c = 0$), one has $\lambda_i = \rho A_{p,i} / V_{w,i}$ and the spectrum $n(\lambda)$ at $S_w = 1$ exactly corresponds to the volume-weighted distribution of inverse pore size.

3. RESULTS AND DISCUSSION

Determining all the eigenvalues and eigenvectors of large sparse matrices is a numerically intensive task. Due to computational constraints, we limited our investigation to pore networks of size 15^3 in 3D (without spatial correlation of pore sizes) and 50^2 in 2D (with and without spatial correlation of pore sizes). To obtain statistically representative results, we studied 10 realizations of each pore network and averaged the results. Key properties of the pore networks used are listed in Table 1.

3.1. Simulation of Drainage and Imbibition Capillary Pressure Curves

Figure 3a illustrates the effect of parameter S_w^* on the simulated drainage and imbibition capillary pressure curves for a value of $\eta = 1.5$ (networks 3D-SS1U and 3D-SS2U). The results from all 10 different realizations are plotted to illustrate the magnitude of variability among different realizations. In all simulations reported here, $\theta_R = 40^\circ$ and $\theta_A = 60^\circ$. It is observed that, as S_w^* increases, the increase of nonwetting phase saturation with capillary pressure (drainage) becomes more gradual. During imbibition, a greater amount of nonwetting fluid is displaced before a large number of piston-type and snap-off events take place. The location of the main leg of the imbibition curve is not affected, but the residual nonwetting-phase saturation is considerably reduced as S_w^* increases.

The effect of parameter η is illustrated in Fig. 3b (networks 3D-SS2U and 3D-SS3U). As can be seen, changing η from 1.5 (rough, but not fractal pore geometry) to 0.5 (fractal pore geometry) results in even more gradual drainage and imbibition curves. Reducing η has negligible effect on the residual nonwetting-phase saturation. In Fig. 3c, we compare the results from network 3D-SS3U (average over 10 realizations) to experimental mercury porosimetry curves of a Berea sandstone sample (36). Note that both drainage and imbibition curves are well reproduced by the model using $\xi = 6$, $\eta = 0.5$, $S_w^* = 0.24$, and the pore and throat size distributions shown in Fig. 1. No further attempt was made to optimize these parameters.

To further verify the simulator's ability to describe quasi-static drainage and imbibition in reservoir rock, we modeled the mercury porosimetry curves obtained experimentally from a North Sea chalk sample (37). Backscatter SEM images of this sample were used to generate a 3D stochastic replica of its

TABLE 1
Network Simulation Parameters

Model	ρ ($\mu\text{m/s}$)	ξ	S_w^*	η	PSD, TSD	Spatial correlation
3D-SS1U	—	6	0.24	0.5	Fig. 1	No
3D-SS2U	—	6	0.24	0.5	Fig. 1	No
3D-SS3U	5,100	6	0.24	0.5	Fig. 1	No
3D-CH1U	—	2	0.30	1.1	Fig. 4a	No
2D-SS3U	5	6	0.24	0.5	Fig. 1	No
2D-SS3C	5	6	0.24	0.5	Fig. 1	Exponential

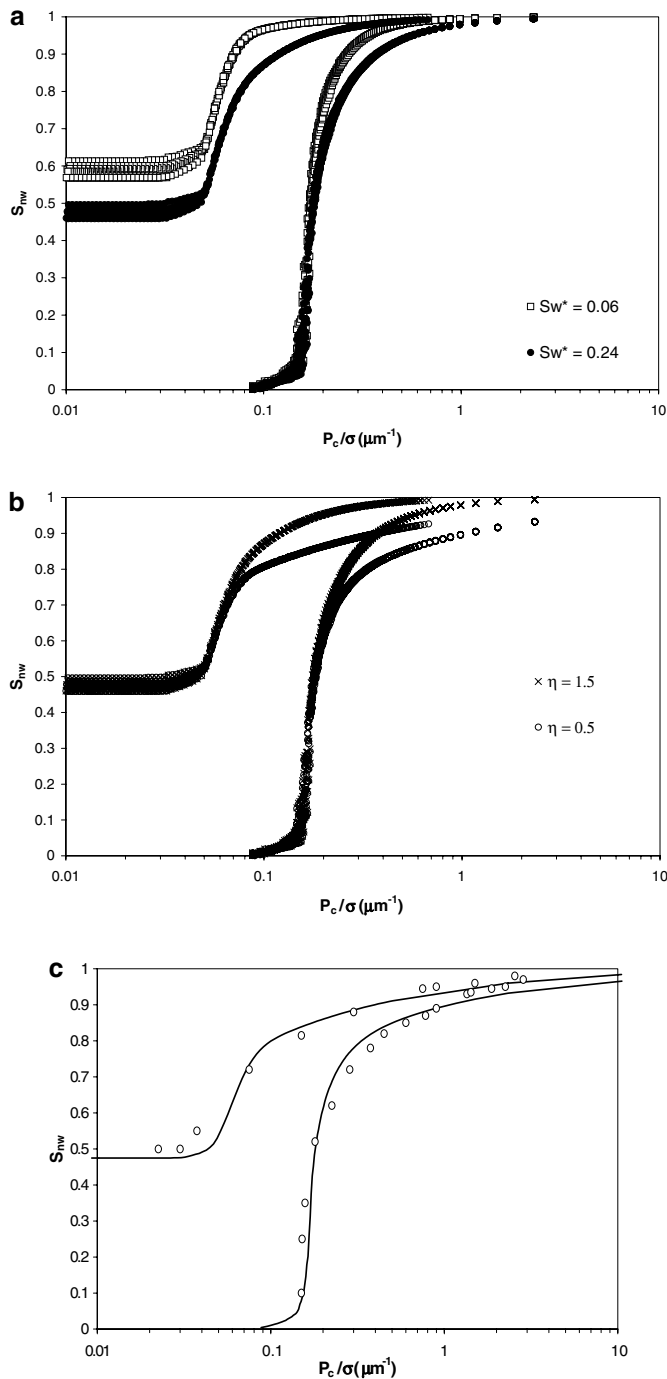


FIG. 3. (a) Effect of parameter S_w^* on simulated drainage and imbibition capillary pressure curves. (b) Effect of parameter η on drainage and imbibition capillary pressure curves. (c) Comparison of simulated and experimental mercury porosimetry curves for Berea sandstone.

microstructure, which was then analyzed using pore space partitioning techniques developed recently (23). The pore and throat size distributions measured in the 3D stochastic replica of the sample are shown in Fig. 4a. For details on the 3D reconstruction and characterization of the chalk microstructure, the reader

is referred to Talukdar *et al.* (37). As shown in Fig. 4b, network simulation of drainage and imbibition using $\xi = 2$, $\eta = 1.1$, and $S_w^* = 0.3$ (network 3D-CH1U) is in very good agreement with the experiment. Again, no rigorous optimization of these parameters was attempted.

It is encouraging that the values of critical pressure ratio ξ that give reasonable agreement with experiment are also in reasonable agreement with theoretical analyses of interfacial stability in pore throats of geometry representative of the media under consideration. A value of $\xi = 6$ for sandstone suggests the presence of short pore throats of anticlastic shape, as might be expected for consolidated packing of grains, whereas a value of $\xi = 2$ for chalk suggests the presence of elongated pore throats. Additionally, the value $\eta = 0.5$ for sandstone corresponds to a surface fractal dimension $D_s = 2.5$, in reasonable agreement with published values (38), whereas pore surfaces in chalk appear to be rough, but not fractal.

It should be noted that the aforementioned results were obtained from pore networks in which throat sizes are correlated to the sizes of adjoining pores, but pores of similar size are not

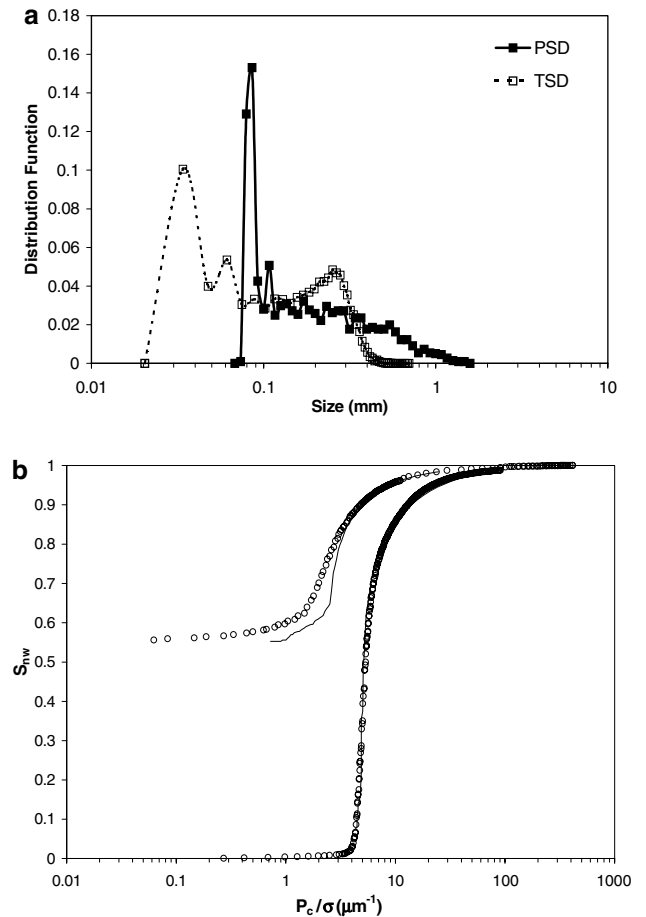


FIG. 4. (a) Number-based pore (PSD) and throat (TSD) size distributions for North Sea chalk (after (37)). (b) Comparison of simulated and experimental mercury porosimetry curves for North Sea chalk.

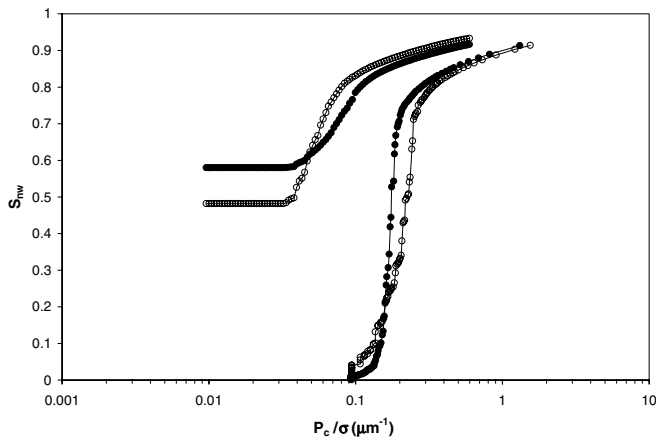


FIG. 5. Effect of spatial correlation on simulated drainage and imbibition capillary pressure curves of 2D pore networks. Solid circles, uncorrelated pore sizes; empty circles, exponentially correlated pore sizes (correlation length = 5 pores).

clustered together (no spatial correlations). In previous studies (14, 15, 22), it was found that spatial correlation among the pores causes the drainage curve to become more gradual, increases the hysteresis between drainage and imbibition and reduces the residual nonwetting-phase saturation. These observations are essentially confirmed by the results shown in Fig. 5 for networks 2D-SS3U and 2D-SS3C. Using spatially correlated networks in the modeling of experimental drainage and imbibition capillary pressure curves would likely result in a modest decrease in the values of S_w^* , η and ξ that fit the experimental data.

3.2. Decay Rate Spectra of Fully Saturated Model Pore Networks

Having established a pore network model capable of describing the drainage and imbibition capillary pressure curves of a real sample, we examined NMR relaxation at $S_w = 1$. Using the parameters corresponding to network 3D-SS3U (see Table 1), we carried out simulations using $\rho = 100 \mu\text{m/s}$ (high surface relaxivity) and $\rho = 5 \mu\text{m/s}$ (low surface relaxivity). These values are roughly at the extremes of reported surface relaxivity values, with $\rho = 5 \mu\text{m/s}$ frequently attributed to sandstones (17, 39). In each case, we compared the true decay rate spectrum to the one obtained if diffusion between the pores is ignored (i.e., $\omega_c = 0$). These comparisons are shown in Fig. 6a ($\rho = 100 \mu\text{m/s}$) and Fig. 6b ($\rho = 5 \mu\text{m/s}$). For $\rho = 100$ and $5 \mu\text{m/s}$, the mean relaxation times are $\tau_o = 0.0684$ and 1.37 s, respectively. The ratio of the two is precisely equal to the ratio of surface relaxivities, as expected. For Berea sandstone samples, Latour *et al.* (6) have reported mean relaxation times in the range 0.2–0.5 s for temperatures in the range 25–175°C. Examination of Fig. 6 reveals that, for a high value of ρ , the effects of pore coupling on the spectrum of decay rates appear to be small, whereas for a low value of ρ , the spectrum is reduced to a sharp peak. The latter result is in qualitative accord with the predictions of McCall *et al.* (11), for situations corresponding to $\omega_c \gg \omega_\rho$. If $\rho = 5 \mu\text{m/s}$ is

representative of sandstones, however, our computations show that magnetization decay should be dominated by diffusive coupling, something which has not been observed experimentally (6).

Remarkably, spatial correlations mask the effects of pore coupling on the decay rate spectra. This is born out by comparing the magnetization decay in 2D pore networks without and with spatial correlations (networks 2D-SS3U and 2D-SS3C, respectively) for a value of $\rho = 5 \mu\text{m/s}$. These results are shown in Fig. 7a for 2D-SS3U and Fig. 7b for 2D-SS3C. In the absence of spatial correlation the results are similar to those obtained from 3D networks (see Fig. 6a), as strong diffusive coupling significantly reduces the breadth of the spectrum. The effect is somewhat weaker in 2D because of reduced pore connectivity, as also noted by McCall *et al.* (11). On the contrary, when pores of similar size are clustered together, exchange of magnetization by diffusion between pores cannot effectively homogenize the kinetics of surface-enhanced relaxation (see Fig. 7b). Note the presence of a number of small peaks in the spectrum shown in Fig. 7b. Coupling is not weak but localized, and therefore its effects are not observable in the presence of spatial structure

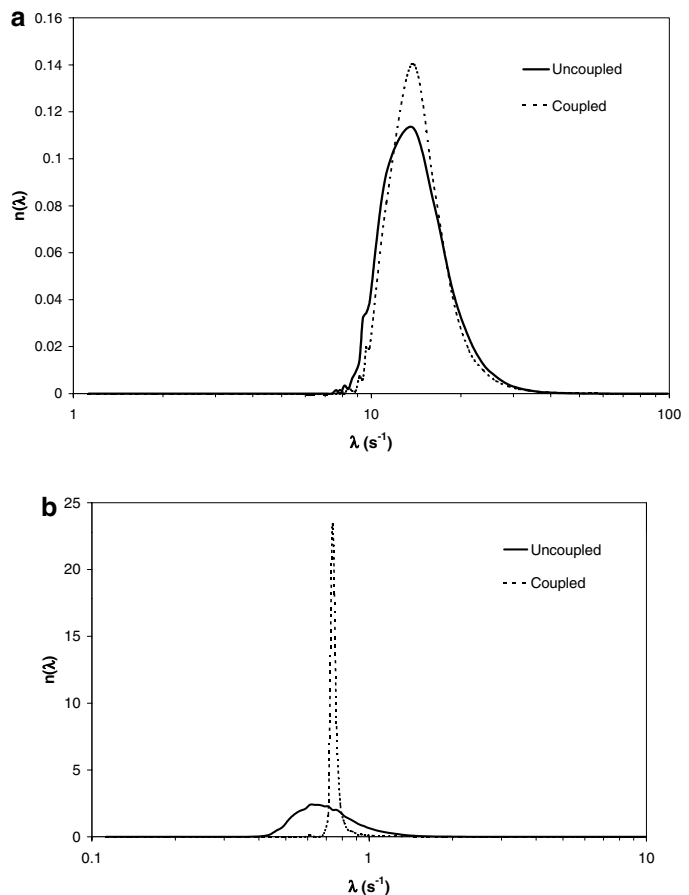


FIG. 6. (a) Magnetization decay spectra for network 3D-SS3U with $\rho = 100 \mu\text{m/s}$ ($\omega_\rho = 15.59$, $\omega_c = 0.98$). (b) Magnetization decay spectra for network 3D-SS3U with $\rho = 5 \mu\text{m/s}$ ($\omega_\rho = 0.78$, $\omega_c = 1.03$).

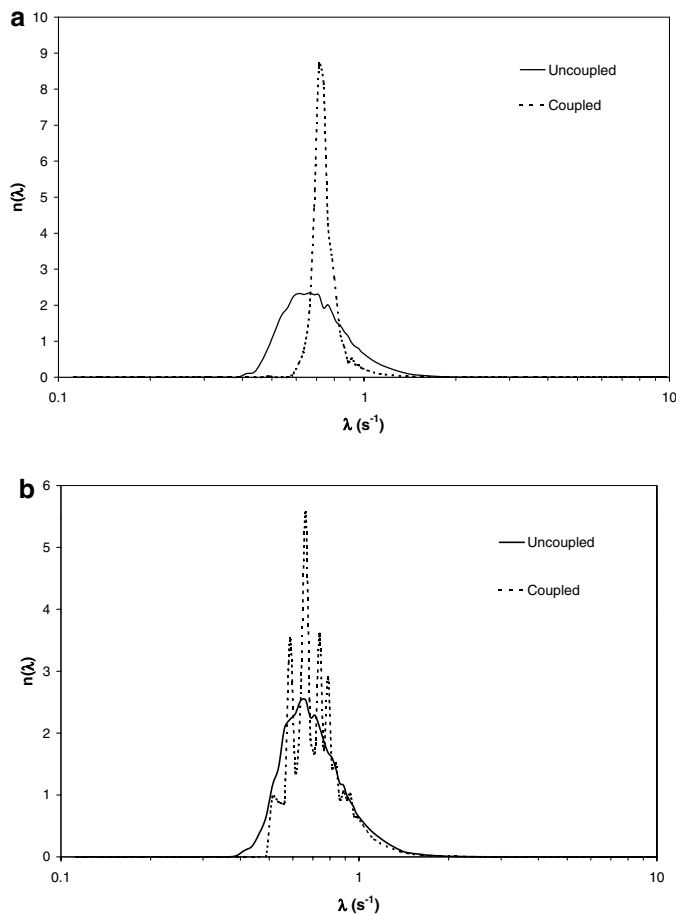


FIG. 7. (a) Magnetization decay spectra for network 2D-SS3U with $\rho = 5 \mu\text{m/s}$ ($\omega_\rho = 0.78$, $\omega_c = 0.98$). (b) Magnetization decay spectra for network 2D-SS3C with $\rho = 5 \mu\text{m/s}$ ($\omega_\rho = 0.78$, $\omega_c = 0.98$).

(i.e., locally homogeneous pore size). On the basis of these observations, we suggest that the broad relaxation rate spectra, obtained from water-saturated reservoir rock samples in numerous experimental studies, constitute evidence, not only of broad pore size distributions but also of correlated heterogeneity at length scales greater than the length scales of diffusion during NMR observation times.

As a further test, we considered the recovery of the known volume-weighted distribution of pore sizes using multiexponential inversion of the computed magnetization data, $M(t)$ (see Eq. [3]). This inversion was accomplished using a simulated annealing algorithm described in the appendix. As shown in Fig. 8, the pore size distribution obtained by multiexponential analysis of the $M(t)$ data agrees very well with the known pore size distribution. However, none of the peaks evident in the spectrum of decay rates is recovered. This is likely a consequence of the ill-conditioned nature of the multiexponential inversion problem (4). It is, thus, not surprising that such peaks are not observed in the analysis of experimental data.

3.3. Decay Rate Spectra of Partially Saturated Model Pore Networks

Network Model 3D-SS3U was used to study NMR relaxation under conditions of drainage and imbibition. This network lacks spatial structure and, on the basis of the observations made in the previous section, it is not entirely representative of Berea sandstone. Since we could not investigate NMR relaxation in large three-dimensional spatially correlated pore networks, we chose to study the problem using Model 3D-SS3U with $\rho = 100 \mu\text{m/s}$. Under such conditions, the effects of pore coupling appear to be weak (see Fig. 6a).

As seen in Fig. 9a, the larger pores are preferentially filled with nonwetting fluid during drainage. Note that we do not consider bulk relaxation effects and therefore no signal from the nonwetting phase is reported in the spectra. The wetting fluid in the invaded pores continues to be in contact with the entire surface of the pores, but its volumetric content decreases during drainage. This results in the gradual disappearance of the slower decay rates as the effective surface to volume ratio of the invaded pores increases, and the spectrum shifts toward higher rates. The remaining slow decay rates present in the spectrum belong to larger pores that are inaccessible by the nonwetting fluid.

During imbibition, the smaller pores are preferentially invaded by the wetting fluid, resulting in the gradual disappearance of the faster decay rates, as shown in Fig. 9b. The spectrum gradually shifts to slower rates as an increasingly greater amount of wetting phase exists in larger pores. As the nonwetting phase saturation decreases, the spectrum of decay rates shown in Fig. 9a for $S_w = 1$ is only partly recovered, due to the presence of residual nonwetting-phase saturation. This is more clearly shown in Fig. 10a. Note the presence of a second peak in the spectrum under conditions of residual nonwetting-phase saturation. Pores containing trapped nonwetting phase also contain

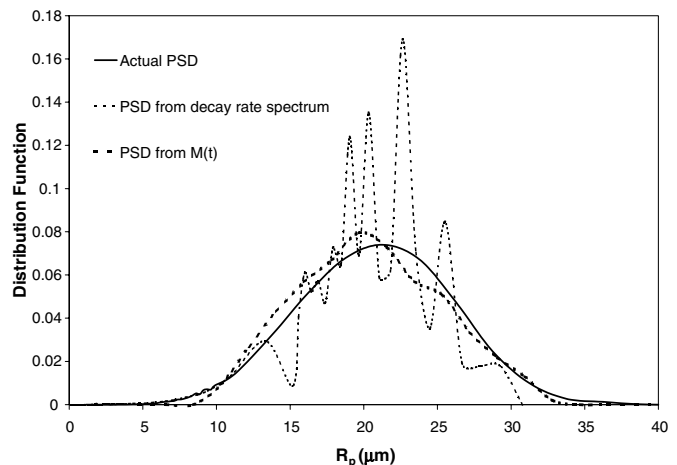


FIG. 8. Recovery of pore size distribution (PSD) from the spectrum $n(\lambda)$ and by inversion of $M(t)$ using a simulated annealing algorithm.

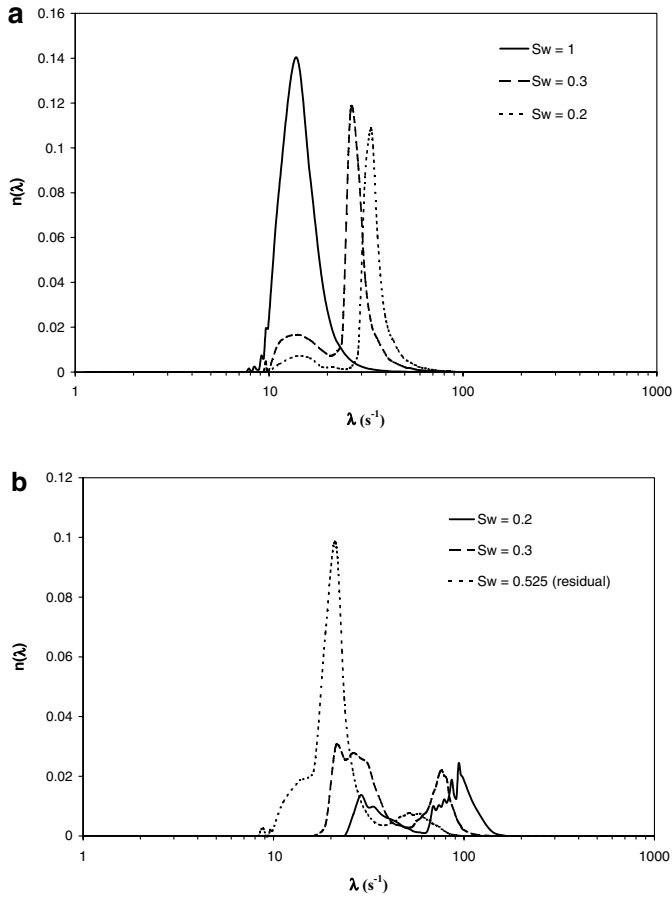


FIG. 9. (a) Magnetization decay spectra of partially saturated pore network 3D-SS3U ($\rho = 100 \mu\text{m/s}$) during drainage ($S_w = 1$, $\omega_p = 15.59$, $\omega_c = 0.98$; $S_w = 0.3$, $\omega_p = 100.49$, $\omega_c = 2.39$; $S_w = 0.2$, $\omega_p = 201.75$, $\omega_c = 2.92$). (b) Magnetization decay spectra of partially saturated pore network 3D-SS3U ($\rho = 100 \mu\text{m/s}$) during imbibition ($S_w = 0.2$, $\omega_p = 85.40$, $\omega_c = 1.30$; $S_w = 0.3$, $\omega_p = 54.24$, $\omega_c = 1.49$; $S_w = 0.525$, $\omega_p = 29.50$, $\omega_c = 1.46$).

a significant amount of wetting phase, whose characteristic relaxation rate is faster than most of the rates appearing in the spectrum at $S_w = 1$. Figure 10b compares NMR relaxation rate spectra at the same saturation ($S_w = 0.3$) for drainage and imbibition. These spectra reflect the markedly different distribution of the wetting phase in each case. During drainage, large pores not accessible to nonwetting phase contribute to the slow-rate end of the spectrum. During imbibition, many of these pores contain trapped nonwetting phase, resulting in the appearance of a second peak corresponding to fast relaxation rates in the spectrum. Clearly, interpretation of decay spectra in terms of pore structure parameters cannot be made without knowledge of the history of saturation change.

On the basis of the above observations, we investigated whether it is possible to identify the pores containing trapped nonwetting phase in the spectrum of decay rates at conditions of residual nonwetting phase saturation. A decomposition of this spectrum assuming that coupling effects are negligible is shown

in Fig. 11a. As seen in this figure, the spectrum is almost perfectly split into two portions. The slower rates are contributed almost exclusively from pores that are fully saturated with the wetting phase. If diffusion is ignored, the area under the peak corresponding to the fastest decay rates measures the fraction of pore volume occupied by wetting phase in pores containing residual nonwetting phase. Note that the slowest of these rates correspond to the largest pores containing residual nonwetting phase.

When diffusion between pores is accounted for, a very different picture emerges by decomposing the decay spectrum (see Fig. 11b). Slow decay rates are contributed by pores that are fully filled with wetting phase, as well as from pores that contain residual nonwetting phase. The same is true of fast decay rates. Large pores containing trapped nonwetting phase exchange magnetization with smaller pores fully saturated with wetting phase and their effective decay rate decreases. Similarly, large pores fully saturated with wetting phase exchange magnetization with smaller pores containing residual nonwetting phase and their effective decay rate increases. Note that these observations concern a case for which surface relaxation

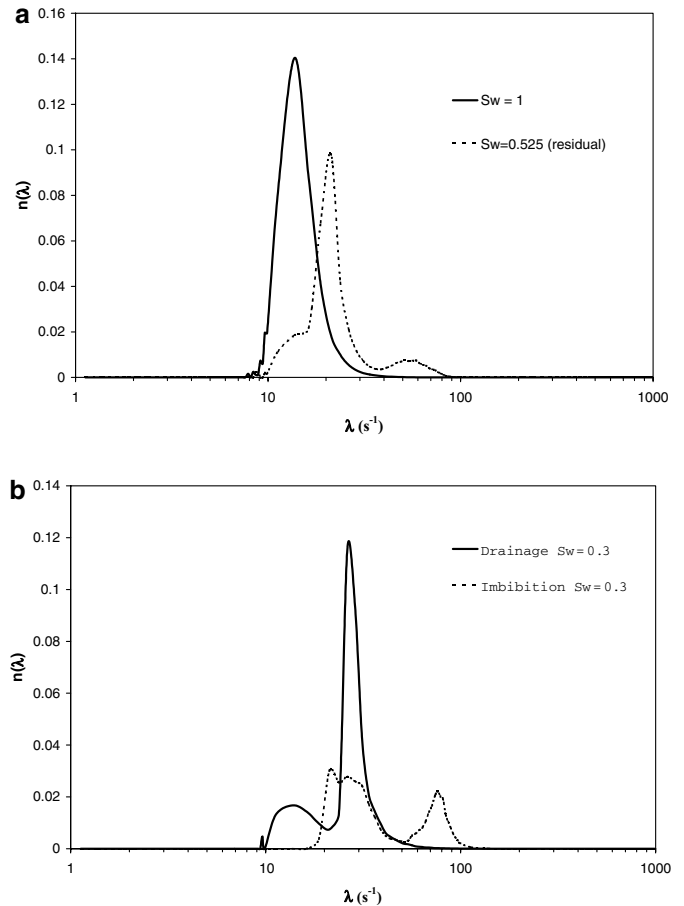


FIG. 10. (a) Comparison of magnetization decay spectra at $S_w = 1$ and $S_w = 0.525$ (residual after imbibition). (b) Comparison of magnetization decay spectra at equal saturation ($S_w = 0.3$) after drainage and imbibition.

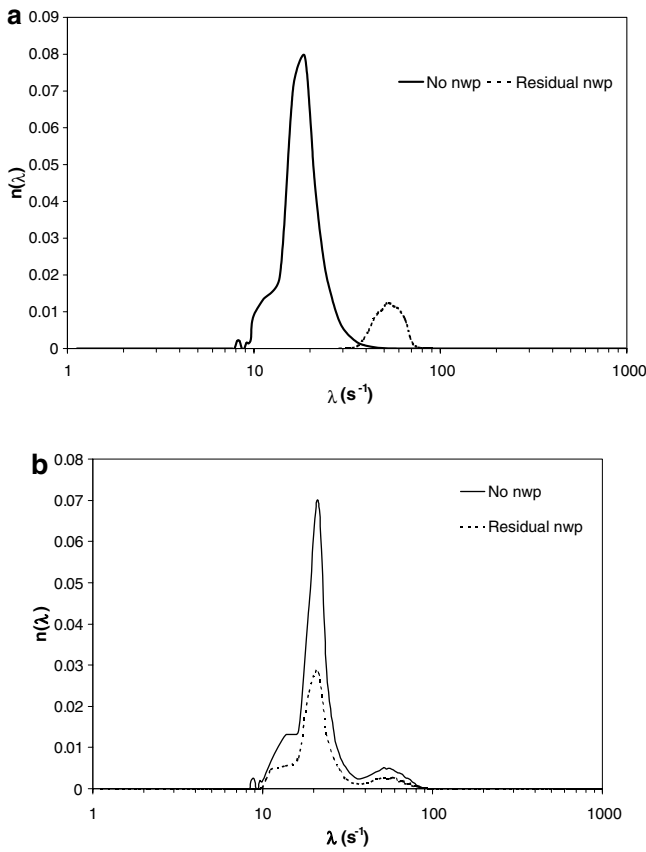


FIG. 11. (a) Decomposition of magnetization decay spectrum at S_{or} conditions in the absence of inter pore diffusion. (b) Decomposition of magnetization decay spectrum at S_{or} conditions in the presence of inter pore diffusion.

is relatively strong ($\rho = 100 \mu\text{m/s}$) and the effects of inter pore diffusion on the spectrum for $S_w = 1$ are apparently weak (see Fig. 6a). Nonetheless, diffusive coupling makes it impossible to distinguish pores containing trapped nonwetting phase in the spectrum. The same qualitative picture was obtained from simulations using Model 2D-SS3C.

3.4. Relationship between Decay Rate Spectrum at $S_w = 1$ and Primary Drainage Capillary Pressure Curve

Assuming fast-diffusion and weak effects of diffusive coupling, Kleinberg (17) suggested that relaxation rate spectra at $S_w = 1$ may be used to estimate primary drainage capillary pressure curves of porous rock. This suggestion is based on the additional assumptions that: (i) all pores of any given size are accessible by mercury at their corresponding entry pressure, and (ii) there is correlation between pore and throat sizes, so that entry pressures, which are dictated by pore throat size, are directly related to the size of adjoining pore bodies. These assumptions suggest a scaling between drainage capillary pressure and relaxation rate, such that $\kappa = P_c/\lambda$ is a constant. We tested this hypothesis using the spectra of decay rates obtained by simulation of the magnetization decay in Models 3D-SS3U ($\rho = 100 \mu\text{m/s}$,

Fig. 6a) and 2D-SS3C ($\rho = 5 \mu\text{m/s}$, Fig. 7b) at $S_w = 1$. The results are shown in Fig. 12. In each case, a good match of the simulated drainage capillary pressure curves is obtained in the approximate range $0.3 < S_w < 1$ with scaling factor $\kappa \approx 1$ (Model 3D-SS3U) and $\kappa \approx 20$ (Model 2D-SS3C). The ratio of

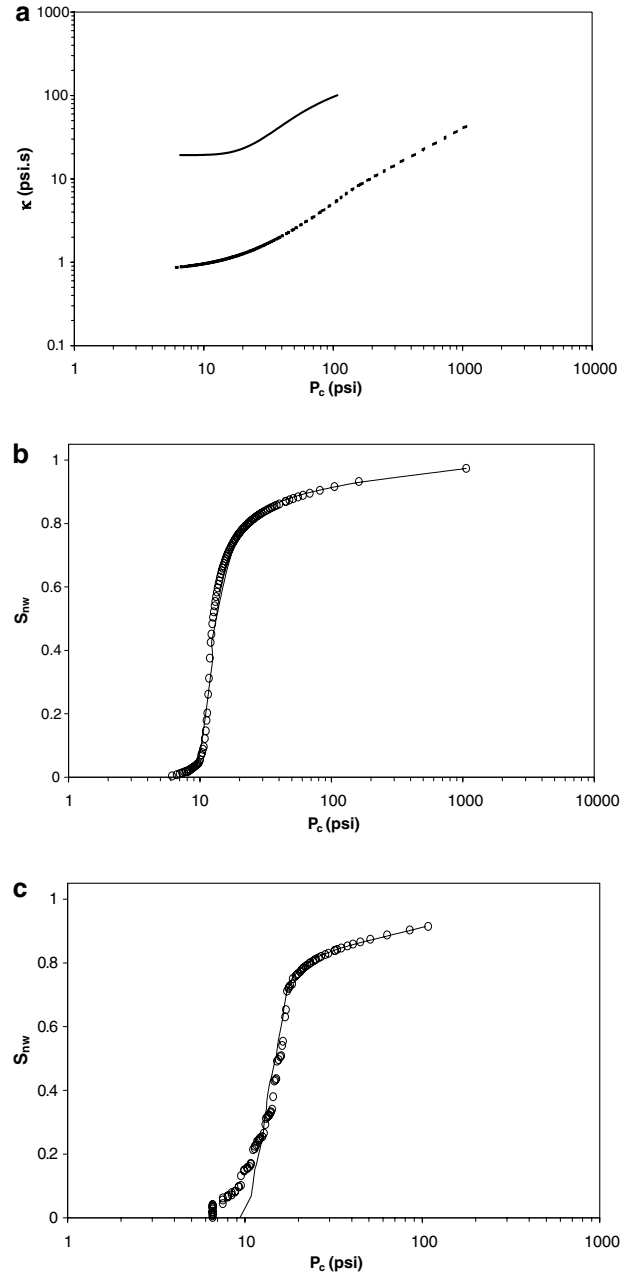


FIG. 12. (a) Scaling factor κ between relaxation rate and capillary pressure. Solid line, Model 3D-SS3U ($\rho = 100 \mu\text{m/s}$, Fig. 6a); broken line, Model 2D-SS3C ($\rho = 5 \mu\text{m/s}$, Fig. 7b). (b) Drainage capillary pressure curve (solid line) constructed by scaling the relaxation rate spectrum at $S_w = 1$ (Model 3D-SS3U, $\rho = 100 \mu\text{m/s}$) to the simulated drainage capillary pressure curve (points). (c) Drainage capillary pressure curve (solid line) constructed by scaling the relaxation rate spectrum at $S_w = 1$ (Model 2D-SS3C, $\rho = 5 \mu\text{m/s}$) to the simulated drainage capillary pressure curve (points).

the two κ values is equal to the inverse of the ratio of the corresponding surface relaxivity values, as expected. For $S_w < 0.3$, however, the scaling factor required to bring the two distributions into coincidence is observed to increase sharply with increasing capillary pressure, not unlike what is often observed experimentally (40). This result is explained by considering that pore space features that retain wetting phase at $S_w < 0.3$ during drainage (i.e., microporosity) are here assumed to be diffusively coupled to the macropores in which they are contained. Within each pore in the network, magnetization is therefore uniform and decays as a single exponential. It is precisely for this reason that fast relaxation rates attributed to microporosity are not seen in the simulated spectra at $S_w = 1$ (see Fig. 6a).

4. CONCLUSIONS

Network modeling was employed to study NMR relaxation under conditions of drainage and imbibition in porous media. A novel aspect of this work is that the model pore networks studied were developed using microstructure information obtained by comprehensive geometric and topological characterization of 3D stochastic replicas of sandstone and chalk samples. The effects of unresolved pore space geometry on fluid distribution were taken into account using a simple but physically meaningful two-parameter model. This enabled us to model quantitatively the measured mercury-air drainage and imbibition capillary pressure curves of Berea sandstone and North Sea chalk samples. Simulations of magnetization evolution in pore networks revealed that, for pore structure and surface relaxation parameters representative of sandstones, spin diffusion between pores results in significant narrowing of the decay rate spectrum, unless spatial correlation among pore sizes is taken into account. This result offers an explanation for the apparently contradictory experimental observation of broad relaxation time distributions under conditions of fast diffusion in water-saturated reservoir rock samples. We therefore suggest that such observations are evidence of correlated heterogeneity in porous rock.

Additionally, we used the simulator to explore the relaxation response of partially saturated porous rock. The results reveal the appearance in the drainage spectra of decay rates faster than any of the rates observed at $S_w = 1$. These rates were seen to correspond to the decay of magnetization in partially drained small pores. We suggest that similar observations in experimental systems may be taken as an indication of diffusive coupling of magnetization between macro- and microporosity. Furthermore, it was observed that relaxation spectra at partial saturation are very sensitive to saturation history (drainage vs imbibition). It was also shown that diffusion of magnetization between pores precludes the identification of pores containing trapped oil in the spectrum of decay rates at S_{or} . Finally, the connections between drainage capillary pressure curves and relaxation spectra at $S_w = 1$ were explored.

NOMENCLATURE

A_p	Surface area of a pore [L^2]
A_t	Cross-sectional area of a throat [L^2]
$A_{w,ij}$	Cross-sectional area occupied by wetting phase within a throat connecting pores i and j [L^2]
D	Self-diffusion coefficient of wetting phase [$L^2 t^{-1}$]
D_s	Surface fractal dimension [—]
f_i	Fraction of pore volume with relaxation time T_i [—]
L_{ij}	Length of diffusion path between pores i and j [L]
M_i	Magnetization in pore i [a.u.]
P_c	Capillary pressure [$ML^{-1} t^{-2}$]
$P_{c,i}^*$	Capillary pressure for nonwetting-phase invasion in pore i [$ML^{-1} t^{-2}$]
R_p	Equivalent pore radius [L]
R_t	Equivalent throat radius [L]
S_w	Wetting phase saturation [—]
S_w^*	Wetting phase saturation in pore upon invasion by the nonwetting phase [—]
$S_{w,ij}$	Wetting phase saturation in throat connecting pores i and j [—]
T_s	Surface relaxation time [t]
T_b	Bulk relaxation time [t]
$V_{p,i}$	Volume of pore i [L^3]
$V_{w,i}$	Volume of wetting phase within pore i [L^3]

Greek Letters

α	Characteristic pore size [L]
η	“Filling” exponent [—]
θ_A	Advancing contact angle [$^\circ$]
θ_R	Receding contact angle [$^\circ$]
λ_n	n th eigenvalue of matrix H [t^{-1}]
λ	Relaxation rate [t^{-1}]
ρ	Surface relaxivity [$L t^{-1}$]
σ	Interfacial tension [$M t^{-2}$]
ξ	Critical pressure ratio [—]
ψ_n	n th eigenvector of matrix H [a.u.]

APPENDIX

Multiexponential Inversion by Simulated Annealing

A multiexponential magnetization decay $M(t)$ may in general be written as

$$M(t) = M_o \sum_i f_i e^{-t/T_i}, \quad [\text{A.1}]$$

where f_i is the fraction of fluid volume relaxing at a rate $1/T_i$ and is constrained to be nonnegative. The established practice is to select a number of T_i values ($N = 10$ to 100), equally spaced on linear or logarithmic scales, and find the corresponding coefficients f_i (4). This problem is known to be ill-conditioned and its solution is not unique when $M(t)$ is noisy or when the mathematical computation has finite precision. To improve the

reproducibility of the results, modern inversion algorithms make use of regularization, a technique that effects a certain degree of smoothing on the resulting distributions (4).

In this work, inversion of simulated magnetization data, $M_{sim}(t)/M_o$, to obtain f_i was achieved using a simple algorithm. First, a number of T_i values ($N=26$), equally spaced in the range $[0.3125, 3.4375]$, were assigned to the midpoints of an equal number of bins. The amplitude of these bins was initialized to $f_i^{(0)}=1/N$, corresponding to an initially uniform distribution of T_i values. Simulated annealing was then implemented to minimize the objective function:

$$E^{(k)} = \sum_j \left(M_{sim}(t_j)/M_o - \sum_i f_i^{(k)} e^{-t_j/T_i} \right)^2. \quad [A.2]$$

At each iteration step k , two bins m and n ($m \neq n$, $f_m^{(k-1)} \geq \delta f$, $f_n^{(k-1)} \geq \delta f$) were randomly chosen, and their amplitudes modified as

$$f_m^{(k)} = f_m^{(k-1)} - \delta f, \quad [A.3]$$

$$f_n^{(k)} = f_n^{(k-1)} + \delta f, \quad [A.4]$$

where $\delta f \ll 1/N$. This modification altered the objective function by an amount $\Delta E^{(k)}$ given by

$$\Delta E^{(k)} = E^{(k)} - E^{(k-1)}. \quad [A.5]$$

At each iteration step k , the modification was accepted with probability $p_a^{(k)}$ given by the classical Metropolis rule,

$$p_a^{(k)} = \begin{cases} 1, & \Delta E^{(k)} < 0 \\ e^{-\Delta E^{(k)}/\psi^{(k)}}, & \Delta E^{(k)} \geq 0 \end{cases}, \quad [A.6]$$

where $\psi^{(k)}$ is a control parameter whose value decreases slowly with the number of iterations. Details on determining the initial value and rate of decrease of $\psi^{(k)}$ are given elsewhere (41). The simulated algorithm terminated when $E^{(k)}$ became smaller than a prescribed error tolerance or when no further reduction of $E^{(k)}$ could be achieved after a set number of iteration steps. Note that the conditions $f_m^{(k-1)} \geq \delta f$ and $f_n^{(k-1)} \geq \delta f$ ensured the nonnegativity of f_i . Also, $\sum_i f_i^{(k)} = 1$ at each iteration step k .

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RELATIONSHIPS BETWEEN NON-WETTING PHASE INVASION AND MAGNETIZATION EVOLUTION IN CONNECTED PORE SYSTEMS AS REVEALED BY NETWORK SIMULATION

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ABSTRACT

Pore network models are used to relate fundamental pore structure parameters (pore body and pore throat size distributions) to the distribution of water at the pore level and the decay of proton magnetization under conditions of primary drainage in water-wet rocks. The simulations reveal conditions under which diffusive coupling between pores has a significant effect on the decay spectra and provide insight into the estimation of primary drainage capillary pressure curves from NMR T_2 distributions. The conversion factor $k = P_c T_2$ required to map a NMR cumulative T_2 distribution at $S_w = 100\%$ onto a primary drainage capillary pressure curve furnishes important information about the pore-to-throat size aspect ratio, a parameter that critically affects recovery efficiency by waterflooding in water-wet media.

INTRODUCTION

With the advent of NMR wireline logging tools, the feasibility of estimating primary drainage capillary pressure curves from the NMR T_2 distributions of water-saturated rock samples has recently attracted significant attention (*e.g.*, Volokitin *et al.*, 1999). In a NMR measurement, hydrogen atoms on water molecules probe the pore space by diffusive motion. The magnetic moment carried by these atoms relaxes due to dipolar interactions with other molecules and with the fluid-solid interface. For a single pore in the fast diffusion limit, the surface relaxation rate is slow compared to the rate of magnetization equilibration by diffusive motion (Brownstein and Tarr, 1979). Consequently, magnetization is approximately uniform across the pore and decays with a characteristic time T_2 that is inversely proportional to the pore's surface area-to-volume ratio:

$$\frac{1}{T_2} = r \frac{A}{V} \quad (1)$$

The proportionality constant r is known as the surface relaxivity and knowledge of its value allows one to convert a rate of surface relaxation to a characteristic pore size R_p (e.g., $A/V = 3/R_p$ for spherical pores). The time evolution of proton magnetization in sedimentary rocks is usually characterized by a spectrum of decay rates (Kenyon, 1997), a fact used to estimate the distribution of pore volume by pore size. Such estimates are invariably made under the assumption that, although in the fast diffusion limit, individual pores are uncoupled from their neighbors. The effect of this assumption has been theoretically investigated by McCall *et al.* (1991). These authors studied network models in which the sizes of pores were broadly distributed, but the pore throat cross-sectional areas and inter-pore diffusion lengths were assumed constant. Such models are not quite representative of the pore structure of reservoir rocks, in which pore throats are known to also be broadly distributed. This is borne out by observation of mercury intrusion curves of reservoir rock samples, which show significant penetration over a wide range of capillary pressures. Nonetheless, McCall *et al.* (1991) suggested that in a system with a broad distribution of pore sizes, in the range typically found in sandstones, pore coupling should be marginally significant. This implies that the spectrum of decay rates obtained by multi-exponential decomposition of total magnetization measurements should reflect the volume density of pore sizes reasonably well. In the limit of “strong enough” diffusive coupling between pores, the entire pore space would behave as a single pore in the fast diffusion limit and the spectrum of decay rates would be a δ -function centered at rA/V , where A and V are the total area and pore volume. What constitutes “strong enough” coupling and how it affects the magnetization decay spectra, especially of partially-saturated porous media, is still not fully understood.

A primary drainage capillary pressure curve provides the distribution of pore volume accessible to a non-wetting phase by capillary pressure. The latter is related to the size of pore throats that are accessible to a non-wetting phase during drainage through the Laplace equation:

$$P_c = \frac{s f(q)}{R_t} \quad (2)$$

where R_t is a characteristic pore throat size and $f(q)$ is a function of the mercury-air advancing contact angle that depends on the geometry of the pore throat cross-section (e.g., $f(q) = 2\cos q$ for cylindrical pore throats).

Even if a NMR T_2 distribution adequately reflects the distribution of pore volume by pore size (i.e., if diffusive coupling effects are insignificant), additional assumptions are necessary to convert this distribution into a drainage capillary pressure curve. Namely, it is necessary to assume that (i) the pore-body to pore-throat size aspect ratio is either constant or known as a function of pore body size, (ii) that non-wetting phase invasion into a pore network is not subject to accessibility constraints and (iii) that, upon invasion by a non-wetting phase, pores are fully drained. Assumption (ii) ignores the fact that

some large pores are invaded at capillary pressures higher than the pressures corresponding to the size of pore throats they are connected to, because they are “shielded” by smaller pores. Assumption (iii) ignores the fact that, upon invasion by the non-wetting phase into a pore, pore corners and roughness features remain occupied by the wetting phase. Thus, only a fraction of the volume of each pore probed by water molecules during a NMR experiment is also probed by a non-wetting phase when the pore is first invaded. The purpose of this paper is to examine the impact of these phenomena on decay rate spectra by means of computer simulation in model pore networks. Pore networks with prescribed pore and throat size distributions and pore coordination number are constructed and primary drainage by a non-wetting phase is simulated using a bond-correlated site percolation algorithm.

NETWORK SIMULATION APPROACH

Construction of model pore networks

The pore structure model used is similar to the one used by Ioannidis and Chatzis (1993) to simulate the permeability, formation factor and drainage capillary pressure curves of several sandstones. It is a simple cubic network of pores and throats of square cross-section (Fig. 1). Pore body sizes are randomly assigned to the sites of the network, whereas pore throat sizes are correlated to the size of the smaller of the two pores each throat connects. Pore body and pore throat sizes are assumed to follow respective shifted Weibull distribution functions (Shapiro and Gross, 1981). These distributions were used for their versatility:

$$f_i(R) = \left(\frac{b}{a}\right) \cdot \left(\frac{R - R_{\min}}{a}\right)^{b-1} \cdot \exp\left[-\left(\frac{R - R_{\min}}{a}\right)^b\right] \quad \begin{array}{l} i = p, b \\ p = \text{pores} \\ t = \text{throats} \end{array} \quad (3)$$

where a , b are parameters of the distributions and $R_{\min}$ is a minimum pore body or pore throat size. Fixing the porosity of the network determines the distance between pores, thus permitting the estimation of pore throat lengths, L_{ij} . These lengths are only used to estimate the resistance to diffusion and fluid flow between adjacent pores; all pore volume is assigned to the pore bodies. Network models containing 15^3 pore bodies are used throughout this work. All results reported are averages over 10 realizations.

Simulation of magnetization evolution in model pore networks

A balance of magnetization M_i may be written for each pore i in the network. The net rate of magnetization decay in each pore is determined by magnetization relaxation in the vicinity of the fluid-matrix interface and by diffusion to and from neighboring pores:

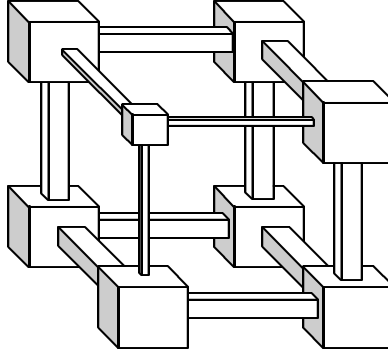


Figure 1. Schematic representation of the geometry of the pore network model.

$$\frac{dM_i}{dt} = -r \cdot A_i \cdot \frac{M_i}{V_i} + D \cdot \sum_j^{NN} \frac{S_{ij}}{L_{ij}} \cdot \left(\frac{M_j}{V_j} - \frac{M_i}{V_i} \right) \quad (4)$$

where A_i , V_i are the surface area and volume of pore i , respectively, D is the diffusion coefficient of water and S_{ij} , L_{ij} are the cross-sectional area and length of connecting pore throats. In dimensionless form, Eq. (4) may be written as follows:

$$\frac{dM_i}{dt} = -w_r \cdot r_i \cdot M_i + w_c \cdot \sum_j^{NN} \frac{s_{ij}}{l_{ij}} \cdot \left(\frac{M_j}{v_j} - \frac{M_i}{v_i} \right) \quad (5)$$

In writing Eq. (5) the following definitions are used:

$$\begin{aligned} w_r &= r \times \frac{\bar{A}}{\bar{V}} = \frac{1}{t_o} & w_c &= \frac{D\bar{S}}{\bar{V}\bar{L}} \\ s_{ij} &= \frac{S_{ij}}{\bar{S}} & l_{ij} &= \frac{L_{ij}}{\bar{L}} \\ v_{ij} &= \frac{V_{ij}}{\bar{V}} & r_i &= \frac{A_i}{V_i} \times \frac{\bar{V}}{\bar{A}} \end{aligned} \quad (6)$$

In the equations above, overbars indicate average quantities. The parameters w_r and w_c represent a characteristic rate of relaxation at the fluid-solid interface and a characteristic rate of molecular diffusion between pores, respectively. Application of Eq. (5) to each pore yields an initial-value, linear ordinary differential equation problem, which is most

efficiently solved by matrix diagonalization methods (McCall *et al.*, 1991). The solution provides the volume-weighted distribution $n(l)$ of decay rates l , hereafter referred to as spectrum of decay rates, where each l is an eigenvalue of the problem. In the absence of coupling between pores, $l_i = rA_i/V_i$ and the spectrum $n(l)$ exactly corresponds to the distribution of pore volume by inverse pore size.

Simulation of primary drainage

Simulation of primary drainage is carried out using a bond-correlated site percolation algorithm described in detail by Ioannidis and Chatzis (1993). It is assumed in the simulations that, once a pore is invaded by the non-wetting phase, a fraction equal to 6% of its volume V_i remains occupied by water. This residual water is also assumed to have access to the entire surface area A_i of the pore. Water diffusion between neighboring pores that are both invaded by the non-wetting phase is assumed to take place through constrictions whose cross-sectional area is 6% of the cross-sectional area S_{ij} of the connecting pore throat. This value corresponds to the water saturation in capillaries of square cross-section at conditions of gas breakthrough (Legait, 1983). Of course, a more realistic treatment of “late pore-filling” may involve other assumptions on the geometry of corners and crevices and should provide residual water saturation values that decrease with increasing capillary pressure in each invaded pore.

RESULTS AND DISCUSSION

Porous media properties

We have considered two rather different pore network models, hereafter referred to as Model 1 and Model 2. Their respective pore body size distributions (PSD) and pore throat size distributions (TSD) are shown in Fig. 2. Model 1 corresponds to a medium with a rather uniform TSD that significantly overlaps the PSD and is more typical of unconsolidated materials. Model 2 corresponds to a medium with a heavily skewed TSD and is more typical of reservoir sandstones (Ioannidis and Chatzis, 1993a).

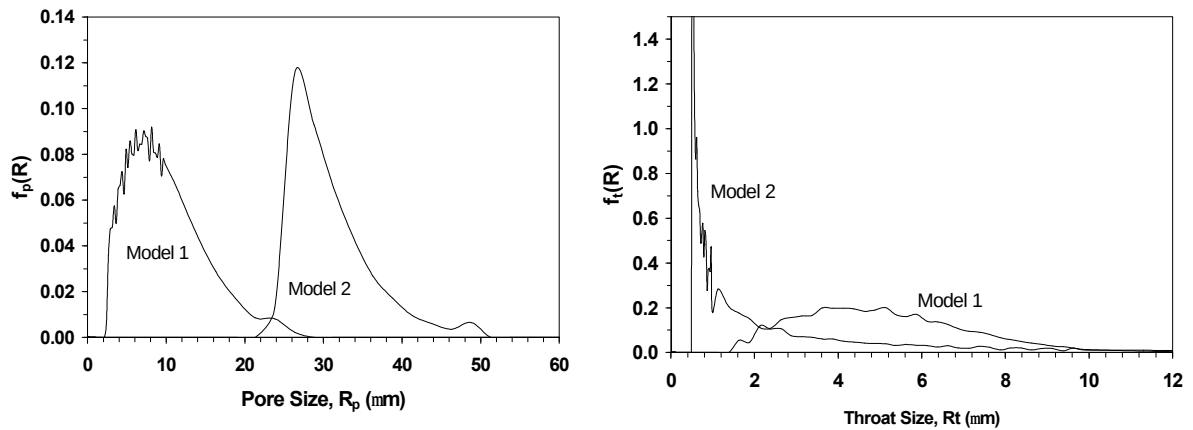


Figure 2. Pore body and pore throat size distributions used in network simulations

Note that the throat size distribution for Model 2 is not completely shown in Fig.2 because there is a long tail to a maximum throat size of 94 μm and it truly has a mean throat size equal to 4 μm . Table 1 lists relevant simulation parameters, including the computed absolute permeability of each model. A value of surface relaxivity typical of sandstones (100 $\mu\text{m/s}$) is used in all computations (Slijkerman and Hofman, 1998; Roberts *et al.*, 1995).

Table 1: Pore network model characteristics and simulation parameters

<i>Parameter</i>	<i>Model 1</i>	<i>Model 2</i>
Mean of PSD (μm)	10	31
Standard deviation of PSD (μm)	5	5
Minimum pore size (μm)	2	25
Maximum pore size (μm)	26.3	98.8
Mean of TSD (μm)	5	4
Standard deviation of TSD (μm)	2	7
Minimum throat size (μm)	1	0.5
Maximum throat size (μm)	13.8	93.8
Porosity (%)	17.8	17.8
Permeability (mD)	459	32.5
Water Diffusivity ($\mu\text{m}^2/\text{s}$)	2500	2500

Decay of proton magnetization during primary drainage

Knowledge of the water distribution at the pore level, obtained by simulation of primary drainage, enables determination of the magnetization decay spectra at different values of water saturation. Pore coupling is insignificant at all saturation levels in Model 2. This is mainly due to the higher frequency of small throat sizes in this model, which result in a much higher diffusion resistance in the network. Decay spectra at various levels of water saturation in this system are shown in Fig. 3. The spectrum at $S_w = 100\%$ reflects the narrow distribution of pore sizes in this system. Primary drainage results in gradual disappearance of slow decay rates from the spectrum as water is being drained from the larger pores. Further reduction of S_w causes the appearance of fast rates in the spectrum, associated with the presence of residual water in the pore corners, which are not present in the spectrum at $S_w = 100\%$. Unlike Model 2, the effects of diffusive pore coupling in Model 1 are evident at all saturation levels. As shown in Fig. 4(a), diffusion between pores at $S_w = 100\%$ narrows the spectrum of decay rates and skews it towards larger rates. The effect of coupling is more pronounced at lower levels of water saturation, as shown in Fig. 4(b). The evolution of decay spectra with S_w is illustrated in Fig. 5. As with

Model 2, accounting for the presence of residual water in pores invaded by the non-wetting phase significantly extends the spectrum of decay rates at low S_w .

Drainage capillary pressure curves from T_2 distributions

In order to convert a NMR T_2 distribution at $S_w = 100\%$ to a primary drainage capillary pressure curve, Volokitin *et al.* (1999) suggested the use of a variable $k = P_c/l$ scaling factor, namely one that increases with increasing P_c . The network modeling approach pursued in this study allows us to explore the nature of this conversion and determine the relationship of the scaling factor to the pore and throat size distributions. A variable scaling $k(P_c)$ suggested by Volokitin *et al.* (1999) is given below:

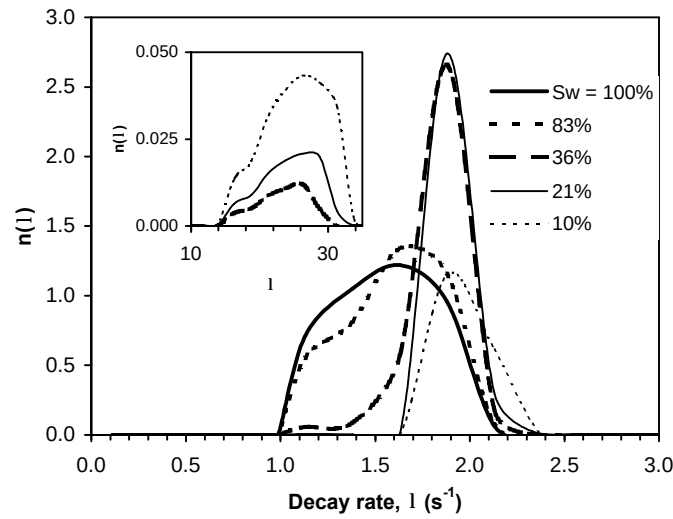


Figure 3. Magnetization decay spectra for Model 2 during primary drainage. Inset shows the fast-rate components of the spectra at low S_w .

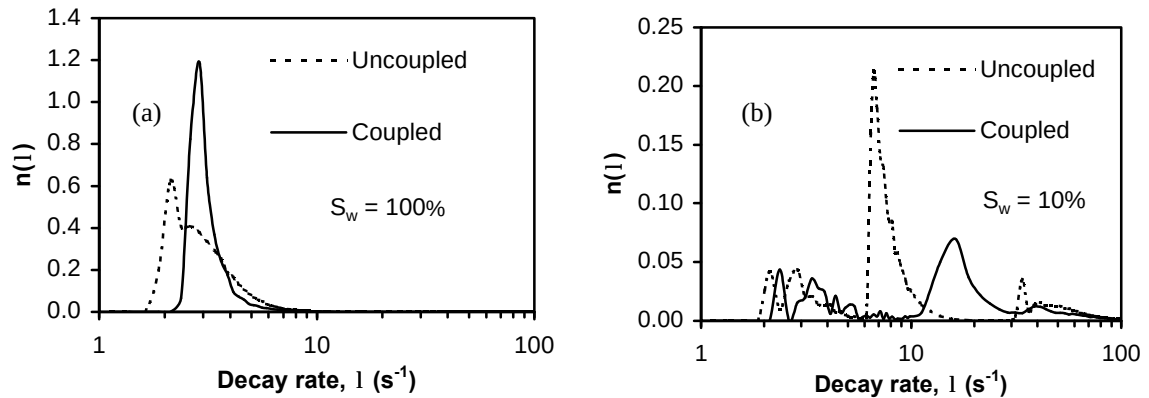


Figure 4. Effect of pore coupling on magnetization decay spectra of Model 1: (a) $S_w = 100\%$, (b) $S_w = 10\%$.

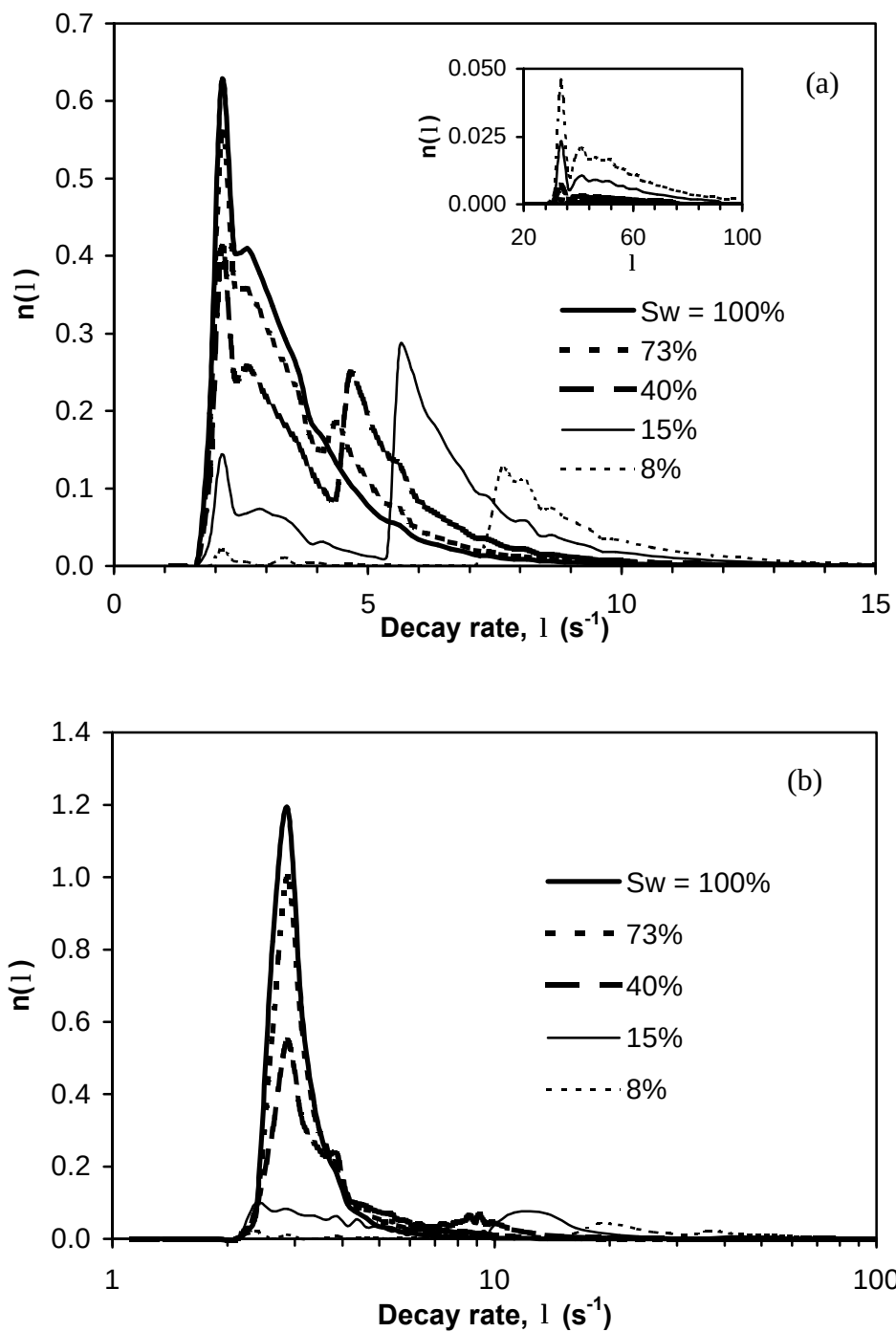


Figure 5. Magnetization decay spectra for Model 1 during primary drainage: (a) pore coupling ignored, (b) pore coupling accounted for.

$$k(P_c) = k_o \left(1 + \frac{A}{B/P_c + 1} \right)^C \quad (7)$$

where A, B, C and k_o are adjustable constants determined by optimization methods. If it can be assumed that the surface relaxivity is independent of pore size, then the value of k reflects an effective pore body-to-pore throat size aspect ratio.

The functions $k(P_c)$ of the form of Eq. (7) which best reproduce the drainage capillary pressure curves of Model 1 and Model 2 are shown in Fig. 6, whereas the simulated drainage capillary pressure curves and those obtained from the NMR T_2 distributions are compared in Fig. 7. A nearly constant scaling factor $k \approx 4.7$ is obtained for Model 1. This model has a rather uniform and relatively narrow TSD (see Fig. 2). Thus, primary drainage in this model takes place within a narrow range of capillary pressures, giving a rather steep drainage capillary pressure curve. Using the value of r for sandstone = 100 $\mu\text{m/s}$, we estimate a pore-to-throat size aspect ratio of 2.2 from $k \approx 4.7$. This value is to be compared with the actual aspect ratio of the smallest pores and throats accessible at breakthrough conditions in Model 1 ($P_c = 13.4$ psi; $S_w = 0.80$), the latter value being equal to 1.5. Even though there is a difference in the magnetization decay spectra for Model 1 (having a pore structure with narrow throat size distribution), no significant difference was found for the k values, as seen from the results show in Figure 6.

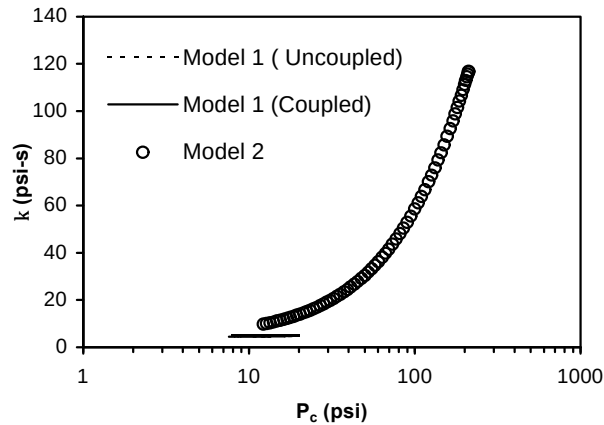


Figure 6. Scaling factor k used to convert NMR decay spectra to primary drainage capillary pressure curves. Data shown correspond to $0.10 < S_w < 0.80$ for Model 1 and $0.10 < S_w < 0.88$ for Model 2.

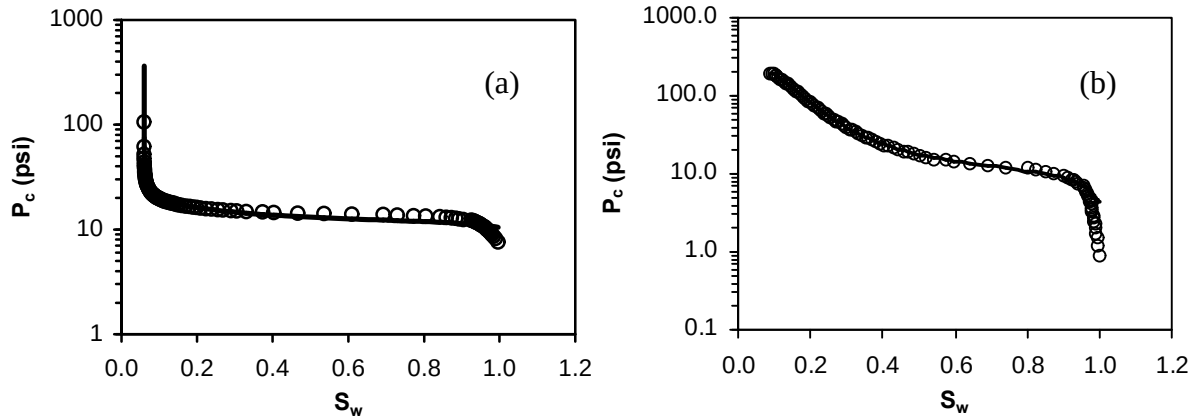


Figure 7. Simulated (points) and NMR-reconstructed (lines) primary drainage capillary pressure curves: (a) Model 1, (b) Model 2.

Model 2 is characterized by a broader TSD that is heavily skewed towards smaller pore throat pore throat sizes (see Fig. 2). This gives rise to a much more gradual capillary pressure curve that extends over a wider capillary pressure range. As shown in Fig. 6, the scaling factor k required to reproduce the capillary pressure curve from the NMR decay spectrum at $S_w = 100\%$ (Fig. 3) is not constant in this case. The aspect ratio of the smallest pores and throats accessible at breakthrough conditions in Model 2 ($P_c = 11.3$ psi; $S_w = 0.88$) is equal to 2.7. This value is to be compared with the aspect ratio of 4.3, estimated using κ (at $P_c = 11.3$ psi) = 9.3. An estimated higher aspect ratio is consistent with the pore and throat size distributions of this model (Fig. 2). However, the apparent aspect ratio calculated from the k values at breakthrough conditions is higher than the actual aspect ratio of the pores and throats involved. This discrepancy reflects the effect of limited pore accessibility near the percolation threshold. The results indicate that scaling NMR T_2 distributions to primary drainage capillary pressure curves provides information about an apparent pore body-to-pore throat size aspect ratio. This is an important pore structure parameter affecting the magnitude of residual non-wetting phase saturation after imbibition (Ioannidis and Chatzis, 1993b).

CONCLUSIONS

1. The NMR response of partially saturated porous media during drainage displacements, can be efficiently simulated using network models of pore structure with pore and throat size distributions. The same type of network models have been used in previous studies of our group to predict a host of petrophysical properties dependent on fluid distributions at the pore scale. In this paper, we provide a useful approach to relate NMR T_2 distributions from partially saturated porous media to drainage capillary pressure curves and corresponding fluid distributions.

2. In the fast diffusion limit, pore coupling can be significant in media with a broad pore size distribution when the throat size distribution and inter-pore length are taken into account. In such cases, pore coupling is more pronounced at lower wetting phase saturations.
3. For the cases studied, simulations confirm that the conversion of NMR T_2 distributions to primary drainage capillary pressure curves provides information about the size aspect ratio of pore throats and pore bodies, which may be usefully correlated to recovery efficiency.

NOMENCLATURE

T_2 – NMR relaxation time (s)
 A_i – Surface area of pore i (μm^2)
 V_i – Volume of pore i (μm^3)
 S_{ij} – Cross sectional area of throat connecting pores i and j (μm^2)
 L_{ij} – Diffusion length between pores i and j (μm)
 R_p – Characteristic pore size (μm)
 R_t – Characteristic throat size (μm)
 $R_{\min}$ – Minimum pore or throat size (μm)
 D – Diffusion coefficient ($\mu\text{m}^2/\text{s}$)
 M_i – Normalized magnetization of pore i
 S_w – Water saturation
 P_c – Capillary pressure (psi)
 α – Weibull location/spread parameter
 β – Weibull kurtosis/skewness parameter
 ρ – Surface relaxivity ($\mu\text{m}/\text{s}$)
 σ – Surface tension (N/m)
 ω_p – Characteristic rate of relaxation at the fluid-solid interface (1/s)
 ω_c – Characteristic rate of molecular diffusion between pores (1/s)
 λ – Relaxation rate (1/s)
 κ – Scaling factor between λ and P_c (psi's)

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PORE NETWORK SIMULATION OF LOW-FIELD NMR RELAXOMETRY UNDER CONDITIONS OF DRAINAGE AND IMBIBITION: EFFECTS OF PORE STRUCTURE AND SATURATION HISTORY

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ABSTRACT

Nuclear magnetic resonance (NMR) relaxometry tests, carried out on partially saturated porous media, convey information on the distribution of fluids within the pore space. Proper interpretation of this information requires a means to isolate the effects of pore structure (pore and throat shape and size distributions, spatial correlations, etc.) wettability and saturation history. A pore network simulator of NMR relaxation that can clarify the effect of these factors on observed T2 distributions is developed in this work. The pore network model can be calibrated with pore structure parameters (pore volume and throat size distributions, pore connectivity) determined from analysis of stochastic 3D replicas of the pore space generated from thin-section image information. The presence of irregular and, possibly, fractal pore surfaces is accounted for through a "late pore filling" model calibrated to match mercury porosimetry data. The network model is then used to simulate the water distribution and proton magnetization decay under conditions of drainage and imbibition. The results obtained reveal: (a) conditions under which diffusive coupling between pores has a significant effect on the decay spectra, (b) contributions to the magnetization decay spectra from water trapped at the irregular pore surfaces, and (c) contributions to the magnetization decay spectra from (i) water in large pores which are inaccessible by the non-wetting phase during drainage and (ii) water in pores containing oil trapped during imbibition.

INTRODUCTION

With the increasing use of NMR techniques in the field and laboratory, there is a need to clarify how the NMR response of partially saturated reservoir rock depends on pore structure parameters, wettability and displacement history. Pore network simulation has been used extensively in the past to study the relationship of the aforementioned factors on capillary pressure, electrical resistivity and relative permeability. In recent years, pore network simulation has been endowed with predictive power, as a result of advances in (a) 3D stochastic reconstruction from thin-section image statistical information (Ioannidis *et al.*, 1999; Liang *et al.*, 2000; Bekri *et al.*, 2000) and (b) grain deposition/diagenesis modeling (Bakke and Øren, 1997). If pore network models are to be further developed as tools for petrophysical evaluation, then their extension to include predictions of the NMR response of reservoir rock should be considered. This task was recently undertaken by Chang *et al.* (2000) and is further pursued in the present contribution.

In a NMR measurement, hydrogen atoms on water molecules probe the pore space by diffusive motion. The magnetic moment carried by these atoms relaxes due to dipolar interactions with other molecules and with the fluid-solid interface. For a single pore in the fast diffusion limit, the surface relaxation rate is slow compared to the rate of magnetization equilibration by diffusive motion (Brownstein and Tarr, 1979). Consequently, magnetization is approximately uniform across the pore and decays with a characteristic time T_2 that is inversely proportional to the pore's surface area-to-volume ratio:

$$\frac{1}{T_2} = \rho \frac{A}{V} \quad (1)$$

The proportionality constant ρ is known as the surface relaxivity and knowledge of its value allows one to convert a rate of surface relaxation to a characteristic pore size R_p (e.g., $A/V = 3/R_p$ for spherical pores). The time evolution of proton magnetization in sedimentary rocks is usually characterized by a spectrum of decay rates (Kenyon, 1997), a fact used to estimate the distribution of pore volume by pore size. Such estimates are invariably made under the assumption that, although in the fast diffusion limit, individual pores are uncoupled from their neighbors. The effect of this assumption has been theoretically investigated by McCall *et al.* (1991). These authors studied network models in which the sizes of pores were broadly distributed, but the pore throat cross-sectional areas and inter-pore diffusion lengths were assumed constant. Such models are not representative of the pore structure of reservoir rocks, in which pore throat sizes and inter-pore lengths are also broadly distributed. McCall *et al.* (1991) suggested that in a system with a broad distribution of pore sizes, in the range typically found in sandstones, pore coupling should be marginally significant. This implies that the spectrum of decay rates obtained by multi-exponential decomposition of total magnetization measurements should reflect the volume density of pore sizes reasonably well. This conclusion is not generally true, especially at conditions of partial saturation (Chang *et al.*, 2000).

A primary drainage capillary pressure curve provides the distribution of pore volume accessible to a non-wetting phase by capillary pressure. The latter is related to the size of pore throats that are accessible by a non-wetting phase during drainage through the Laplace equation:

$$P_c = \frac{s f(\theta)}{R_t} \quad (2)$$

where R_t is a characteristic pore throat size and $f(\theta)$ is a function of the receding contact angle that depends on the geometry of the pore throat cross-section (e.g., $f(\theta) = 2\cos\theta$ for cylindrical pore throats). Under imbibition conditions Eq. (2) also governs the piston-type retraction of non-wetting phase from pore bodies and pore throats. Under such conditions, the angle θ is the advancing contact angle. The difference between receding and advancing contact angles (contact angle hysteresis) is a cause of hysteresis between the drainage and imbibition capillary pressure curves. Quasistatic imbibition is controlled by the interplay between two main types of capillary instability: piston-type retraction of fluid interfaces from pore bodies and pore throats and snap-off of non-wetting phase threads in pore throats (Ioannidis and Chatzis, 1993a). The collective outcome of these pore-level phenomena is the establishment of a residual non-wetting phase saturation, the magnitude of which is determined by the geometry and connectivity of the pore space and by wettability.

The mercury-air drainage and imbibition curves are a strong signature of the geometry, connectivity and spatial arrangement of pore bodies and pore throats within a rock sample. Their prediction is, therefore, a rigorous test on a network model's ability to represent the porous microstructure of real rock. The task of developing pore network models which honor experimental mercury porosimetry data is, however, highly nontrivial (Tsakiroglou and Payatakes, 2000). A main difficulty arises from the fact that real pores have an irregular, rough and often fractal geometry. When a non-wetting fluid first invades a pore in real rock, the amount of wetting phase that remains within this pore is considerably larger than would be predicted by assuming that the pore has a regular, albeit angular, cross-section (Ioannidis and Chatzis, 1993b). Neglect of this fact imparts considerable uncertainty on network model calculations, certainly regarding volumetric calculations and possibly regarding calculations of electrical resistivity and relative permeability.

In this work we propose a simple "late pore filling" model to account for the rough or fractal pore geometry. Use of this model permits a description of proton magnetization evolution at conditions of partial saturation in a manner that is consistent with image analysis and mercury porosimetry data. This model is the new feature in a mixed-percolation algorithm that is employed to simulate primary drainage and imbibition in model pore networks. At each level of wetting phase saturation along the drainage and imbibition path, the associated NMR problem is solved using a matrix diagonalization method (Chang *et al.*, 2000; McCall *et al.*, 1991). Results are presented which illustrate the effects of pore structure and saturation history on magnetization evolution.

METHODOLOGY

Construction of model pore networks

The pore structure model used is similar to the one used by Ioannidis and Chatzis (1993b) to simulate the permeability, formation factor and drainage capillary pressure curves of several sandstones. It is a simple cubic network of pores and throats of square cross-section (Fig. 1). Pore body sizes are randomly assigned to the sites of the network, whereas pore throat sizes are correlated to the size of the smaller of the two pores each throat connects. Pore body and pore throat sizes are assumed to follow respective Weibull distribution functions (Shapiro and Gross, 1981):

$$f_i(R) = \left(\frac{\beta}{a}\right) \cdot \left(\frac{R - R_{\min}}{a}\right)^{\beta-1} \cdot \exp\left[-\left(\frac{R - R_{\min}}{a}\right)^{\beta}\right] \quad \begin{array}{l} i = p, b \\ p = \text{pores} \\ t = \text{throats} \end{array} \quad (3)$$

where α , β are parameters of the distributions and $R_{\min}$ is a minimum pore body or pore throat size. The inter-pore lengths are also distributed in the network according to a Weibull distribution and are positively correlated to the size of adjoining pores. These lengths are only used to estimate the resistance to diffusion and fluid flow between adjacent pores; all pore volume is assigned to the pore bodies. The network model thus constructed is a realistic approximation of pore networks in 3D stochastically reconstructed porous media (Liang *et al.*, 2000; Ioannidis and Chatzis, 2000). Network models containing 15^3 pore bodies are used throughout this work. All results reported are averages over 10 realizations.

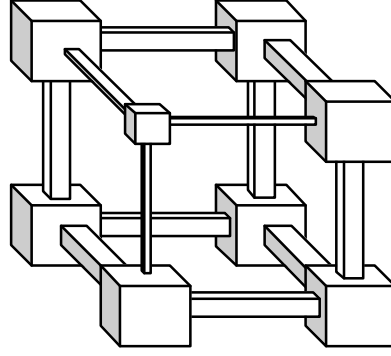


Figure 1: Schematic representation of the geometry of the pore network model.

Simulation of magnetization evolution in model pore networks

A balance of magnetization M_i may be written for each pore i in the network. The net rate of magnetization decay in each pore is determined by magnetization relaxation in the vicinity of the fluid-matrix interface and by diffusion to and from neighboring pores:

$$\frac{dM_i}{dt} = -\omega_p \cdot A_i \cdot \frac{M_i}{V_i} + D \cdot \sum_j^{NN} \frac{S_{ij}}{L_{ij}} \cdot \left(\frac{M_j}{V_j} - \frac{M_i}{V_i} \right) \quad (4)$$

where A_i , V_i are the surface area and volume of pore i , respectively, D is the diffusion coefficient of water and S_{ij} , L_{ij} are the cross-sectional area and length of connecting pore throats. In dimensionless form, Eq. (4) may be written as follows:

$$\frac{dM_i}{dt} = -\omega_p \cdot r_i \cdot M_i + \omega_c \cdot \sum_j^{NN} \frac{s_{ij}}{l_{ij}} \cdot \left(\frac{M_j}{v_j} - \frac{M_i}{v_i} \right) \quad (5)$$

In writing Eq. (5) the following definitions are used:

$$\begin{aligned} \omega_p &= \rho \cdot \frac{\bar{A}}{\bar{V}} = \frac{1}{\tau_o} & \omega_c &= \frac{D\bar{S}}{\bar{V}\bar{L}} \\ s_{ij} &= \frac{S_{ij}}{\bar{S}} & l_{ij} &= \frac{L_{ij}}{\bar{L}} \\ v_{ij} &= \frac{V_{ij}}{\bar{V}} & r_i &= \frac{A_i}{V_i} \cdot \frac{\bar{V}}{\bar{A}} \end{aligned} \quad (6)$$

In the equations above, overbars indicate average quantities. The parameters ω_p and ω_c represent a characteristic rate of relaxation at the fluid-solid interface and a characteristic rate of molecular diffusion between pores, respectively. Application of Eq. (5) to each pore yields an initial-value, linear ordinary differential equation problem, which is most efficiently solved by matrix diagonalization. The solution provides the volume-weighted distribution $n(\lambda)$ of decay rates λ , hereafter referred to as the spectrum of decay rates, where each λ is an eigenvalue of the problem. In the absence of coupling between pores,

$\lambda_i = \rho A_i / V_i$ and the spectrum $n(\lambda)$ exactly corresponds to the distribution of pore volume by inverse pore size.

Simulation of primary drainage and primary imbibition

Simulation of primary drainage is carried out using a bond-correlated site percolation algorithm described in detail by Ioannidis and Chatzis (1993b). The capillary pressure for piston-type non-wetting phase invasion into a pore throat is calculated from Eq. (2). For pore throats of rectangular cross-section the function $f(\theta)$ is (Legait, 1983):

$$f(\theta) = \frac{\theta + \cos^2 \theta - \pi/4 - \sin \theta \cos \theta}{\cos \theta - \sqrt{\pi/4 - \theta + \sin \theta \cos \theta}} \quad (7)$$

where θ is the receding contact angle. (i.e., $f(\theta) = 1.5274$ when $\theta = 40^\circ$). The capillary pressure for piston-type retraction of non-wetting phase from a pore body is also calculated by equation (2), where the advancing contact angle is used to calculate $f(\theta)$. The critical pressure ratio (CPR) is defined as the ratio of capillary pressure for piston-type retraction from a pore throat to the critical capillary pressure for snap-off (Li and Wardlaw, 1986) and is a function of pore throat geometry and contact angle. Knowledge of this parameter enables the calculation of the capillary pressure for snap-off as:

$$P_{c, \text{snap-off}} = \frac{s f(\theta_a) / R_t}{\text{CPR}} \quad (8)$$

A mixed percolation algorithm is used to simulate quasistatic imbibition. This algorithm has been detailed elsewhere (Ioannidis and Chatzis, 1993a).

Late pore filling model

We propose that upon invasion of the non-wetting fluid into each individual pore at capillary pressure P_c^* , a fraction S_w^* of the volume of the pore is occupied by the wetting fluid. The fraction of the volume of the pore that contains wetting fluid at capillary pressure $P_c > P_c^*$, is assumed to vary according to the following expression (Brooks and Corey, 1964):

$$\frac{S_w}{S_w^*} = \left(\frac{P_c^*}{P_c} \right)^\eta \quad (9)$$

The fraction S_w^* and late pore filling exponent η are model parameters. For $0 < \eta < 1$ late pore filling corresponds to filling of a surface fractal with dimension $2 < D_s < 3$, whereas for $\eta > 1$ the pore behaves as a rough, but not fractal, object. The parameter S_w^* may be identified with the fraction of total pore volume contributed by length scales not explicitly represented in network models. The value of P_c^* is different for each pore and depends on the pore throat size and pore accessibility. During primary imbibition, and for as long as piston-type retraction or disconnection of the non-wetting fluid in a pore has not taken place, the wetting fluid content of a pore increases according to Eq. (9).

RESULTS AND DISCUSSION

Simulation parameters

Pore structure parameters used in the simulations are typical of sandstone. Simulation parameters are listed in Table 1, whereas the pore size, throat size, and inter-pore length size distributions are depicted in Figure 2. Surface relaxivity values ranging from 5 $\mu\text{m/s}$ to greater than 100 $\mu\text{m/s}$ have been reported for sandstones (Howard *et al.*, 1993; Slijkerman and Hofman, 1998; Roberts *et al.*, 1995; Kleinberg, 1996). A surface relaxivity value of 100 $\mu\text{m/s}$ is used in all computations reported here. A critical pressure ratio (CPR) equal to 6 was used in all simulations. Although the value of CPR can be estimated from knowledge of contact angle and pore throat geometry (Ioannidis and Chatzis, 1993), CPR is treated as an adjustable parameter in our simulations.

Capillary pressure curves including late pore filling

Figure 3 illustrates the effect of parameter S_w^* on the simulated drainage and imbibition capillary pressure curves for a value of $\eta = 1.5$ (rough, but not fractal pores). Results from 10 different realizations are plotted to illustrate the magnitude of variability. As the amount of wetting fluid remaining in pores upon penetration by the non-wetting fluid (S_w^*) increases, the increase of S_{nw} with increasing capillary pressure becomes much more

Table 1 : Simulation parameters.

<i>Parameter</i>	<i>value</i>
Mean of PSD (μm)	15.6
Standard deviation of PSD (μm)	5.6
Minimum pore size (μm)	1
Mean of TSD (μm)	5.6
Standard deviation of TSD (μm)	2.8
Minimum throat size (μm)	0.1
Mean of LSD (μm)	9.45
Standard deviation of LSD (μm)	5.67
Minimum diffusion length (μm)	1
Water diffusivity ($\mu\text{m}^2/\text{s}$)	2500
Surface relaxivity, ρ ($\mu\text{m/s}$)	100
Critical pressure ratio	6
Receding contact angle	40°
Advancing contact angle	60°
S_w^*	0.06 & 0.24
η	1.5 & 0.5

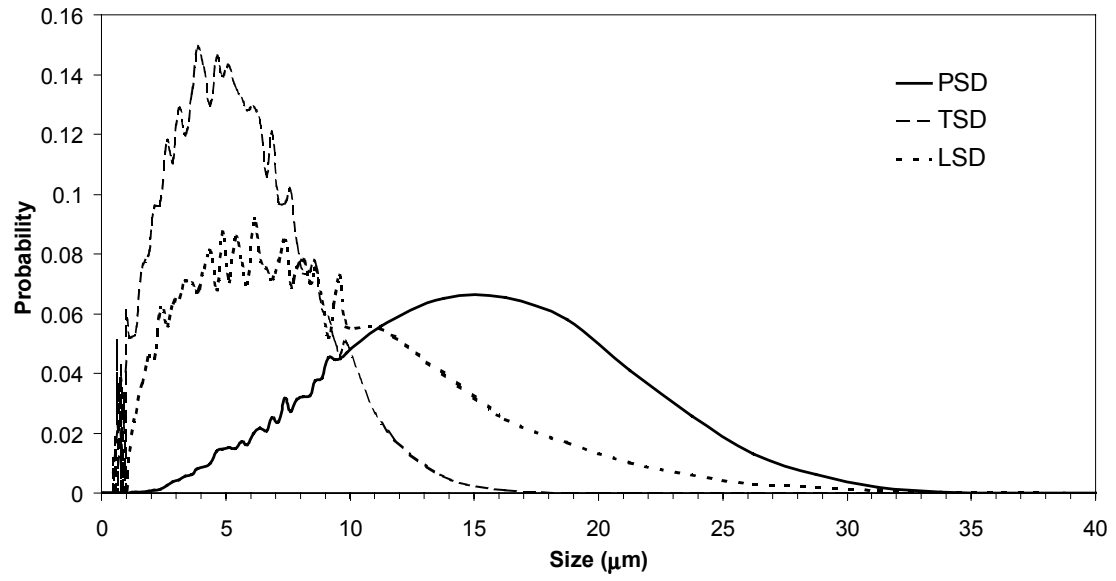


Figure 2: Pore size, throat size and inter-pore length distributions used in simulations.

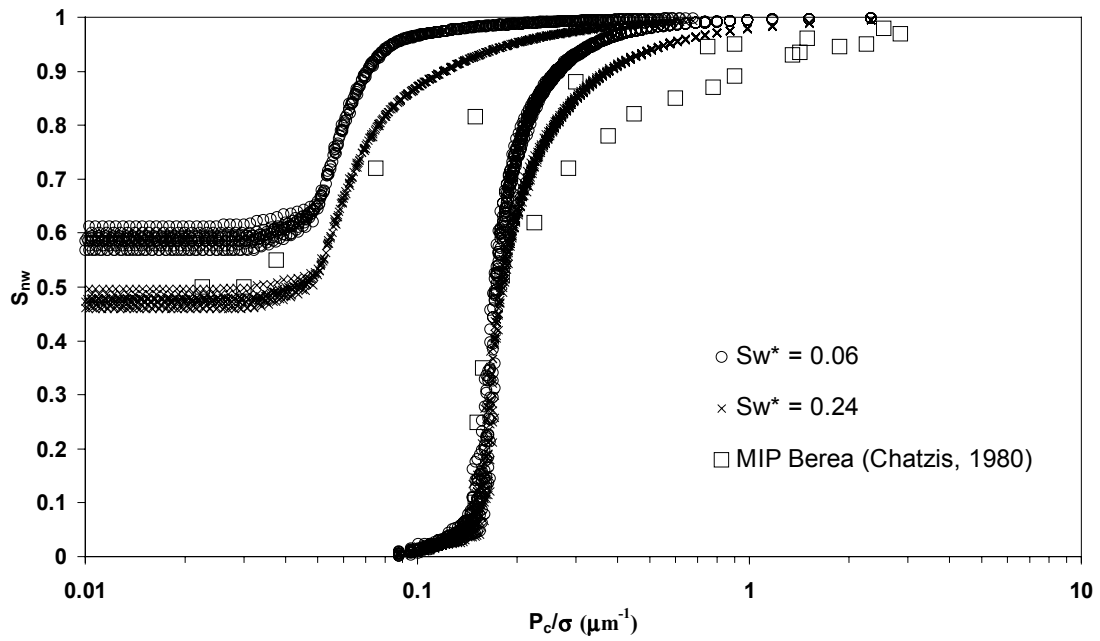


Figure 3: Effect of parameter S_w^* on simulated capillary pressure curves ($\eta = 1.5$).

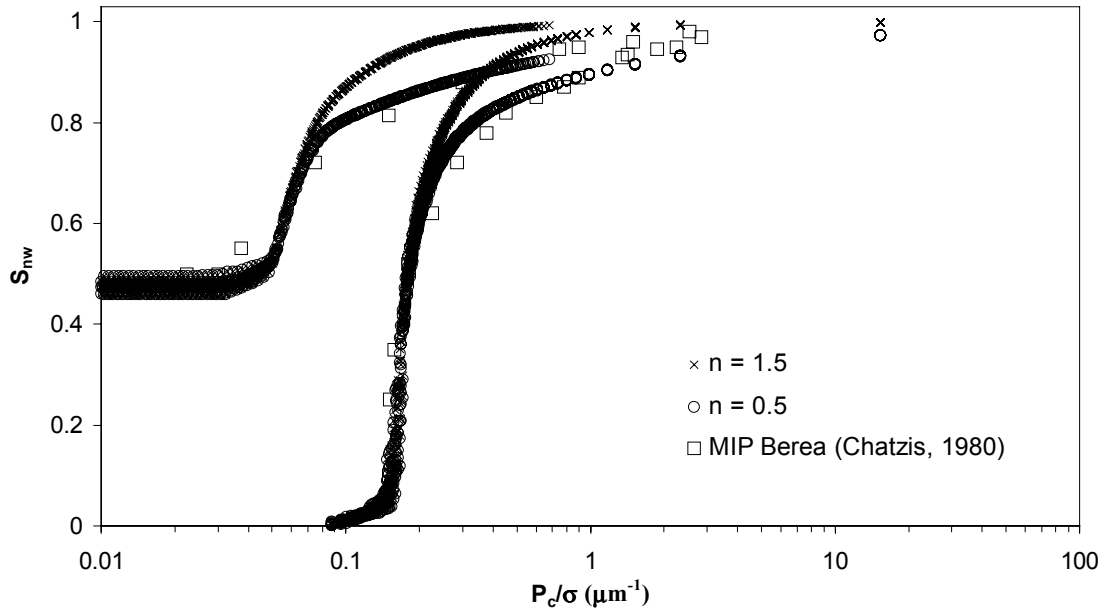


Figure 4: Effect of parameter η on simulated capillary pressure curves ($S_w^* = 0.24$).

gradual. During imbibition, a greater amount of non-wetting fluid is displaced before a large number of piston-type retraction events take place. The main leg of the imbibition curve is not affected, but the residual non-wetting phase saturation is considerably reduced. The effect of parameter η is illustrated in Figure 4. As can be seen, changing η from 1.5 (non-fractal pore geometry) to 0.5 (fractal pore geometry) results in even more gradual drainage and imbibition curves. Changing η has no effect on the residual non-wetting phase saturation. The results, which are clearly representative of experimental mercury porosimetry curves of sandstones, demonstrate the flexibility of the late pore filling model. This model provides a means of capturing pore structure effects from unresolved length scales (e.g., microporosity). Its application in the network modeling of electrical resistivity of partially saturated porous media will be addressed in a forthcoming communication.

Effect of saturation history on the NMR response

The spectra of magnetization decay rates for given water content during drainage and imbibition are compared in Figure 5. These results were obtained using $S_w^* = 0.24$ and $\eta = 1.5$ and may be explained as follows. During drainage, the larger pores are preferentially filled with nonwetting fluid. Note that we do not consider bulk relaxation effects and therefore no oil signal is reported in the simulations. The wetting fluid in the invaded pores continues to be in contact with the entire surface of the pore, but its volumetric content decreases. This results in the gradual disappearance of the slower decay rates (long T_2 values) as the effective surface to volume ratio of the invaded pores increases (Fig. 5(a)). The remaining slow decay rates present in the spectrum belong to larger pores that are inaccessible by the non-wetting fluid.

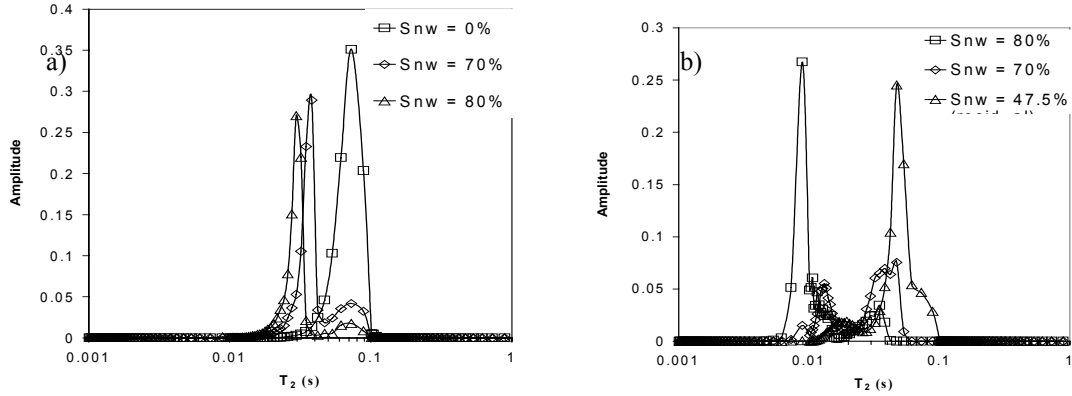


Figure 5: Spectra of proton magnetization decay rates during (a) drainage and (b) imbibition.

During imbibition, the smaller pores are preferentially invaded by the wetting fluid, resulting in gradual disappearance of the faster decay rates (short T_2 values), as shown in Figure 5(b). The spectrum gradually shifts to slower rates as an increasingly greater amount of water exists in larger pores. As the nonwetting phase saturation decreases, the spectrum of decay rates shown in Figure 5(a) for $S_{nw} = 0\%$ is only partly recovered due to the presence of residual nonwetting phase saturation. This is more clearly shown in Figure 6 (b). Note the presence of a second peak in the spectrum at conditions of residual nonwetting phase saturation. Pores containing trapped non-wetting phase also contain a significant amount of water whose characteristic relaxation rate is faster (short T_2 values) than any of the rates appearing in the spectrum from the water-saturated medium. This is because the water in pores containing trapped oil is in contact with the entire pore surface. Figure 6(a) compares NMR relaxation rate spectra at the same saturation ($S_{nw} = 0.7$) for drainage and imbibition. These spectra reflect the markedly different distribution of water in each case. Clearly, interpretation of these spectra in terms of pore structure parameters cannot be made in the absence of knowledge regarding the saturation history. During drainage, pores not accessible to non-wetting phase contribute to the slow-rate end of the spectrum. During imbibition, many of these pores contain trapped oil resulting in the appearance of a second peak corresponding to fast relaxation rates in the spectrum.

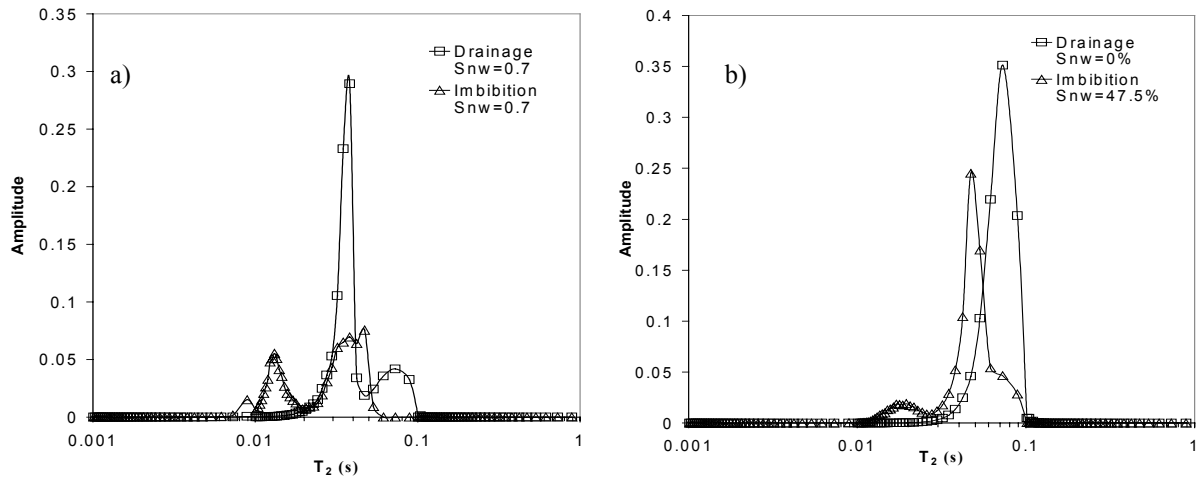


Figure 6: Effects of saturation history on the NMR spectra.

CONCLUSIONS

A pore network model capable of simulating the NMR response of porous rock under conditions of drainage and imbibition was developed. This model takes into account, in a physically based manner, drainage and imbibition associated with pore space features not explicitly accounted for in the pore and throat size distributions. As a result the model is capable of simulating experimentally measured drainage and imbibition capillary pressure curves. Simulations of NMR relaxation reveal rich behavior that is explained in terms of the pore scale mechanisms of drainage and imbibition. The model has all the features necessary for establishing quantitative relationships between the NMR response of partially saturated rock and petrophysical properties of paramount importance such as resistivity index and relative permeability.

NOMENCLATURE

T_2 – NMR relaxation time (s)
 A_i – Surface area of pore i (μm^2)
 V_i – Volume of pore i (μm^3)
 S_{ij} – Cross sectional area of throat connecting pores i and j (μm^2)
 L_{ij} – Diffusion length between pores i and j (μm)
 R_p – Characteristic pore size (μm)
 R_t – Characteristic throat size (μm)
 $R_{\min}$ – Minimum pore or throat size (μm)
 D – Diffusion coefficient ($\mu\text{m}^2/\text{s}$)
 M_i – Normalized magnetization of pore i
 S_w – Wetting phase saturation
 S_w^* – Wetting phase saturation in a single pore at non-wetting phase invasion
 P_c – Capillary pressure (Pa)
 P_c^* – Capillary pressure for non-wetting phase invasion (Pa)
CPR – Critical Pressure Ratio (scaling factor between piston type withdrawal and snap-off pressures for pore throats)
 D_s – Fractal dimensionality of pore space
 α – Weibull location/spread parameter
 β – Weibull kurtosis/skewness parameter
 ρ – Surface relaxivity ($\mu\text{m}/\text{s}$)
 σ – Surface tension (N/m)
 ω_p – Characteristic rate of relaxation at the fluid-solid interface (1/s)
 ω_c – Characteristic rate of molecular diffusion between pores (1/s)
 λ – Relaxation rate (1/s)
 η – Late pore filling exponent

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Magnetization evolution in connected pore systems

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The equations governing magnetization evolution in fluid-filled pore systems are developed. Particular attention is given to the effect of coupling between pores. In pore systems with a range of pore sizes (and/or decay rates), the evolution of the total dipole moment is described exactly by a spectrum of decay rates, leading to multiple-exponential decay. We study this spectrum of decay rates as a function of coupling strength between pores, using perturbation theory, effective-medium theory, and matrix diagonalization. The spectrum of decay rates evolves from the individual pore decay-rate distribution, at zero coupling, to a δ -function distribution, at infinite coupling. The development of the spectrum of decay rates between the two extremes of weak and strong coupling is discussed. The three methods employed to study this problem are in good agreement. The effect of coupling between pores is important in NMR measurements when comparing different porous materials and in the study of the temperature dependence of characteristic decay rates in these materials.

I. INTRODUCTION

Porous-media systems consist of matrix, pore fluid, and probe particle. For example, in water saturated fused glass beads, the glass matrix is complementary to the pore space which is filled with weakly ionized water. In an electrical measurement the probe particle is an ion, e.g., Na^+ , which probes the pore space as a participant in the current carrying process. In an NMR measurement, the probe particle is the H atom on a water molecule, the carrier of a H nuclear magnetic moment. This molecule explores the pore space by diffusive motion. The magnetic moment it carries relaxes due to dipolar interactions with other molecules and interactions at the pore-fluid-matrix interface. The magnetization evolves by diffusive motion and relaxation processes. It responds to features of the geometry of the pore space. We are interested in the geometrical information which may be garnered from NMR studies of magnetization evolution.

Brownstein and Tarr,¹ in a classic work, have discussed magnetization evolution in independent pores. Our interest is in magnetization evolution in coupled-pore systems. We take the characteristics of the diffusive motion and the relaxation mechanisms to be known and concern ourselves with geometry. The geometry of the pore system we wish to study is that of a lattice of pores (e.g., sites characterized by volume and surface) connected by throats (e.g., bonds characterized by cross-sectional area). See Fig. 1. Thus, we describe the pore space with a set of geometrical parameters: V the volume of a pore; A the surface area of a pore; S the cross-sectional area of a

throat; and l the distance between the center of adjacent pores. In addition, we define a pore size a , where $a = V/A$.

The behavior of magnetization in a pore space is controlled by four rates.

(1) $w_B = 1/T_{1,B}$, the bulk relaxation rate, an intrinsic property of the pore fluid.

(2) $w_p = \rho/a$, the relaxation rate in the vicinity of the fluid-matrix interface. [$\rho = d(1/\tau_w - 1/T_{1,B})$], where d is the thickness of the fluid layer next to the matrix walls

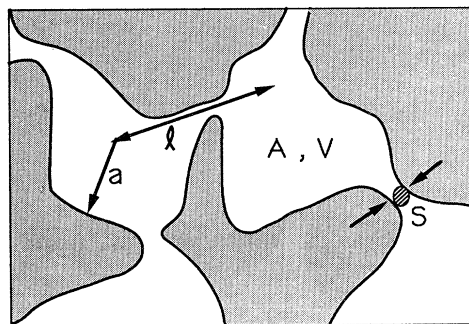


FIG. 1. Schematic of a porous-medium cross-sectional area. The porous medium is characterized by large voids, each assigned a volume V and surface area A , connected by narrow channels of cross-sectional area S . The distance between pores is l and $a = V/A$.

which is influenced by the matrix and τ_w^{-1} is the relaxation rate of fluid in d . Thus, ρ has units of a velocity and characterizes how well the matrix relaxes the fluid in contact with it.]

(3) $w_D = D/a^2$, the rate of molecular diffusion to the fluid-matrix interface (D is the bulk fluid diffusion coefficient).

(4) $w_c = SD/Vl$, the rate of molecular diffusion between pores, the inter pore coupling rate.

The effect of a finite value for w_B can be accounted for exactly² and simply in terms of the solution when $w_B \equiv 0$, thus we will neglect w_B in our discussion. Also, note $w_c < w_D$.

The work of Brownstein and Tarr¹ (BT) on single pores corresponds to the weak coupling limit $w_c \rightarrow 0$. These authors describe the behavior of the magnetization in the slow diffusion limit $w_p \gg 10w_D$, the crossover from slow diffusion to fast diffusion at $w_p \approx w_D$, and the behavior of the magnetization in the fast diffusion limit $w_p \ll w_D$. See Fig. 2. In the fast diffusion limit, the surface relaxation rate is slow compared to the rate of diffusive magnetization equilibration; the magnetization is approximately uniform across a pore and the decay is single exponential. Coupling between pores can be important in the fast diffusion regime.

In Fig. 2 we show that BT have discussed the behavior of the magnetization in going from point 1 to 2 in rate space. In this paper we develop methods of studying the behavior of the magnetization in going from point 2 to point 3 in this space. The BT methodology is valid here, in principle, but cumbersome in describing this evolution, in practice. It seems to be the case²⁻⁴ that sedimentary rocks correspond to a point somewhere along the 2→3 line of Fig. 2. Individual pores would be in the fast diffusion limit were they uncoupled from their neighbors and the spectrum of decay rates is conventionally interpreted as a spectrum of pore sizes. If the coupling w_c is "strong enough," however, then the entire pore space becomes a single pore in the fast diffusion limit and the de-

cay spectrum narrows to a δ function at the rate $w_p = \rho A/V$, where A and V are the total area and pore volume of the system. Decay in this limit is strictly single exponential, $M(t) = M(0)e^{-w_p t}$, where M is the total dipole moment of the system. A goal of this paper is to understand this evolution and what constitutes "strong enough" coupling. Since w_c is proportional to the diffusion constant, it can be increased by a factor of ~ 20 simply by superheating the sample past the boiling point of water. Thus, we feel that temperature dependent decay spectra should prove to be a powerful probe of the physics of magnetization evolution.

Section II contains a description of various measures of magnetic behavior that concern us. The model porous medium we study is described. The equations of motion of the magnetization in this porous medium are discussed. In Sec. III, we develop perturbation theory solutions to the equations of motion in both the weak and strong coupling limits (points 2 and 3 in Fig. 2). Section IV contains a description and results of an effective medium theory treatment of the equations of motion. We discuss numerically exact solutions to the equation of motion for several small system realizations of the model porous medium in Sec. V. Finally, in Sec. VI we compare the three methods we have employed to study the behavior of magnetization and discuss the relevance of our results to physical systems.

II. MAGNETIZATION EQUATIONS AND POROUS MEDIA MODEL

In a typical magnetization measurement, the time evolution of the total dipole moment is probed, i.e., $M(t)$ versus t . The total dipole moment may decay as a single exponential, a multiple exponential,³ a stretched exponential,² etc. To examine the content of a decay curve in a model independent way, we define the mean lifetime:

$$\tau_m = \frac{\int_0^\infty t (dM/dt) dt}{\int_0^\infty (dM/dt) dt} = \frac{\int_0^\infty M(t) dt}{M(0)}, \quad (1)$$

where $M(t)$ is the total dipole moment of the system at time t . Note that τ_m is directly related to the Fourier-Laplace transform of M , $\tilde{M}(s, k)$,

$$\tau_m = \frac{1}{M(0)} \lim_{s \rightarrow 0} \lim_{k \rightarrow 0} \tilde{M}(s, k). \quad (2)$$

This quantity is discussed by Torquato⁵ in the context of permeability and by Wilkinson, Johnson, and Schwartz⁶ in the context of magnetization decay in isolated pores. Given that the magnetization evolution involves both diffusion and decay, it is reasonable to expect to represent (M) in the form

$$\frac{M(t)}{M(0)} = \int_0^\infty d\lambda n(\lambda) e^{-\lambda t}, \quad (3)$$

where $n(\lambda)$ is the spectrum of decay rates. In fact $n(\lambda)$ can be expressed uniquely as $n(\lambda) = \sum_n I_n \delta(\lambda - \lambda_n)$, where λ_n is the eigenvalue and I_n its amplitude, as discussed by BT. Then,

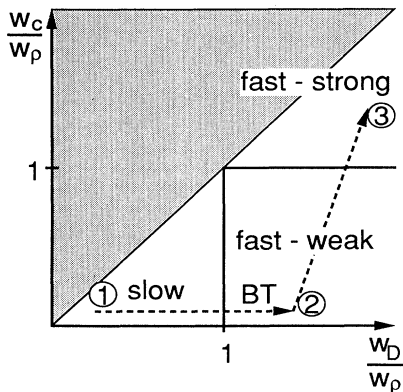


FIG. 2. Rate space for magnetization decay in pore systems. Brownstein and Tarr discuss evolution from slow to fast diffusion, 1→2. We discuss evolution from weak to strong coupling, 2→3.

$$\tau_m = \int_0^\infty d\lambda \frac{1}{\lambda} n(\lambda) \equiv \langle \lambda^{-1} \rangle, \quad (4)$$

i.e., τ_m is the appropriately weighted average of λ^{-1} . Thus, τ_m , the $n = -1$ moment of $n(\lambda)$, favors small λ or slow decay rates.

At early time, $t \rightarrow 0$, the decay of the total dipole moment is given by

$$\frac{M(t)}{M(0)} = 1 - \frac{t}{\tau_0}, \quad (5)$$

where

$$\frac{1}{\tau_0} = \int_0^\infty d\lambda \lambda n(\lambda) \equiv \langle \lambda \rangle. \quad (6)$$

The short-time behavior provides a measure of the $n = 1$ moment of $n(\lambda)$. Given that surface effects are the dominant mechanism for relaxation, magnetization decay at the very shortest times is proportional to the total matrix surface to total fluid volume in the system;⁷ i.e., $1/\tau_0 = \rho A/V = w_\rho$.

In the strong coupling limit, e.g., $D \rightarrow \infty$, the entire system can be viewed as a single pore with total dipole moment decay given by its surface to volume ratio; i.e., $1/\tau_m = \rho A/V$. Thus,

$$\lim_{D \rightarrow \infty} \frac{1}{\tau_m} = \frac{1}{\tau_0}. \quad (7)$$

We will focus our attention on the quantities τ_m , τ_0 , and $n(\lambda)$. Through calculations of $M(t)$ in terms of $n(\lambda)$ we hope to learn $n(\lambda)$ (and therefore τ_m and τ_0) and its relationship to the geometry of the pore space.

While the fluid-saturated rocks in which we are primarily interested do not contain a regular lattice of pores, they do contain a multiply connected system of pores far from the percolation threshold. In addition, the magnetization evolution is governed by diffusive processes which are independent of the precise connectivity of the lattice. Thus, to have a specific model of a pore system, we choose a hypercubic lattice of pores and throats. We identify pores with sites in a lattice and throats with bonds between sites. Each site is characterized by a volume V_i and a surface area A_i . Each bond is characterized by a cross-sectional area S_{ij} and a length l_{ij} . Thus, in a hypercubic lattice model in d dimension, we admit N possible values of V and A , and dN possible values of l and S , where N is the number of sites. For example, in a particularly simple model we might take $A_i = 4\pi R_i^2$, $V_i = 4\pi R_i^3/3$, $l_{ij} = \bar{l}$, and $S_{ij} = \bar{S}$, where R_i is a pore radius.

Following Cohen and Mendelson,⁴ we assume that each pore is in the fast diffusion limit (or would be if it were uncoupled from the rest of the system) and that all of the lengths l_{ij} are the same. We are led to Eq. (6.15a) of Cohen and Mendelson⁴ for the dipole moment in pore i ,

$$\frac{dM_i}{dt} = -w_\rho r_i M_i + w_c \sum_j^{NN} s_{ij} \left[\frac{M_j}{v_j} - \frac{M_i}{v_i} \right] \equiv - \sum_j \mathcal{B}_{ij} M_j, \quad (8)$$

where $r_i = A_i \bar{V} / \bar{A} V_i$, $s_{ij} = S_{ij} / \bar{S}$, $v_i = V_i / \bar{V}$, $w_c = D \bar{S} / l \bar{V}$, $w_\rho = \rho \bar{A} / \bar{V}$ (note, $w_\rho \equiv 1/\tau_0$) and the sum is over all sites j which are nearest neighbors (NN) to i . Overbars indicate averages.

We seek solutions to Eq. (8) subject to the initial condition at $t=0$ of uniform magnetization, $M_i(0) = \bar{m} V_i$, where $\bar{m}$ is the magnetization. Since $s_{ij} = s_{ji}$, but $v_i \neq v_j$ (usually) the operator $\mathcal{B}_{ij}$ is not symmetric in the interchange of i and j . Equation (8) is symmetrized by using $M_i = \sqrt{v_i} \theta_i$; then

$$\frac{d\theta_i}{dt} = -\mathcal{H}_{ij} \theta_j, \quad (9)$$

where

$$\mathcal{H}_{ij} = \left[w_\rho r_i + w_c \sum_k^{NN} \frac{s_{ik}}{v_i} \right] \delta_{ij} - w_c \frac{s_{ij}}{\sqrt{v_i v_j}}, \quad (10)$$

and $s_{ij} = 0$ for j not a nearest neighbor of i . Given our initial condition we find that the total dipole moment $M(t) = \sum_i M_i$ can be expressed as

$$\frac{M(t)}{M(0)} = \langle \theta(0) | \theta(t) \rangle, \quad (11)$$

where

$$|\theta(0)\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} \sqrt{v_1} \\ \sqrt{v_2} \\ \vdots \\ \sqrt{v_N} \end{bmatrix} \quad (12)$$

and $\langle \theta(0) | \theta(0) \rangle = 1$. Expanding $|\theta\rangle$ in terms of the complete set of eigenstates of $\mathcal{H}$,

$$|\theta(t)\rangle = \sum_n a_n(t) |\psi_n\rangle, \quad (13)$$

where

$$\mathcal{H} |\psi_n\rangle = \lambda_n |\psi_n\rangle \quad (14)$$

[in the present case the matrix, Eq. (10), is real symmetric, the $|\psi_n\rangle$ are a complete set of orthonormal states and the λ_n are positive definite] we find

$$|\theta(t)\rangle = \sum_n \langle \psi_n | \theta(0) \rangle e^{-\lambda_n t} |\psi_n\rangle \quad (15)$$

and

$$\frac{M(t)}{M(0)} = \sum_n |\langle \theta(0) | \psi_n \rangle|^2 e^{-\lambda_n t}. \quad (16)$$

Upon comparison with Eq. (3) we have

$$n(\lambda) = \sum_n |\langle \theta(0) | \psi_n \rangle|^2 \delta(\lambda - \lambda_n). \quad (17)$$

The rates λ_n are the eigenvalues of $\mathcal{H}$. The amplitude at each rate λ_n is the overlap of $|\psi_n\rangle$, from Eq. (14), with $\langle \theta(0) |$, the state created in experiment by making the magnetization uniform at $t=0$. In the next three sections of this paper we will describe three methods for developing solutions to Eqs. (16) and (17).

III. PERTURBATION THEORY FOR STRONG AND WEAK COUPLINGS

We wish to find perturbation theory solutions to Eq. (9), given the Hamiltonian in Eq. (10). The ratio w_c/w_ρ determines whether the system is in the weak coupling limit (near point 2 in Fig. 2) or the strong coupling limit (near point 3 in Fig. 2). Therefore, we rewrite Eq. (9) as

$$\frac{d|\theta\rangle}{d\tau} = -(\mathcal{H}^{(0)} + \epsilon\mathcal{V})|\theta\rangle, \quad (18)$$

where ϵ is the small parameter and the quantities τ , $\mathcal{H}^{(0)}$, and $\mathcal{V}$ are chosen according to the region of interest.

Strong coupling:

$$\begin{aligned} \epsilon &= \frac{w_\rho}{w_c}, \quad \tau = w_c t, \\ \mathcal{H}_{ij}^{(0)} &= \sum_k \frac{s_{ik}}{v_i} \delta_{ij} - \frac{s_{ij}}{\sqrt{v_i v_j}}, \\ \mathcal{V}_{ij} &= r_i \delta_{ij}. \end{aligned} \quad (19)$$

Weak coupling:

$$\begin{aligned} \epsilon &= \frac{w_c}{w_\rho}, \quad \tau = w_\rho t, \quad \mathcal{H}_{ij}^{(0)} = r_i \delta_{ij}, \\ \mathcal{V}_{ij} &= \sum_k \frac{s_{ik}}{v_i} \delta_{ij} - \frac{s_{ij}}{\sqrt{v_i v_j}}. \end{aligned} \quad (20)$$

Using standard Schrödinger perturbation theory⁸ we may investigate either the strong or the weak coupling limit. We assume we know the eigenvalues E_n and eigenkets $|n\rangle$ which satisfy

$$\mathcal{H}^{(0)}|n\rangle = E_n|n\rangle, \quad (21)$$

and look for solutions to

$$(\mathcal{H}^{(0)} + \epsilon\mathcal{V})|\psi_n\rangle = \lambda_n|\psi_n\rangle \quad (22)$$

in terms of E_n and $|n\rangle$. To second order in the perturbation,

$$\lambda_n = E_n + \epsilon\langle n|\mathcal{V}|n\rangle + \epsilon^2 \sum_{k(\neq n)} \frac{|\langle k|\mathcal{V}|n\rangle|^2}{E_n - E_k}, \quad (23)$$

and to first order

$$|\psi_n\rangle = c_n \left[|n\rangle + \epsilon \sum_{k(\neq n)} |k\rangle \frac{\langle k|\mathcal{V}|n\rangle}{E_n - E_k} \right] \quad (24)$$

Here c_n is given by

$$c_n = 1 - \frac{\epsilon^2}{2} \sum_{k(\neq n)} \frac{|\langle k|\mathcal{V}|n\rangle|^2}{(E_n - E_k)^2}, \quad (25)$$

correct to second order in the perturbation. From Eq. (16) the normalized total dipole moment is given by

$$\frac{M(t)}{M(0)} = - \sum_n |\langle \theta(0)|\psi_n\rangle|^2 e^{-\lambda_n \tau}. \quad (26)$$

The problem at hand is to solve Eq. (21) for $|n\rangle$ and E_n and to use the results to find $M(t)$.

We consider the strong coupling case first. By inspection we see that the ground state of $\mathcal{H}^{(0)}$, with energy $E_0=0$, is equivalent to the $t=0$ solution for $|\theta\rangle$ described in Sec. II, i.e., $|0\rangle = |\theta(0)\rangle$, where $|\theta(0)\rangle$ is given in Eq. (12). Since $|\theta(0)\rangle$ is orthogonal to all nonzero $|n\rangle$ and $\langle 0|\mathcal{V}|0\rangle=1$ we can write the normalized total dipole moment in the strong coupling limit, using Eqs. (23)–(26), in the form

$$\begin{aligned} \frac{M(t)}{M(0)} &= \left[1 - \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \right] \exp \left[- \left[1 - \epsilon \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k} \right] w_\rho t \right] \\ &+ \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \exp \left[- \left[E_k w_c + w_\rho \langle k|\mathcal{V}|k\rangle + \epsilon w_\rho \sum_{m(\neq k)} \frac{|\langle m|\mathcal{V}|k\rangle|^2}{E_k - E_m} \right] t \right]. \end{aligned} \quad (27)$$

correct to second order in the perturbation. As expected, for infinitely strong coupling between pores ($\epsilon=0$) the magnetization is single exponential with decay rate w_ρ reflecting the total surface to volume ratio in the system. The first correction to the decay rate (second order in the perturbation, first order in ϵ) decreases it below w_ρ . Note that the first correction to the amplitude is second order in the small parameter $\epsilon = w_\rho/w_c$. The multiple exponential spectrum due to second-order corrections consists of small amplitude contributions at very high rates. As $w_c \rightarrow \infty$, the dominant term in the decay rate is $E_k w_c$; thus we expect the decay rates to increase in proportion to w_c . These fast rates correspond to the transfer of magnetization between pores due to high coupling. From Eq. (17)

$$\begin{aligned} n(\lambda) &= \left[1 - \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \right] \delta \left[\lambda - \left[1 - \epsilon \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k} \right] w_\rho \right] \\ &+ \epsilon^2 \sum_{k(\neq 0)} \frac{|\langle k|\mathcal{V}|0\rangle|^2}{E_k^2} \delta \left[\lambda - \left[E_k w_c + w_\rho \langle k|\mathcal{V}|k\rangle + \epsilon w_\rho \sum_{m(\neq k)} \frac{|\langle m|\mathcal{V}|k\rangle|^2}{E_k - E_m} \right] \right], \end{aligned} \quad (28)$$

a δ function of almost unit amplitude and small amplitude contributions at rates increasing in proportion to w_c . Note, to zeroth order in the perturbation, $\tau_m = \tau_0 = 1/w_\rho$ as expected [see Eq. (7)]. This perturbation expansion of

the spectrum of rates is valid when a magnetic moment has had time to explore enough of the pore space to be influenced by the average geometry of the system, as opposed to isolated sections, before decaying due to relaxa-

tion at the walls. In the next section, in connection with effective medium theory, we present a simple self-consistent perturbation theory (SCPT) which has some computational advantages.

We now turn to the weak coupling case. Here, the unperturbed Hamiltonian $\mathcal{H}^{(0)}$ is diagonal, so we find $E_n = r_n$ and $|n\rangle$ is a vector with 1 in the n th place and 0 elsewhere. The matrix elements $\langle k|\mathcal{V}|n\rangle$ can be evaluated analytically; we find

$$\langle k|\mathcal{V}|n\rangle = \sum_m \frac{s_{mn}}{v_n} \delta_{kn} - \frac{s_{kn}}{\sqrt{v_k v_n}}. \quad (29)$$

The total dipole moment to first order can be written explicitly in the weak coupling limit:

$$\frac{M(t)}{M(0)} = \sum_n \frac{v_n}{N} \left[1 - \frac{2\epsilon}{v_n} \sum_k \frac{s_{kn}}{r_n - r_k} \right] \times \exp \left[- \left[r_n w_\rho + w_c \sum_k \frac{s_{kn}}{v_n} \right] t \right]. \quad (30)$$

When the pores are decoupled ($\epsilon=0$, $w_c=0$) the total dipole moment contains a spectrum of decay rates corresponding to the spectrum of surface to volume ratios of the individual pores. The amplitudes of the exponential contributions are proportional to the volumes of the pores. The first-order correction for weak coupling changes the amplitudes up or down depending on the relative size of nearest-neighbor decay rates, and increases the decay rates by a factor proportional to the coupling rate w_c . The spectrum of rates is

$$n(\lambda) = \sum_n \frac{v_n}{N} \left[1 - \frac{2\epsilon}{v_n} \sum_k \frac{s_{k,n}}{r_n - r_k} \right] \times \delta \left[\lambda - \left[r_n w_\rho + w_c \sum_k \frac{s_{k,n}}{v_n} \right] \right]. \quad (31)$$

In this case, the zeroth order lifetimes τ_m and τ_0 are not equivalent: $\tau_m = \sum_n v_n / N w_\rho r_n$ and $\tau_0 = N / (w_\rho \sum_n v_n r_n)$. The former weights long times, the latter weights short times.

The perturbation theory expressions found for the total dipole moment and spectrum of rates in the strong and weak coupling limits are useful benchmarks for more elaborate calculations which allow us to explore the full range of coupling strength.

IV. EFFECTIVE MEDIUM THEORY

Effective medium theory (EMT) has been described previously for calculations of electrical conduction in resistor networks,⁹ as well as for electronic structure calculations where it is known as the coherent potential approximation (CPA).¹⁰ Random values of conductance are replaced by an average value, chosen such that when one resistor is changed back to its original value, the effect on the conductance is on average zero. EMT works best for very simply problems. Equation (8), the equation of motion for the dipole moment at site i , contains bond disorder, S_{ij} , as well as two kinds of site disorder, A_i and V_i .

To illustrate the application of EMT we examine the physical system in an extreme approximation. We take all of the cross-sectional areas equal and all of the volumes equal, i.e., $s_{ij}=1$ or 0 and $v_i=1$. We retain disorder only in the pore decay rates, e.g., due to variation of ρ from pore to pore ($r_i = \rho_i / \bar{\rho}$).

It is particularly convenient to work in the Laplace transform domain. Before we describe EMT, we review some exact relationships. From Eqs. (16) and (17) we have

$$\frac{\tilde{M}(s)}{M(0)} = \int_0^\infty \frac{n(\lambda)}{s+\lambda} d\lambda. \quad (32)$$

Since $\tilde{M}(s)$ is analytic everywhere except on the negative real axis, the contour integral which takes one from $\tilde{M}(s)$ to $M(t)$ can be wrapped around the negative real axis with the result that Eq. (3) is reproduced. Furthermore, $n(\lambda)$ can be obtained from Eq. (32) by

$$n(\lambda) = \frac{1}{\pi} \lim_{s \rightarrow -\lambda - i\epsilon} \text{Im} \left[\frac{\tilde{M}(s)}{M(0)} \right]. \quad (33)$$

If all the r_i are equal to a common value $\bar{r}$, then $n(\lambda) = \delta(\lambda - w_\rho \bar{r})$ and $\tilde{M}(s)/M(0) = [s + w_\rho \bar{r}]^{-1}$. When the values of r_i are not all equal, we define the complex valued function $\bar{r}(s)$ such that

$$\frac{\tilde{M}(s)}{M(0)} = \frac{1}{s + w_\rho \bar{r}(s)}. \quad (34)$$

The goal of EMT is to determine the optimal $\bar{r}(s)$, analogous to the self-energy in quantum mechanics (see below).

Following the EMT prescription we replace all decay rates r_i by an effective rate $\bar{r}$. We seek to find $\bar{r}$ such that changing the decay rate at one site, $i=0$, back to its original value does not change the total dipole moment, on average. Taking the Laplace transform of Eq. (8), using $r_i = \bar{r} + \Delta r \delta_{i0}$, where $\Delta r = r - \bar{r}$, we find

$$(s + w_\rho \bar{r}) \tilde{M}_i = \bar{m} \bar{V} - w_\rho \Delta r \tilde{M}_0 \delta_{i0} + w_c \sum_j^{\text{NN}} (\tilde{M}_j - \tilde{M}_i), \quad (35)$$

where $\bar{m}$ is the initial magnetization. We solve for the dipole moment on the zeroth site using a Fourier series transform and find

$$\tilde{M}_0 = \frac{\bar{m} \bar{V}}{s + w_\rho \bar{r}} \left[1 - \frac{w_\rho I(s, \bar{r}) \Delta r}{1 + w_\rho I(s, \bar{r}) \Delta r} \right], \quad (36)$$

where

$$I(s, \bar{r}) = \frac{1}{\pi^d} \int_0^\pi d\phi_1 \cdots \int_0^\pi d\phi_d \frac{1}{s + w_\rho \bar{r} + 4w_c \sum_{i=1}^d \sin^2 \frac{\phi_i}{2}}, \quad (37)$$

and d is the dimension of the hypercubic lattice. [We often use $I(s)$ as shorthand for $I(s, \bar{r})$ when there is no ambiguity.]

We impose the condition that at site zero the dipole moment averaged over the distribution of rates $p(r)$ is

the same as the total dipole moment of the model system, i.e., a system made up entirely of rates $\bar{r}$. This condition is imposed at finite s . Thus $\bar{r}$ is a function of s . We require

$$\int dr p(r) \frac{r - \bar{r}(s)}{1 + w_p I(s)[r - \bar{r}(s)]} = 0, \quad (38)$$

where $p(r)$ is the distribution of rates r . Equation (38) is a single-site approximation. Given a distribution of rates we can find an $\bar{r}$ for each s . Calculations of $n(\lambda)$, $M(t)$, τ_m , and τ_0 are straightforward, once we have $\bar{r}(s)$.

We note that if we write out the Hamiltonian in Eq. (10) with v_i and s_{ij} independent of i and j (the extreme approximation under discussion) we see that $\mathcal{H}_{ij}$ corresponds to a tight-binding Hamiltonian with site disorder r_i . Therefore, there is an isomorphism between the two problems:

$$\begin{aligned} s &\leftrightarrow z, \\ w_p r_i &\leftrightarrow \epsilon_i, \\ w_c &\leftrightarrow t, \\ I &\leftrightarrow F_0, \\ w_p \bar{r} &\leftrightarrow \Sigma, \\ n(\lambda) &\leftrightarrow A(\mathbf{k}=0, E). \end{aligned} \quad (39)$$

The quantities on the right are those defined in Ref. 11. Equation (38) is the CPA equation. Note in particular that the spectral density relevant to this problem, $n(\lambda)$, is not the total density of states, but rather it is the $\mathbf{k}=0$ component only, reflecting the condition that the system was initially magnetized uniformly.

A calculation of $n(\lambda)$ in EMT, requires a solution to the integral $I(s, \bar{r})$. Equation (37) can be solved in closed form in the case of a one-dimensional lattice,

$$I_1(s, \bar{r}) = \frac{1}{4w_c \sqrt{z} \sqrt{z+1}}, \quad (40)$$

$$I_3(s, \bar{r}) = \frac{1}{\pi^2 w_c} \int_0^{\pi/2} dx \int_0^{\pi/2} dy \frac{1}{\sqrt{z + \sin^2 x + \sin^2 y} \sqrt{z + \sin^2 x + \sin^2 y + 1}} \quad (41)$$

by brute force. Equation (37) can also be rewritten as a single integral in time over a modified Bessel function of complex argument raised to the dimensional power.¹¹ We did not pursue this method here.

Our first choice of decay rate distribution is the bimodal distribution given by

$$p(r) = \sin^2(\pi r), \quad 0 \leq r \leq 2. \quad (42)$$

In Figs. 3 and 4 we show $n(\lambda)$ calculated in EMT (the solid line) for the one-dimensional and three-dimensional cases, respectively. The spectra of decay rates were calculated for various values of w_c , $0.01 \leq w_c/w_p \leq 100$ in Fig. 3 and $0.01 \leq w_c/w_p \leq 0.2$ in Fig. 4. To find $n(\lambda)$ we

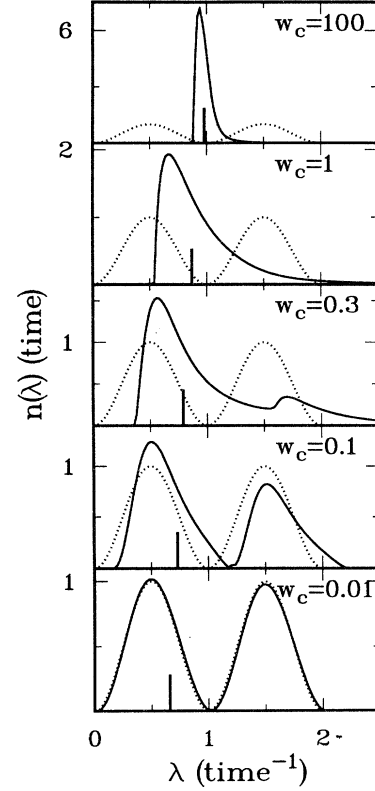


FIG. 3. One-dimensional spectra of decay rates from EMT. The distribution of individual pore decay rates is given by Eq. (42) (dotted lines). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by short vertical lines. As $w_c \rightarrow \infty$, $\tau_m \rightarrow \tau_0$.

where $z = (s + w_p \bar{r})/4w_c$. In higher dimension, the integral must be evaluated numerically. For the three-dimensional case, one integration can be done by hand and we did the remaining two-dimensional integral:

use the following procedure.

(1) Choose $w_p = 1$, $\epsilon = 0.001$ and $p(r)$ given in Eq. (42). We choose ϵ small enough that smaller choices produce no visible change in $n(\lambda)$.

(2) For each $s = -\lambda - i\epsilon$, numerically minimize the absolute value of the left-hand side of Eq. (38) with respect to $\bar{r}$.

(3) Use the resultant $\bar{r}(-\lambda - i\epsilon)$ and Eqs. (33) and (34) to find $n(\lambda)$.

The results shown in Figs. 3 and 4 confirm those expected from perturbation theory. In the weak coupling limit $n(\lambda)$ is the same as the original distribution of rates. In the strong coupling limit $n(\lambda)$ approaches a δ function

with a tail on the high rate side. Note that the rates in $n(\lambda)$ extend beyond the $\lambda=2$ limit of $p(r)$ as w_c increases. The mean lifetime τ_m , whose inverse is marked with a vertical line on the figures, approaches $\tau_0=1$ in the limit of infinite coupling. The most striking difference between the one- and three-dimensional cases is that the spectrum of decay rates narrows for much lower values of the coupling ratio in three dimensions than in one dimension. The degree of connectedness of the system, providing more ways for magnetic moments to move from one pore to another, has a strong effect on the resulting spectrum of decay rates.

Because of the connection with the quantum mechanical problem we can bring to bear on the problem set by Eqs. (9)–(17), any of a variety of approximate methods designed for electron and phonon problems. As an illustration we sketch in Appendix A the treatment of Eqs. (9)–(17) using the self-consistent perturbation theory (SCPT) formalism of Ehrenreich and Schwartz.¹⁰ The outcome of this treatment is a self-consistent equation that provides $\bar{r}$,

$$\bar{r} = \langle r \rangle - w_\rho \langle (r - \langle r \rangle)^2 \rangle I(s, \bar{r}(s)), \quad (43)$$

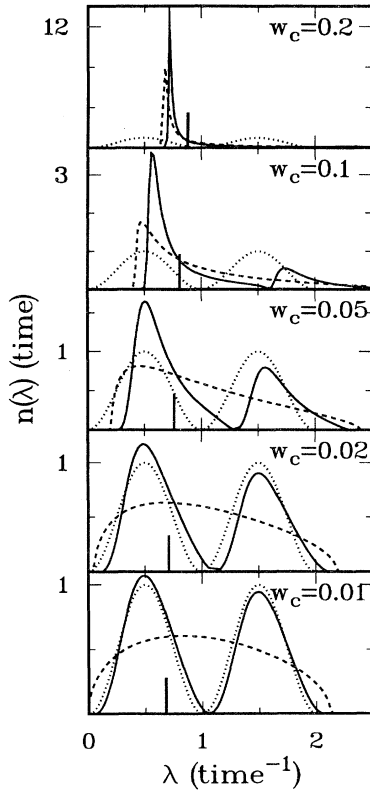


FIG. 4. Three-dimensional spectra of decay rates from EMT and SCPT. The EMT results are the solid lines; the SCPT results are the dashed lines. The distribution of individual pore decay rates is given by Eq. (42) (dotted lines). The coupling rate varies; $w_\rho = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} from EMT is marked by short vertical lines.

where $I(s, \bar{r}(s))$ is given by Eq. (37), for use in Eqs. (33) and (34). Equation (43) can also be found from approximate treatment of the EMT condition, Eq. (38). In Fig. 4 we show $n(\lambda)$ versus λ with $\bar{r}(s)$ from Eq. (43) (the dashed line) for the three-dimensional case. The distribution of decay rates is given by Eq. (42). Since Eq. (43) is dependent on the first and second moments of the rate distribution and an integral determined by the crystal structure only, SCPT captures the width of the spectrum of rates, but not the details of the exact $n(\lambda)$. The approximate treatment of EMT, Eq. (43), is in good agreement with its exact implementation for $\langle (\Delta r)^2 \rangle / w_c \ll 1$ ($w_c > 0.2$ in our example).

EMT is a valuable method for studying the spectrum of decay rates in a simple system. In this case, the system contains variation in pore values of ρ only. We are also interested in situations in which there are variations in pore radius and interpore throat size. Having more than one varying quantity in an EMT calculation leads to ambiguities in the choice of conditions and their meaning. Instead of expanding the discussion of EMT calculations we turn to matrix diagonalization.

V. MATRIX DIAGONALIZATION

In principle, the total dipole moment and spectrum of rates $n(\lambda)$ can be found exactly for any given porous system by diagonalizing the Hamiltonian in Eq. (10). In practice, computation time and memory constraints require the model porous medium systems we study by this method to be orders of magnitude smaller than any experimentally studied piece of material. However, for these small systems we may examine the consequences of variations in volume, surface area, and cross-sectional area.

We use the following procedure to find $n(\lambda)$.

(1) Diagonalize the Hamiltonian $\mathcal{H}_{ij}$ to find the eigenvalues and eigenvectors of the matrix.

(2) Find the overlap of each eigenvector $|\psi_n\rangle$ with $|\theta(0)\rangle$ given in Eq. (12). The square of this overlap is the amplitude of the spectrum of rates at the rate equal to the corresponding eigenvalue λ_n [see Eq. (17)].

In order to compensate for our finite-size systems we use periodic boundary conditions in the Hamiltonian and average $n(\lambda)$ over many realizations, typically 500 or 1000 for systems of $8 \times 8 \times 8 = 512$ pores. The results of matrix diagonalization (MD) in several cases are described in the following.

(1) The spectra of decay rates $n(\lambda)$ shown in Fig. 5 are for a one-dimensional lattice of pores. The pore volumes and throat cross-sectional areas are constant; the individual pore decay rates vary with the rate distribution given by Eq. (42). The lattice size is 512 pores and the results are averaged over 500 realizations. The coupling ratio w_c/w_ρ varies from 0.01 to 100; $w_\rho = 1$. The dashed curves are the EMT results shown in Fig. 3; we find the spectra of decay rates very well approximated by EMT. When $w_c = 10^{-2}$, the spectrum of decay rates is that of the individual pore decay rates. As w_c increases, the spectrum of decay rates collapses such that the majority of the amplitude is in a sharp peak at $\lambda = 1$. Although

high rate amplitudes are decreasing, the maximum rate is increasing proportional to w_c for large w_c/w_p as predicted from perturbation theory.

(2) In Fig. 6 we show $n(\lambda)$ for a three-dimensional lattice of pores. The pore volumes and throat cross-sectional areas are constant; the individual pore decay rates vary with the rate distribution given by Eq. (42). The lattice size is $8 \times 8 \times 8 = 512$ pores and the results are averaged over 500 realizations. The coupling ratio range is $0.01 \leq w_c/w_p \leq 0.2$; $w_p = 1$. Each pore has six nearest neighbors rather than two as in the one-dimensional case. This increased connectivity causes the spectrum of rates to collapse to a δ function much more rapidly with increasing w_c than for the one-dimensional case. The width of the spectrum of decay rates is already very narrow when $w_c = 0.2$, although $\tau_m^{-1} < w_p$. Although the EMT results of Fig. 4 are plotted in Fig. 6, they are almost not visible. The EMT approximation is known to improve as dimension is increased.

The previous examples have focused on cases where the decay rates vary between pores, but are independent of volume. In a physical system in the fast diffusion regime, $w_D/w_p > 1$, we expect an individual pore decay rate to be determined by the surface to volume ratio of the pore. In addition, the spectrum of decay rates measured

experimentally in uncoupled pores is the volume weighted distribution of individual pore decay rates. That is, larger pores will contribute more signal to a measurement since they contain more magnetic moments. Making the assumption that all pores are spherical we can assign volumes and decay rates to individual pores given a distribution of radii a_i ,

$$v_i = \frac{a_i^3}{\langle a^3 \rangle}, \quad r_i = \frac{1}{a_i} \frac{\langle a^3 \rangle}{\langle a^2 \rangle}, \quad (44)$$

where $\langle a^n \rangle$ is the average of a_i^n . Allowing v_i to vary introduces substantial variation in the coupling terms of the Hamiltonian $\mathcal{H}$, even if there is no variation in cross-sectional area s_{ij} .

(3) In Fig. 7 we show $n(\lambda)$ for a three-dimensional lattice of pores. The throat cross-sectional areas are constant; the pore radii are chosen from the distribution

$$p(a) = \frac{A}{a^5} \sin^2 \left[\frac{2\pi a_{\min}}{a} \right], \quad a_{\min} \leq r < \infty, \quad (45)$$

where $a_{\min}$ and A are chosen such that $\langle a \rangle = 1$ and the spectrum of rates $n(\lambda)$ is given by Eq. (42) when the pores are uncoupled. The lattice size is $8 \times 8 \times 8 = 512$ pores and the results are averaged over 1000 realizations.

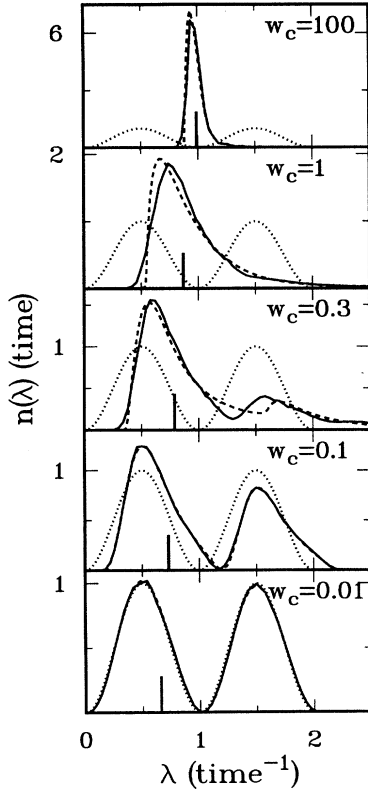


FIG. 5. One-dimensional spectra of decay rates from MD. The distribution of pore decay rates is given by Eq. (42) (dotted line). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The EMT results of Fig. 3 are the dashed lines.

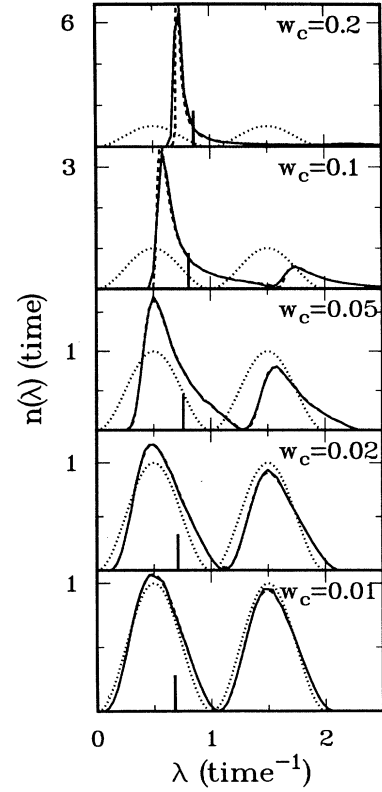


FIG. 6. Three-dimensional spectra of decay rates from MD. The distribution of pore decay rates is given by Eq. (42) (dotted line). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The dashed lines are the EMT results of Fig. 4.

The coupling ratio range is $10^{-2} \leq w_c/w_p \leq 10^2$; $w_p=1$. Note that the distribution of pore radii is heavily weighted toward small pores. In order that the volume weighted distribution of rates at zero coupling have equal size peaks above and below $\lambda=1$, 95% of the pores have decay rates in the range $1 < \lambda < 2$. Thus we can think of the large pores (small decay rates) as dilute disorder in a system of very small pores. Initially, as w_c is increased, the small pores couple to each other forming a narrow peak in the decay spectrum at $\lambda \approx 1.5$, while the large pores remain relatively unaffected. This is due to the comparatively large number of small pores and also to the inverse volume effect in the coupling terms of the Hamiltonian [see Eq. (10)]. The effective coupling of large pores to their neighbors is much smaller than that of small pores. This feature is retained even to $w_c=10^2$; $n(\lambda)$ still has finite amplitude at small λ in contrast to the results of Fig. 6.

(4) In Fig. 8 we show $n(\lambda)$ for a three-dimensional lattice of pores. The throat cross-sectional areas are constant; the pore radii are chosen from a log-normal distribution

$$p(a) = \frac{1}{\sigma a \sqrt{2\pi}} \exp \left[-\frac{(\ln a - u_0)^2}{2\sigma^2} \right], \quad (46)$$

where $u_0 = -\frac{1}{4}$ and $\sigma^2 = \frac{1}{2}$, values which make the average pore radius one. Experimental measurements on rocks show the pore size distribution may vary over an order of magnitude or more.³ The distribution of Eq. (46) is an attempt to mimic a very broad spectrum such as might be seen in rocks. The lattice size $8 \times 8 \times 8 = 512$ pores and the results are averaged over 1000 realizations. The coupling ratio range is $0 \leq w_c/w_p \leq 100$; $w_p=1$. The raggedness of the results decreases slowly with increased number of realizations and increased bin size; each smooth curve is constructed using 50 bins spaced evenly on a logarithmic scale from 10^{-1} to 10^1 . The small λ part of the spectrum has substantial amplitude even for large coupling rates, due to the inverse volume effect in the coupling term of the Hamiltonian. Decay due to coupling is enhanced for small pores and decreased for large pores when volume effects are included in the coupling term of the Hamiltonian.

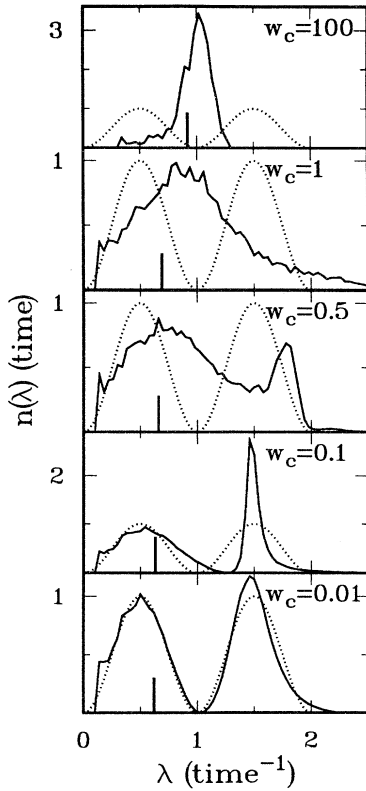


FIG. 7. Spectra of decay rates from MD when rates and volumes are correlated. The distribution of pore radii is given by Eq. (45). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The dotted lines are the uncoupled pore result, Eq. (42).

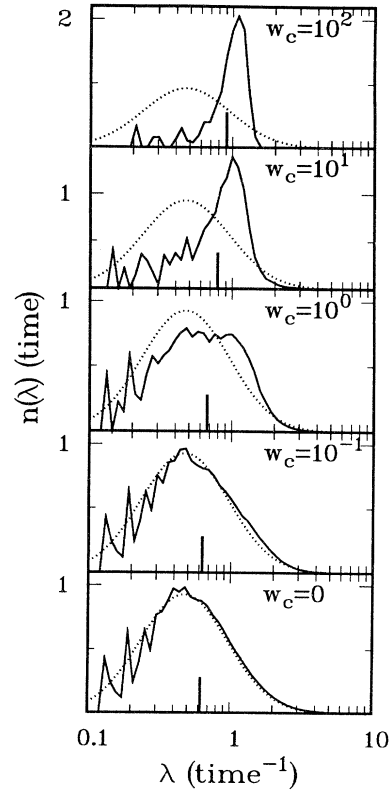


FIG. 8. Spectra of decay rates from MD for a broad distribution of pore radii. The distribution of pore radii is given by Eq. (46). The coupling rate varies; $w_p = \tau_0^{-1} = 1$. The mean decay rate τ_m^{-1} is marked by vertical lines. The dashed lines are $n(\lambda)$ for the uncoupled system.

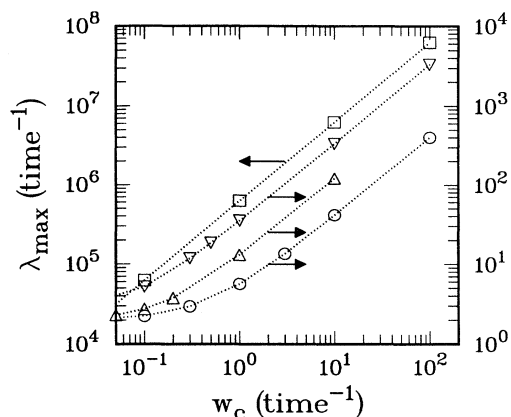


FIG. 9. Maximum decay rate as a function of coupling rate. The maximum decay rates are shown for the MD results of Figs. 5 (○) and 6 (△) for which the distribution of pore decay rates is given by Eq. (42), and Figs. 7 (□) and 8 (▽) for which the distributions of pore radii are given by Eqs. (45) and (46), respectively. $\lambda_{\max} \propto w_c$ as predicted by perturbation theory in the strong coupling limit.

One of the predictions of perturbation theory in the strong coupling limit [see Eq. (28)] is that the small amplitude contributions to $n(\lambda)$ are at rates which increase in proportion to the coupling rate; $\lambda \propto w_c$. We show this to be the case in Fig. 9, where the maximum decay rate $\lambda_{\max}$ is plotted against the coupling rate w_c for all four of the MD examples. Increased coupling causes very high frequency modes of very low amplitude as the magnetization flux from one pore to its neighbors increases. For the special cases in which only the pore rates vary, it is simple to show from Eq. (19) that $\lim_{w_c \rightarrow \infty} \lambda_{\max} = 4dw_c$, where d is the dimension. This is borne out in Fig. 9.

MD is a powerful method for studying the magnetization equation for connected pore systems. Any quantity appearing in the equation may be varied. We chose to study systems where only the individual pore rate varied, confirming our EMT results, and where both rate and volume vary. There are many more options to consider: the throat cross-sectional areas between pores are not all the same in a physical system, and the surface area and volume of a pore are not necessarily strictly related. MD is the most accurate method we discuss for studying the effects of a variety of geometrical features on the total dipole moment spectrum of decay rates.

VI. CONCLUSIONS

We have developed equations governing the magnetization evolution in fluid-filled connected pore systems. The evolution of the total dipole moment can be described by a spectrum of decay rates which reflects the decay rates of individual pores as well as the strength of the coupling between pores. We have used three methods, perturbation theory, effective-medium theory, and matrix diagonalization, to study the evolution of a

particular system's decay-rate spectrum as the coupling between pores is increased. The results of all three methods are in good agreement. The important qualitative features appearing in all three methods of solution are (1) when the coupling between pores is zero or very weak, the spectrum of decay rates is that of the individual pores. (2) As the coupling is increased, the range of decay rates increases, shifting to high rates. (3) For very strong coupling, the spectrum of decay rates narrows to a peak at a rate determined by the ratio of the total surface to total volume in the pore system.

The mean lifetime τ_m is a single parameter which characterizes a spectrum of decay rates in a model independent way. In Fig. 10 we summarize our results for τ_m as a function of the coupling rate w_c for the various situations we have encountered. In all cases, τ_0 is unity. The EMT and MD results for similar distribution and dimension are in good agreement. The upper pair of curves are from the three-dimensional MD calculations using log-normal [Eq. (46)] and bimodal [Eq. (45)] distribution of pore sizes. The central pair of curves is from the one-dimensional MD and EMT calculations using the bimodal distribution of pore decay rates given by Eq. (42). The lower triplet is from the three-dimensional MD, EMT, and SCPT calculations using the bimodal distribution of pore decay rates given by Eq. (42). The most notable feature of this figure is that the crossover from uncoupled to strongly coupled behavior occurs over approximately two decades in the coupling strength.

In order to relate our theoretical study to physical systems we must make some estimate of the coupling between pores. Two examples of porous media with widely different properties are sandstone and sol-gel glass. Both

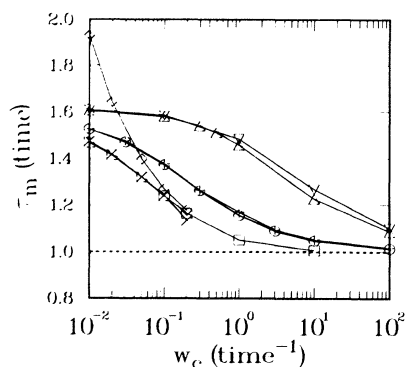


FIG. 10. Mean lifetime versus coupling strength. In the strong coupling limit, $\tau_m \rightarrow \tau_0 = 1.0$. The lines are a guide for the eye. The mean lifetime is plotted for all the cases previously discussed. For distribution of pore decay rates given by Eq. (42): the one-dimensional cases in Figs. 3 (+) and 5 (○), and the three-dimensional cases in Fig. 4 (×, EMT; ◇, SCPT), and 6 (□). For the three-dimensional cases using the distribution of pore radii given in Eq. (45) and shown in Fig. 7 (△), and using the distribution of pore radii given in Eq. (46) and shown in Fig. 8 (▽).

systems have been studied extensively using nuclear magnetic resonance to measure the decay of the total dipole moment in water-saturated samples. Using data for various sandstones from Kenyon *et al.*,³ we estimate a typical porosity of 20%, pore diameter of 80 μm , and throat diameter of 30 μm . A typical mean T_1 is 250 ms; we take this to be approximately τ_0 . The data were taken at room temperature, where the diffusion coefficient of water is $D = 2.5 \times 10^{-5} \text{ cm}^2/\text{s}$. These numbers imply a coupling strength $w_c/w_p < 0.1$. Kenyon *et al.* have decomposed their total dipole moment measurements into spectra of decay times which are one or more orders of magnitude wide. The decomposition is dependent on the assumption that the pores are uncoupled, i.e., $w_c \approx 0$. Given our small estimated coupling strength and the very broad distribution of pore sizes typically found in sandstones, the assumption that a spectrum of decay rates reflects the density of pore sizes in the rocks is probably a good one. Data on sandstones correspond to the $w_c \leq 0.1$ spectra in Fig. 8.

The sol-gel glasses characterized by Bhattacharja, *et al.*¹² have very different characteristics from the sandstones just discussed. The typical porosity in these systems is 85%, a representative pore diameter (assuming spherical pores) is 0.50 μm , and an average throat diameter is 0.24 μm . The magnetization decay measured for these water-saturated silica glasses was single exponential with a decay time $T_1 \approx 100$ ms at room temperature. These numbers imply a coupling strength $w_c/w_p = 10^3$, which is certainly in agreement with the single exponential result. In such highly coupled pore systems, we would expect the spectrum of decay rates to be a δ function, particularly in sol-gel glasses where the distribution of pore sizes is narrow.

An obvious question is whether there exist physical systems in which a range of coupling strength between pores can be explored. A possible method of attack is to change the diffusion coefficient of the saturating fluid by raising or lowering the temperature. In this way the coupling strength between pores may be changed by an order of magnitude or more. On the other hand, if the porous sample used in such an experiment has a very broad spectrum of pore sizes, the extra range in coupling may not make an experimentally evident change in the spectrum of decay rates. In the future we hope to compare our results with relevant experiments exploring coupling between pores.

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APPENDIX: SELF-CONSISTENT PERTURBATION THEORY IN THE STRONG COUPLING LIMIT

Here we employ a perturbation theory in the strong coupling limit appropriate to the problem in which there is a decay rate distribution whose width is small compared to the coupling parameter w_c . We write the Hamiltonian as the sum of an unperturbed term describing a perfect tight-binding crystal and a perturbation term due to small fluctuations in rates:

$$\mathcal{H} = \mathcal{H}^{(0)} + \mathcal{V}, \quad (\text{A1})$$

where

$$\mathcal{H}_{ij}^{(0)} = (w_p r^{(0)} + x w_c) \delta_{ij} - w_c \delta_{ij_{nn}} \quad (\text{A2})$$

and

$$\mathcal{V}_{ij} = \Delta r_i \delta_{ij}. \quad (\text{A3})$$

Here x is the number of nearest neighbors, j_{nn} is a nearest neighbor to i , $r^{(0)}$ is an arbitrary dimensionless rate describing the perfect crystal, and $\Delta r_i = r_i - r^{(0)}$. The average Green function defines a self-energy, Σ , through the expression

$$\langle G \rangle = \left\langle \frac{1}{z - \mathcal{H}^{(0)} - \mathcal{V}} \right\rangle \equiv \frac{1}{z - \mathcal{H}^{(0)} - \Sigma}. \quad (\text{A4})$$

Expanding Σ to second order in $\mathcal{V}$ we find

$$\Sigma \approx \langle \mathcal{V} \rangle + \langle \mathcal{V} G^{(0)} \rangle - \langle \mathcal{V} \rangle G^{(0)} \langle \mathcal{V} \rangle, \quad (\text{A5})$$

where $G^{(0)}$ is the Green function for a perfect tight-binding crystal. In the magnetization problem, the self-energy describes the change in an effective rate of decay from the reference point $r^{(0)}$. Thus, inserting the perfect crystal Green function, we find the $\bar{r}$ of Eq. (34) can be written

$$\bar{r} = \langle r \rangle - w_p \langle (r - \langle r \rangle)^2 \rangle I(s, r^{(0)}), \quad (\text{A6})$$

where $I(s, r^{(0)})$ is given by Eq. (37). The method of SCPT suggests we replace $r^{(0)}$ in the integral $I(s, r^{(0)})$ with $\bar{r}$. We then have a self-consistent equation for $\bar{r}$ which is independent of the reference value $r^{(0)}$ in the original problem [Eq. (43)]. In this formalism the value of $\bar{r}$ is given by the average and the second moment of the distribution of rates and an integral which depends only on the crystal structure. It is straightforward, but slightly tedious, to show that Eqs. (43) and (28) are equivalent to first order in ϵ .

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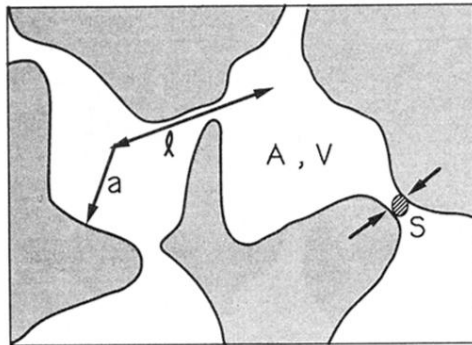


FIG. 1. Schematic of a porous-medium cross-sectional area. The porous medium is characterized by large voids, each assigned a volume V and surface area A , connected by narrow channels of cross-sectional area S . The distance between pores is l and $a = V/A$.

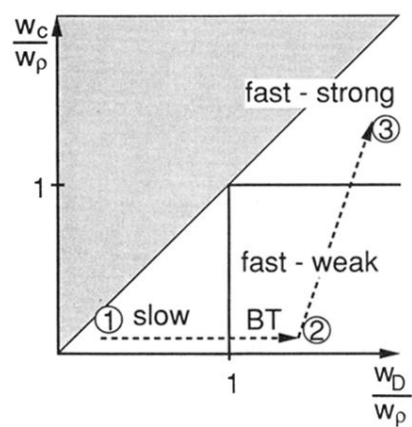


FIG. 2. Rate space for magnetization decay in pore systems. Brownstein and Tarr discuss evolution from slow to fast diffusion, $1 \rightarrow 2$. We discuss evolution from weak to strong coupling, $2 \rightarrow 3$.